Recovering Relativity and the Standard Model from Observer Overlap Consistency

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Abstract

Observer-Patch Holography starts from a finite screen and one rule: neighboring observer patches must agree on their overlap. The compact argument takes that rule to the main structural branch of the theory.

Finite repair and modular geometry lead to the three-dimensional observer-facing spatial chart, the Einstein branch, and receipt-conditional compact gauge reconstruction. On the realized branch this gives the Standard Model quotient, exact hypercharge and charge quantization, three colors, three generations, the Maxwell sector, and a Yang–Mills repair-gap route with its transfer conditions. The same fixed-point stack also carries the de Sitter capacity relation, the fine-structure and particle branches, the weak scale, and the zero Higgs naturality defect on their stated branches.

1 Introduction

Physics is usually presented in two boxes. One box contains spacetime and gravity. The other contains the particle world. This paper asks whether both can be recovered from a smaller starting point: a finite screen whose neighboring observer patches must agree on their overlaps.

The computation in this paper uses fixed-point language, but OPH does not define simulation as a bare fixed point. A conventional frame-rendering computation would evolve surrogate universe states $U_t \to U_{t+1} \to \cdots$. OPH instead studies a readback-and-repair operator on observer-facing world candidates and asks for stable normal forms W satisfying $\mathcal{T}(W) = W$. The companion consensus paper makes the firewall explicit: Definition `def:oph-simulation` requires recovery-derived endogenous update, nontrivial quotient-readable records, overlap repair with schedule-independent normal form, elimination of a proper candidate basin in the selected boundary/sector fiber, and implementation/clock closure; Theorem `thm:oph-simulation` proves those clauses for the selected finite packet under its stated branch and record hypotheses and gives a generic fixed-point/variational counterexample [3]. The repair schedule is a theorem device for selecting quotient normal forms; it is not a one-to-one clock for cosmic time. In the fundamental description there is no global timeline that renders spacetime contents tick by tick; history is the internal readout of the selected normal form.

The scope of observer time is correspondingly narrow. The recovered-core theorem in this paper gives a schedule-independent terminal quotient normal form and, on the BW branch, a modular parameter for the extracted observer-facing cap pair. It does not by itself prove that worker counters, repair iterations, queue positions, packet latencies, or wall-clock timestamps define observer history or observer clock time. Scheduler-independent observer-readable histories require a separate history-augmented quotient, semantic event identities independent of executor metadata, a descended global observer registry, observer-algebra extraction, and an operational clock-instrument calibration. Those are certificate gates instead of implementation field names.

That certified consistency requirement drives a long structural chain. It explains why the Lorentz group appears, why the low-energy spacetime description follows a Jacobson-type Einstein branch, why the gauge structure is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6},$$

why the observed three colors and three generations are realized, why strong/color, weak-isospin, hypercharge, and electric-charge assignments take their observed form, what fixes the electroweak and flavor scales, and what the framework can and cannot say about the cosmological constant and possible dark-sector continuations.

This paper presents the compact reconstruction program based on Observer-Patch Holography (OPH). Physical data are organized patchwise: each observer has access only to a patch algebra, neighboring observers must agree on overlaps, and the reconstruction is driven by those consistency conditions. Under the five axioms and the branch hypotheses and theorems stated below, that structure feeds a Lorentzian modular-phase classification, a Jacobson-type Einstein branch, receipt-conditional compact gauge reconstruction in a bosonic internal-gauge sector, color and electroweak charge quantization, and the low-energy architecture fixed on the realized Minimal Admissible Realization (MAR) gauge branch. Finite horizon/collar record statements are structural support inside the horizon-screen branch. Black-hole spectroscopy, Page-curve/island, evaporation, QNM/ringdown, dark-sector, and particle-family continuations belong to companion or continuation surfaces.

The paper separates exact structural theorems, scaling-limit branches, quantitative outputs, and phenomenological continuations. Several longer arguments are written in compressed proof form so the support boundary stays readable.

The framework is presented as a reconstruction program in which general relativity and the Standard Model appear as effective sectors derived from the stated quantum-algebraic axiom set, explicit refinement constructions, and the controlled BW scaling theorem. This theory-of-everything framing is a claim about recovering the observed effective universe from that basis and its declared branch hypotheses, not a demand that every mathematical ingredient be derived from a blank starting point. In the gauge lane, the logical split is explicit: overlap and holonomy data classify fixed-stage transportable sectors and obstruction routing; on a cofinal tail carrying the compact-gauge refinement receipt, Doplicher–Roberts/Tannaka reconstruction yields a compact group from their tensor-generated bosonic colimit category; MAR plus the realized one-Higgs chiral matter package selects the Standard Model quotient on the realized branch.

Main Results

The results emphasized in this paper are ordered below by a combination of impact and defensibility.

1. A Jacobson-type Einstein branch is recovered from observer-overlap modular geometry, the controlled BW scaling theorem on the prime geometric subnet, and the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family; in the null modular bridge, quasi-local propagation, exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line control, the

resulting weak tail generator, and the half-sided modular pair are supplied internally by the null-interval structure and the scaling-limit geometric action, after which Borchers–Wiesbrock yields the explicit positive null-translation generator on its Stone domain together with the affine half-line modular relation and the half-line generator/charge identification with the local null-stress charge on that same family; the Einstein side then uses the internal small-ball bridge from the geometric cap generator together with the bounded-interval kernel discharged in Lemma 6.135, and the later tensor upgrade states its all-directions/all-reference-states conditions explicitly on that scaling branch. Three packets sharpen the entry into this branch: a quotient-intrinsic geometry producer derives the screen S^2, its conformal/cap structure, and the finite cap-normal BW certificate from repair normal forms on a computable-receipt branch, with an underdetermination no-go showing confluence alone selects no topology, dimension, framing, or normalization (§6.1.2); the null net is proved standard with a derived half-sided inclusion, a Markov modular-locality theorem replacing the earlier global-locality inference (with an explicit counterexample boundary), Möbius bounded-interval covariance, and a four-translation assembly with future-cone spectrum (§6.2.1); and a conditional Lorentzian event manifold is constructed from repaired records, an event base of signature $(-+++)$ over which H^3 is strictly the frame fiber, under explicit population/separation/frame/cocycle receipts, with countermodels showing $\dim H^3 = 3$ promotes nothing by itself (§6.3). The closure packets complete the chain: a local conserved stress tensor is constructed from the directional modular charges by null tomography (with a countermodel showing per-direction generators alone admit no rank-two source), the generalized-entropy first law is repaired to carry the edge/center term exactly and its stationarity is extended to the coupled changing-stress class by the exact multiplier identity, one uniform scaling family replaces the per-radius diagonal choice, a vacuum-reference receipt converts first-variation constancy into the absolute semi-classical equation $G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$ with the structural coupling, and the entire chain is composed into one typed branch-entry theorem with a no-hidden-geometry audit; realized-branch nonemptiness is a machine-checkable gate and is open work in progress (§6.6).

2. The exact Standard Model gauge structure

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

is recovered on the receipt-certified realized MAR-admissible branch of the bosonic compact-gauge theorem; together with the one-Higgs package it fixes the structural electroweak force content $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_Q$ and the $W^\pm/Z/A_Q$ generator and connection directions. The abelian connection has $F_Q = dA_Q$. After independently assuming the low-energy Maxwell action and a nonzero kinetic coefficient, its variation also gives $dF_Q = 0$ and $d * F_Q = g_Q^2 * J_Q$, i.e. Maxwell's equations after canonical electromagnetic normalization. The structural part is a realized-branch gauge result with explicit refinement-receipt, matter-package, and admissibility inputs; the action, numerical W/Z masses, and mixing are additional quantitative-branch data.

3. On the controlled compact-gauge branch, OPH derives the four-dimensional Euclidean Yang–Mills form from compact-gauge holonomy data and the local MaxEnt/Gibbs continuum limit. On a finite weighted quotient, observation-fiber resampling is the orthogonal conditional-expectation projector, with the noncircular matrix-recognition criterion proved in Ref. [1]. A concrete collar repair kernel has that status only after its independently extracted transition matrix passes that receipt. A uniform positive repair gap transfers to the compact-gauge Hamiltonian only on a branch with the multiresolution continuum, reflection-positivity, transfer/intertwiner, and nontriviality certificate. On that certified branch the Yang–Mills gap is the

repair gap for every compact simple group carried by the OPH branch. Without that certificate, the finite repair construction remains a regulator mechanism instead of a Clay-admissible four-dimensional construction.

4. On the realized one-generation chiral matter plus one-Higgs package, anomaly constraints and Yukawa invariance fix the hypercharge lattice, the realized color triplet fixes $N_c = 3$, and the same one-Higgs quark branch then fixes $N_g = 3$ from CKM phase counting, weak-sector asymptotic freedom, and MAR.

5. The quantitative implementation used here exposes two fixed-point closure coordinates, the local pixel fixed point $P_\star$ and the cosmic record-capacity fixed point $N_{\text{CRC}} = F(N_{\text{CRC}})$, plus a selected no-$G$ scale certificate expressed as $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, or equivalently $B_\star = 3\pi/\ell_\star^2$. Approximate measurements may identify the observed branch, because the OPH universe is a closed fixed structure read from inside. The exact coordinates must be emitted as unique endpoints of the declared closure maps.

6. Downstream matter-sector continuations, including the charged-lepton exact centered-readback / common-shift frontier, are visible but explicitly outside this SM/GR derivation paper's recovered-core theorem package.

7. The same Einstein branch closes globally at the cosmic record-capacity fixed point, tying the dimensionless cosmological-constant relation to global screen capacity instead of local vacuum energy; the SI scale display is supplied separately.

8. Exclusion of gauge-mediated proton decay follows from product-group structure; any stronger proton-lifetime claim is UV/EFT/QCD-dependent.

9. Dark-sector, heuristic baryogenesis continuations, string, and spectroscopy branches are kept visible only as phenomenological continuations or, where additional ansätze are involved, conjectural phenomenology.

10. A bare overlap net is a finite constraint code: codewords are the globally overlap-consistent states $C = \Phi^{-1}(0)$. On the declared finite quotient branch, overlap repair induces a total local map $\text{locRep}_\lambda : Q \to Q$ and a total idempotent global repair map $\text{Rep}_\lambda : Q \to Q$ that is boundary-preserving, schedule-independent, and lands in C_Q. Strict descent gives termination, while atomic conflict-component transactions plus repair completeness give confluence. Same-boundary uniqueness additionally requires a preserved boundary/sector map whose consistent quotient fiber has a unique extension; the layered finite carrier proves $H_B \wedge H_{\text{fib}}$, and on the functional selected-fiber branch multiple same-boundary candidate interiors are nontrivially eliminated while all surviving candidates share one quotient normal form. Settled-form hashes are evidence records for quotient equality, not selectors among physically distinct endpoints. The same package induces refinement-limit normal-form and holonomy classes on separated cofinal refinement systems with compatible projections, and makes reconciliation commute with coarse-graining up to explicitly declared normal-form and obstruction defects. QECC distance, min-cut resilience, exponential convergence, BFT wall-clock liveness, and hardware speedup are separate certified branches, not consequences of a generic overlap graph.

11. The fixed-cutoff topological UV package closes on the ordinary branch, the central-defect branch, and the genuinely noncentral branch through a compact crossed-module higher-gauge collar theorem; the controlled continuum BW/geometric lift is supplied by Theorem 6.13, while

the realized compact-gauge branch is supplied by the compact-gauge witness and physical-UV landing theorem below.

What This Paper Contributes

The compact paper is the technical bridge from observer overlap to the familiar structural laws. Its ingredients are deliberately recognizable. Lyapunov descent and Newman's lemma handle finite repair. Bisognano–Wichmann modular theory supplies the modular-geometric template. Jacobson's 1995 and 2016 arguments supply the entropy-to-Einstein comparison target [18, 19]. Doplicher–Roberts/Tannaka reconstruction supplies the compact group reconstruction language.

The OPH addition is the observer-patch route into those tools. Repair is a recovery-derived move on visible overlaps, with holonomy marking the place where local agreement fails globally. The Lorentz branch is read from controlled cap modular flow after the finite collar and recovery controls are carried through the scaling limit. The Einstein branch is Jacobson-type, and OPH supplies the fixed-cap generalized-entropy stationarity theorem, null-stress bridge, internal small-ball bridge, carried remainders, and the all-directions tensor upgrade on its stated branch.

The gauge result is equally specific. On a cofinal tail carrying the compact-gauge refinement receipt, compact gauge structure is reconstructed from the tensor-generated zero-obstruction transportable sector category and its forgetful fiber; MAR then selects the realized Standard Model quotient on the declared one-Higgs chiral matter package. The same branch fixes exact hypercharge, charge quantization, the realized color triplet, and the generation count. The cosmological term is handled globally: the local Einstein branch leaves the metric term open, and the global screen-capacity fixed point closes it on the de Sitter capacity row. This is the dark-energy result carried by the compact paper.

The later quantitative rows use the structural branch as their carrier. They add $P_\star$, N_{CRC}, and the selected scale certificate as fixed-point/readback coordinates with declared uniqueness tests. Failure of one quantitative row does not erase the recovered core unless that row is a parent of the theorem being used.

D. Matscheko's OPHProofChain Lean 4 audit [5] supplies an independent machine-checked finite-algebra companion for the parts of this paper that are genuinely finite: quotient repair and same-boundary reconstruction criteria, the finite binary boundary-reconstruction carrier, finite modular flow with the KMS boundary condition and matrix-algebra uniqueness, the Einstein-branch tensor algebra and λ-constancy step, the hypercharge/$\mathbb{Z}_6$ algebra, the scalar channel bridge, the collar-channel bookkeeping, twelve-port surface bookkeeping, selected numerical cage checks, and the finite QBFT quorum-overlap caveat. That audit does not discharge the physical or scaling-limit branch hypotheses themselves: BW/geometric identification, generalized-entropy stationarity, MAR, collar instantiation, P's source branch, record-to-gravity calibration, and the continuum transfer assumptions remain exactly the declared inputs tracked in the dependency table.

Jacobson comparison and reduction limit

The gravity result is Jacobson-type by design. The intellectual debt is explicit: Jacobson's 1995 argument uses local Rindler horizons and the Clausius relation; Jacobson's 2016 argument uses small geodesic balls and fixed-volume entanglement equilibrium. OPH adds the observer-patch machinery that gets to that route from finite observer-visible data.

Dimension	Jacobson 1995	Jacobson 2016	OPH
Basic region	local Rindler horizon	small geodesic ball	observer-visible screen cap or local diamond
Entropic premise	Clausius relation	fixed-volume entanglement equilibrium	derived fixed-cap generalized-entropy stationarity on the realized cap-label-preserving family
Modular input	Unruh/Rindler structure	small-ball modular input	controlled cap modular flow and null bridge
UV handling	continuum thermodynamics	continuum QFT	finite collar, recovery map, refinement limit
Error handling	thermodynamic approximation	first-order and matter caveats	explicit collar and derivative remainders on the stated branch
Tensor upgrade	all local horizons	all ball directions	overlap-supplied local directions and reference-state clauses
Cosmological term	integration ambiguity	reference-state term	separate cosmic record-capacity closure
Claimed OPH delta	foundational result imported	foundational result imported	finite observer-patch route, regulator/collar control, and separate global closure

Proposition 1.1 (Reduction to the Jacobson small-ball limit)**.** *On the type-I, exact-Markov, vanishing-collar-remainder, maximally symmetric small-cap limit of the OPH Einstein branch, the fixed-cap stationarity relation of Theorem 6.99 reduces to Jacobson's fixed-volume small-ball entanglement-equilibrium relation.*

Proof. The type-I clause turns the cap modular generator into the special operator form used by the small-ball modular Hamiltonian. Exact Markovity removes the recovery defect in the collar split. The vanishing-collar-remainder assumption removes the OPH finite-cutoff correction terms carried by the null bridge and the small-ball transport. The maximally symmetric small-cap limit identifies the OPH cap-label-preserving MaxEnt variation with the fixed-volume ball variation. Under those identifications, the OPH fixed-cap generalized-entropy stationarity equation has the same first variation as Jacobson's entanglement-equilibrium relation. The subsequent local quadratic polarization/tensor step is the standard upgrade listed in the D5 row. □

Proposition 1.2 (OPH delta relative to the Jacobson limit)**.** *Relative to the reduction limit of Proposition 1.1, the OPH branch adds controlled modular extraction, finite-cutoff collar recovery, explicit remainder bookkeeping, observer-overlap direction completion, and an independent global capacity closure for the cosmological term.*

Proof. The added clauses are exactly the hypotheses and outputs of Theorem 6.13, Theorem 6.73, Theorem 6.99, Lemma 6.101, Theorem 6.107, and Theorem 6.151. Removing those clauses leaves the Jacobson-type limit described above; keeping them is the OPH-specific branch. □

Famous recovered formulas by support status

The compact paper uses the following public-facing map. Core theorem results, corollaries, and inherited limits are separated by their branch conditions.

Recovered formula or output	Support status	Display	Boundary
Lorentz transformations and light cone	core theorem	$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$	prime geometric BW branch
Einstein field equation	core theorem	$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$	E0 dependency discharge; local branch leaves Λ open
Newton-Poisson limit	inherited limit	$\nabla^2\Phi = 4\pi G\rho$, $\ddot{\mathbf{x}} = -\nabla\Phi$	weak field, slow motion
Maxwell equations	conditional corollary	$F_Q = dA_Q,\ dF_Q = 0,\ d{*}F_Q = g_Q^2 {*}J_Q$	electromagnetic connection branch plus explicit Maxwell action/current hypothesis
Yang–Mills equations	conditional corollary	$F = dA + A \wedge A,\ DF = 0,\ D{*}F = g^2 {*}J$	compact-gauge connection branch plus explicit Yang–Mills action/current hypothesis
Bekenstein–Hawking / de Sitter area law	global closure	$N_{\mathrm{scr}} = A/(4\ell_P^2) = 3\pi/(G\Lambda)$	capacity fixed point
Charge quantization	core theorem	$Q = T_3 + Y$, observable color singlets have $Q \in \mathbb{Z}$	fractional quarks are confined colored fields
Classical carrier-mode poles	conditional action theorem	$K_X^{\mathrm{phys}} \propto (\omega^2 - c_\star^2 \lvert\mathbf{k}\rvert^2)\Pi_X$	Maxwell, perturbative pure-Yang–Mills, or pure-Einstein action/phase receipt; quantum particle gate separate
Colors and generations	core theorem	$N_c = 3, \quad N_g = 3$	flavor masses are separate continuations

On that same fixed-cutoff consensus surface, the support split is sharp: normal-form computation is a finite-state decidable problem with the Lyapunov step bound from the consensus paper, the only automatic approximate-stability inputs are the collar-local splice and record estimates carried there. Long-run noisy approximate consensus is theorem-level only on the separate fair-block contraction branch, where the implementation supplies $(\lambda, \varepsilon, A, \beta, L)$ for distance to the exact quotient normal-form set. Computational expressiveness for growing patch-net families is a separate consensus-paper complexity boundary instead of a recovered-core dependency. The same firewall applies to coding language: the default code is a finite constraint code, while topological-code distance/min-cut and Knill–Laflamme resilience require explicit topological-code and error-model certificates. The imported consensus package is the fixed-cutoff theorem stack: the quotient repair operators locRep_λ and Rep_λ for the declared recovery-derived repair law after transactional acceptance; asynchronous confluence, with termination separated from the atomic conflict-component local diamond and repair-completeness clauses; the layered $H_B \wedge H_{\mathrm{fib}}$ carrier; the functional selected-fiber branch-elimination theorem; the cycle-obstruction / higher-gauge defect package, gauge-quotient descent, observable-level confluence on the declared physical observable algebras, the refinement-limit normal-form / holonomy theorem, the repair-morphism criterion that discharges exact normal-form naturality, the distributed one-universe realization theorem for admissible worker presentations of one global carrier, the controlled coarse-graining / reconciliation square, the no-free-min-cut boundary for bare graphs, and the record-algebra theorem on the observer-accessible surface. The imported consensus branch conditions are repair completeness, validation-complete transactional read sets with exact descent and preserved boundary/sector data, and the stated Petz support/CPTP clause where that branch is used. The scaling-limit bridge from that fixed-cutoff patch-net package to the Lorentz, Einstein, and compact-gauge branches is separate; the consensus paper supplies the intermediate inverse-limit normal-form and holonomy theorem once the cofinal refinement projections commute with finite-stage normal forms and holonomy maps, and it supplies the approximate RG/reconciliation comparison once the chosen coarse-graining channel has controlled normal-form and obstruction defects.

2 Overview of Results

This section summarizes the support levels used throughout. The table below is the formal dependency map for this SM/GR derivation paper. The recovered core contains fixed-cutoff overlap repair and collar structure, the Lorentz/null-modular/Einstein branch, receipt-conditional bosonic compact gauge reconstruction, and the realized-branch Standard Model quotient with exact hypercharge, structural electroweak force content, $N_c = 3$, and $N_g = 3$ under the explicit matter-package and admissibility inputs. The quantitative closures contain the screen-capacity closure of the same Einstein branch and the pixel-driven electroweak/gauge-coupling branch. The continuations contain flavor details beyond the stated theorem surfaces, dark-sector proposals, heuristic baryogenesis branches, spectroscopy, hadrons, and string/worldsheet effective-description branches. The canonical five-axiom basis, the quantitative quantities listed next, the controlled BW scaling theorem, and the fixed-stage/receipt-conditional gauge chain used below are fixed in Section 3. Each row names a node in the reconstruction program, records its immediate internal parents, separates imported standard mathematics from declared external inputs, and states the resulting support level.

Theorem key for the table. For the recovered core, the scaling/BW step is Theorem 6.13, the controlled BW scaling theorem. The compact-gauge nontriviality step is handled by the compact-gauge witness and physical-UV landing theorem. The fixed-cap generalized-entropy stationarity result is Theorem 6.99. Transportability and the fixed-cutoff bosonic category are theorem-produced by Theorems 5.22 and 5.26; refinement/fiber descent is Theorem 7.3 conditional on the explicit receipt of Definition 7.2.

Table 3: Dependency checklist for the OPH SM/GR reconstruction program.

Node	Output	Immediate internal ingredients	Standard mathematics used	Branch-local inputs / external data	Support level
D1	Total local and global quotient repair maps $\mathrm{locRep}_\lambda, \mathrm{Rep}_\lambda : Q \to Q$; unique schedule-independent normal form from each fixed initial physical quotient state; boundary-conditioned uniqueness under a preserved boundary/sector map with unique consistent extension; layered finite $H_B \wedge H_{\mathrm{fib}}$ carrier; nontrivial same-boundary branch elimination on the functional selected-fiber branch; observable-level confluence on the declared fixed-cutoff physical algebras for the recovery-derived repair relation, inverse-limit normal-form / holonomy classes on separated cofinal refinement systems, distributed one-universe realization for admissible worker presentations of a single global carrier, and controlled reconciliation/coarse-graining compatibility	overlap-consistency problem on a finite patch net together with the declared fixed-cutoff collar recovery proposal, its touched-overlap local-fit eligibility contract, transactional snap-shot/read/write validation with boundary/sector preservation and exact descent, support-local disjoint commutation, restriction-compatible union-collar gluing for canonical conflict-component payloads on the physical quotient, the fixed-point / quotient-local physical observable algebras carried by that same collar package, cofinal refinement projections compatible with finite-stage normal forms and holonomy maps, a distributed physical projection from authoritative worker state to the monolithic quotient, and a coarse-graining channel with declared normal-form and obstruction defects	local-to-global well-founded descent; local-diamond completion from atomic conflict-component commits using validation-complete read sets; Newman's lemma; layered functional boundary reconstruction; rooted functional extension for selected fibers; inverse limits of separated finite quotient systems; reachability invariance of the quotient normal form under projected repair paths, physical stutters, and certified rollbacks; pseudometric control of commuting coarse-graining diagrams	repair completeness; stated Petz support/CPTP control where that branch is used; for same-boundary uniqueness, preservation of the boundary/sector map and at most one consistent quotient extension in the fiber, or the layered/functional selected-fiber theorem with explicit obstruction/ambiguity gates; repair-morphism normal-form naturality, holonomy cochain naturality, visible separation, explicit coarse-graining defect bounds for the refinement system, and distributed certificates for global carrier, partition, cut interfaces, linearized commits, restart roots, final monolithic readout, and normal-form hash evidence only after quotient equality is proved	Phase I structural theorem
D2	Collar Markov/entropy split, finite-range Gibbs conditional-mixing theorem, and G-dictionary setup (Theorems 5.3, 5.9; Propositions 5.33, 5.34, 5.40; Definition 5.41)	Axioms 3.5–3.10 plus the strong conditional Gibbs mixing premise off the exact central-interface branch	HJPW Markov structure theorem; matrix conditional-log-density mixing; Petz and Fawzi–Renner recovery theory	the scalar CMI bound has explicit uniform constants and a boundary-count prefactor; ordinary two-point clustering does not imply the required conditional mixing; the scaling limit requires $\delta/\xi - \log \lvert\partial C\rvert_{\mathrm{UV}} \to +\infty$, not merely $\delta/\ell_{\mathrm{UV}} \to \infty$; exact block-factorized identities require exact Markovity plus the Markov-split alignment hypothesis (Definition 5.10; both derived under the declared central-interface collar clause of Axiom 3.7, Theorem 5.15); the $A/(4G)$ step is the coarse-grained dictionary of Definition 5.41	conditional theorem with exact central-interface specialization
D3a	Regularized support-visible modular transport on fixed cap-local algebras	Axioms 3.5–3.10, collar Markov/recovery package, and Theorem 6.2	weak-$*$/GNS extraction; modular theory with regularized finite generators	fixed-collar comparison errors, cutoff schedule, and multiresolution reference tower; no full finite-algebra spectral floor is assumed	theorem

Table 3: Dependency checklist for the OPH SM/GR reconstruction program.

Node	Output	Immediate internal ingredients	Standard mathematics used	Branch-local inputs / external data	Support level
D3b	Finite cap-normal and support-order certificate	D3a plus a cofinal nondegenerate cap mesh	spherical cap geometry; de Sitter cap-normal compactness; finite support-order reflection	BW-framed caps, support-order faithfulness, mixed-GNS Cauchy residuals, and support covariance; this certificate is not produced by bare finite consensus inside D3, and on the producer branch it is produced from the computable incidence/mesh/cross-ratio/normalization receipts (Theorem 6.31), whose selection is proved underdetermined by confluence (Theorem 6.32)	Theorem 6.6 hypothesis; produced on the D3h receipt branch
D3c	Oriented cross-ratio and framed-cap rigidity	D3b	Möbius/cross-ratio rigidity on S^2	held-out separated quartets and an orientation witness; fitted anchors alone carry no validation value	theorem under certificate
D3d	Geometric 2π-KMS normalization	D3b–D3c	KMS uniqueness for faithful normal states	independently normalized geometric parameter $h_{\frown C}(z) \mapsto e^{-s} h_{\frown C}(z)$, finite strip bounds, and wrong-β separation	theorem under certificate
D3e	Support-visible BW cap automorphism $\sigma_t = \alpha_{\lambda_{\frown C}(2\pi t)}$ (Theorem 6.13)	D3a–D3d	Bisognano–Wichmann modular template at automorphism level	type-I generator form includes $K_C = 2\pi B_C + Z_C$; in the generic non-type-I case the automorphism identity is the theorem	corollary
D3f	Exact cap-normal Lorentz/H^3 chart: $S^2 \simeq \mathbb{P}\mathcal{N}^+$, $\mathrm{Cap}^{\mathrm{or}}_{\mathrm{round}}(S^2) \simeq dS^{\mathrm{cap}}_3$, $n_{gC} = \Lambda_g n_C$, and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ (Corollaries 6.40, 6.46)	D3e plus oriented round-cap extraction and time orientation	projective future-null-cone geometry; $\mathrm{SL}(2,\mathbb{C})$ Hermitian-matrix realization; Lorentz homogeneous-space and hyperboloid geometry	a cap determines an H^3 plane/half-space, not a preferred observer point; population, neutral bulk, physical R_H, stress, and Einstein dynamics remain separate gates	exact conditional geometric theorem
D3g	Conditional record-populated H^3 localization (Corollary 6.64)	D3f plus record-conditioned modular cap responses, point-local source factorization, compact source domain, quantitative cap frame, bounded error, and residual optimization	hyperbolic cap half-spaces, finite conditioned frames, finite ε-nets, and residual inverse-stability estimates	point locality is a branch hypothesis or finite receipt; noisy finite uniqueness requires $\Delta_{\mathrm{loc}} > 0$; species, stress, neutral bulk, and Einstein entry are not implied	conditional theorem
D3h	Quotient-intrinsic geometry producer and dimension-selection boundary: support-visible incidence complex, topology/conformal/cap production, certificate production from computable receipts, and the confluence-underdetermination no-go (Lemma 6.26; Theorems 6.29, 6.30, 6.31, 6.32)	D1 normal forms plus the spherical incidence, disk/mesh, modular cross-ratio, and 2π-normalization receipts	classification of closed combinatorial surfaces; Radó uniqueness; Möbius cross-ratio rigidity on S^2	the receipts are decidable finite-stage predicates on normal forms, not declared geometry; their selection is not implied by confluence (explicit T^2, $\partial\Delta^4$, wedge, and wrong-β countermodels); S^2 and its conformal class are derived only on the receipt branch	producer theorem plus underdetermination no-go

Table 3: Dependency checklist for the OPH SM/GR reconstruction program.

Node	Output	Immediate internal ingredients	Standard mathematics used	Branch-local inputs / external data	Support level
D4	Null modular bridge to T_{kk}, including exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line families, the weak tail generator, the derived half-sided modular inclusion, the explicit positive half-line null-translation generator on its Stone domain with affine half-line modular relations, and the exact half-line generator/charge identification with the local null-stress charge (Propositions 6.66, 6.69; Corollaries 6.67, 6.71; Theorem 6.70, Lemma 6.72; Theorem 6.73)	D2+D3 and Axioms 3.5–3.10	Borchers–Wiesbrock positive-generator theorem for standard half-sided inclusions; Stone's theorem; distributional differentiation on half-lines	the theorem-local null-cut center transfer, the inherited left/right strip-split package used for the spatial-collar-type tensor decomposition, and the exact-or-controlled Markov hypotheses of Proposition 6.66 and Corollary 6.67; quasi-local propagation and endpoint-Lipschitz control are supplied internally by Axiom 3.7 on the local finite-constraint branch; the geometric scaling action on the null half-line blow-up net derives the half-sided modular pair; bounded-interval formulas are discharged by Lemma 6.135; null-net standardness, the derived half-sided inclusion, Möbius bounded-interval covariance, Markov modular locality with its counterexample boundary, and the four-translation assembly are supplied by Theorems 6.77–6.84, with the Cyc, NTI, weak-additivity, MI, and kernel-residual receipts explicit	Phase I bridge theorem plus D4-standardness packet
D4b	Conditional Lorentzian event manifold: quotient-intrinsic event classes, conditional four-dimensional atlas with signature $(-{+}{+}{+})$, causal order and time orientation, H^3 strictly as frame fiber, tetrads/metric/connection/curvature readout, glued modular clocks, and countermodels (§6.3)	D3f–D3h and D4 outputs plus the population, separation, rank-four frame, and overlap-cocycle receipts E1–E4	bi-Lipschitz atlas arguments; convex-cone/quadric signature classification; Rademacher/Carathéodory regularity	E1–E4 with E4′/smooth upgrade, the MI/assembly branch, absolute conformal scale, stable causality, and the record-Cauchy receipt remain explicit; countermodels show $\dim H^3 = 3$ promotes nothing by itself	conditional construction theorem
D4c	Local conserved stress tensor from modular charges: null tomography, constructed $\langle T_{ab} \rangle$ with symmetry/locality/covariance/common normalization, generator/charge identification, weak Ward identity, ambiguity classification (Theorems 6.114, 6.116; Proposition 6.118)	D4/D4b outputs plus the MI/assembly branch, kernel-residual receipt, and universal-coupling receipt UC	null polarization/tomography of symmetric tensors; least-squares stability	ϕg_{ab} and improvement ambiguities explicit; UC is a physical-identification receipt; countermodel: per-direction generators without MI linearity admit no rank-two source; scalar CMI is never promoted	conditional construction theorem
D4d	Bulk/edge/central first law with the edge term carried exactly, algebraic type-I boundary statement, and MaxEnt stationarity extended to the coupled changing-stress class via the exact multiplier identity $dS/dt = \lambda = 2\pi$ (Theorems 6.119, 6.120; Proposition 6.121)	central-interface branch of Axiom 3.7, type-I generator form $K_C = 2\pi B_C + Z_C$, edge normalization $z_\alpha = \log d_\alpha$	finite first law $\delta S = \langle K \delta\rho \rangle$; MaxEnt envelope/Legendre identity	the Axiom-4 leading term and edge normalization are named branch axioms; shape/null-cut and metric variations are not covered; countermodels: wrong coefficient or wrong edge weights break stationarity by computable defects	exact finite theorem plus conditional bridge

Table 3: Dependency checklist for the OPH SM/GR reconstruction program.

Node	Output	Immediate internal ingredients	Standard mathematics used	Branch-local inputs / external data	Support level
D5	Jacobson-type Einstein branch: rest-frame relation plus tensor upgrade (Theorem 6.99, Lemma 6.101, Theorems 6.105, 6.106, 6.107)	D3+D4 and Axioms 3.5–3.10	fixed-volume area-variation identity for small geodesic balls; local quadratic-polarization argument for the tensor upgrade	Theorem 6.99, which fixes the admissible fixed-cap MaxEnt variation class and derives generalized-entropy stationarity on the realized cap-label-preserving MaxEnt family; the internal small-ball bridge of Lemma 6.101, which uses the geometric cap generator together with the D4 half-line generator/charge identification and Lemma 6.135; Theorem 6.105; locally Lorentzian $d = 4$ scaling regime, supplied conditionally by the D4b event-manifold packet under receipts E1–E4 (§6.3); small-ball constancy assumptions; Lemma 6.136 for $o(\ell^4)$ control; all local directions/reference states for the tensor upgrade via Lemma 6.137; the uniform scaling family, coverage, and absolute base condition are supplied by Theorems 6.122–6.125 under the VR/UC receipts, and the full chain is composed in Theorem 6.127 with realized-branch nonemptiness open (Remark 6.129)	Phase I scaling-limit theorem package plus composed branch-entry theorem
D6	Local/global theorem stack for Λ: null-invisible metric ambiguity, screen-capacity closure of the same Einstein branch, the de Sitter entropy relation with scale-certified static-patch display, and the cosmic record-closure fixed point on the observed branch, plus a legacy capacity-level neutrino side estimate (Theorem 6.151)	D5 local Einstein recovery leaves a $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation, static-patch radius/time formulas after scale certification, finite self-closing normal-form density/readback map, and dimensional analysis	cosmic record-capacity fixed point $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$ and its observed de Sitter entropy readout	Phase II zero-input global closure
D7	Construction of the refinement-stable bosonic sector category plus compact gauge reconstruction (Theorem 7.3, Theorem 7.6, Theorem 7.7), the conditional four-dimensional Euclidean Yang–Mills form (Theorem A.22), and the conditional compact-gauge repair-gap theorem (Theorem A.37)	Axioms 3.5–3.10 plus the compact-gauge refinement receipt	directed-colimit descent for monoidal C^*-categories; Doplicher–Roberts / Tannaka reconstruction; holonomy-to-curvature expansion; finite-stage conditional expectations and spectral comparison for commuting collar projections	Theorems 5.22 and 5.26 on the ordinary or central-defect bosonic zero-obstruction branch, plus Definition 7.2 for Theorem 7.3; the compact-gauge witness theorem supplies realized MAR-admissible data, while D8–D9 contain Standard Model selection; the Yang–Mills statements additionally require the finite cylinder/projective extraction theorem and the renormalized Yang–Mills, finite ground-state-transform/cross-fiber, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts; the genuinely noncentral fixed-cutoff branch remains separate	Phase I conditional structural theorem
D8	Product gauge structure up to finite quotient (Lemmas 7.10–7.15, Theorem 7.16)	D7 + Axiom 3.11	compact Lie representation classification; Schur's lemma	the same ordinary or central-defect bosonic branch as D7, together with a connected positive-dimensional Lie admissible class, one connected abelian factor, and faithful action on the minimal coupled carrier	Phase I realized-branch theorem

Table 3: Dependency checklist for the OPH SM/GR reconstruction program.

Node	Output	Immediate internal ingredients	Standard mathematics used	Branch-local inputs / external data	Support level
D9	Realized Standard Model quotient, hypercharges, structural electroweak force content, $N_c = 3$, $N_g = 3$, and product-group corollaries (Theorem 7.36, Theorem 7.17, Corollaries 7.20, 7.21, 7.28, Proposition 7.24)	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting; stabilizer computation for a nonzero neutral Higgs vacuum vector	realized one-generation chiral matter plus one Higgs package, with the CP-capability and weak-sector UV clauses contained in Axiom 3.11; no extra post-MAR selector is used	Phase I realized-branch theorem/corollary chain
D10	Forward gauge-coupling closure and declared electroweak readout of the integrated D10 quantitative-closure package	D9 + pixel ratio P	printed RG evolution, matching, and scheme-conversion conventions	pixel constraint and the first-principles forward transmutation solve $\mathcal{F}(\alpha_U; P) = 0$; the fixed-cutoff edge heat-kernel / Casimir theorem on the microphysics surface together with the compact-group / Peter–Weyl lift used on the D10 lane; printed beta-function, threshold, and scheme-conversion conventions	Phase II quantitative-closure sector
D12	Charged-lepton exact centered-readback / common-shift frontier, strong-CP branch, texture, H^3 record-worldline stitch certificates, dark-sector, heuristic baryogenesis continuations, black-hole spectroscopy and evaporation/ringdown bridges, proton-spin, proton-lifetime estimates beyond the gauge-channel exclusion, controlled large-N_{edge} string/worldsheet effective descriptions, conjectural critical-superstring extensions, and other continuations	various subsets of D6, D9, and D10	branch-specific EFT and phenomenological manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, discrete texture choices, declared H^3 atlas and cross-boundary record-assignment certificates, dark-sector response assumptions, discrete-horizon assumptions, physical black-hole bridge objects for exterior time, radiation entropy, flux closure, radiative quotient, and asymptotic readout, a distinct large-N_{edge} regime with fixed $\tau = tN_{\text{edge}}$ window and uniform genus-remainder control, or additional worldsheet/CFT assumptions	Phase III phenomenological continuation branch

The recovered core of this SM/GR derivation paper is D1–D5 together with D7–D9. Within that package, D7 is the bosonic sector-category construction plus compact-gauge reconstruction stage, while D8–D9 are realized-branch results that keep the realized one-generation chiral matter plus one-Higgs package and MAR admissibility inputs explicit. D6 sits outside that recovered core because the cosmological-capacity closure is the separate global completion of the Einstein branch. D5 leaves the metric term undetermined, D6 closes that same branch at the cosmic record-capacity fixed point $N_{\text{CRC}} = F(N_{\text{CRC}})$, and the same D6 stack emits the de Sitter entropy relation and static-patch display once the selected no-G scale certificate supplies $\ell_\star^2$. The density $\log|\Omega_N^{\text{sc}}| - N$ is the finite-count representation of the same target after division by the full screen Hilbert-space dimension e^N. Informally, that point is where the universe reads back its own boundary without deficit or slack. D10 is the integrated quantitative-closure branch, and D12 collects phenomenological continuations. The modular-anomaly dark-sector branch belongs to D12: it imports the D6 static-patch scale only as an IR benchmark and does not derive MOND/RAR-style response laws, a controlled galaxy-scale source/response law, or galaxy/cluster phenomenology. The branch data used downstream are $P_\star$, N_{CRC}, and the selected scale certificate $\gamma_\star$ or $B_\star$; descendants use their branch-appropriate readouts.

Exceptional finite-symmetry sidecar. The finite $E_8/\mathrm{Spin}(8)$ triality certificate belongs beside D7 as an algebraic representation-closure sidecar, not as a new recovered-core input. It records an exact matrix-level $A_8 \subset E_8$ root subsystem, the even subgroup $\mathrm{Alt}(9) \subset W(A_8) \cong \mathrm{Sym}(9)$, a nonsplit $2\cdot\mathrm{Alt}(9)$ lift in $\mathrm{Spin}(8)$, and a positive half-spin image that preserves an E_8 lattice. The vector and half-spin presentations have different mod-2 orbit fingerprints on $E_8/2E_8 \setminus \{0\}$, namely $\{9, 36, 84, 126\}$ and $\{120, 135\}$, and are identified only after adjoining the outer triality automorphism. This supports finite exceptional representation bookkeeping on the compact-gauge side. It does not select the realized Standard Model quotient, add a force or particle, prove OPH, close a heterotic critical-edge gate, or count as hardware evidence.

Cosmology continuations above D6 obey the same firewall. The flat-FLRW reading is allowed only as a clocked-FLRW holonomy identification statement: spatial curvature is read from the spatial Levi–Civita connection on a declared clock slice, and zero curvature is the zero visible spatial holonomy branch. Selection of that branch is separate Phase III data: a direct no-curvature theorem, a conditional cosmological-minimal-holonomy selector, or an explicit flat-branch assumption. MAR acts on low-energy gauge/matter packages and is not a cosmological flatness selector. Conditional screen-spectrum continuation is theorem-level only after the cosmology companion supplies the geometric collar-volume scalar $q_r = \Pi_{\geq 2,r}(1/3)\log(J_{X,r}/\bar{J}_{X,r})$, a normalized positive scalar precision operator K_r, a first-principles quotient ensemble, the expected scalar release energy

$$E_r^{\mathrm{src}} = \frac{1}{2}\mathbb{E}_{\mu_r}[q_r^{\mathsf{T}} K_r q_r], \qquad A_r = 2E_r^{\mathrm{src}}/d_r,$$

refinement convergence, and an operator-derived or otherwise proved tilt. The resulting screen-level covariance is C_ℓ^q, not C_ℓ^{TT}. A primordial curvature spectrum follows only after separate source-stress, radial-window, null-space, forward-residual, and scale-bridge certificates convert A_q to A_ζ. The special $P_\star/48$ tilt is a theorem-level value only if the edge-center reserve, half-collar scalar projection, and reserve-to-RG Ward identity pass. This package is a Phase III conditional continuation, not a recovered-core output. Full inflation replacement, physical TT/TE/EE scalar or parity transfer, H_0/S_8 prediction, dark/anomaly growth kernels, and baryogenesis require their finite covariant Boltzmann-source, transfer, frozen-likelihood, source-provenance, pooled-reducer, and no-data-use contracts and are not supported by the D6 capacity relation.

Corollary 2.1 (Cosmological physical-scale continuation boundary)**.** *The controlled Lorentz branch and the H^3 observer-facing spatial chart determine the kinematic target and spatial dimension of the recovered core. They do not, by themselves, construct a populated finite FLRW spatial metric, source-shell embedding, comoving Laplacian, physical wavenumber k, scale-factor history, or freezeout surface. The constants $P_\star$, N_{CRC}, and $\ell_\star$ do not identify one implementation patch with one physical cell; neutral quotient pseudometrics are not automatically physical spatial metrics. Physical CMB promotion is therefore a Phase III conditional claim and must cite the finite physical scale-bridge theorem: imported frozen FLRW geometry gives only the `CONDITIONAL_PHYSICAL` tier, while the `OPH_NATIVE_PHYSICAL` tier also requires a quotient-derived* $\mathsf{CosmoGeomRead}_r$ *and source embedding theorem. The aggregate physical scale-bridge receipt is required in addition to the finite source, transfer, and likelihood receipts.*

Corollary 2.2 (Physical cosmology promotion conjunction)**.** *Physical CMB, BAO, lensing, growth, RSD, and S_8 claims require*

$$\mathsf{CMB\_READY} = \mathsf{SOURCE} \wedge \mathsf{SCALE} \wedge \mathsf{PARENT} \wedge \mathsf{KERNEL} \wedge \mathsf{INIT} \wedge \mathsf{TRANSFER} \wedge \mathsf{FREEZE} \wedge \mathsf{LIKE}.$$

The scale term is the conjunction of `PHYSICAL_SPATIAL_K_RECEIPT`, `SCREEN_TO_PHYSICAL_K_ASSOCIATION_RECEI`
`SOURCE_ANGULAR_SECTOR_RECEIPT`, `CALIBRATED_A_EVOLUTION_RECEIPT`, `PHYSICAL_MODE_FREEZEOUT_MAP_RECEI`

`PHYSICAL_FREEZEOUT_SURFACE_RECEIPT`, a common primordial/anomaly mode-basis receipt, nonempty cross-receipt identity, and no-post-hoc-calibration receipt. Here PARENT *includes finite packet kinematics, mass shell, finite packet stress readout, variational/moment stress agreement, reaction-channel four-momentum, explicit recipient stress for nonzero exchange, exchange-current closure, total stress closure, local-frame and carrier-quotient invariance, cosmological gauge invariance, finite domain of dependence, subluminal characteristics, retarded response, response stability, refinement, and CDM-limit recovery receipts. A visually successful TT curve or a source table with ρ_A, $\rho_{A,\mathrm{eq}}$, and B_A does not instantiate this conjunction. Before the active fiber, conserved-sector decomposition, physical clock, and common-parent response pole are certified, the transition number is only $\gamma_{\text{repair step}}$, not Γ_{rec}. While the conjunction is false, the correct class is diagnostic baseline or conditional OPH prediction, not recovered-core physics.*

3 Five Axioms, the Phase-II Pixel Fixed Point, the Capacity Target, the Scale Certificate, and Theorem Checklist

This section states the theorem checklist and dependency map for this SM/GR derivation paper. The basis used below consists of the five core axioms, the quantitative quantities listed next, the controlled BW scaling theorem, and the fixed-stage/receipt-conditional bosonic gauge-reconstruction chain. This is an algebraic-information basis: the screen-net, state, trace/probability, and generalized-entropy structures are starting ingredients of the OPH framework. The purpose is to test whether this basis supports a coherent theory-of-everything reconstruction of the observed effective universe. On the fixed-cutoff local finite-constraint MaxEnt branch, the local-Gibbs form, quasi-local dynamics, Lieb–Robinson propagation control, and endpoint-Lipschitz interval control are internal consequences. Any Dobrushin/local-Gibbs/mixing language used below names only a sufficient fixed-cutoff collar-recoverability mechanism on that branch. It is not an infinite-volume uniqueness claim for the refinement-limit theory. What Axiom 3.7 fixes under refinement is the realized state-side MaxEnt branch itself. Theorem 6.2 fixes the operational/geometric split at fixed cutoff, and Theorem 6.13 fixes the controlled scaling-limit cap automorphism, its 2π normalization, and the Lorentz-facing geometric cap action on that subnet. Corollary 6.71 then derives the null half-sided modular pair after the null blow-up step, and Lemma 6.72 promotes that pair to the explicit positive null-translation generator on its Stone domain with the affine half-line modular relation. Theorem 6.99 internalizes the fixed-cap generalized-entropy stationarity step for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family. The compact-gauge realized branch is supplied by Theorem 7.35. The bounded-interval transport/projective branch is a branch-local construction only where those interval formulas are invoked. Gluing-side path-independent transport is Theorem 5.22; refinement-limit compact-gauge reconstruction additionally requires Definition 7.2. On the central branch the combined zero-obstruction condition is $[z]_\Sigma = 0$ together with trivial sector holonomy after strictification. On the genuinely noncentral branch, the higher associator must be strictifiable, $o^{(2)}_\Sigma = 0$, and at least one allowed strict G_Σ-valued 1-cocycle representative must have trivial represented holonomy. A nonzero $o^{(2)}_\Sigma$ is routed to the fixed-cutoff higher-gauge sector, while a strictifiable orbit for which every strict representative has residual loop holonomy is excluded from ordinary compact-group reconstruction; the full orbit q_Σ does not canonically select one ordinary H^1 class.

The formulation uses two dimensionless closure coordinates together with one selected no-G

scale certificate:

$$P \equiv a_{\rm cell}/\ell_\star^2, \tag{1}$$

$$N_{\rm CRC} \equiv \log\dim \mathcal{H}_{\rm tot}, \tag{2}$$

$$\gamma_\star \equiv \frac{\ell_\star \nu_{\rm Cs}}{c}, \qquad B_\star \equiv \frac{3\pi}{\ell_\star^2}. \tag{3}$$

In the quantitative implementation used here, P is the local particle-physics scale variable fixed by the outer/inner closure program summarized in the synthesis paper *Observers Are All You Need* [2]; this compact paper uses it only as the quantity carried by the forward particle-physics map. That closure writes the outside detuning as

$$P = \varphi + \alpha_{\rm in}(P)\sqrt{\pi},$$

where the inner side is the electromagnetic observation scale emitted by the same cell on the declared quantitative branch. The OPH-plus-empirical comparison coordinate is

$$\alpha^{-1}(0) = 137.035999177(21), \qquad \alpha \simeq 0.00729735256433, \qquad P_C \simeq 1.6309682094.$$

Here P_C is defined from the measured CODATA/NIST endpoint. A source-only fixed point requires the OPH-QCD endpoint map and its interval certificate; the comparison coordinate does not supply that missing map. The source computation runs in the order

$$P \mapsto M_U(P) \mapsto \alpha_U(P) \mapsto \alpha_i(m_Z; P) \mapsto a_0(P) \mapsto A_T(P),$$

where $A_T(P) = \alpha_{\rm em}^{-1}(0; P)$ is the Ward-projected Thomson endpoint. The realized cell solves $P = \varphi + \sqrt{\pi}/A_T(P)$. The fine-structure value is forced on this branch because the same local pixel must satisfy the outer entropy-detuning equation and the inner electromagnetic endpoint equation with no remaining free local scale. The source-only diagnostic trunk, before the low-energy hadronic endpoint transport is supplied, emits

$$\alpha_{\rm root}^{-1} = 136.99483516462164945794999458578 7\ldots.$$

For comparison bookkeeping, adding the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel, $\alpha_U(P_C) = 0.041124336195630495$, to the source/root value gives the mixed-provenance no-hadron diagnostic

$$A_{\alpha_U}^{\rm fp} = 136.994835164621649457949994585787\ldots + 0.041124336195630495 = 137.03595950081727995294999458578$$

An observer inside the realized branch does not measure either no-hadron source-side value as the Thomson-limit fine-structure constant. A low-energy measurement probes the dressed $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme endpoint matching have been transported to $q^2 = 0$. At that mixed comparison coordinate, the CODATA/NIST-minus-diagnostic bookkeeping gap is

$$\Delta_{\rm H,req}^{\rm fp} = 137.035999177 - 137.035959500817279952949994585787\ldots = 0.000039676182720047050005414213\ldots.$$

Normalized by the comparison-pixel value $\alpha_U(P_C) = 0.041124336195630495$, this correction is

$$\Delta_{\rm H,req}^{\rm fp} = \alpha_U(P_C)\,(C_{24,Q} - 1), \qquad C_{24,Q} = 1.000964785973232625384951114070247 5\ldots.$$

The full root-to-CODATA difference

$$\Delta^{\mathrm{fp}}_{\mathrm{H,cal}} = 137.035999177 - \alpha^{-1}_{\mathrm{root}} = 0.0411640123783505420500054142 13\ldots = \alpha_U(P_C)C_{24,Q}$$

is bookkeeping; it includes the comparison-pixel $\alpha_U(P_C)$ contribution and is not a source-emitted hadronic correction. The proximity to α_U is a branch-scale clue rather than an identity or source-only endpoint prediction. The 24-register count explains the scale. It does not determine the nonconstant Ward-projected hadronic spectral functional. A source-only fine-structure endpoint requires an OPH-QCD hadronic backend, not a fitted scalar residual. Its minimum receipt bundle must expose the QCD quotient ensemble, source QCD parameter map, Euclidean slab/vacuum-transfer construction, hadronic Hilbert quotient, Ward-normalized electromagnetic current ledger, positive two-current spectral export $d\rho^{(2)}_Q$, same-scheme remainder Ξ_Q, and no-target-leak source DAG. The two-current measure is only the marginal needed for running-α and HVP transport; it does not by itself close HLbL or rare-decay long-distance amplitudes, which require the higher-point and transition spectral sectors of the same backend. Concretely, the source-only fine-structure endpoint requires `WARD_Q_SPECTRAL_MEASURE_RECEIPT`, `J24Q_SOURCE_JACOBI_RECEIPT`, `OMEGAQ_NORMALIZATION_RECEIPT`, `XI_SAME_SCHEME_REMAINDER_RECEIPT`, `NO_THOMSON_TARGET_LEAK_DAG`, `FULL_PIXEL_CONTRACTION_INTERVAL`, and `ALPHA_PREDICTION_BEATS_DIRECT_MEASUREMENT` under the direct-measurement interval adopted for that release. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. The technical endpoint table in Section 9 records the source-side diagnostic trunk and residual package. $\gamma_\star$ is the primary dimensionless scale certificate on the observation-located branch interval, with $B_\star$ as its SI curvature display. It is not solved from P, from the capacity N_{CRC}, from measured G, or from $\Lambda = 3\pi/(GN)$. Once the certificate is supplied, $\ell_\star^2 = 3\pi/B_\star$, ℓ_P^2 is the usual SI Planck-area display of $\ell_\star^2$, and $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$. The no-$G$ burden of this scale certificate is the clock hierarchy. In the synthesis paper this is stated as the OPH clock-hierarchy theorem. Its first upstream record is the no-measured-input $\alpha_U(P_\star)$ proof record, where

$$\Phi_U(P,a) = \bar\ell_{\mathrm{SU(2)}}(t_2(P,a)) + \bar\ell_{\mathrm{SU(3)}}(t_3(P,a)) - \frac{P}{4}$$

has a unique source zero in the declared interval, with frozen representation cutoffs, beta packet, matching prescription, threshold packet, and renormalization convention. The hierarchy artifact records

$$\begin{aligned} I_U &= [0.041123336195630494,\ 0.041125336195630496],\\ \alpha_U(P_\star) &= 0.041124336195630495, \end{aligned}$$

a Krawczyk image strictly inside I_U, and a derivative enclosure contained in $[-10.995768, -10.985284]$. On the declared graph there is no directed path from measured W/Z, measured $\alpha_s(m_Z)$, measured $\sin^2\theta_W$, measured v, measured G, Planck-area data, or measured Λ into Φ_U, α_U, or $v/E_\star$. This solves the local electroweak hierarchy problem on the stated formula stack; it supplies only the first component of the clock hierarchy below. The cesium gap $\varepsilon_{\mathrm{Cs}} = \hbar\omega_{\mathrm{Cs}}/E_\star$ must then factor through the first-principles chain

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}^{133\mathrm{Cs}}_{\mathrm{QCD/nuc}} + \mathcal{R}^{133\mathrm{Cs}}_{\mathrm{atom}}.$$

Here $\mathcal{R}_\alpha$ emits the electromagnetic endpoint used by the atomic Hamiltonian, $\mathcal{R}_e^{\mathrm{abs}}$ emits the absolute electron mass ratio, the cesium QCD/nuclear branch emits the source-side $^{133}\mathrm{Cs}$ nuclear packet, and the atomic branch emits the hyperfine spectral gap. The combined dependency graph may contain no path from measured G, Planck area/mass/time, measured low-energy electroweak data, measured Λ, $3\pi c^3/(\hbar G)$, or an equivalent calibrated scale. Failure of that dependency test makes the numerical G row calibration. Passing the dependency test makes the SI gravity row inherit the no-G condition of the emitted clock gap. Printed digits beyond the source interval bounds are calibration checksum digits. N_{CRC} enters the cosmological-capacity branch. The capacity normalization uses the Gibbons–Hawking de Sitter entropy [20]: if

$$N_{\mathrm{patch}} = \left(\frac{r_{\mathrm{dS}}}{\ell_P}\right)^2$$

denotes the bare horizon area ratio, then

$$N_{\mathrm{scr}} = S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2} = \pi N_{\mathrm{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

Using the Planck-2018 cosmological benchmark [21] gives $r_{\mathrm{dS}} \simeq 1.66 \times 10^{26}$ m, $N_{\mathrm{patch}} \simeq 1.05 \times 10^{122}$, $N_{\mathrm{scr}} \simeq 3.31 \times 10^{122}$, and $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$. Any retained capacity-level neutrino side estimate on that input is legacy bookkeeping. The separate weighted-cycle construction has target-informed template-candidate status.

The same screen-capacity normalization supplies the global repair-tick lemma used in the hierarchy continuation. At $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$, let $\rho = \ell/r_{\mathrm{CRC}}$, so the horizon sits at $\rho = 1$ and the local cell at $\rho_\star = \ell_\star/r_{\mathrm{CRC}} = (N_{\mathrm{CRC}}/\pi)^{-1/2}$. The global repair cycle is the scale-free readback transport attached to the fixed point. The fixed-point equation supplies its closure. With the Observers-synthesis readback map, modeled by the D6 area-law count at the effective delivery resolution ρ_{read} through the D6-consistent counting identification, the readback reconstructs capacity $F(N) = \pi/\rho_{\mathrm{read}}^2$, so the fixed-point equation $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}}) = \pi/\rho_\star^2$ forces $\rho_{\mathrm{read}} = \rho_\star$, i.e. $G_N(1) = \rho_\star$ exactly, and

$$G_N(\rho) = \left(\frac{N_{\mathrm{CRC}}}{\pi}\right)^{-1/2} \rho.$$

The 24-tick count is the doubled observer-visible product-adjoint repair spectrum,

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24.$$

The factor 2 is the reversible write/verify orientation. The $\mathfrak{su}(5)$ adjoint has the same single-orientation integer for a different support because it contains X/Y mixed gauge channels excluded by the OPH product branch. With $G_N = g_N^{24}$, the one-tick homogeneous normal form is

$$g_N(\rho) = \left(\frac{N_{\mathrm{CRC}}}{\pi}\right)^{-1/48} \rho, \qquad |g_\star'| = \left(\frac{N_{\mathrm{CRC}}}{\pi}\right)^{-1/48}.$$

The integer 24 is branch architecture for this lemma.

The joint source branch uses the product space

$$X = I_P \times I_x, \qquad x = \log N,$$

and the product-separated source map

$$\mathcal{J}(P, x) = (\Gamma(P), \widehat{C}(x)),$$

where Γ is the local pixel closure map and $\widehat{C}(x) = \log F(e^x)$ is the global capacity readback map. If the component certificates give

$$|\Gamma(P) - \Gamma(Q)| \le q_P |P - Q|, \qquad |\widehat{C}(x) - \widehat{C}(y)| \le q_N |x - y|, \qquad q_P, q_N < 1,$$

then $\mathcal{J}$ is a contraction in the product metric with constant $q = \max\{q_P, q_N\}$. It has a unique stable joint fixed point $(P_\star, \log N_{\mathrm{CRC}})$. A source branch with genuine cross-feedback must instead provide a weighted-sup derivative certificate: for derivative bounds a, b, c, d, there must exist $r > 0$ with

$$\max\{a + b/r,\ d + rc\} < 1.$$

Without that coupled-map certificate, the product theorem is the closed branch and coupled feedback remains residual branch freedom.

On the selected exact branch, the RG/Higgs naturality square is also closed. Let Q_s be the source hierarchy quotient and Q_H the Higgs/electroweak quotient, and define

$$\rho_{sH}([x]_s) = \left[P(x), N(x), \left(\frac{N(x)}{\pi}\right)^{-P(x)/12}, \Pi_{HT} F_{D11} F_{D10}(P(x), N(x)), 0\right]_H.$$

For the source and Higgs normal-form maps n_s, n_H, and for the corresponding obstruction maps h_s, h_H with induced map χ_{sH}, the selected branch has

$$\rho_{sH} n_s = n_H \rho_{sH}, \qquad \chi_{sH} h_s = h_H \rho_{sH}.$$

Thus

$$\varepsilon_H = \max\{\varepsilon^n_{sH}, \varepsilon^h_{sH}\} = 0, \qquad \varepsilon_H \in [0, 0].$$

Measured v, W/Z, Higgs/top, G, Planck-area, and Λ data are not inputs to this naturality bound. The electroweak tick-projection bridge is closed by

$$\Pi_{\mathrm{EW}}(P, N) = \frac{24\pi}{\alpha_U(P)\log(N/\pi)}, \qquad \Pi_{\mathrm{EW}}(P_\star, N_{\mathrm{CRC}}) = 4P_\star \iff B_{\mathrm{EW}}(P_\star, N_{\mathrm{CRC}}) = 0,$$

where

$$B_{\mathrm{EW}}(P, N) = \alpha_U(P)\log(N/\pi) - \frac{6\pi}{P}.$$

The exact bridge target is

$$N_{\mathrm{EW}}(P_\star) = \pi \exp\left[\frac{6\pi}{P_\star \alpha_U(P_\star)}\right] = 3.5323546226929906511187512962330547600462 \times 10^{122}.$$

Equivalently, the log-capacity map

$$\mathcal{C}_{\mathrm{EW}}(P, x) = (1-\lambda)x + \lambda \frac{6\pi}{P\alpha_U(P)}$$

is a contraction at $\lambda = 1/2$, with unique fixed point $x_\star = 6\pi/(P_\star \alpha_U(P_\star))$ and $N^{\mathrm{EW}}_{\mathrm{CRC}} = \pi e^{x_\star}$. Thus $B_{\mathrm{EW}}(P_\star, N^{\mathrm{EW}}_{\mathrm{CRC}}) = 0$. The rounded 3.31×10^{122} capacity label is diagnostic. The finite readback-resolution certificate defines

$$F_r(N) = \mathrm{Cap}_{\mathrm{read}}\left(\mathrm{Obs}(\mathrm{nf}_{r,N}(\mathfrak{U}_{r,N}))\right), \qquad \rho_{\mathrm{read}}(r, N) = \sqrt{\pi/F_r(N)}.$$

On the selected finite branch the observer-facing capacity register has one positive central capacity atom, so ρ_{read} is a singleton and $\rho_{\mathrm{read}}(r, N_{\mathrm{CRC}}) \to (N_{\mathrm{CRC}}/\pi)^{-1/2}$ in the positive-root refinement limit. Together with the representation-to-spectrum count $m_{\mathrm{rep}} = 24$, read as the 24-slot oriented repair register supplied by reversible write/check orientation, this closes the local/global hierarchy-resonance theorem on the selected branch.

This is the reverse-engineering protocol used for the quantitative branch. Observed values may name a neighborhood of the branch: P near the electromagnetic endpoint, N_{CRC} near the de Sitter entropy capacity, and $\gamma_\star$ near the SI scale branch. The paper then asks whether the OPH readback map is a self-map on that neighborhood and whether Banach contraction, derivative sign, strict concavity, or an equivalent uniqueness certificate selects one fixed point. Only that selected point is counted as an OPH output; the observational seed is branch location, not a fitted constant.

The companion compact proof uses the same convention as a compression estimate. A quantitative row enters that estimate only when the declared source map has no dependency path from the measured target or a calibrated proxy. If p_i upper-bounds the conditional accidental hit probability of row i after previous accepted rows, then

$$P_{\mathrm{acc}} \leq \prod_i p_i.$$

Twelve one-percent source-certified rows give $P_{\mathrm{acc}} \leq 10^{-24}$, and twenty give 10^{-40}. The numerical bound is secondary to the theorem stack: the same two closure coordinates organize the observer readout, gravity/gauge reconstruction, hierarchy bridge, dark energy, dark-sector budget, product-group proton-decay exclusion, particle inventory, and string-vacuum selector.

Theorem 3.1 (What the two dimensionless closures determine)**.** *Let the local pixel closure be*

$$P_\star = \frac{a_{\mathrm{cell}}}{\ell_\star^2},$$

and let the global capacity closure on the de Sitter branch be

$$N_\star = \frac{A_{\mathrm{dS}}}{4\ell_\star^2} = \frac{3\pi}{\Lambda_\star \ell_\star^2}.$$

Write

$$\mathcal{B}_\ell := \Lambda_\star N_\star$$

for the scale product called B_ℓ in the compact scale note. Then the two closures imply

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\mathrm{cell}} = \frac{3\pi P_\star}{N_\star},$$

and

$$\mathcal{B}_\ell \ell_\star^2 = 3\pi, \qquad \mathcal{B}_\ell a_{\mathrm{cell}} = 3\pi P_\star.$$

These are dimensionless invariants of the OPH geometry.

Proof. The capacity closure gives

$$N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2},$$

so

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}.$$

Multiplying by $P_\star = a_{\mathrm{cell}}/\ell_\star^2$ gives

$$\Lambda_\star a_{\mathrm{cell}} = P_\star \Lambda_\star \ell_\star^2 = \frac{3\pi P_\star}{N_\star}.$$

Since $\mathcal{B}_\ell = \Lambda_\star N_\star$,

$$\mathcal{B}_\ell \ell_\star^2 = \Lambda_\star N_\star \ell_\star^2 = 3\pi.$$

Multiplying by $P_\star$ gives

$$\mathcal{B}_\ell a_{\mathrm{cell}} = 3\pi P_\star.$$

□

Theorem 3.2 (Scale underdetermination by the two closures)**.** *No formula using only $P_\star$, $N_\star$, pure dimensionless constants, and exact display constants $c, \hbar$ can determine $\mathcal{B}_\ell$ in m^{-2}, $\ell_\star^2$ in m^2, or G_{SI} in SI units.*

Proof. For any $\lambda > 0$, rescale all lengths by $L \mapsto \lambda L$. Then

$$\ell_\star^2 \mapsto \lambda^2 \ell_\star^2, \qquad a_{\mathrm{cell}} \mapsto \lambda^2 a_{\mathrm{cell}}, \qquad \Lambda_\star \mapsto \lambda^{-2}\Lambda_\star.$$

The ratios

$$P_\star = \frac{a_{\mathrm{cell}}}{\ell_\star^2}, \qquad N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2}$$

are unchanged, but

$$\mathcal{B}_\ell = \Lambda_\star N_\star \mapsto \lambda^{-2}\mathcal{B}_\ell.$$

Thus $P_\star$ and $N_\star$ determine the invariant product $\mathcal{B}_\ell \ell_\star^2 = 3\pi$, while the SI value of $\mathcal{B}_\ell$, the SI area $\ell_\star^2$, and

$$G_{\mathrm{SI}} = \frac{c^3}{\hbar}\ell_\star^2$$

change with the length scale. □

Theorem 3.3 (Selected OPH scale certificate)**.** *Let $N_\star$ be the OPH cosmic record-capacity fixed point*

$$N_\star = F(N_\star),$$

and let the selected no-G scale branch supply

$$\gamma_\star \equiv \frac{\ell_\star \nu_{\mathrm{Cs}}}{c} = 4.955974336548419463665778243369643187948431970582 5 \times 10^{-34},$$

with $\nu_{\mathrm{Cs}} = 9{,}192{,}631{,}770\,\mathrm{s}^{-1}$. Then

$$\ell_\star^2 = \left(\frac{\gamma_\star c}{\nu_{\mathrm{Cs}}}\right)^2 = 2.6122803023742777788734776921545100146120267686 6 \times 10^{-70}\,\mathrm{m}^2.$$

Equivalently, the same certificate has the SI curvature display

$$B_\star := \frac{3\pi}{\ell_\star^2} = \begin{matrix}3.607873914680321576051841480160172547687708307217185382124 2\\ 0701987898719888 86\end{matrix} \times 10^{70}\,\mathrm{m}^{-2}.$$

On the de Sitter branch this is the same scale product

$$B_\star = \Lambda_\star N_\star,$$

so the OPH area coordinate is

$$\ell_\star^2 = \frac{3\pi}{B_\star}.$$

Proof. The clock-ratio definition gives

$$\ell_\star = \frac{\gamma_\star c}{\nu_{\mathrm{Cs}}},$$

and hence the displayed value of $\ell_\star^2$. The curvature display is only a re-expression of the same area coordinate:

$$B_\star = \frac{3\pi}{\ell_\star^2}.$$

The de Sitter capacity identity is

$$N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2}.$$

Multiplying by $\Lambda_\star \ell_\star^2$ gives

$$\Lambda_\star N_\star \ell_\star^2 = 3\pi.$$

Since $B_\star = \Lambda_\star N_\star$ on the selected branch, division by $B_\star$ gives

$$\ell_\star^2 = \frac{3\pi}{B_\star}.$$

□

Corollary 3.4 (Newton coupling after the scale certificate)**.** *The local OPH gravity branch gives*

$$a_{\mathrm{cell}} = P_\star \ell_\star^2, \qquad \bar{\ell}_{\mathrm{shared}} = \frac{P_\star}{4}.$$

Therefore

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}} = \frac{P_\star \ell_\star^2}{4(P_\star/4)} = \ell_\star^2.$$

The SI display is

$$G_{\mathrm{SI}} = \frac{c^3}{\hbar}\ell_\star^2.$$

Consequently, the two OPH closures determine the complete dimensionless geometry. The local edge law then identifies G_{geom} with the OPH area quantum $\ell_\star^2$. The numerical SI value of G is fixed only after the selected no-G scale certificate supplies $\gamma_\star$, equivalently $B_\star = 3\pi/\ell_\star^2$. Observed branch values can locate the scale branch in reverse-engineering mode. The rounded display $N_{\mathrm{scr}} \simeq 3.31 \times 10^{122}$ is only a capacity-scale benchmark; the exact G row uses the fixed-point value $N_\star$ together with the selected scale certificate. Using only the rounded capacity with the Planck-2018 curvature benchmark would move the Newton row at the 8.4×10^{-4} level, so it is not a precision certificate.

Axiom 3.5 (Screen Net)**.** *Physical data is organized on a horizon screen S^2 carrying a net of local algebras*

$$P \longmapsto \mathcal{A}(P)$$

for connected patches $P \subset S^2$, with isotony

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

Axiom 3.6 (Overlap Consistency)**.** *For overlapping patches $P_1 \cap P_2 \neq \varnothing$, the local states induced on the shared algebra agree:*

$$\omega_{P_1}|_{\mathcal{A}(P_1 \cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1 \cap P_2)}.$$

Axiom 3.7 (Local MaxEnt and Refinement Stability)**.** *At the regulator scale ℓ_{UV}, the realized branch is selected by maximizing entropy subject to the finitely many homogeneous global-sum constraints*

$$\Big\langle \sum_x O_a(x) \Big\rangle = C_a, \qquad a = 1, \dots, N_{\mathrm{con}},$$

built from a finite list of gauge-invariant local densities $O_a(x)$, each supported in a ball of radius $O(\ell_{\mathrm{UV}})$. The constrained functionals are the N_{con} global sums, together with the finitely many optional global conserved charges, not one independent constraint per regulator cell; there is therefore one Lagrange multiplier λ_a per density label a, and the multiplier count is cutoff-independent by construction. The per-cell specification $\langle O_a(x)\rangle = c_a(x)$ with cell-dependent multipliers $\lambda_a(x)$ defines a strictly larger family whose dimension grows with the number of cells; it is admitted only as a near-equilibrium approximation regime and is never used in the refinement or branch-stability arguments of this paper. The axiom additionally asserts a refinement-closure clause*: under the admissible refinement channels of this framework, the coarse-grained realized state again lies in the exponential family generated by the same finite density list at the coarser scale. Coarse-graining a finite-range Gibbs family generically generates additional interactions, so closure is a substantive renormalization condition on the realized branch, not a bookkeeping consequence of reusing the operator labels a; the closure defect, the induced projection, and its trace-norm residual bound are supplied by Definition 3.8 and Lemma 3.9, and what remains assumed rather than proved is that the defect vanishes along the realized branch. Granting closure, the realized states at different cutoffs belong to one common $(N_{\mathrm{con}} + N_{\mathrm{glob}})$-dimensional MaxEnt family. The realized low-energy branch is the refinement-stable branch of that family; symmetry-allowed relevant operators are therefore held at zero on that branch only by symmetry or because they lie in the explicitly retained constraint family. This is a statement about branch persistence, not a universal entropy-ordering theorem for arbitrary phases away from that branch.*

Quotient-ensemble selector boundary. Axiom 3.7 controls the realized state-side branch, while the consensus normal-form map controls closure. Neither statement by itself chooses probability weights on terminal quotient normal forms: every distribution supported on normal forms is fixed by deterministic settling. A finite OPH quotient ensemble therefore additionally requires either an intrinsic quotient base weight m_r and action S_r, with

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-S_r(q)},$$

or an intrinsic projective prior ν_r whose normal-form pushforward is declared physical. Exact refinement compatibility for $s \succeq r$ requires the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

with α_{sr} independent of q. Seed noise, repair jitter, worker partitioning, and conventional free-field or lattice-gauge baselines are implementation or calibration data until this quotient-ensemble selector, or a separate transfer-matrix vacuum gate, is supplied.

Derived local-Gibbs branch. On the finite regulator realization of Axiom 3.5, the standard finite-dimensional Lagrange-multiplier argument applied to Axiom 3.7 gives

$$\omega_{\ell_{\mathrm{UV}}}(\lambda) = Z_{\ell_{\mathrm{UV}}}(\lambda)^{-1} \exp\!\big(-K_{\ell_{\mathrm{UV}}}(\lambda)\big),$$

with

$$K_{\ell_{\mathrm{UV}}}(\lambda)=\sum_x\sum_{a=1}^{N_{\mathrm{con}}}\lambda_a O_a(x)+\sum_b \mu_b Q_b,$$

where the Q_b are the finitely many optional global conserved-charge constraints. Because the constraints are the homogeneous global sums, the exponent carries one cell-independent multiplier λ_a per density label, so the multiplier-space dimension is $N_{\mathrm{con}}+N_{\mathrm{glob}}$ on every lattice and matches the count of independent constraints there; per-cell constraints would instead produce cell-dependent multipliers $\lambda_a(x)$ and a parameter space growing with the number of cells. The local-density part of the logarithm is therefore a quasi-local UV generator. Locality of the optional Q_b terms requires the separate condition in the next paragraph.

For the finite-range collar theorem, each retained Q_b must itself be a sum of uniformly bounded finite-range densities, or it must be fixed and central on the superselection sector under study. An otherwise nonlocal charge term remains valid MaxEnt data but falls outside Definition 5.1; the Lagrange-multiplier argument alone does not turn such a term into a local interaction.

Derived quasi-local propagation and interval control. Fix the regulator graph metric coming from cell adjacency and let $\tau_t^{\ell_{\mathrm{UV}}}$ denote the automorphism group generated by $K_{\ell_{\mathrm{UV}}}(\lambda)$, or more generally by any branch generator lying in the norm-closed algebra generated by the same bounded-support local densities. Standard Lieb–Robinson estimates for finite-range interactions on the finite regulator net then give constants $C,\xi,v_{\ell_{\mathrm{UV}}}<\infty$ such that for local observables A_X, B_Y, [9]

$$\big\|[\tau_t^{\ell_{\mathrm{UV}}}(A_X),B_Y]\big\|\le C\,\|A_X\|\,\|B_Y\|\,\min(|X|,|Y|)\,e^{-(d(X,Y)-v_{\ell_{\mathrm{UV}}}|t|)/\xi}.$$

So finite-velocity quasi-local propagation is branch-internal: it is the local finite-constraint MaxEnt branch written in propagation form, not an extra selector.

The same locality statement yields the bounded-interval endpoint-Lipschitz control used by the compact-gauge branch in the null modular chain. Whenever later sections use a fixed-cutoff interval or collar generator on this branch, it is the corresponding restriction of the same local density list, with the central endpoint term separated off exactly as in the later edge-center bookkeeping. Changing an interval I to I' then alters only an $O(|I'\Delta I|)$ collar of local terms. Hence on any fixed local-energy-bounded domain one has

$$\big|\langle\psi,(K[I']-K[I])\phi\rangle\big|\le C_{\psi,\phi,I_{\max}}\,|I'\Delta I|,$$

which is precisely the endpoint-Lipschitz / finite-variation control used by the compact-gauge branch in the null modular chain. This paper does *not* separately derive a second independent microscopic generator beyond this quasi-local branch generator; if a UV completion carries one, the only later requirement is that it lie in the same bounded-support algebraic closure so that the same propagation constants control the selected branch.

This local-Gibbs/Lieb–Robinson package, together with any Dobrushin-type estimate used by the compact-gauge branch, should be read only as fixed-cutoff collar-local recoverability, support control, and carried-error bookkeeping on the selected branch. One may impose such estimates on a chosen finite collar model without deciding the refinement-limit gauge phase. The package is therefore not a proof that the refinement limit lands in a trivial thermodynamic phase and not a proof that the realized branch is nontrivial. Refinement persistence and bosonic fiber descent for zero-obstruction sectors follow from Theorem 7.3 only on a cofinal tail carrying the compact-gauge refinement receipt. The local mixing estimate does not supply that receipt.

Internal refinement notion. Choose any family of refinement channels $\Phi_{\ell\to L}$ compatible with Axiom 3.7, including its refinement-closure clause. By that closure clause, the regulator-scale realized states are parameterized by one common finite-dimensional multiplier space λ, and on the realized branch one may write

$$\Phi_{\ell\to L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell\to L}(\lambda))$$

for an induced map $R_{\ell\to L}$ on that multiplier space. The existence of $R_{\ell\to L}$ as a self-map of the same finite-dimensional space is exactly the closure clause at work: interactions generated by coarse-graining outside the retained density list must vanish on the realized branch or be absorbed into it. Definition 3.8 and Lemma 3.9 below make this quantitative: the moment-matching I-projection onto the coarser family exists and is unique on the finite regulator, the closure clause is the statement that its relative-entropy defect vanishes along the realized branch, and a nonzero defect degrades expectation values by an explicit trace-norm residual; what this paper assumes rather than proves is the vanishing of that defect. The refinement-stable realized branch is therefore the persistent trajectory or invariant subset selected inside this finite-dimensional space, not an imported regularity package. This state-side refinement notion is enough to compare realized states across cutoffs. Any Dobrushin/local-mixing hypothesis used by the compact-gauge branch is logically separate: it supplies a collar estimate on a fixed finite-dimensional model, not information about the refinement-limit gauge phase. Whenever later sections speak of a "refinement-stable directed colimit" of zero-obstruction edge sectors, the monoidal-refinement/fiber clause is Theorem 7.3 together with Definition 7.2; Axiom 3.7 supplies only the realized state branch along which a certified sector witness may persist.

Definition 3.8 (Closure defect and induced refinement map)**.** *Fix regulator scales ℓ (finer) and L (coarser) with their finite realized algebras, let $\mathcal{E}_L = \{\omega_L(\lambda')\}$ be the homogeneous exponential family generated by the retained global-sum densities and optional global charges at scale L, and let $\Phi_{\ell\to L}$ be an admissible refinement channel. For fine-scale multipliers λ set $\sigma := \Phi_{\ell\to L}(\omega_\ell(\lambda))$ and define the* closure defect

$$\varepsilon_{\ell\to L}(\lambda) := \inf_{\lambda'} D(\sigma \,\|\, \omega_L(\lambda')),$$

with D the quantum relative entropy. When the infimum is attained at a unique λ^*, set $R_{\ell\to L}(\lambda) := \lambda^*$.

Lemma 3.9 (I-projection residual bound)**.** *On the finite regulator realization, assume the constrained operators $S_c \in \{\sum_x O_a(x)\}_{a=1}^{N_{\mathrm{con}}} \cup \{Q_b\}_{b=1}^{N_{\mathrm{glob}}}$ at scale L, together with $\mathbf{1}$, are linearly independent. Then: (i) $\lambda' \mapsto D(\sigma\|\omega_L(\lambda'))$ is smooth and strictly convex; for faithful σ the infimum is attained at a unique $\lambda^* = R_{\ell\to L}(\lambda)$, characterized by moment matching $\langle S_c\rangle_{\omega_L(\lambda^*)} = \langle S_c\rangle_\sigma$ for every constrained operator; (ii) $\left\|\sigma - \omega_L(R_{\ell\to L}(\lambda))\right\|_1 \le \sqrt{2\,\varepsilon_{\ell\to L}(\lambda)}$, so the family reproduces $\langle B\rangle_\sigma$ up to $\|B\|\sqrt{2\varepsilon_{\ell\to L}(\lambda)}$ for every bounded observable B; (iii) the refinement-closure clause of Axiom 3.7 is exactly the statement $\varepsilon_{\ell\to L}(\lambda) = 0$ along the realized branch, in which case $\sigma = \omega_L(R_{\ell\to L}(\lambda))$ and the induced map of the internal refinement notion above is this moment-matching I-projection.*

Proof. With $\log\omega_L(\lambda') = -\sum_c \lambda'_c S_c - \log Z_L(\lambda')\,\mathbf{1}$,

$$D(\sigma\|\omega_L(\lambda')) = -S(\sigma) + \sum_c \lambda'_c \langle S_c\rangle_\sigma + \log Z_L(\lambda').$$

$\log Z_L$ is smooth, and its Hessian is the Duhamel (Kubo–Mori) covariance matrix of the S_c in $\omega_L(\lambda')$, which is positive definite exactly when the S_c and $\mathbf{1}$ are linearly independent; this gives strict convexity, hence uniqueness of any minimizer, and the vanishing-gradient condition is the displayed moment matching. Attainment for faithful σ is finite-dimensional exponential-family duality: the moment map $\lambda' \mapsto (\langle S_c \rangle_{\omega_L(\lambda')})_c$ is a diffeomorphism onto the relative interior of the achievable moment set, which contains the moment vector of every faithful state [24]. Part (ii) is the quantum Pinsker inequality $D(\rho\|\tau) \geq \frac{1}{2}\|\rho - \tau\|_1^2$ [25] evaluated at λ^*. Part (iii) follows because the attained infimum vanishes iff $D(\sigma\|\omega_L(\lambda^*)) = 0$ iff $\sigma = \omega_L(\lambda^*)$, by strict positivity of relative entropy off the diagonal. □

A finite two-lattice acceptance check implementing this counting and projection package—the constraint/multiplier count against the displayed $(N_{\mathrm{con}} + N_{\mathrm{glob}})$-dimensional family, the moment-matching projection, the residual bound, a generic channel with strictly positive closure defect, and an exactly closed subfamily—is included with the repository sources (`code/maxent/`).

Central-interface collar clause (declared branch input). For every collar cut Σ used by the literal exact-Markov identities of this manuscript, the retained constraint family of Axiom 3.7 is declared to be *central-interface* in the sense of Theorem 5.15: every retained density whose support meets both half-collars of Σ acts through the boundary-charge (flux) functions in $\pi_L(Z(C^*(\widehat{K}_\Sigma)))$, while all remaining terms are one-sided $\widehat{K}_\Sigma$-invariant operators. This is an explicit axiom-level input of the declared branch, on the same footing as the refinement-closure clause above: it is *not* derived from overlap consistency. Under this clause, Theorem 5.15 derives the Markov-split alignment hypothesis (Definition 5.10) and exact collar Markovianity for the MaxEnt reference states, so the MSA-conditioned identities of this manuscript hold on the declared branch without a separate state hypothesis. Lattice-gauge-type regulators satisfy the clause manifestly, their interface energy being a function of the conserved flux; the Bell-pair state of Remark 5.11 violates it and marks the failure boundary. Whether overlap-consistent repair *forces* this clause—that is, excludes non-central cross-cut couplings rather than merely not selecting them—remains an open derivation target (Remark 5.16).

Axiom 3.10 (Recoverable Generalized Entropy)**.** *A generalized entropy functional exists on caps,*

$$S_{\mathrm{gen}}(C) = S_{\mathrm{bulk}}(C) + \langle L_C \rangle,$$

where L_C is a positive edge-center entropy functional. In the semiclassical scaling branch, its leading coarse-grained contribution is identified with $A(\partial C)/(4G)$. The functional obeys quantum focusing on null generators, and collar tripartitions have small CMI with controlled recovery maps [23, 27, 28, 29].

The small-CMI clause is exact on the declared central-interface branch by Theorem 5.15. On the finite-range noncommuting branch it has the explicit envelope of Theorem 5.3 only when the uniform strong conditional mixing premise is supplied. Outside those branches, recoverability remains an axiom-level input rather than a consequence of local Gibbs form.

Axiom 3.11 (Minimal Admissible Realization)**.** *Among realized sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are loop-coherent, anomaly-free, refinement-stable with light chiral matter, single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, intrinsically quark-sector CP-capable, and weak-sector UV-completable on that same one-Higgs branch, the realized package is the lexicographically minimal one under*

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}}, N_{\mathrm{nonab}}, N_c, N_g).$$

$\mathfrak{S}$ *is the sector package on which MAR acts; it is not the bare tensor category alone. MAR is therefore an explicit structural-economy axiom on admissible realized low-energy branches, not a theorem derived from the preceding axioms.* χ_{cpl} *is the coupled carrier dimension: the smallest unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is stronger than the abstract minimal faithful representation dimension.*

Definition 3.12 (MAR realization space and order)**.** *Let* $\mathfrak{A}_{\mathrm{MAR}}$ *be the set of isomorphism classes of finite low-energy sector packages*

$$\mathfrak{S} = (G^0, \mathcal{R}_{\mathrm{light}}, H, \mathcal{Y}, \mathcal{F})$$

on the ordinary or central zero-obstruction bosonic branch, together with the explicit one-Higgs chiral matter package used below. A package is MAR-admissible exactly when it satisfies the six predicates in Axiom 3.11. Two packages are physically equivalent when a compact-group isomorphism and a fiber-compatible symmetric monoidal equivalence preserve the observer-visible representations, Yukawa invariants, anomaly polynomial, normalized hypercharge lattice, and one-Higgs branch, modulo generation relabeling, charge-conjugation convention, gauge-center quotienting, implementation hiding, and inert ancillary stabilization. MAR orders packages lexicographically by $C(\mathfrak{S}) \in \mathbb{N}^4$ *and then quotients ties by this physical equivalence.*

Proposition 3.13 (Well-founded MAR minima)**.** *Every nonempty MAR-admissible class has at least one MAR-minimal package. The minimal packages are exactly those whose complexity vector is the lexicographically least element of* $C(\mathfrak{A}) \subseteq \mathbb{N}^4$ *for the chosen nonempty admissible class* $\mathfrak{A}$*.*

Proof. Lexicographic order on $\mathbb{N}^4$ is well-founded: minimize the first coordinate, then the second on that fiber, then the third, then the fourth. Each step minimizes a nonempty subset of $\mathbb{N}$. ☐

Remark 3.14 (Meaning of MAR uniqueness)**.** *MAR uniqueness means uniqueness of the observer-visible low-energy package modulo Definition 3.12. It is not uniqueness of a microscopic regulator representative. The later D8–D9 lemmas prove that, within the connected positive-dimensional Lie admissible class with one connected abelian factor and faithful action on the minimal coupled carrier, all MAR-minimal representatives realize the same connected SM gauge image, normalized hypercharge lattice, structural electroweak force content,* $N_c = 3$*, and* $N_g = 3$*.*

Remark 3.15 (Recovered-core theorem sources)**.** *The scaling/BW step is Theorem 6.13, the support-visible BW scaling theorem. The fixed-cutoff realized presentation, the local-Gibbs form, quasi-local Lieb–Robinson propagation, endpoint-Lipschitz interval control, the induced finite-dimensional refinement branch, the fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, and the operational/geometric split of Theorem 6.2 are established internally from Axioms 3.5–3.7. Transportability and the fixed-cutoff bosonic category are supplied by Theorems 5.22 and 5.26; the refinement/fiber ladder additionally requires Definition 7.2 before Theorem 7.3 applies. The realized compact-gauge witness is Theorem 7.35. The genuinely noncentral fixed-cutoff branch is controlled by Theorems 5.17–5.20:* $o^{(2)}_{\Sigma} = 0$ *means the orbit admits one or more strict* G_{Σ}*-valued* 1*-cocycle representatives and enters the compact-group transportable case only if at least one has trivial represented loop holonomy;* $o^{(2)}_{\Sigma} \neq 0$ *remains a higher-gauge sector and is not itself the ordinary compact-group reconstruction theorem. The full orbit* q_{Σ} *need not determine a unique ordinary* H^1 *class because higher-gauge edge changes can relate distinct strict representatives. Fermionic signs and chirality belong to the later fermionic/super-Tannakian matter lift. The* *classification/selection split is packaged in Theorem 7.8, and the observer-visible selected endpoint is Theorem 7.36.*

Remark 3.16 (Log convention)**.** *Unless an explicit base is written, logarithms and entropies in this paper use natural logs (nats). Base-2 logarithms are written as* $\log_2$ *and quoted in bits.*

Dependency DAG

The table below is the dependency map for this SM/GR derivation paper. The "Immediate OPH ingredients" column records only parents internal to the OPH program. Imported standard mathematics, branch-local constructions, and external inputs are stated separately instead of hidden inside the word "derived."

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Branch-local inputs / external data	Status
D1	Theorems 4.8, 4.11, 4.14, 4.15, and Corollary 4.16: total quotient repair maps $\mathrm{locRep}_\lambda, \mathrm{Rep}_\lambda : Q \to Q$, unique quotient normal form and schedule independence from a fixed initial quotient state, same-boundary uniqueness under unique consistent extension, layered $H_B \wedge H_{\mathrm{fib}}$, and nontrivial selected-fiber branch elimination	overlap-consistency problem on a finite patch net together with the declared fixed-cutoff collar recovery proposal, its touched-overlap local-fit eligibility contract, transactional snapshot/read/write validation, boundary/sector preservation, and restriction-compatible union-collar gluing for canonical conflict-component payloads on the physical quotient	local-to-global well-founded descent; local-diamond completion from atomic conflict-component commits with validation-complete read sets; Newman's lemma; layered functional boundary reconstruction; rooted functional extension in selected fibers	repair completeness; stated Petz support/CPTP control where that branch is used; for same-boundary uniqueness, boundary/sector preservation and either at most one consistent quotient extension in the boundary fiber or the layered/functional selected-fiber theorem; obstruction/ambiguity gates when no unique quotient endpoint exists; normal-form hashes as equality receipts, not selectors	structural theorem
D2	Theorems 5.3, 5.9, Propositions 5.33, 5.34, 5.40, Definition 5.41: collar Markov/entropy and conditional Gibbs-recovery layer	Axioms 3.5–3.10 plus strong conditional matrix mixing off the exact branch	matrix conditional-log-density mixing; HJPW Markov structure theorem; Petz and Fawzi–Renner recovery theory	the nonexact CMI bound requires the uniform finite-range and strong-mixing premises together with $\delta/\xi - \log \lvert\partial C\rvert_{\mathrm{UV}} \to +\infty$; ordinary clustering is insufficient; exact Markov identities require exact Markovity plus the Markov-split alignment hypothesis (Definition 5.10; both derived under the declared central-interface collar clause, Theorem 5.15); the $A/(4G)$ identification is the coarse-grained dictionary of Definition 5.41	conditional theorem with exact central-interface specialization

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Branch-local inputs / external data	Status
D3a	Regularized support-visible modular transport on fixed cap-local algebras	Axioms 3.5–3.10, collar Markov/recovery package, and Theorem 6.2	weak-$*$/GNS extraction; modular theory with regularized finite generators	fixed-collar comparison errors, cutoff schedule, and multiresolution reference tower; no full finite-algebra spectral floor is assumed	theorem
D3b	Finite cap-normal and support-order certificate	D3a plus a cofinal nondegenerate cap mesh	spherical cap geometry; de Sitter cap-normal compactness; finite support-order reflection	BW-framed caps, support-order faithfulness, mixed-GNS Cauchy residuals, and support covariance; this certificate is not produced by bare finite consensus inside D3, and on the producer branch it is produced from the computable incidence/mesh/cross-ratio/normalization receipts (Theorem 6.31), whose selection is proved underdetermined by confluence (Theorem 6.32)	Theorem 6.6 hypothesis; produced on the D3h receipt branch
D3c	Oriented cross-ratio and framed-cap rigidity	D3b	Möbius/cross-ratio rigidity on S^2	held-out separated quartets and an orientation witness; fitted anchors alone carry no validation value	theorem under certificate
D3d	Geometric 2π-KMS normalization	D3b–D3c	KMS uniqueness for faithful normal states	independently normalized geometric parameter $h_{\widehat{C}}(z) \mapsto e^{-s}h_{\widehat{C}}(z)$, finite strip bounds, and wrong-β separation	theorem under certificate
D3e	Theorem 6.13: support-visible BW cap automorphism $\sigma_t = \alpha_{\lambda_{\widehat{C}}(2\pi t)}$	D3a–D3d	Bisognano–Wichmann modular template at automorphism level	type-I generator form includes $K_C = 2\pi B_C + Z_C$; in the generic non-type-I case the automorphism identity is the theorem	corollary
D3f	Corollaries 6.40, 6.46: exact cap-normal Lorentz/H^3 chart, including $S^2 \simeq \mathbb{P}\mathcal{N}^+$, $\mathrm{Cap}^{\mathrm{or}}_{\mathrm{round}}(S^2) \simeq dS_3^{\mathrm{cap}}$, $n_{gC} = \Lambda_g n_C$, and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$	D3e plus oriented round-cap extraction and time orientation	projective future-null-cone geometry; $\mathrm{SL}(2,\mathbb{C})$ Hermitian-matrix realization; Lorentz homogeneous-space and hyperboloid geometry	a cap determines an H^3 plane/half-space, not a preferred observer point; population, neutral bulk, physical R_H, stress, and Einstein dynamics remain separate gates	exact conditional geometric theorem
D3g	Conditional record-populated H^3 localization (Corollary 6.64)	D3f plus record-conditioned modular cap responses, point-local source factorization, compact source domain, quantitative cap frame, bounded error, and residual optimization	hyperbolic cap half-spaces, finite conditioned frames, finite ε-nets, and residual inverse-stability estimates	point locality is a branch hypothesis or finite receipt; noisy finite uniqueness requires $\Delta_{\mathrm{loc}} > 0$; species, stress, neutral bulk, and Einstein entry are not implied	conditional theorem

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Branch-local inputs / external data	Status
D3h	Lemma 6.26; Theorems 6.29, 6.30, 6.31, 6.32: quotient-intrinsic geometry producer (incidence complex, produced S^2 topology, conformal/cap structure, certificate production) and the confluence-underdetermination no-go	D1 normal forms plus the spherical incidence, disk/mesh, modular cross-ratio, and 2π-normalization receipts	classification of closed combinatorial surfaces; Radó uniqueness; Möbius cross-ratio rigidity on S^2	receipts are decidable finite-stage predicates on normal forms; their selection is not implied by confluence (explicit T^2, $\partial\Delta^4$, wedge, and wrong-β countermodels); S^2 and its conformal class are derived only on the receipt branch	producer theorem plus underdetermination no-go
D4	Propositions 6.66, 6.69; Corollaries 6.67, 6.71; Theorem 6.70, Lemma 6.72; Theorem 6.73: null modular bridge to T_{kk}, including exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line families, the weak tail generator, the derived half-sided modular inclusion, the explicit positive half-line null-translation generator on its Stone domain with affine half-line modular relations, and the exact half-line generator/charge identification with the local null-stress charge	D2+D3 and Axioms 3.5–3.10	Borchers–Wiesbrock positive-generator theorem for standard half-sided inclusions; Stone's theorem; distributional differentiation on half-lines	the theorem-local null-cut center transfer, the inherited left/right strip-split package used for the spatial-collar-type tensor decomposition, and the exact-or-controlled Markov hypotheses of Proposition 6.66 and Corollary 6.67; quasi-local propagation and endpoint-Lipschitz control are supplied internally by Axiom 3.7 on the local finite-constraint branch; the geometric scaling action on the null half-line blow-up net derives the half-sided modular pair; bounded-interval formulas are discharged by Lemma 6.135; null-net standardness, the derived half-sided inclusion, Möbius bounded-interval covariance, Markov modular locality with its counterexample boundary, and the four-translation assembly are supplied by Theorems 6.77–6.84, with the Cyc, NTI, weak-additivity, MI, and kernel-residual receipts explicit	bridge theorem plus D4-standardness packet
D4b	§6.3 (Theorems 6.90, 6.92, 6.94, 6.95; Propositions 6.93, 6.96): conditional Lorentzian event manifold — quotient-intrinsic event classes, conditional four-dimensional atlas with signature $(-{+}{+}{+})$, causal order/time orientation, H^3 strictly as frame fiber, tetrads/metric/connection/curvature readout, glued modular clocks, countermodels	D3f–D3h and D4 outputs plus receipts $\mathsf{E1}$–$\mathsf{E4}$	bi-Lipschitz atlas arguments; convex-cone/quadric signature classification; Rademacher/Carathéodory regularity	$\mathsf{E1}$–$\mathsf{E4}$ with $\mathsf{E4}'$/smooth upgrade, MI/assembly branch, absolute conformal scale, stable causality, and record-Cauchy receipt remain explicit; countermodels show $\dim H^3 = 3$ promotes nothing by itself	conditional construction theorem

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Branch-local inputs / external data	Status
D4c	Local conserved stress tensor from modular charges: null tomography, constructed $\langle T_{ab}\rangle$ with symmetry/locality/covariance/conservation, normalization, generator/charge identification, weak Ward identity, ambiguity classification (Theorems 6.114, 6.116; Proposition 6.118)	D4/D4b outputs plus the MI/assembly branch, kernel-residual receipt, and universal-coupling receipt UC	null polarization/tomography of symmetric tensors; least-squares stability	ϕg_{ab} and improvement ambiguities explicit; UC is a physical-identification receipt; countermodel: per-direction generators without MI linearity admit no rank-two source; scalar CMI is never promoted	conditional construction theorem
D4d	Bulk/edge/central first law with the edge term carried exactly, algebraic type-I boundary statement, and MaxEnt stationarity extended to the coupled changing-stress class via the exact multiplier identity $dS/dt = \lambda = 2\pi$ (Theorems 6.119, 6.120; Proposition 6.121)	central-interface branch of Axiom 3.7, type-I generator form $K_C = 2\pi B_C + Z_C$, edge normalization $z_\alpha = \log d_\alpha$	finite first law $\delta S = \langle K \delta\rho \rangle$; MaxEnt envelope/Legendre identity	the Axiom-4 leading term and edge normalization are named branch axioms; shape/null-cut and metric variations are not covered; countermodels: wrong coefficient or wrong edge weights break stationarity by computable defects	exact finite theorem plus conditional bridge

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Branch-local inputs / external data	Status
D5	Theorem 6.99, Lemma 6.101, Theorems 6.105, 6.106, 6.107: Jacobson-type Einstein branch via a rest-frame relation plus tensor upgrade	D3+D4 and Axioms 3.5–3.10	fixed-volume area-variation identity for small geodesic balls; local quadratic-polarization argument for the tensor upgrade	Theorem 6.99, which fixes the admissible fixed-cap MaxEnt variation class and derives generalized-entropy stationarity on the realized cap-label-preserving MaxEnt family; the internal small-ball bridge of Lemma 6.101, which uses the geometric cap generator together with the D4 half-line generator/charge identification and Lemma 6.135; Theorem 6.105; locally Lorentzian $d = 4$ scaling regime, supplied conditionally by the D4b event-manifold packet under receipts E1–E4 (§6.3); small-ball constancy assumptions; Lemma 6.136 for $o(\ell^4)$ control; all local directions/reference states for the tensor upgrade via Lemma 6.137; the uniform scaling family, coverage, and absolute base condition are supplied by Theorems 6.122–6.125 under the VR/UC receipts, and the full chain is composed in Theorem 6.127 with realized-branch nonemptiness open (Remark 6.129)	scaling-limit theorem package plus composed branch-entry theorem
D6	Theorem 6.151: local/global theorem stack for Λ, namely the null-invisible metric ambiguity, the screen-capacity closure of the same Einstein branch, the de Sitter entropy relation with scale-certified static-patch display, and the cosmic record-closure readback fixed point, plus a legacy neutrino side estimate	D5 local Einstein recovery, which leaves the $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation, static-patch radius/time formulas after scale certification, finite self-closing normal-form counting/readback, and dimensional analysis	cosmic record-capacity fixed point $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$ and its observed de Sitter entropy readout	zero-input global closure

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Branch-local inputs / external data	Status
D7	Theorem 7.3, Theorem 7.6, Theorem 7.7: refinement-stable bosonic sector category and compact gauge reconstruction; Theorem A.22: conditional four-dimensional Euclidean Yang–Mills form; Theorem A.37: conditional support-visible compact-gauge Yang–Mills repair gap	Axioms 3.5–3.10 plus the compact-gauge refinement receipt	directed-colimit descent for monoidal C^*-categories; Doplicher–Roberts / Tannaka reconstruction; holonomy-to-curvature expansion; finite-stage conditional expectations and spectral comparison for commuting collar projections	Theorems 5.22 and 5.26 on the ordinary or central-defect bosonic zero-obstruction branch, plus Definition 7.2 for Theorem 7.3; Theorem 7.35 supplies realized MAR-admissible witness data, while D8–D9 contain Standard Model selection; the Yang–Mills statements additionally require the finite cylinder/projective extraction theorem and the renormalized Yang–Mills, finite ground-state-transform/cross-fiber, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts; the genuinely noncentral fixed-cutoff branch remains separate	conditional structural theorem
D8	Lemmas 7.10–7.15, Theorem 7.16: product gauge structure up to finite quotient	D7 + Axiom 3.11	compact Lie representation classification; Schur's lemma	the same ordinary or central-defect bosonic branch as D7, together with a connected positive-dimensional Lie admissible class, one connected abelian factor, and faithful action on the minimal coupled carrier	realized-branch theorem
D9	Theorem 7.36, Theorem 7.17, Corollaries 7.20, 7.21, 7.28, Proposition 7.24: realized Standard Model quotient, hypercharges, structural electroweak force content, $N_c = 3$, $N_g = 3$, and product-group corollaries	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting; stabilizer computation for a nonzero neutral Higgs vacuum vector	realized one-generation chiral matter plus one Higgs package, with the CP-capability and weak-sector UV clauses contained in Axiom 3.11; no extra post-MAR selector is used	realized-branch theorem/corollary chain
D10	forward gauge-coupling closure and declared electroweak readout of Section 7	D9 + pixel ratio P	printed RG evolution, matching, and scheme-conversion conventions	pixel constraint and the source-only forward transmutation solve $\mathcal{F}(\alpha_U; P) = 0$; the fixed-cutoff edge heat-kernel / Casimir theorem on the microphysics surface together with the compact-group / Peter–Weyl lift used on the D10 lane; printed beta-function, threshold, and scheme-conversion conventions	quantitative-closure sector

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Branch-local inputs / external data	Status
D12	Sections 8–10 continuations: charged leptons, H^3 record-worldline stitch certificates, strong CP, proton spin/lifetime, and controlled string/worldsheet effective-description branches	various subsets of D6, D9, and D10	branch-specific EFT and phenomeno-logical manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, texture choices, declared H^3 atlas and ID-independent cross-boundary assignment certificates, dark-sector response assumptions, discrete-horizon assumptions, a distinct large-N_{edge} regime with fixed $\tau = tN_{\text{edge}}$ window and uniform genus-remainder control, or additional worldsheet/CFT assumptions	phenomeno-logical continuation

The dependency DAG records the immediate internal parents, imported mathematics, branch-local constructions, and external data for each node.

Theorem 3.17 (Summary theorem for the recovered relativity-plus-Standard-Model core)**.** *This theorem packages the recovered-core nodes D1–D5 and D7–D9 of the dependency DAG above; the gravity-side chain through D3h, D4, D4b, D4c, and D4d is additionally composed, with partitioned inputs and a no-hidden-geometry audit, in the branch-entry theorem 6.127. Node D6 sits outside this recovered-core theorem because the cosmic record-capacity fixed point is the separate global closure of the Einstein branch recovered in item (iii); D10 and D12 are outside this theorem's scope. Assume Axioms 3.5–3.10, Axiom 3.11, a cofinal tail carrying the compact-gauge refinement receipt of Definition 7.2, and the hypotheses of Theorems 4.8, 4.11, 6.13, 6.73, 6.106, 7.3, 7.6, 7.7, 7.16, and 7.17, together with Corollaries 7.21, 7.20, and Proposition 7.24. Then, on the support-visible scaling branch of Theorem 6.13:*

(i) overlap repair admits a unique schedule-independent normal form from each fixed initial physical quotient state, and from fixed boundary/sector data when the consistent quotient extension in that fiber is unique;

(ii) cap modular flow on the extracted prime geometric cap pair is geometric and yields the connected Lorentz group

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1);$$

the associated observer-facing spatial chart is

$$H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3), \qquad \dim H^3 = 3;$$

(iii) the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family together with the null modular bridge and the bounded-interval kernel of Lemma 6.135 yield the Jacobson-type rest-frame relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle$$

at the cap center in the diamond rest frame; if that rest-frame relation holds for all local directions and reference states in the scaling regime, then the null-to-tensor upgrade gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle;$$

(iv) on the ordinary or central-defect bosonic zero-obstruction branch with the stated refinement receipt, the edge-sector category reconstructs a compact gauge group, and the realized connected gauge structure has the form

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ*.*

After the hypercharge lattice, color-count, generation-count, and trivial-action quotient steps, the finite quotient is fixed to $\Gamma = \mathbb{Z}_6$*, so the realized gauge structure on the realized MAR-admissible branch is*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

with the exact Standard Model hypercharge lattice on the realized matter package. The same package also fixes the structural electroweak force content: the weak $\mathrm{SU}(2)_L$ *factor, the hypercharge* $\mathrm{U}(1)_Y$ *factor, the nonzero neutral-Higgs vacuum-vector stabilizer* $\mathrm{U}(1)_Q$*, and the* $W^{\pm}/Z/A_Q$ *generator and connection directions. Here the count statements are internal to the same realized branch: the D8 minimal coupled-carrier theorem stack fixes the color triplet and hence* $N_c = 3$*; CKM phase counting and weak-sector asymptotic freedom then bound* N_g *on that same one-Higgs branch; MAR fixes the smallest realized value* $N_g = 3$*; and Witten's* $\mathrm{SU}(2)$ *anomaly is retained only as a consistency check on the resulting triplet-doublet package.*

Proof. Items (i)–(iii) are collected from Theorem 4.8, Corollaries 6.40 and 6.46, Theorem 6.106, and Theorem 6.107. On the gauge side, Theorem 5.22 supplies the strict zero-obstruction transport criterion and Theorem 5.26 constructs the fixed-cutoff bosonic collar-sector categories. Given the compact-gauge refinement receipt, Theorem 7.3 constructs the monoidal refinement transport and compatible forgetful fibers. Theorem 7.6 descends those data to the refinement-limit category and fiber functor, and compact gauge reconstruction is then Theorem 7.7. Theorem 7.8 separates this classification stage from the MAR selection stage. The realized nontrivial branch is supplied by Theorem 7.35. The product gauge structure up to finite quotient is Theorem 7.16; and the final identification of the exact Standard Model quotient and structural electroweak content is Theorem 7.36, Theorem 7.17 together with Corollaries 7.20, 7.21, 7.28, and Proposition 7.24. □

Theorem 3.17 therefore packages only the recovered relativity-plus-Standard-Model core: D1–D5 together with D7–D9. D6 is not a detached second gravity lane: it is the global closure of the same Einstein branch at the cosmic record-capacity fixed point. The quantitative-closure branch and phenomenological continuations are outside the theorem's scope.

Usually Postulated Elsewhere, Recovered or Continued Here with Explicit Status

Structure	Common treatment	OPH treatment
Lorentz kinematics and spatial dimension	background symmetry or starting axiom	scaling-limit branch from geometric modular flow on screen caps; the rest-space chart is $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ and has dimension 3
Einstein dynamics	fundamental field equation	first-variation relation with tensor upgrade from generalized entropy, null modular data, and derived fixed-cap generalized-entropy stationarity
Gauge group	model input	construction of the refinement-limit bosonic edge-sector category, then compact group reconstruction from it; exact SM quotient selected on the realized MAR-admissible branch with the realized one-generation/one-Higgs package
Hypercharge lattice	matter-assignment input	solved from anomaly cancellation and Yukawa invariance on the realized one-generation chiral matter plus one-Higgs package
Color and generation count	empirical input	color triplet fixed by the D8 minimal coupled carrier; generation count fixed by CKM phase counting, weak-sector asymptotic freedom, and MAR; Witten parity retained as a consistency check
Flavor hierarchy unit	model-dependent small parameter	phenomenological ansatz $\varepsilon = 1/6$ motivated by a uniform $\mathbb{Z}_6$ center-label ensemble
Charged-lepton continuation	Koide-style phenomenological fit	exact centered readback plus the closed common-shift no-go above $\widehat{C}_e^{\mathrm{cand}}$; any $\delta = 2/9$ phase relation is continuation-only
Cosmological constant	local vacuum-energy puzzle	dimensionless global screen-capacity relation fixed by $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$, with $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$ and the SI display supplied by the selected scale certificate

4 Observer Overlap Consistency as a Fixed-Point Problem

The discrete overlap problem is the mathematical skeleton of the framework.

Definition 4.1 (Patch Net)**.** *Let $G = (V, E)$ be a finite connected graph. Each vertex $i \in V$ carries a finite local state space S_i. For each edge $e = \{i, j\}$ let I_e be an interface alphabet and let*

$$\pi_{i,e} : S_i \to I_e, \qquad \pi_{j,e} : S_j \to I_e$$

be interface projections. The global state space is

$$\Sigma = \prod_{i \in V} S_i,$$

and the consistency set is

$$C = \{s \in \Sigma : \pi_{i,e}(s_i) = \pi_{j,e}(s_j) \text{ for all } e = \{i, j\} \in E\}.$$

This finiteness is a regulator-level assumption for the discrete patch-net normal-form theorem; it is not the continuum limit statement.

Define the inconsistency potential

$$\Phi(s) = \sum_{e=\{i,j\}\in E} w_e\, d_e\big(\pi_{i,e}(s_i), \pi_{j,e}(s_j)\big),$$

with $w_e > 0$ and $d_e(a,b) = 0$ iff $a = b$.

Let

$$q_{\mathrm{phys}} : \Sigma \to Q$$

be the physical quotient map identifying hidden representatives with the same observer-facing overlap data, and write

$$C_Q := q_{\mathrm{phys}}(C).$$

Primitive collar recoveries are required to descend to quotient proposals on Q. The accepted repair relation on Q is the transactional relation defined below. When no hidden representative quotient is present, take $Q = \Sigma$.

Remark 4.2 (Default code claim)**.** *At this level the overlap net is a finite constraint code and nothing stronger: the codewords are the states in C, equivalently the zero set $C = \Phi^{-1}(0)$. A graph min-cut, by itself, does not determine the distance of this code, since the same graph can carry trivial constant readouts with distance 1 or repetition constraints with distance $|V|$. QECC distance, topological-code resilience, Knill–Laflamme correction, exponential convergence, and BFT wall-clock liveness are therefore read only on the corresponding certified branches, not from the bare overlap graph.*

Definition 4.3 (Recovery-derived local repair law)**.** *A law λ is a family of local repair maps*

$$T_i^{\lambda} : \Sigma \to \Sigma$$

changing only patch i or a bounded neighborhood of i. On the fixed-cutoff collar branch these are not free rewrite primitives: each local update is only a proposal read from exact Markov splice or a declared Petz/Fawzi–Renner recovery move on a collar chart around i, then lifted back to the finite patch presentation. A proposal becomes physical only after the quotient transaction below validates its read snapshot, write support, preserved boundary/sector data, and strict descent. Parenthesization-independent union-collar gluing supplies candidate aggregate payloads; it does not by itself authorize two overlapping one-site commits.

Definition 4.4 (Transactional quotient repair branch)**.** *Fix a boundary/sector/holonomy record map $B : Q \to \mathcal{B}$ and a well-founded exact measure $\mu : Q \to (W, \prec)$, such as a lexicographic integer vector $(N_{\mathrm{hard}}, \Phi_Q, N_{\mathrm{unresolved}})$. A prepared repair transaction is a tuple*

$$\tau = (R_\tau, W_\tau, \sigma_\tau, p_\tau)$$

of a read set, write set, read snapshot, and quotient payload. It may commit at $x \in Q$ only when the snapshot is current on R_τ, the payload changes no register outside W_τ, B is preserved, and

$$\mu(\mathrm{Apply}_\tau(x)) \prec \mu(x).$$

A stale or aborted transaction is not a rewrite step. The read set is validation-complete for the measure: whenever a term of μ can change because a written register changes, that term's whole *support is read. For seam potentials this means reading both endpoints of every seam whose score can change.*

At a state x, form the conflict graph of enabled primitive repair proposals, with $\tau\#\sigma$ when either write set intersects the other's read-or-write set. Each connected component K is replaced by exactly one canonical aggregate transaction τ_K, computed on the union collar/component support. Primitive members of K do not commit separately. Write $x \Rightarrow y$ for a successful aggregate commit.

Definition 4.5 (Quotient repair operators)**.** *For a fixed finite transactional quotient repair branch, let A be the finite set of accepted aggregate transactions, each with a domain $D_a \subseteq Q$, equipped with a fixed total order $\prec_{\mathsf{A}}$. For $x \in Q$, write*

$$\mathsf{A}(x) = \{a \in \mathsf{A} : x \in D_a\}$$

for the enabled aggregate transactions at x. The local quotient repair map is

$$\mathrm{locRep}_\lambda(x) := \begin{cases} a_{\min}(x), & \mathsf{A}(x) \neq \varnothing, \\ x, & \mathsf{A}(x) = \varnothing, \end{cases}$$

where $a_{\min}$ is the $\prec_{\mathsf{A}}$-least enabled transaction. Iterating

$$x_0 = x, \qquad x_{n+1} = \mathrm{locRep}_\lambda(x_n)$$

reaches a least fixed stage because every nontrivial accepted step strictly descends in the finite set $\mu(Q)$. Define

$$\mathrm{Rep}_\lambda(x) := x_{N(x)}.$$

Proposition 4.6 (Local quotient repair closure)**.** *On any branch satisfying boundary preservation, exact descent, and repair completeness,*

$$\mathrm{locRep}_\lambda : Q \to Q$$

is total, preserves B, and either fixes x or strictly lowers μ. It satisfies

$$\mathrm{locRep}_\lambda(x) = x \iff x \in C_Q.$$

Proof. Finiteness and the fixed order on A give the least enabled transaction whenever one exists; if none exists, the definition returns x. Boundary preservation and exact descent are the transaction acceptance tests. Strict descent excludes a nontrivial fixed move, and repair completeness identifies states with no enabled aggregate transaction with C_Q. □

Lemma 4.7 (Validation support for local mismatch measures)**.** *Suppose the exact measure has local finite supports,*

$$\mu(x) = \sum_{a \in A} \mu_a(x|_{S_a}), \qquad S_a \subseteq D.$$

For a transaction writing $W \subseteq D$, only terms with $S_a \cap W \neq \varnothing$ can change. Thus the descent test is snapshot-local once the read set contains

$$R^\mu(W) := \bigcup_{a : S_a \cap W \neq \varnothing} S_a.$$

For an OPH edge mismatch potential this says that a seam transaction must read both endpoints of every overlap whose score may change under the write.

Proof. If $S_a \cap W = \varnothing$, the transaction leaves every argument of μ_a unchanged, so that term cancels in the pre/post comparison. Every remaining term is determined by the payload and the restriction to $R^\mu(W)$. The edge-mismatch statement is the specialization in which each term is supported on the two endpoint registers of one overlap. $\square$

Theorem 4.8 (Transactional quotient normal-form uniqueness)**.** *Assume:*

(i) *finite transactional descent: $x \Rightarrow y$ implies $\mu(y) \prec \mu(x)$;*

(ii) *atomic conflict-component commits as in Definition 4.4;*

(iii) *repair completeness: terminal quotient states for $\Rightarrow$ are exactly C_Q.*

Then every initial quotient state $x \in Q$ has a unique terminal normal form

$$N_Q(x) = \mathrm{Rep}_\lambda(x) \in C_Q,$$

and $N_Q(x)$ is independent of asynchronous repair order. The map

$$\mathrm{Rep}_\lambda : Q \to Q$$

is total, boundary-preserving, and idempotent, and

$$\mathrm{Rep}_\lambda(x) = x \iff x \in C_Q.$$

Proof. Well-founded descent gives termination. For a one-step peak $y \Leftarrow x \Rightarrow z$, let the two commits come from conflict components K and L. If $K = L$, canonical aggregation gives the same aggregate payload, hence $y = z$. If $K \neq L$, the components have disjoint read/write supports. Snapshot validity, unchanged-outside-write validation, and validation-complete read sets preserve each transaction after the other commits; applying them in either order gives the same quotient state. Thus the aggregate relation has the local diamond. Newman's lemma [8] says that a terminating locally confluent rewrite system is confluent. If two terminal states y, z are reachable from the same x, confluence gives a common descendant w. Since y and z are terminal, $y = w = z$. Repair completeness identifies the terminal state with an element of C_Q, so $N_Q : Q \to C_Q$ is well defined and schedule-independent. The canonical iteration in Definition 4.5 is one valid accepted repair execution, hence $\mathrm{Rep}_\lambda(x) = N_Q(x)$. Boundary preservation follows by induction along accepted transactions. Idempotence and the fixed-point characterization follow from Proposition 4.6, since every element of C_Q has no enabled aggregate transaction. $\square$

Remark 4.9 (Confluence is an extra obligation)**.** *The descent argument above proves termination only. OPH does not infer uniqueness from termination. Uniqueness and order-independence enter through the atomic conflict-component local diamond on the physical quotient, Newman's lemma, and repair completeness. Receipts named seam descent, atomic commit, distributed local diamond, repair completeness, and normal-form hash are evidence contracts for this theorem surface, not substitutes for these hypotheses and not selectors among physically distinct minimizing quotient states. If two accepted schedules from the same initial state terminate in different observer-facing quotient normal forms, with no declared holonomy/higher-gauge obstruction and no mere hidden-representative difference, then the proposed repair law fails the fixed-cutoff consensus criterion.*

Corollary 4.10 (Schedule Independence)**.** *If observables factor through Q and are evaluated on $N_Q(q_{\mathrm{phys}}(s))$, the resulting physical law is independent of update order from the fixed initial quotient state $q_{\mathrm{phys}}(s)$.*

Theorem 4.11 (Boundary-conditioned uniqueness)**.** *Let*

$$B : Q \to \mathcal{B}$$

record fixed external boundary data, conserved charge, root packet, holonomy sector, or task input. Assume B is preserved by accepted repairs:

$$x \Rightarrow y \implies B(x) = B(y).$$

If for each $b \in \mathcal{B}$ the consistent quotient fiber

$$C_b := \{x \in C_Q : B(x) = b\}$$

has at most one element, then all initial quotient states with boundary value b settle to the same observer-facing normal form. Equivalently, $B(x) = B(x')$ implies

$$N_Q(x) = N_Q(x').$$

Proof. By Theorem 4.8, $N_Q(x)$ and $N_Q(x')$ exist and lie in C_Q. Boundary preservation along repair sequences gives $B(N_Q(x)) = B(x)$ and $B(N_Q(x')) = B(x')$. If $B(x) = B(x') = b$, then both normal forms lie in C_b. The unique consistent extension assumption says C_b has at most one element, hence the two normal forms are equal. □

Remark 4.12 (Generic cross-source criterion)**.** *For an observation-preserving repair relation whose normal forms are exactly the consistent states, Ref. [1] proves that agreement, modulo any declared silent equivalence, of normal endpoints reached from every pair of same-boundary sources is equivalent to injectivity of the induced boundary map on the consistent quotient. This cross-source criterion is distinct from same-source confluence and from liveness. In this paper, Theorem 4.8 supplies the same-source endpoint result, while Theorem 4.11 applies only after the consistent boundary fiber is proved singleton. The layered and functional results discharge that premise only on their named finite carrier branches.*

Definition 4.13 (Layered functional boundary carrier)**.** *A layered functional boundary carrier is a finite directed layered graph*

$$V = L_0 \sqcup L_1 \sqcup \cdots \sqcup L_D, \qquad D \geq 2,$$

with boundary layer L_0, finite alphabets A_v, nonempty parent sets

$$P(v) \subseteq L_0 \sqcup \cdots \sqcup L_{d-1} \qquad (v \in L_d,\ d \geq 1),$$

and deterministic local rules

$$F_v : \prod_{u \in P(v)} A_u \to A_v.$$

Let

$$Q = \prod_{v \in V} A_v, \qquad B(a) = a|_{L_0}.$$

For boundary data b, define $E(b) \in Q$ recursively by $E(b)_v = b_v$ on L_0 and

$$E(b)_v = F_v((E(b)_u)_{u \in P(v)}) \qquad (v \in L_d,\ d \geq 1).$$

Optional cross-check predicates may be added. A state is consistent when all functional equations and cross-check predicates pass; let C_Q be the consistent state set. A boundary b is admissible when $E(b) \in C_Q$. The layer repair map R_d rewrites exactly layer L_d to the displayed functional value and fixes all other layers.

Theorem 4.14 (Layered carrier proves $H_B \wedge H_{\text{fib}}$)**.** *For a layered functional boundary carrier, set*

$$a^{(0)} = a, \qquad a^{(d)} = R_d(a^{(d-1)}), \quad d = 1, \ldots, D,$$

and let $b = B(a)$. Then

$$B(a^{(d)}) = b \qquad (d = 0, \ldots, D).$$

If b is admissible, then $a^{(D)} = E(b) \in C_Q$, and

$$C_Q \cap B^{-1}(b) = \{E(b)\}.$$

Consequently, on any accepted quotient repair presentation whose aggregate transactions include the layer repairs and whose boundary value $b = B(x)$ is admissible,

$$\mathrm{Rep}_\lambda(x) = E(B(x)).$$

Proof. Each R_d writes only L_d with $d \geq 1$, so no stage changes B. Induct on d: after stage d, every vertex in $L_0 \sqcup \cdots \sqcup L_d$ agrees with $E(b)$, because all parents of a layer-d vertex lie in earlier layers. Thus $a^{(D)} = E(b)$, and admissibility puts it in C_Q. If $c \in C_Q \cap B^{-1}(b)$, the same induction applied to the functional equations for c gives $c = E(b)$. The final equality follows because Theorem 4.8 makes $\mathrm{Rep}_\lambda(x)$ a boundary-preserving element of C_Q. □

Theorem 4.15 (Functional selected-fiber uniqueness)**.** *Let a same-boundary quotient fiber $Q_b := B^{-1}(b)$ be presented on a rooted finite packet tree T with root r. Suppose the boundary fixes the root value $u_r = \beta(b)$, and for each non-root vertex v there is a deterministic extension map*

$$f_v : X_{p(v)} \times \mathcal{B} \to X_v.$$

Define u_b recursively by $u_v = f_v(u_{p(v)}, b)$. If all non-tree overlaps, sector checks, and holonomy checks pass on u_b, then $C_b = \{u_b\}$. If any such check fails, then $C_b = \varnothing$.

Proof. Tree edges force each vertex value recursively from the root and b, so any consistent state in the fiber must equal u_b on every tree vertex. The remaining equations are exactly the non-tree overlap, sector, and holonomy checks. If they pass, u_b is the single consistent extension. If one fails, no tree-compatible candidate satisfies all consistency equations. □

Corollary 4.16 (Nontrivial branch elimination)**.** *Assume Theorem 4.8, boundary preservation, and Theorem 4.15. If $|Q_b| \geq 2$ and $C_b = \{u_b\}$, then every candidate state in Q_b normalizes to u_b, while $Q_b \setminus C_b$ is nonempty and is eliminated by repair. Surviving candidates therefore share one quotient normal form. If the selected fiber is empty the branch is obstructed, and if a union solver finds two physically distinct consistent quotient endpoints the branch is ambiguous instead of hash-selected.*

Theorem 4.17 (Cycle Obstruction / Holonomy Criterion)**.** *Let A be an abelian group. For each oriented edge $e : u \to v$ assign $b_e \in A$ and consider the affine overlap equations*

$$x_v - x_u = b_e.$$

A global solution exists if and only if the signed sum of b_e vanishes on every cycle.

Proof. Summing the edge equations around a cycle gives necessity. For sufficiency, fix a root and define each x_v by summing labels along a path from the root to v. Vanishing cycle sums make this independent of the chosen path. □

Definition 4.18 (Physical repair law and representative lift)**.** *Suppose a local gauge group $\Gamma = \prod_i \Gamma_i$ acts on Σ while leaving all interface data invariant. Write*

$$q : \Sigma \to \Sigma/\Gamma, \qquad q(s) = [s].$$

A physical repair law is the family of local quotient maps induced by the recovery-derived collar updates just described:

$$\overline{T}_i^\lambda : \Sigma/\Gamma \to \Sigma/\Gamma$$

on the overlap-invariant quotient. A representative repair family is any family

$$T_i^\lambda : \Sigma \to \Sigma$$

with

$$q \circ T_i^\lambda = \overline{T}_i^\lambda \circ q.$$

Proposition 4.19 (Representative lifts descend to the quotient)**.** *For any representative repair family of Definition 4.18,*

$$q(T_i^\lambda(\gamma \cdot s)) = q(T_i^\lambda(s)) \qquad \forall\, i,\ \forall\, \gamma \in \Gamma,\ \forall\, s \in \Sigma.$$

In particular the repair relation descends to Σ/Γ without any extra gauge-covariance axiom.

Proof. Because $q(\gamma \cdot s) = q(s)$,

$$q(T_i^\lambda(\gamma \cdot s)) = \overline{T}_i^\lambda(q(\gamma \cdot s)) = \overline{T}_i^\lambda(q(s)) = q(T_i^\lambda(s)).$$

So gauge-equivalent inputs induce the same repaired physical state. □

Proposition 4.20 (Gauge Quotient)**.** *Under Definition 4.18, Proposition 4.19, and Theorem 4.8 with $Q = \Sigma/\Gamma$, the normal-form map is a quotient map:*

$$\overline{N}_\lambda : \Sigma/\Gamma \to q(C), \qquad [s] \mapsto N_Q([s]).$$

Hence gauge-invariant observables are unique on the quotient, and the fixed-cutoff physical observable algebra has representative-independent terminal expectations on that carrier.

Proof. By Proposition 4.19, every repair step has quotient image determined only by the current orbit. Iterating, the quotient image of any repair sequence depends only on the initial orbit. Theorem 4.8 gives a unique terminal quotient normal form $N_Q([s])$ from that orbit. This makes $\overline{N}_\lambda$ well-defined. □

Corollary 4.21 (Repair respects gauge)**.** *Define the quotient-valued representative repair map*

$$\mathrm{Rep}_\lambda^\Sigma := \mathrm{Rep}_\lambda \circ q : \Sigma \to \Sigma/\Gamma.$$

Then for every $\gamma \in \Gamma$ and $s \in \Sigma$,

$$\mathrm{Rep}_\lambda^\Sigma(\gamma \cdot s) = \mathrm{Rep}_\lambda^\Sigma(s).$$

Consequently every physical observable $M : \Sigma/\Gamma \to Y$ has the same repaired value on gauge-equivalent representatives.

Proof. The quotient map satisfies $q(\gamma \cdot s) = q(s)$. Applying Rep_λ gives the displayed equality, and applying M gives the observable statement. □

On the fixed-cutoff quantum lift, the same quotient-local carrier does more than fix the terminal orbit: for every declared fixed-cutoff physical algebra, the terminal expectation functional is the same on any two representative lifts whose regional collar data lie in one quotient-local glued state, even when the microscopic representatives differ by gauge or sector relabelings on that carrier.

Theorem 4.22 (Refinement-limit consensus classes)**.** *Let R be a directed cofinal refinement set. For each $r \in R$, let $Q_r = \Sigma_r/\Gamma_r$ be the finite physical quotient state space, let $n_r : Q_r \to Q_r$ be the finite quotient normal-form map, and let $h_r : Q_r \to \mathcal{H}_r$ be the finite holonomy or higher-gauge obstruction map. Suppose that for $r \preceq s$ there are restriction maps*

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r$$

forming directed inverse systems and satisfying

$$\rho_{sr} n_s = n_r \rho_{sr}, \qquad \chi_{sr} h_s = h_r \rho_{sr}.$$

Assume visible separation: compatible families in the inverse limits are equal whenever they agree on a cofinal subset of finite stages. Then

$$n_\infty((x_r)_r) = (n_r(x_r))_r, \qquad h_\infty((x_r)_r) = (h_r(x_r))_r$$

define a unique schedule-independent refinement-limit normal-form class and a refinement-limit holonomy class. The finite normal forms and holonomies converge to those classes in the inverse-limit topology. A nonzero limiting holonomy has a finite-stage witness, and agreement of the normal-form and holonomy projections on a cofinal tail gives the same refinement-limit consensus class.

Proof. Compatibility of the x_r gives $\rho_{sr}(x_s) = x_r$. Normal-form naturality then gives

$$\rho_{sr}(n_s(x_s)) = n_r(\rho_{sr}(x_s)) = n_r(x_r),$$

so the normal forms form a compatible inverse-limit family. Holonomy naturality gives the same calculation for $h_s(x_s)$. Finite-stage schedule independence comes from Theorem 4.8; visible separation upgrades agreement of all cofinal finite projections to uniqueness of the inverse-limit class. A nonzero inverse-limit holonomy must have a nonzero projection at some finite stage by visible separation. □

Proposition 4.23 (Coarse-graining / reconciliation compatibility)**.** *Let $r \preceq s$ be two refinement stages in the D1 consensus system, with coarse-graining maps*

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r.$$

Equip Q_r and $\mathcal{H}_r$ with the pseudometrics used for macroscopic readout at stage r. If the chosen coarse-graining channel has normal-form and obstruction defects ε^n_{sr} and ε^h_{sr}, meaning

$$d^Q_r(\rho_{sr} n_s(x), n_r \rho_{sr}(x)) \le \varepsilon^n_{sr}, \qquad d^{\mathcal{H}}_r(\chi_{sr} h_s(x), h_r \rho_{sr}(x)) \le \varepsilon^h_{sr}$$

for every $x \in Q_s$, then reconciling at stage s and coarse-graining to r gives the same macroscopic normal-form and obstruction readout as coarse-graining first and reconciling at r, up to

$$\max\{\varepsilon^n_{sr}, \varepsilon^h_{sr}\}.$$

In the exact separated cofinal system of Theorem 4.22, these defects are zero. If the defects vanish on cofinal tails, the two procedures define the same macroscopic inverse-limit consensus class.

Proof. The displayed inequalities are exactly the two components of the claimed product-readout bound. Exact naturality in Theorem 4.22 is the zero-defect case. Cofinal vanishing gives convergence of every fixed coarse cylinder value in the inverse-limit topology. □

This removes the extra gauge-covariance axiom and keeps repair on its actual quotient-local carrier: recovery dynamics on fixed-cutoff collars. On the declared fixed-cutoff branch, the theorem-local branch condition is repair completeness, together with the Petz support/CPTP clause where that branch is used. The touched-overlap local-fit contract supplies Lyapunov Φ-descent on accepted moves, and the support-local commutation and union-collar compatibility package belongs to the declared repair law itself. Stability of that package under refinement or branch change is handled by the compatibility clauses of Theorem 4.22 when a separated cofinal refinement system is supplied. Proposition 4.23 is the corresponding RG-facing statement: it does not say that arbitrary coarse-graining maps commute with repair, only that the selected coarse-graining branch commutes with reconciliation to the extent that the displayed defects are controlled.

Definition 4.24 (Quotient-visible neutral geometry certificate)**.** *For each shard s, let Σ_s be the raw record space and let Γ_s be the presentation groupoid generated by declared gauge-representative changes, local port relabelings, and other nonphysical presentation moves. Write*

$$q_s : \Sigma_s \to \overline{Q}_s := \Sigma_s/\Gamma_s, \qquad X_s := n_s(\overline{Q}_s)$$

for the terminal visible chart obtained by applying the schedule-independent normal-form map. Geometry is read only on the X_s, never from raw rows or intermediate repair states.

A neutral-bulk atlas consists of interface domains $U_{st} \subseteq X_s$, $U_{ts} \subseteq X_t$, and bijections

$$\tau_{ts} : U_{st} \to U_{ts}$$

with $\tau_{ss} = \mathrm{id}$, $\tau_{st} = \tau_{ts}^{-1}$, and $\tau_{us} = \tau_{ut}\tau_{ts}$ whenever the composite is defined. On graph-shaped shard systems the certificate also records zero closed-path holonomy. A channel registry $\mathcal{C}$ assigns to each channel c a metric space (Z_c, d_c), a weight $w_c > 0$, and local features $F_{s,c} : D_{s,c} \subseteq X_s \to Z_c$. The channel is quotient-visible when domain membership and values are transported by τ:

$$x \in D_{s,c} \iff \tau_{ts}x \in D_{t,c}, \qquad F_{t,c}(\tau_{ts}x) = F_{s,c}(x).$$

Approximate certificates replace the last equality by a reported channel defect $\eta_{ts,c}$. The evidence bundle must report the corresponding path-sum transport bound.

Theorem 4.25 (Quotient chart descent and neutral metric)**.** *Assume the atlas laws of Definition 4.24. Let*

$$Q_{\mathrm{vis}} := \left(\bigsqcup_s X_s\right) / \sim_\tau$$

where $x \sim_\tau y$ when an admissible interface path transports x to y. Then $\sim_\tau$ is an equivalence relation, the canonical maps $\iota_s : X_s \to Q_{\mathrm{vis}}$ satisfy $\iota_t\tau_{ts} = \iota_s$, and zero closed-path holonomy makes each ι_s injective. Every compatible family of maps $G_s : X_s \to Y$ factors uniquely through Q_{vis}.

In particular, each quotient-visible feature $F_{s,c}$ descends to a unique partial feature

$$F_c : Q_{\mathrm{vis}} \dashrightarrow Z_c.$$

For a declared complete geometry channel set $\mathcal{C}_\star$, let

$$Q_\star = \{x \in Q_{\mathrm{vis}} : F_c(x) \text{ exists for all } c \in \mathcal{C}_\star\}$$

and, for $p \geq 1$, define

$$d_{\mathrm{neu}}(x,y) = \left[\sum_{c \in \mathcal{C}_\star} w_c \, d_c(F_c(x), F_c(y))^p \right]^{1/p}.$$

Then d_{neu} is independent of shard representative and is a pseudometric on $Q_\star$. It is a metric on $Q_\star/\equiv_F$, where $x \equiv_F y$ means all declared channels agree, and it is a metric directly on $Q_\star$ only if those channels jointly separate points.

Proof. Reflexivity, symmetry, and transitivity of $\sim_\tau$ follow from the identity, inverse, and composition atlas laws. The canonical-map identity is the definition of the generated quotient. If $\iota_s(x) = \iota_s(y)$, then y is obtained from x by a closed path; zero holonomy gives $x = y$. The universal property follows by defining $G([x]) = G_s(x)$ and using interface compatibility along a connecting path. Applying this channel by channel gives feature descent. The metric formula is well defined by that descent. Nonnegativity and symmetry are channelwise, and the triangle inequality is Minkowski's inequality applied to the weighted channel distances. Zero distance is exactly $\equiv_F$; quotienting by that relation, or proving joint separation, supplies identity of indiscernibles. □

Remark 4.26 (Missingness and presentation invariance)**.** *Pairwise "compare whatever channels overlap" is not a metric policy: three points can satisfy $d(A,B) = 0$, $d(B,C) = 0$, and $d(A,C) > 0$ when each pair shares a different channel. OPH therefore permits only complete-case comparison, a fixed missing symbol whose mask is itself quotient-visible, or a train-only imputation map labelled as a metric on imputed representations. Gauge changes, port relabelings, observer-row permutations, repair schedules, and shard partitions preserve the neutral metric only when they induce a bijection Θ on Q_{vis} and channel isometries I_c with*

$$F'_c(\Theta x) = I_c(F_c(x)).$$

Then the displayed product formula gives $d'(\Theta x, \Theta y) = d(x,y)$. A common-refinement proof of partition neutrality is the same statement with Θ induced by equal fibers over the refinement. A refinement-limit neutral metric additionally requires a cofinally vanishing tail modulus for the finite-stage distances; it is not automatic from running more cells.

Proposition 4.27 (Finite Euclidean and batch-held-out certificates)**.** *For a finite sample with distance matrix D, set*

$$H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \qquad B = -\frac{1}{2}H(D^{\circ 2})H.$$

The sampled metric is exactly Euclidean iff $B \succeq 0$, and the smallest exact Euclidean dimension is $\operatorname{rank}(B)$. Noisy Euclidean claims must report negative-eigenvalue mass, positive/effective rank, and held-out stress; a low-dimensional plot is not a certificate.

Statistical claims split independent generative batches before preprocessing:

$$\mathcal{B}_{\mathrm{train}} \sqcup \mathcal{B}_{\mathrm{val}} \sqcup \mathcal{B}_{\mathrm{test}}.$$

All chart alignment, scaling, channel weights, imputation, graph construction, dimension selection, and thresholds are fitted on train/validation data only. If a bounded batch-level test loss $\ell(\widehat{\theta}; B) \in [0,1]$ is evaluated once on independent test batches, then, conditioning on training and validation,

$$\left| \widehat{R}_{\mathrm{test}} - R(\widehat{\theta}) \right| \leq \sqrt{\frac{\log(2/\delta)}{2m_{\mathrm{test}}}}$$

with probability at least $1 - \delta$. Shared seeds, shard batches, boundary conditions, trajectory families, duplicates, descendants, or repeated test-set inspection block this certificate.

Definition 4.28 (Ancilla stabilization)**.** *Fix a finite-cutoff OPH realization*

$$\mathfrak{U} = (\{\mathcal{H}_P, \mathcal{A}(P), \omega_P\}_{P\in\mathcal{P}}, \Gamma, T^\lambda).$$

Choose finite-dimensional ancillary factors K_P with product state $\eta = \bigotimes_{P\in\mathcal{P}} \eta_P$. The associated ancilla stabilization is the realization

$$\mathfrak{U}^\eta = (\{\mathcal{H}_P^\eta, \mathcal{A}^\eta(P), \omega_P^\eta\}_{P\in\mathcal{P}}, \Gamma, (T^\lambda)^\eta),$$

with

$$\mathcal{H}_P^\eta := \mathcal{H}_P \otimes K_P, \qquad \mathcal{A}^\eta(P) := \mathcal{A}(P) \otimes \mathbf{1}_{K_P}, \qquad \omega_P^\eta := \omega_P \otimes \eta_P,$$

and lifted repair dynamics acting trivially on the ancillas.

Proposition 4.29 (Ancilla-stable UV underdetermination)**.** *Let $\mathfrak{U}^\eta$ be an ancilla stabilization of a finite-cutoff OPH realization $\mathfrak{U}$. Then:*

1. *observable expectations on the physical subalgebras are unchanged:*

$$\omega_P^\eta(a \otimes \mathbf{1}_{K_P}) = \omega_P(a) \qquad \forall\, a \in \mathcal{A}(P);$$

2. *the interacting local MaxEnt branch on the physical subalgebra is unchanged, since*

$$\omega^\eta_{\ell_{\mathrm{UV}}}(\lambda) = \omega_{\ell_{\mathrm{UV}}}(\lambda) \otimes \eta;$$

3. *for every collar split $A : B : D$,*

$$I(A : D \mid B)_{\rho\otimes\eta} = I(A : D \mid B)_\rho,$$

   so the Fawzi–Renner remainder $r_{\mathrm{FR}}(\varepsilon)$ and the collar Markov modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ are unchanged;

4. *if the ancillas are inert under repair, then they remain hidden representative data and the quotient normal form on physical observables is identical:*

$$N_{Q^\eta}\Big(q^\eta_{\mathrm{phys}}(s \otimes k)\Big) = N_Q(q_{\mathrm{phys}}(s))\,;$$

5. *if the ancillas carry only trivial neutral sectors, the MAR-selected realized sector package is unchanged.*

Hence $\mathfrak{U}$ and $\mathfrak{U}^\eta$ are OPH-indistinguishable although they are different microscopic regulator realizations.

Proof. Item 1 is immediate from the definition of $\mathcal{A}^\eta(P)$ and ω_P^η. Item 2 is the product-state form of the same fixed-cutoff MaxEnt branch. Item 3 follows from additivity of entropy for product ancillas, which cancels in the conditional mutual-information combination and therefore leaves both r_{FR} and δ^{M} unchanged. Item 4 holds because the lifted repair maps do not act on the ancillary factors. Item 5 is immediate when the ancillas are neutral and carry no extra realized sector data. □

Definition 4.30 (OPH-stable UV equivalence)**.** *Two finite-cutoff realizations $\mathfrak{U}$ and $\mathfrak{U}'$ are* OPH-stably equivalent*, written*

$$\mathfrak{U} \sim_{\mathrm{OPH}} \mathfrak{U}',$$

if after finite ancilla stabilizations they are related by local gauge-covariant *-isomorphisms intertwining the observable patch net, overlap maps, local states, and repair dynamics on the physical subalgebras.

Corollary 4.31 (Unique physical UV branch only modulo OPH-stable equivalence)**.** *Under the OPH axiom language, the UV invariant determined by the theory is the class* $[\mathfrak{U}]_{\mathrm{OPH}}$*, not a unique microscopic regulator presentation. Literal microscopic UV uniqueness is therefore not an OPH invariant.*

Proof. Propositions 4.19 and 4.20 fix the schedule-independent physical branch on the quotient, while Proposition 4.29 shows that inert ancillary refinements leave every OPH observable invariant. So the physical branch is fixed only modulo $\sim_{\mathrm{OPH}}$, not at the level of one microscopic representative. □

These three results encode much of the eventual physical interpretation. Objectivity becomes confluence, gauge symmetry becomes quotient invariance induced by the physical overlap algebra, and stable defects are represented by nontrivial overlap holonomy classes. The physically relevant uniqueness statement is therefore quotient uniqueness together with ancilla-stable equivalence: OPH fixes a unique gauge-invariant physical branch modulo boundary redundancy, implementation hiding, and inert ancillary stabilization, not a unique microscopic regulator presentation.

5 Collars, Edge Centers, and Generalized Entropy

The fixed-point picture explains why overlap consistency produces gauge quotients: once repair is read on the physical overlap algebra, descent to the quotient is automatic. The gravity and consensus arguments later use a more specific collar fact: after edge-center completion, interior observables become insensitive to compatible exterior substitutions, and modular additivity becomes exact in the Markov normal form. The point of this section is therefore twofold. First, at fixed regulator scale, we derive the collar block decomposition from overlap consistency itself on the ordinary or central-defect branch and then state its genuinely noncentral higher-gauge replacement. Second, we state precisely when the approximate recoverability clause of Axiom 3.10 is allowed to converge to the exact HJPW normal form used by the later spatial and null collar theorems.

The sharp claim boundary is as follows. Small conditional mutual information always gives a constructive recovered comparison state with $O(\varepsilon^{1/2})$ observable error. It does *not* by itself give a universal one-shot trace-norm bound to an exact Markov state. The exact Markov normal form is recovered either when $I(A : D \mid B) = 0$ holds literally, or for a controlled family on one fixed finite-dimensional collar model, or after pullback to such a model, where $\varepsilon \to 0$. On that fixed collar one gets a collar-dependent modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$ measuring distance to the exact Markov set, and this is the error that must be carried into later exact splice or modular-additivity identities. A separate boundary, independent of any modulus, is that the HJPW normal form factorizes the state over a *state-dependent* decomposition of the middle system: identifying its factors with the preselected edge-center factors $b^{\alpha}_{L}, b^{\alpha}_{R}$ is the Markov-split alignment hypothesis (Definition 5.10), which exact Markovity does not imply (Remark 5.11) and which is carried explicitly wherever the preselected factors are used. On the declared central-interface branch both alignment and exact collar Markovianity are derived rather than assumed (Theorem 5.15).

The quantitative route off that exact branch needs a stronger premise than ordinary covariance clustering. The definitions below isolate the premise, keep all constants visible, and separate the finite receipt from the scaling statement.

Definition 5.1 (Finite-range collar Gibbs family)**.** *At regulator scale* $\ell > 0$*, let* $G_\ell = (V_\ell, E_\ell)$ *be the cell-adjacency graph, with* $\dim \mathcal{H}_x \leq q$ *and interaction degree at most* Δ_0*. In each fixed* *superselection sector, let*

$$\rho_\ell = Z_\ell^{-1} e^{-\beta H_\ell}, \qquad H_\ell = \sum_{X \subset V_\ell} \Phi_\ell(X) + H_\ell^{\mathrm{cen}},$$

where H_ℓ^{cen} is central and therefore scalar on the chosen sector. Assume constants $r_0, J_0 < \infty$, independent of the cut and of ℓ, such that

$$\operatorname{diam}_{G_\ell}(X) \le r_0, \qquad \sup_{x \in V_\ell} \sum_{X \ni x} \|\beta \Phi_\ell(X)\| \le J_0.$$

A noncentral all-to-all constraint term fails this finite-range premise; it cannot be hidden in H_ℓ^{cen}.

For a cut $C_\ell \subset V_\ell$, let

$$\partial_{r_0}^{\mathrm{UV}} C_\ell := \{x \in V_\ell : d_{G_\ell}(x, C_\ell) \le r_0, d_{G_\ell}(x, C_\ell^c) \le r_0\}, \qquad |\partial C_\ell|_{\mathrm{UV}} := |\partial_{r_0}^{\mathrm{UV}} C_\ell|.$$

Let $A_\delta : B_\delta : D_\delta$ be a tripartition in which B_δ shields $A_\delta \subset C_\ell$ from $D_\delta \subset C_\ell^c$ by m_ℓ graph layers, with $m_\ell > r_0$, and define the resolved physical collar width by $\delta := m_\ell \ell$.

Definition 5.2 (Strong conditional Gibbs exponential mixing)**.** *For the faithful Gibbs state of Definition 5.1, embed all marginal logarithms in the tripartite algebra and define the matrix conditional log-density defect*

$$\mathbf{J}_\rho(A : D \mid B) := \log \rho_{ABD} + \log \rho_B - \log \rho_{AB} - \log \rho_{BD}.$$

The family has strong conditional Gibbs exponential mixing with constants $\kappa < \infty$ and $\zeta > 0$ when, for every admitted cut and collar, it has a boundary-anchor expansion

$$\mathbf{J}_\rho(A_\delta : D_\delta \mid B_\delta) = \sum_{z \in \partial_{r_0}^{\mathrm{UV}} C_\ell} E_{z,\ell,\delta}, \qquad \|E_{z,\ell,\delta}\|_\infty \le \kappa e^{-(m_\ell - r_0)/\zeta}.$$

The constants are uniform in ℓ, the cut, the boundary condition, and the collar in the declared family. This is a conditional or matrix mixing condition. A two-point estimate of the form

$$|\omega(XY) - \omega(X)\omega(Y)| \le K\|X\|\|Y\|e^{-d(\operatorname{supp} X, \operatorname{supp} Y)/\zeta}$$

does not imply Definition 5.2 for a general noncommuting Gibbs state. Local Gibbs form and ordinary exponential clustering therefore cannot discharge this premise by themselves.

Theorem 5.3 (Collar CMI decay from finite-range conditional Gibbs mixing)**.** *Under Definitions 5.1 and 5.2,*

$$\boxed{I(A_\delta : D_\delta \mid B_\delta)_{\rho_\ell} \le \kappa |\partial C_\ell|_{\mathrm{UV}} e^{-(m_\ell - r_0)/\zeta} \le c|\partial C_\ell|_{\mathrm{UV}} e^{-\delta/\xi_\ell}}$$

with the explicit choices

$$\xi_\ell := \zeta \ell, \qquad c := \kappa e^{r_0/\zeta}.$$

Thus c and ξ_ℓ/ℓ depend only on the declared uniform mixing constants and the finite interaction range; q, Δ_0, J_0, β enter only through an independently proved or received pair (κ, ζ).

Proof. Every finite Gibbs state is faithful, and so are its marginals. Expanding the four entropy terms inside the common tripartite trace gives the exact identity

$$I(A : D \mid B)_\rho = \mathrm{Tr}[\rho_{ABD}\, \mathbf{J}_\rho(A : D \mid B)]\,.$$

Strong subadditivity gives nonnegativity. The boundary expansion, trace-norm duality, and the triangle inequality give

$$I(A_\delta : D_\delta \mid B_\delta)_{\rho_\ell} \le \|\mathbf{J}_{\rho_\ell}(A_\delta : D_\delta \mid B_\delta)\|_\infty \le \kappa |\partial C_\ell|_{\mathrm{UV}} e^{-(m_\ell - r_0)/\zeta}.$$

Since $\delta = m_\ell \ell$, the last expression equals $\kappa e^{r_0/\zeta} |\partial C_\ell|_{\mathrm{UV}} e^{-\delta/(\zeta\ell)}$, which is the displayed bound. $\square$

Corollary 5.4 (Sharp double scaling and recovery error)**.** *Let $\ell_n \downarrow 0$ and $\delta_n \downarrow 0$ lie in one family with the uniform constants of Theorem 5.3. Then*

$$\Gamma_n := \frac{\delta_n}{\xi_{\ell_n}} - \log |\partial C_{\ell_n}|_{\mathrm{UV}} \longrightarrow +\infty \quad \Longrightarrow \quad I(A_{\delta_n} : D_{\delta_n} \mid B_{\delta_n}) \longrightarrow 0.$$

If, for constants $a, p, \ell_0 > 0$,

$$|\partial C_\ell|_{\mathrm{UV}} \le a(\ell_0/\ell)^p, \qquad \delta(\ell) \ge \zeta(p + \eta)\ell \log(\ell_0/\ell) \quad (\eta > 0),$$

then $\delta(\ell) \to 0$, $\delta(\ell)/\ell \to \infty$, and

$$I(A_\delta : D_\delta \mid B_\delta) \le ca(\ell/\ell_0)^\eta \longrightarrow 0.$$

For a smooth cut on the two-dimensional screen, $p = 1$. The Fawzi–Renner map may be chosen with the certified trace-distance error

$$r_{\mathrm{FR}}(\ell, \delta) \le 2\sqrt{1 - e^{-\varepsilon_{\ell,\delta}}} \le 2\sqrt{\varepsilon_{\ell,\delta}}, \qquad \varepsilon_{\ell,\delta} := c|\partial C_\ell|_{\mathrm{UV}} e^{-\delta/\xi_\ell}.$$

Proof. Taking logarithms of the theorem bound gives

$$\log I \le \log c + \log |\partial C_\ell|_{\mathrm{UV}} - \delta/\xi_\ell = \log c - \Gamma.$$

This tends to $-\infty$ under the first condition. The polynomial boundary estimate and the displayed schedule give $\Gamma \ge \eta \log(\ell_0/\ell) - \log a$, which proves the explicit rate. The recovery estimate is Proposition 5.33 applied with the theorem envelope. $\square$

Remark 5.5 (The ratio $\delta/\ell \to \infty$ is not the rate condition)**.** *The bare double-scaling ratio does not force the boundary-prefactored bound to vanish. Set $\ell_n = \ell_0 e^{-n^2}$, $\delta_n = \zeta n \ell_n$, and $|\partial C_{\ell_n}|_{\mathrm{UV}} = e^{n^2}$. Then $\ell_n, \delta_n \to 0$ and $\delta_n/\ell_n = \zeta n \to \infty$, while the theorem envelope is proportional to $e^{n^2 - n}$. The condition in Corollary 5.4, or direct verification of the complete product, is load-bearing.*

Definition 5.6 (Finite collar-decay receipt)**.** *A theorem-aligned finite-stage receipt records*

- R1. *the regulator graph hash, local dimensions, β, interaction supports and norms, r_0, J_0, Δ_0, Gibbs reconstruction residual, and the treatment of central sector terms;*
- R2. *the tripartition hash, graph-layer or geodesic collar width, ℓ, $|\partial C|_{\mathrm{UV}}$, and the separator check $m_\ell > r_0$;*
- R3. *direct regional entropies and CMI in nats, their numerical error bound and spectral floor, together with the matrix defect norm and the boundary-expansion tail where it is computed;*

R4. *predeclared κ, ζ, boundary-condition and held-out-cut coverage, the log envelope*

$$L_{\ell,\delta} := \log \kappa + \log |\partial C|_{\mathrm{UV}} - (m_\ell - r_0)/\zeta,$$

its slack against the measured upper error bar, and the Fawzi–Renner trace-error envelope;

R5. *on a tower, one common cap and interaction family, uniform constants, and the rate margin $\Gamma = \delta/(\zeta\ell) - \log |\partial C|_{\mathrm{UV}}$.*

A fitted correlation length from the same CMI rows is diagnostic. A finite list of increasing rate margins is a scaling proxy. The cofinal conclusion requires an analytic schedule such as the one in Corollary 5.4, or an independent uniform proof; no finite sample proves a limit. A local random-triplet or diagonal packet CMI is likewise diagnostic and does not substitute for the regional density-matrix receipt.

Remark 5.7 (Claim status of the collar theorem)**.** *On the declared central-interface branch, Theorem 5.15 gives $I(A : D \mid B) = 0$ exactly. On a general noncommuting finite-range branch, Theorem 5.3 is conditional on the strong matrix mixing premise of Definition 5.2. It proves recoverability with constants and the sharp scaling schedule; it does not derive strong conditional mixing from bare finite OPH consensus, ordinary two-point clustering, or local Gibbs form. The scalar CMI and its recovery error are not promoted to a rank-two stress tensor or a dark-sector source.*

Proposition 5.8 (Derived regulator gluing datum)**.** *Fix a regulator-scale finite patch cover of a cap neighborhood and let R be any finite union of regulator cells. Assume only the finite patch-net presentation implicit in Definition 4.1 together with overlap consistency on common interfaces. Then:*

1. *each regulator cell i with finite local state space S_i can be Hilbertized as $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{|S_i|}$, so the extended algebra before overlap quotienting is the finite type-I algebra*

$$\widetilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R := \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i;$$

2. *for each connected boundary component $\Sigma \subset \partial R$, after choosing a reference cut presentation $\tilde{\mathcal{H}}_\Sigma$, every overlap-consistent change of local chart along Σ is implemented by a unitary on $\tilde{\mathcal{H}}_\Sigma$, and the compact closure of the subgroup generated by all such recharting unitaries is a compact boundary redundancy group*

$$K_\Sigma \subset U(\tilde{\mathcal{H}}_\Sigma);$$

before the choice of reference chart, the same data form a compact unitary groupoid of overlap-preserving transitions;

3. *if triple-overlap defects are central, the projective composition law of item 2 lifts to a direct action of a compact central extension $\widehat{K}_\Sigma$; a genuinely noncentral defect is the only obstruction to reducing the transition system to an ordinary compact group action;*

4. *in the invariant-state realization of the quotient, the chart-independent endomorphisms of the lifted presentation form the boundary-invariant algebra*

$$\mathcal{A}_{\mathrm{inv}}(R) = \widetilde{\mathcal{A}}(R)^{\widehat{K}_{\partial R}}, \qquad \widehat{K}_{\partial R} := \prod_{\Sigma \subset \partial R} \widehat{K}_\Sigma,$$

with the convention $\widehat{K}_\Sigma = K_\Sigma$ whenever no central extension is needed; the physical state space is the corresponding invariant subspace, and on collars the sector-preserving algebra induced on that subspace is the block-diagonal algebra used in Theorem 5.9.

Items 1 and the lifted fixed-point presentation of item 4 are the fixed-cutoff realized data used by the compact-gauge branch; they are not extra axioms beyond the finite regulator presentation plus overlap redundancy.

Proof sketch. Item 1 is the Hilbertization of finite sets. For item 2, each overlap-consistent recharting acts on a finite-dimensional matrix algebra, and every *-automorphism of such an algebra is inner. After transporting all local overlap presentations to a chosen reference chart, each recharting is therefore implemented by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group, which yields K_Σ; without fixing the reference chart one has the corresponding compact unitary groupoid. Item 3 is the usual projective-versus-central-extension lift for a central 2-cocycle. Item 4 records the lifted fixed-point presentation: chart-independent endomorphisms of the unreduced boundary data are exactly the fixed points of the derived boundary action, while the collar theorem uses the invariant-state realization and the sector-preserving algebra induced on the matched-cut subspace. If the defect is genuinely noncentral, the quotient is well defined, but the correct bookkeeping object is the crossed-module or higher-gauge data instead of an ordinary compact group. □

Theorem 5.9 (Derived edge-center collar decomposition and exact Markov normal form)**.** *Let A-B-D be a collar tripartition realized at fixed regulator scale on the ordinary or central-defect branch of Proposition 5.8. Write $B = B_L \cup B_R$ with common interface $\Sigma = \partial C$, and let $\widehat{K}_\Sigma$ be the derived compact boundary action attached to that cut, with $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. In the invariant-state realization,*

$$\mathcal{H}_B = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma}.$$

Decompose the left boundary data as

$$\tilde{\mathcal{H}}_{B_L} \cong \bigoplus_\alpha W_\alpha \otimes \mathcal{H}_{b_L^\alpha},$$

with W_α irreducible $\widehat{K}_\Sigma$-modules. Because the right half-collar carries the inverse overlap transport across the same cut, it decomposes contragrediently:

$$\tilde{\mathcal{H}}_{B_R} \cong \bigoplus_\beta W_\beta^* \otimes \mathcal{H}_{b_R^\beta}.$$

Then

$$\mathcal{H}_B \cong \bigoplus_\alpha \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha},$$

and the sector-preserving collar algebra

$$\mathcal{A}_{\mathrm{EC}}(B) := \bigoplus_\alpha \mathcal{B}(\mathcal{H}_{b_L^\alpha}) \otimes \mathcal{B}(\mathcal{H}_{b_R^\alpha})$$

has center

$$Z(\mathcal{A}_{\mathrm{EC}}(B)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

If, in addition, the reference state on $A \cup B \cup D$ is exact Markov,

$$I(A : D|B)_\rho = 0,$$

and satisfies the Markov-split alignment hypothesis of Definition 5.10 below, then the state decomposes as

$$\rho_{ABD} = \bigoplus_\alpha p_\alpha\, \rho^{(\alpha)}_{Ab_L^\alpha} \otimes \rho^{(\alpha)}_{b_R^\alpha D}.$$

The same formula holds in any limit state that is exact Markov and split-aligned on this fixed collar model. Thus the collar-center block decomposition is forced by overlap consistency plus the derived regulator package on the ordinary or central-defect branch; exact Markov factorization over the preselected edge factors is an additional state condition—exact Markovity together with *split alignment—and neither part is a consequence of the block decomposition alone. Exact Markovity by itself does not imply alignment (Remark 5.11).*

Proof. By Proposition 5.8, one may realize the physical collar states as the $\widehat{K}_\Sigma$-invariant subspace of $\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R}$. Complete reducibility of finite-dimensional unitary representations gives the displayed decomposition of $\tilde{\mathcal{H}}_{B_L}$, and the right half-collar carries the inverse transport law across the same cut, hence the contragredient decomposition of $\tilde{\mathcal{H}}_{B_R}$. Therefore

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*) \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

Taking $\widehat{K}_\Sigma$-invariants gives

$$\mathcal{H}_B \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

By Schur's lemma,

$$(W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

which yields the displayed block decomposition. The sector-preserving collar algebra therefore has the displayed block-diagonal form, so its center is generated by the block projectors. Within each block, observables from $A \cup B$ act only on the left factor and observables from $B \cup D$ act only on the right factor. If the state is exact Markov, the HJPW structure theorem [30] gives a blockwise factorized normal form over *some* decomposition $\mathcal{H}_B \cong \bigoplus_j \mathcal{H}_{\hat{b}_L^j} \otimes \mathcal{H}_{\hat{b}_R^j}$ of the conditioning system. That conclusion is existential: the decomposition it produces is state-dependent and need not agree with the preselected edge-center factors b_L^α, b_R^α constructed above (Remark 5.11). The Markov-split alignment hypothesis of Definition 5.10 states exactly that the HJPW decomposition can be chosen to be the edge-center decomposition; under that hypothesis the displayed normal form over b_L^α, b_R^α follows. The same formula is used in the idealized recoverability limit when exact identities are taken literally, with alignment carried as part of the limit hypothesis. ☐

Definition 5.10 (EC-aligned exact Markov states and the Markov-split alignment hypothesis)**.** *Fix one collar model with the edge-center decomposition $\mathcal{H}_B \cong \bigoplus_\alpha \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}$ of Theorem 5.9. A state σ_{ABD} is* EC-aligned exact Markov *if*

$$\sigma_{ABD} = \bigoplus_\alpha q_\alpha \, \sigma^{(\alpha)}_{Ab_L^\alpha} \otimes \sigma^{(\alpha)}_{b_R^\alpha D}$$

with respect to the preselected edge factors; write $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ for the set of such states. Every EC-aligned exact Markov state satisfies $I(A : D \mid B) = 0$. A reference state is said to satisfy the Markov-split alignment hypothesis *(MSA) on this collar model when it lies in $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$; equivalently, when one HJPW decomposition of $\mathcal{H}_B$ for the state can be chosen to be the edge-center decomposition itself, with the factor coupled to A equal to b_L^α and the factor coupled to D equal to b_R^α in every sector. The intended mechanism for verifying MSA in a concrete collar model is a commuting square of state-preserving conditional expectations onto the one-sided edge algebras $\bigoplus_\alpha \mathcal{B}(\mathcal{H}_{b_L^\alpha}) \otimes \mathbf{1}$*

and $\bigoplus_\alpha \mathbf{1} \otimes \mathcal{B}(\mathcal{H}_{b_R^\alpha})$*; Proposition 5.12 proves that this mechanism, a modular-splitting condition, and a blockwise entropic condition are all equivalent to MSA, and Theorem 5.15 derives MSA from the regulator package on the declared central-interface branch. Off that branch, wherever an exact identity below uses the preselected factors, MSA is an explicit hypothesis.*

Remark 5.11 (Exact Markovity does not imply alignment: a Bell-pair counterexample)**.** *The HJPW theorem is existential, and the inclusion* $\mathfrak{M}^{\mathrm{EC}}_{A:B:D} \subseteq \mathfrak{M}_{A:B:D}$ *into the exact Markov set of Definition 5.30 is strict in general. Take four qubits ordered* A, B_L, B_R, D *with a single center block,* $\mathcal{H}_{b_L} = \mathcal{H}_{B_L}$*,* $\mathcal{H}_{b_R} = \mathcal{H}_{B_R}$*, and*

$$\rho = \Phi_{AB_R} \otimes \Phi_{B_L D},$$

where Φ *denotes a maximally entangled Bell pair. Direct computation gives* $S(AB) = S(BD) = \ln 2$*,* $S(B) = 2\ln 2$*, and* $S(ABD) = 0$*, hence* $I(A : D \mid B)_\rho = 0$*: the state is exactly Markov. But* $I(A : B_R)_\rho = 2\ln 2 \neq 0$*, whereas any state of the EC-aligned form* $\rho_{Ab_L} \otimes \rho_{b_R D}$ *has* $I(A : B_R) = 0$*. For this state the HJPW decomposition is the transposed split, pairing* A *with* B_R *and* D *with* B_L*. Consequently no vanishing modulus of the form* $\delta^{\mathrm{M}}(\varepsilon)$ *can control the distance to* $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$*: this state has* $I(A : D \mid B) = 0$ *at fixed nonzero trace distance from the EC-aligned class. Small conditional mutual information therefore never implies approximate EC-aligned factorization by itself; alignment must be hypothesized or derived separately.*

The alignment hypothesis is not merely a name: it admits equivalent operational characterizations that make it checkable state by state and localize exactly what an axiom-side derivation must supply. Define the one-sided collar algebras extended by the exterior regions,

$$\mathcal{M}_L := \bigoplus_\alpha \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_{b_L^\alpha}) \otimes \mathbf{1}_{b_R^\alpha D}, \qquad \mathcal{M}_R := \bigoplus_\alpha \mathbf{1}_{Ab_L^\alpha} \otimes \mathcal{B}(\mathcal{H}_{b_R^\alpha} \otimes \mathcal{H}_D).$$

Elements of $\mathcal{M}_L$ and $\mathcal{M}_R$ commute, both algebras contain the center projectors P_α, and $\mathcal{M}_L \cap \mathcal{M}_R = \bigoplus_\alpha \mathbb{C}\, P_\alpha$.

Proposition 5.12 (Operational characterizations of Markov-split alignment)**.** *Let* ρ_{ABD} *be a faithful state on the collar model of Theorem 5.9. The following are equivalent:*

1. $\rho \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$*, i.e.* ρ *satisfies MSA (Definition 5.10);*
2. (modular splitting) $\log \rho \in \mathcal{M}_L + \mathcal{M}_R$*;*
3. (conditional expectation / commuting square) *there exists a* ρ*-preserving conditional expectation* $E_L : \mathcal{B}(\mathcal{H}_{ABD}) \to \mathcal{M}_L$*; equivalently, one onto* $\mathcal{M}_R$*; in that case the pair* (E_L, E_R) *forms a commuting square over the center algebra* $\bigoplus_\alpha \mathbb{C}\, P_\alpha$*;*
4. (entropic) $[\rho, P_\alpha] = 0$ *for all* α*, and* $I(Ab_L^\alpha : b_R^\alpha D)_{\rho^{(\alpha)}} = 0$ *for every block with* $p_\alpha > 0$*, where* $\rho^{(\alpha)} := p_\alpha^{-1} P_\alpha \rho P_\alpha$*.*

Each condition implies $I(A : D \mid B)_\rho = 0$*. Items 1 and 4 are equivalent without faithfulness. The Bell-pair state of Remark 5.11 violates item 4, since* $I(Ab_L : b_R D) \geq I(A : B_R) = 2\ln 2 > 0$ *on its single block, as the general theory requires.*

Proof. $1 \Leftrightarrow 4$: mutual information vanishes iff the state is a product across the named cut, so item 4 says precisely that ρ is block-diagonal with each block a product across $(Ab_L^\alpha) : (b_R^\alpha D)$, which is the aligned normal form. No faithfulness is used.

$1 \Rightarrow 2$: for the aligned normal form, $\log \rho = \bigoplus_\alpha \big[(\log p_\alpha) P_\alpha + \log \rho^{(\alpha)}_{Ab_L^\alpha} \otimes \mathbf{1} + \mathbf{1} \otimes \log \rho^{(\alpha)}_{b_R^\alpha D}\big]$, and the central term lies in $\mathcal{M}_L \cap \mathcal{M}_R$.

$2 \Rightarrow 1$: write $\log\rho = K_L + K_R$ with $K_L \in \mathcal{M}_L$, $K_R \in \mathcal{M}_R$; these commute, are block-diagonal, and have blockwise form $K_L = h_L^\alpha \otimes \mathbf{1}$, $K_R = \mathbf{1} \otimes h_R^\alpha$. Hence $\rho = \bigoplus_\alpha e^{h_L^\alpha} \otimes e^{h_R^\alpha}$, which after normalization is the aligned form.

$1 \Rightarrow 3$: define the blockwise slice map

$$E_L(x) := \bigoplus_\alpha \Big[(\mathrm{id}_{Ab_L^\alpha} \otimes \omega_R^{(\alpha)})(P_\alpha x P_\alpha) \Big] \otimes \mathbf{1}_{b_R^\alpha D}, \qquad \omega_R^{(\alpha)} := \mathrm{Tr}[\,\cdot\,\rho_{b_R^\alpha D}^{(\alpha)}].$$

This is a unital completely positive idempotent onto $\mathcal{M}_L$ with the bimodule property, and $\rho \circ E_L = \rho$ follows from the blockwise product form. The analogous E_R exists, and $E_L E_R = E_R E_L$ is the expectation onto $\bigoplus_\alpha \mathbb{C}\,P_\alpha$ determined by the block weights, which is the asserted commuting square.

$3 \Rightarrow 2$: by Takesaki's theorem [26] (see also [27, 28]), a ρ-preserving conditional expectation onto $\mathcal{M}_L$ exists iff the modular group $\sigma_t^\rho = \mathrm{Ad}\,\rho^{it}$ satisfies $\sigma_t^\rho(\mathcal{M}_L) = \mathcal{M}_L$ for all t. The flow σ_t^ρ then permutes the finitely many minimal projectors of $Z(\mathcal{M}_L) = \bigoplus_\alpha \mathbb{C}\,P_\alpha$ and is norm-continuous with $\sigma_0^\rho = \mathrm{id}$, so it fixes each P_α; hence $[\rho, P_\alpha] = 0$ and $\rho = \bigoplus_\alpha p_\alpha \rho^{(\alpha)}$. Fix a block and regard $\rho^{(\alpha)}$ as a faithful state on $\mathcal{H}_{Ab_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha D}$. Differentiating $\sigma_t^\rho(x \otimes \mathbf{1}) \in \mathcal{B}(\mathcal{H}_{Ab_L^\alpha}) \otimes \mathbf{1}$ at $t = 0$ gives $[\log\rho^{(\alpha)}, x \otimes \mathbf{1}] \in \mathcal{B}(\mathcal{H}_{Ab_L^\alpha}) \otimes \mathbf{1}$ for every x. Expand $\log\rho^{(\alpha)} = \sum_i a_i \otimes b_i$ with $\{b_i\}$ linearly independent and $b_0 = \mathbf{1}$; then $[\log\rho^{(\alpha)}, x \otimes \mathbf{1}] = \sum_i [a_i, x] \otimes b_i$, and membership in $\mathcal{B}(\mathcal{H}_{Ab_L^\alpha}) \otimes \mathbf{1}$ forces $[a_i, x] = 0$ for all x and all $i \neq 0$. Hence $a_i \in \mathbb{C}\mathbf{1}$ for $i \neq 0$, so $\log\rho^{(\alpha)} = a_0 \otimes \mathbf{1} + \mathbf{1} \otimes c$ for some self-adjoint c, which is the blockwise modular splitting; summing blocks gives item 2.

Finally, item 1 implies $I(A:D \mid B)_\rho = 0$ by the direct blockwise computation of Theorem 5.9, and the Bell-pair state violates item 4 by monotonicity of mutual information under partial trace. □

Corollary 5.13 (Axiom-side reduction of the alignment problem)**.** *On the local MaxEnt branch the reference state is Gibbs, $\omega \propto e^{-H_{\mathrm{eff}}}$, and Proposition 5.12 (item 2) gives: ω satisfies MSA on a given collar iff*

$$H_{\mathrm{eff}} = H_L + H_R + H_Z, \qquad H_L \in \mathcal{M}_L, \quad H_R \in \mathcal{M}_R, \quad H_Z \in \bigoplus_\alpha \mathbb{C}\,P_\alpha,$$

i.e. iff every coupling between the two half-collars acts through the edge-center sector label. Deriving MSA from the axioms is therefore equivalent *to proving that overlap-consistent repair forces all cross-cut terms of H_{eff} to be central on the declared branch. This holds manifestly in lattice-gauge-type regulators, where the interface energy is a function of the conserved flux through the cut (a central observable), and fails for generic non-central cross-cut couplings, of which the Bell-pair state is the extreme case. Theorem 5.15 below turns this into a derivation of MSA on the declared central-interface branch; off that branch, MSA remains an explicit hypothesis wherever it is used.*

The reduction of Corollary 5.13 can be discharged for a structurally characterized class of regulator packages. The relevant descent computation is representation-theoretic and deserves separate statement.

Lemma 5.14 (Descent of one-sided invariants and boundary charges)**.** *Work in the invariant-state realization of Proposition 5.8, with $\tilde{\mathcal{H}}_{B_L} \cong \bigoplus_\alpha W_\alpha \otimes \mathcal{H}_{b_L^\alpha}$, $\tilde{\mathcal{H}}_{B_R} \cong \bigoplus_\beta W_\beta^* \otimes \mathcal{H}_{b_R^\beta}$, and $\mathcal{H}_B = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}$. Then:*

1. *every $\widehat{K}_\Sigma$-invariant operator $\tilde{x}_L$ on $\mathcal{H}_A \otimes \tilde{\mathcal{H}}_{B_L}$ has the form $\bigoplus_\alpha \mathbf{1}_{W_\alpha} \otimes x_L^\alpha$ with $x_L^\alpha \in \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_{b_L^\alpha})$; the operator $\tilde{x}_L \otimes \mathbf{1}$ preserves the invariant subspace and acts on it as $\bigoplus_\alpha x_L^\alpha \otimes \mathbf{1}_{b_R^\alpha D} \in \mathcal{M}_L$. The mirror statement holds for right-sided invariants and $\mathcal{M}_R$;*

2. *every element $\tilde{z}$ of the image $\pi_L(Z(C^*(\widehat{K_\Sigma})))$ of the center of the group (C^*-)algebra under the left boundary action—in particular every function of the quadratic Casimir of $\widehat{K_\Sigma}$, i.e. every function of the flux through Σ—acts on $W_\alpha \otimes \mathcal{H}_{b_L^\alpha}$ as $\chi_\alpha(\tilde{z})\,\mathbf{1}$ by the central character χ_α, hence descends on $\mathcal{H}_B$ to the central operator $\bigoplus_\alpha \chi_\alpha(\tilde{z})P_\alpha \in \bigoplus_\alpha \mathbb{C}\,P_\alpha$. For Casimir/flux functions the same central operator is obtained through the right action, because the Casimir eigenvalue agrees on W_α and its contragredient W_α^* and the two boundary charges are matched blockwise on the invariant subspace.*

Proof. Item 1 is the commutation theorem plus Schur's lemma: the commutant of the unitary action $\mathbf{1}_A \otimes (\bigoplus_\alpha \pi_\alpha \otimes \mathbf{1}_{b_L^\alpha})$ is $\bigoplus_\alpha \mathbf{1}_{W_\alpha} \otimes \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_{b_L^\alpha})$ (up to the obvious factor reordering), because the π_α are pairwise inequivalent irreducibles, so no cross-isotypic intertwiners exist and within each isotypic component the commutant of $\pi_\alpha \otimes \mathbf{1}$ is $\mathbf{1}_{W_\alpha} \otimes$ (full multiplicity algebra). Tensoring with $\mathbf{1}$ on the right half-collar commutes with the diagonal action, so the invariant subspace is preserved, and on $(W_\alpha \otimes W_\alpha^*)^{\widehat{K}_\Sigma} \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}) \cong \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}$ the induced action is $x_L^\alpha \otimes \mathbf{1}$. Item 2: central elements of the group algebra act in any irreducible π_α by scalars $\chi_\alpha(\tilde{z})$, so the left action descends to $\bigoplus_\alpha \chi_\alpha(\tilde{z})P_\alpha$. For a Casimir/flux function the contragredient module W_α^* carries the same Casimir eigenvalue as W_α, so on the α-block of the invariant subspace the right action reduces to multiplication by the same scalar, which is the displayed central operator. ☐

Theorem 5.15 (MSA and exact collar Markovianity on the central-interface branch)**.** *Call a declared regulator package for the collar tripartition A-B-D* central-interface *if its pre-quotient effective Hamiltonian has the interface normal form*

$$\widetilde{H}_{\mathrm{eff}} = \widetilde{H}_{AL} + \widetilde{H}_{RD} + \widetilde{H}_\Sigma,$$

where $\widetilde{H}_{AL}$ is a $\widehat{K_\Sigma}$-invariant self-adjoint operator supported on $A \cup B_L$, $\widetilde{H}_{RD}$ likewise on $B_R \cup D$, and $\widetilde{H}_\Sigma \in \pi_L(Z(C^*(\widehat{K_\Sigma})))$ is a boundary-charge (flux) function. Then on the physical collar model:

1. *the quotient effective Hamiltonian satisfies $H_{\mathrm{eff}} \in \mathcal{M}_L + \mathcal{M}_R + \bigoplus_\alpha \mathbb{C}\,P_\alpha$;*

2. *every MaxEnt/Gibbs reference state $\omega \propto e^{-H_{\mathrm{eff}}}$ of the package is EC-aligned exact Markov, $\omega \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$: the Markov-split alignment hypothesis* holds as a theorem*, and*

3. *exact collar Markovianity $I(A : D \mid B)_\omega = 0$ is likewise derived rather than assumed.*

Lattice-gauge-type regulators are central-interface: their interface energy is a function of the electric flux through Σ, which is exactly a central boundary-charge function. The Bell-pair state of Remark 5.11 is not *the Gibbs state of any central-interface package: every such Gibbs state is EC-aligned by item 2, whereas the Bell-pair state has $I(A : B_R) = 2\ln 2$ and sits at fixed trace distance from $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$. The counterexample is thus excluded by a checkable structural condition on the regulator, not by fiat.*

Proof. By Lemma 5.14, $\widetilde{H}_{AL}$ descends into $\mathcal{M}_L$, $\widetilde{H}_{RD}$ into $\mathcal{M}_R$, and $\widetilde{H}_\Sigma$ into $\bigoplus_\alpha \mathbb{C}\,P_\alpha \subset \mathcal{M}_L \cap \mathcal{M}_R$, giving item 1. Hence $\log\omega = -H_{\mathrm{eff}} - \log Z\,\mathbf{1} \in \mathcal{M}_L + \mathcal{M}_R$, which is item 2 of Proposition 5.12, so $\omega \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$. Item 3 follows because every EC-aligned state is exact Markov (Proposition 5.12). For the final claims: the electric interface energy of a lattice-gauge regulator is a function of the quadratic Casimir of the boundary action, hence central by Lemma 5.14, item 2. The Bell-pair exclusion is immediate: every Gibbs state of a central-interface package lies in $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ by item 2 (and is faithful, while the Bell-pair state is pure on $A \cup B_R$), whereas every state in $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ has $I(A : B_R) = 0$ and the Bell-pair state has $I(A : B_R) = 2\ln 2$. ☐

Remark 5.16 (Scope of the derivation). *Theorem 5.15 discharges the Markov-split alignment hypothesis, and with it exact collar Markovianity, on the declared central-interface branch: there, every MSA-conditioned identity in this manuscript (blockwise modular cancellation, the exact splice, the null-strip four-term relation on inherited splits, and checkpoint factorization) holds unconditionally for the package's MaxEnt reference states. This manuscript adopts the central-interface normal form as an explicit declared-branch input—the central-interface collar clause stated with Axiom 3.7—so the conditional status of the exact-Markov chain is localized in one named axiom-level clause rather than distributed as a floating state hypothesis. What remains open at the level of the axioms is whether overlap-consistent repair* forces *that clause, i.e. whether non-central cross-cut couplings are excluded rather than merely not selected; outside the declared branch MSA is carried as an explicit hypothesis, with Remark 5.11 marking the failure boundary.*

Theorem 5.17 (Derived higher-gauge cut datum). *Fix a connected cut Σ and a finite regulator chart $\{P_i\}_{i\in I_\Sigma}$ meeting along Σ, with finite nerve N_Σ. On the genuinely noncentral branch, the weak gluing data are implemented by a compact crossed module*

$$\mathbb{K}_\Sigma = (H_\Sigma \xrightarrow{\partial_\Sigma} G_\Sigma, \triangleright)$$

together with maps

$$g_{ij} : P_{ij} \to G_\Sigma, \qquad h_{ijk} : P_{ijk} \to H_\Sigma,$$

obeying

$$g_{ij}g_{jk} = \partial_\Sigma(h_{ijk})\, g_{ik}, \qquad h_{jkl}h_{ijl} = (g_{ij} \triangleright h_{ikl})\, h_{ijk}.$$

The corresponding higher-gauge change system on the cut is the compact semidirect product

$$\mathcal{T}_\Sigma := C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma),$$

with multiplication

$$(\eta, u) \cdot (\eta', u') = (\eta\,(u \triangleright \eta'),\, uu'),$$

and standard coboundary action

$$g_{ij} \mapsto u_i\, \partial_\Sigma(\eta_{ij})\, g_{ij}\, u_j^{-1},$$
$$h_{ijk} \mapsto (u_i \triangleright h_{ijk})\, \eta_{ij}\, (g_{ij} \triangleright \eta_{jk})\, \eta_{ik}^{-1}.$$

The full fixed-cutoff higher gluing datum on the genuinely noncentral branch is therefore the orbit class

$$q_\Sigma = [(g,h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma),$$

which classifies the full crossed-module equivalence orbit of the pair (g,h). It does not in general determine a unique ordinary G_Σ-valued 1-cocycle class after strictification, because the map below need not be injective. Let the natural strict-locus map be

$$\begin{aligned} \iota_\Sigma : \check{H}^1(N_\Sigma, G_\Sigma) &\longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma), \\ [g] &\longmapsto [(g,1)]. \end{aligned}$$

Define the separate $\{0,1\}$-valued strictification obstruction $o^{(2)}_\Sigma(g,h)$ by

$$\begin{aligned} o^{(2)}_\Sigma(g,h) = 0 \quad &\Longleftrightarrow \quad q_\Sigma \in \mathrm{im}(\iota_\Sigma) \\ &\Longleftrightarrow \quad [(g,h)] \text{ has a representative } (g^{\mathrm{str}}, 1) \\ &\qquad\qquad \text{with } g^{\mathrm{str}}_{ij} g^{\mathrm{str}}_{jk} = g^{\mathrm{str}}_{ik}. \end{aligned}$$

and set $o^{(2)}_\Sigma(g,h)=1$ otherwise. Thus $o^{(2)}_\Sigma$ records only whether the higher associator defect can be removed; it is not the full orbit class q_Σ. No quantitative closure/readback datum P, N_{CRC}, or $\ell^2_\star$ enters this theorem package.

Proof sketch. On pair overlaps, every regulator-scale recharting acts on a finite-dimensional matrix algebra and is therefore inner, so it is implemented by a unitary on the cut Hilbert space. On triple overlaps, genuinely noncentral failure of strict composition is recorded by unitary associators instead of scalars. These 1- and 2-morphisms form a compact unitary 2-group of rechartings. Skeletal strictification of a compact 2-group is equivalent to crossed-module data, which yields the displayed compact $\mathbb{K}_\Sigma$. Because the nerve is finite, $C^1(N_\Sigma, H_\Sigma)$ and $C^0(N_\Sigma, G_\Sigma)$ are finite products of compact groups, hence $\mathcal{T}_\Sigma$ is compact. The displayed formulas are the standard crossed-module coboundary action, and the orbit class is exactly the nonabelian Čech 2-class of the defect data; see [42] for an explicit higher-lattice-gauge realization of the same crossed-module structures. □

Theorem 5.18 (Higher-gauge edge-center completion)**.** *Let $B = B_L \cup B_R$ be a fixed-cutoff collar around a connected cut Σ on the genuinely noncentral branch of Theorem 5.17. Then, as unitary $\mathcal{T}_\Sigma$-modules,*

$$\widetilde{\mathcal{H}}_{B_L} \cong \bigoplus_{\lambda\in\Lambda_\Sigma} W_\lambda \otimes \mathcal{H}_{b_L^\lambda}, \qquad \widetilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\mu\in\Lambda_\Sigma} W^*_\mu \otimes \mathcal{H}_{b_R^\mu},$$

for a finite set Λ_Σ of irreducible unitary representations of $\mathcal{T}_\Sigma$ that actually occur. Hence the physical higher-gauge collar space is

$$\mathcal{H}^{2g}_B := (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_{\lambda\in\Lambda_\Sigma} \mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda},$$

and the collar algebra and its center are

$$\mathcal{A}_{2g}(B) \cong \bigoplus_{\lambda\in\Lambda_\Sigma} \mathcal{B}(\mathcal{H}_{b_L^\lambda}) \otimes \mathcal{B}(\mathcal{H}_{b_R^\lambda}), \qquad Z(\mathcal{A}_{2g}(B)) = \bigoplus_{\lambda\in\Lambda_\Sigma} \mathbb{C}\,\mathbf{1}_\lambda.$$

Proof sketch. Because $\mathcal{T}_\Sigma$ is compact and the half-collar spaces are finite-dimensional, both $\widetilde{\mathcal{H}}_{B_L}$ and $\widetilde{\mathcal{H}}_{B_R}$ split orthogonally into irreducible unitary $\mathcal{T}_\Sigma$-modules with finite multiplicity. The right side sees inverse transport across the same cut, hence the contragredient module. Therefore

$$\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\lambda,\mu} (W_\lambda \otimes W^*_\mu) \otimes (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\mu}).$$

Taking $\mathcal{T}_\Sigma$-invariants leaves

$$(W_\lambda \otimes W^*_\mu)^{\mathcal{T}_\Sigma} \cong \mathrm{Hom}_{\mathcal{T}_\Sigma}(W_\mu, W_\lambda) \cong \begin{cases} \mathbb{C}, & \lambda = \mu, \\ 0, & \lambda \neq \mu, \end{cases}$$

by Schur's lemma. This yields the matched block decomposition and the center formula. □

Theorem 5.19 (Higher-gauge Markov collar and carried errors)**.** *Let ρ_{ABD} be a faithful state on a collar whose middle region B carries the higher-gauge block decomposition of Theorem 5.18. If*

$$I(A : D \mid B)_\rho = 0,$$

and ρ satisfies the Markov-split alignment hypothesis for that block decomposition (Definition 5.10, with α replaced by λ), then

$$\rho_{ABD} = \bigoplus_{\lambda\in\Lambda_\Sigma} p_\lambda\, \rho_{Ab_L^\lambda} \otimes \rho_{b_R^\lambda D}.$$

If instead $I(A : D \mid B)_\rho \le \varepsilon$ on one fixed faithful collar model, then the same carried Fawzi–Renner and fixed-collar Markov errors as in the ordinary branch continue to hold:

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}, \qquad 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Proof sketch. Theorem 5.18 shows that the higher-gauge collar algebra is a finite direct sum of type-I tensor blocks. HJPW depends only on that finite-dimensional algebraic structure, not on whether the block label arose from an ordinary compact group, a central extension, or a crossed-module gauge system; as in the ordinary branch, HJPW supplies a state-dependent decomposition, and the identification of its factors with the preselected higher-gauge edge factors is the carried Markov-split alignment hypothesis. Likewise, the Fawzi–Renner remainder and the fixed-collar Markov modulus are finite-dimensional entropy and recovery statements and are blind to the origin of the block label. □

Theorem 5.20 (Higher-gauge defect strictification)**.** *With the genuinely noncentral cut data, full orbit class q_Σ, and strictification obstruction $o^{(2)}_\Sigma$ of Theorem 5.17, the following hold:*

1. *q_Σ is invariant under all local rechartings and higher-gauge changes $(\eta, u) \in \mathcal{T}_\Sigma$ and classifies the full crossed-module gluing orbit.*
2. *$o^{(2)}_\Sigma$ is an invariant property of that orbit and vanishes if and only if the higher associator defect is removable.*
3. *When $o^{(2)}_\Sigma = 0$, the orbit contains at least one strict representative $(g^{\mathrm{str}}, 1)$, and each such representative is a genuine G_Σ-valued 1-cocycle. Different strict representatives in the same orbit can be related by H_Σ-valued edge changes and need not have the same ordinary H^1 class or represented holonomy.*

Thus $o^{(2)}_\Sigma = 0$ is the strictification condition for the genuinely noncentral branch. Strict endpoint-only transport additionally requires that at least one allowed strict representative have trivial holonomy on the relevant collar-sector representation, as stated in Theorem 5.22; this existential property is invariant on the full orbit.

Proof sketch. Crossed-module coboundaries change representatives while preserving the full orbit q_Σ. By definition, $o^{(2)}_\Sigma = 0$ exactly when that orbit meets the strict subset $(g^{\mathrm{str}}, 1)$. The surviving edge data then form a genuine G_Σ-valued 1-cocycle. The natural map ι_Σ need not be injective: an H_Σ-valued edge change can alter the ordinary G_Σ-holonomy through ∂_Σ. Therefore no individual strict representative's holonomy is assigned to q_Σ. What is invariant is whether the orbit contains a strict representative whose holonomy is trivial on the chosen sector, and associator strictification alone does not imply that existence. □

Definition 5.21 (Overlap-transport groupoid and strict transportability)**.** *Fix a connected cut Σ and a finite regulator charting nerve N_Σ. The* overlap-transport groupoid $\Pi_1^{\mathrm{ov}}(N_\Sigma)$ *has the regulator cut charts as objects and words in overlap-preserving rechartings as morphisms, modulo insertion and deletion of immediate inverse rechartings. On the ordinary or central branch, an oriented edge $i \to j$ is represented by a unitary recharting U_{ij} on the cut presentation, with $U_{ji} = U_{ij}^{-1}$ after choosing the inverse chart. On the genuinely noncentral branch, the same notation denotes the G_Σ-part of the crossed-module data of Theorem 5.17, with the H_Σ-valued associators retained as* 2*-morphisms.*

A collar charge at chart i is a simple edge-center summand, equivalently an irreducible boundary module $W_\alpha^{(i)}$ together with its multiplicity/intertwiner block in the collar decomposition. Write $\mathcal{C}_\alpha^{(i)}$ for that full finite-dimensional transported charge block, including both pieces. For a path

$$p = (i_0 \to i_1 \to \cdots \to i_m)$$

define the transported charge by applying the composite recharting

$$U_p := U_{i_{m-1}i_m} \cdots U_{i_0 i_1}$$

to the boundary module and its intertwiner block. Transport is strictly path-independent *when for any two overlap paths p, p' with the same endpoints the induced transport functors on the collar charge blocks are naturally equal after the allowed local recharting gauge changes. Equivalently, every closed overlap path acts trivially on the sector class and on its transported intertwiner block, instead of only projectively or up to a noncentral 2-morphism.*

After a central or crossed-module defect has been strictified, write U^{str} for a resulting genuine edge 1-cocycle. On the full collar charge block $\mathcal{C}_\alpha$, it induces the ordinary sector-holonomy representation

$$\mathrm{Hol}_{\alpha,U^{\mathrm{str}}} : \pi_1(|N_\Sigma|, i) \longrightarrow U(\mathcal{C}_\alpha), \qquad [\ell] \longmapsto U_\ell^{\mathrm{str}}\big|_{\mathcal{C}_\alpha}.$$

On the noncentral branch, U^{str} ranges over all strict representatives in the crossed-module orbit; no unique ordinary holonomy is assigned to q_Σ. The phrase zero-obstruction sector *below means the combined condition: the relevant triangle/higher defect is strictifiable and a strictification can be chosen for which $\mathrm{Hol}_{\alpha,U^{\mathrm{str}}}$ is the trivial representation. Vanishing of the central triangle class, or removal of the noncentral associator alone, is not called strict transportability.*

Theorem 5.22 (TransportabilityFromOverlapGluing)**.** *Assume the fixed-cutoff regulator gluing package of Proposition 5.8 on the ordinary or central branch, and assume the crossed-module cut datum of Theorem 5.17 on the genuinely noncentral branch. Then path-independent transportability is the following combined theorem-level criterion, not a separate DHR input.*

(i) ***Ordinary branch.*** *If the overlap rechartings are strict on triple overlaps, then transport of a collar charge along an overlap path is computed by the path composite U_p. It is strictly path-independent if and only if every closed overlap loop has trivial holonomy on the collar-sector block. In the tree-cover case this condition is automatic; on a cover with loops it is exactly the ordinary loop-coherence obstruction.*

(ii) ***Central branch.*** *Suppose the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ, the overlap transports act trivially on Z_Σ, and the only failure of strict triple-overlap composition is central:*

$$U_{ij}U_{jk} = z_{ijk}U_{ik}, \qquad z_{ijk} \in Z_\Sigma.$$

Let $[z]_\Sigma \in \check{H}^2(N_\Sigma, Z_\Sigma)$ denote the resulting central triangle-defect class, including the central multipliers accumulated by elementary triangle moves between overlap paths. Strict path-independent transport of a collar charge α exists if and only if

(a) *$[z]_\Sigma = 0$, so a central 1-cochain a_{ij} strictifies the edge system to $U_{ij}^{\mathrm{str}} = a_{ij}^{-1}U_{ij}$; and*

(b) *one such strictification has $\mathrm{Hol}_{\alpha,U^{\mathrm{str}}}([\ell]) = \mathbf{1}_{\mathcal{C}_\alpha}$ for every closed overlap loop ℓ.*

If $[z]_\Sigma \neq 0$, a projective triangle defect remains. If $[z]_\Sigma = 0$ but every allowed strictification fails condition (b)*, the edge transports can be made into genuine* 1*-cocycles but retain ordinary loop path dependence.*

(iii) ***Genuinely noncentral branch.*** *Let $q_\Sigma = [(g,h)]$ be the full crossed-module orbit and $o^{(2)}_\Sigma$ its higher-associator strictification obstruction from Theorem 5.20. Strict path-independent transport of an ordinary collar charge α exists if and only if $o^{(2)}_\Sigma = 0$ and there is a crossed-module strictification for which the resulting genuine G_Σ-valued* 1*-cocycle has trivial holonomy on $\mathcal{C}_\alpha$. When $o^{(2)}_\Sigma \neq 0$, there is no gauge in which all path comparisons are ordinary equalities: the fixed-cutoff sector is instead a higher-gauge sector labelled by q_Σ, and transport is higher transport in the crossed-module* 2*-groupoid instead of ordinary path-independent collar transport (the categorical input to DHR-type constructions, not by itself net-level DHR transportability). When $o^{(2)}_\Sigma = 0$ but every allowed strictification has nontrivial represented G_Σ-holonomy, the higher defect is gone but ordinary loop path dependence remains.*

Thus overlap gluing constructs the transport operation. The complete obstruction to strict path independence is ordinary sector holonomy on the ordinary branch, the pair consisting of $[z]_\Sigma$ and the existence of a trivial-holonomy strictification on the central branch, and the pair consisting of $o^{(2)}_\Sigma$ and the existence of a trivial-holonomy strict representative on the genuinely noncentral branch.

Proof. For a path $p = (i_0 \to \cdots \to i_m)$, Definition 5.21 gives the transport functor by composing the finite-dimensional recharting implementers. This construction uses only the derived overlap unitary transition system of Proposition 5.8; no transportability assumption is invoked.

On the ordinary branch, strict triple-overlap coherence says that elementary replacements of the form $(i \to j \to k) \rightsquigarrow (i \to k)$ do not change the composite recharting. Any two paths with the same endpoints in the finite nerve differ, after inserting or deleting immediate backtracks, by a finite sequence of such elementary moves together with closed loop moves. Backtracks contribute the identity because $U_{ji} = U_{ij}^{-1}$. Therefore the only possible path dependence is the ordinary loop holonomy. The transport is strictly path-independent exactly when that loop holonomy acts trivially on the collar-sector block.

On the central branch, the same elementary triangle replacement changes the composite by the central multiplier z_{ijk}. For a finite sequence of elementary moves between two paths, the total discrepancy is the product of the corresponding z's over the filling 2-chain. The fixed-coefficient and trivial-action hypotheses identify every multiplier in Z_Σ; the quadruple-overlap coherence identity is then precisely the untwisted Čech cocycle identity, so this product depends only on the class $[z]_\Sigma$. If $[z]_\Sigma = 0$, choose a central 1-cochain a_{ij} with $z_{ijk} = a_{ij}a_{jk}a_{ik}^{-1}$. Replacing U_{ij} by $a_{ij}^{-1}U_{ij}$ kills the central multiplier on every triangle, so all elementary path moves preserve the transport functor. The ordinary argument then gives strict path independence exactly when every remaining closed-loop holonomy of this strictified 1-cocycle acts trivially on the sector block. Conversely, strict path-independent transport makes every elementary triangular comparison trivial after an allowed recharting gauge change, so z is a coboundary, and it also makes every closed-loop action trivial. This proves both necessary conditions.

On the genuinely noncentral branch, elementary triangle comparisons are non-scalar H_Σ-valued 2-morphisms h_{ijk}, and the consistency of different fillings of a path homotopy is the crossed-module cocycle identity of Theorem 5.17. A local higher-gauge change (η, u) changes representatives by the crossed-module coboundary action displayed there, while preserving the full class q_Σ and the strictifiability property $o^{(2)}_\Sigma$ by Theorem 5.20. If $o^{(2)}_\Sigma = 0$, the data are equivalent to at least one strict representative: the h_{ijk} can be gauged away and $g_{ij}g_{jk} = g_{ik}$ holds. Each such G_Σ-valued 1-cocycle defines an ordinary path-composite transport, which is endpoint-only exactly when its represented loop holonomy is trivial. Hence ordinary strict transport exists precisely when at least one allowed strict representative passes that test. Conversely, strict ordinary path-independent transport gives a representative in which every triangular 2-comparison is the identity, hence $o_\Sigma^{(2)} = 0$, and every closed-loop action on the sector is trivial. If $o_\Sigma^{(2)} \neq 0$, Theorem 5.20 says the higher defect is not removable, so no strict ordinary transport functor can exist; if $o_\Sigma^{(2)} = 0$ alone, only existence of a strict representative has been proved. □

Remark 5.23 (Triangle-free cycle acceptance test)**.** *Let $N_\Sigma = C_n$ be a cycle nerve with $n \geq 4$ and no filled 2-simplices. There are no triple-overlap constraints, so every central 2-cocycle datum and every higher associator datum is vacuously trivial: $[z]_\Sigma = 0$ and $o_\Sigma^{(2)} = 0$. On a full charge block $\mathcal{C}_\alpha = \mathbb{C}^2$, put the identity transport on all but one oriented cycle edge and put $V = \mathrm{diag}(-1, 1)$ on the remaining edge, with inverse transports on reverse edges. The ordered holonomy is the non-scalar unitary $V \neq \mathrm{id}$, so no central edge rephasing can make it the identity.*

For a genuinely noncentral instance, take the crossed module

$$H_\Sigma = \mathrm{SU}(2) \hookrightarrow G_\Sigma = \mathrm{U}(2)$$

with conjugation action and the standard action on $\mathcal{C}_\alpha$. Its band is $G_\Sigma/\partial H_\Sigma \cong \mathrm{U}(1)$, detected by the determinant. The cycle has $o_\Sigma^{(2)} = 0$, but its band holonomy is $\det V = -1$. An H_Σ-valued edge change has determinant one and a vertex-frame change only conjugates the based loop, so no allowed strict representative has trivial holonomy. The two routes between the endpoints of the distinguished edge therefore induce different transports. Conditions (ii)(b) *and* (iii) *reject the respective central and noncentral assignments even though the central triangle data and higher associator data are trivial.*

Corollary 5.24 (Transportability from overlap gluing)**.** *The ordinary/central compact-gauge branch may invoke only those finite-cutoff collar sectors that satisfy the zero-obstruction conclusion of Theorem 5.22. The condition is the combined strict-transport corollary of overlap gluing: after any central or higher defect is strictified, at least one allowed resulting ordinary sector holonomy must be trivial. If the genuinely noncentral obstruction $o_\Sigma^{(2)}$ is nonzero, the branch is a higher-gauge fixed-cutoff sector rather than an ordinary compact-group DR reconstruction sector; if $o_\Sigma^{(2)} = 0$, it has one or more strict G_Σ-valued 1-cocycle representatives and enters the transportable case only when at least one has trivial sector holonomy.*

Definition 5.25 (Fixed-cutoff zero-obstruction collar sectors)**.** *Fix a regulator cutoff r and a connected overlap collar $B = B_L \cup B_R$ around a cut Σ on the ordinary or central-defect branch. Let $\widehat{K}_{\Sigma,r}$ be the compact boundary action supplied by the fixed-cutoff overlap gluing theorem, with $\widehat{K}_{\Sigma,r} = K_{\Sigma,r}$ on the ordinary branch. Edge-center completion gives*

$$\widetilde{\mathcal{H}}_{B_L} \cong \bigoplus_{\alpha \in A_{\Sigma,r}} W_\alpha \otimes \mathcal{H}_{b_L^\alpha}, \qquad \widetilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\alpha \in A_{\Sigma,r}} W_\alpha^* \otimes \mathcal{H}_{b_R^\alpha},$$

and hence

$$\mathcal{H}_B = (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\widehat{K}_{\Sigma,r}} \cong \bigoplus_{\alpha \in A_{\Sigma,r}} \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}.$$

The minimal central projector onto the α-summand is denoted P_α. Before selecting seeds, choose one common stagewise strict edge 1-cocycle U_r^{str}: on the ordinary branch this is the given strict *system; on the central branch it is one allowed strictification after $[z]_\Sigma = 0$; and a strictified genuinely noncentral datum may enter only after $o^{(2)}_\Sigma = 0$ and the choice of one allowed strict G_Σ-valued representative. A* visible seed charge relative to U_r^{str} *is a pair (P_α, W_α) whose full transported charge block has trivial holonomy for this same representative on every closed overlap loop. The construction does not union seeds that require mutually incompatible strictifications. The full orbit q_Σ classifies the allowed crossed-module representatives but does not select a unique ordinary H^1 class; choosing U_r^{str} is therefore part of the fixed-stage ordinary sector datum. Let S_r be the finite set of visible seed carriers. Define*

$$\mathsf{Sect}_r^{\mathrm{bos}} := \langle \mathbf{1}, S_r, S_r^* \rangle_{\oplus,\otimes,\ \textit{subobjects},\ \textit{isomorphism}} \subset \mathrm{Rep}_{\mathrm{fd}}(\widehat{K}_{\Sigma,r})$$

to be the replete full additive Karoubi rigid tensor subcategory generated by those carriers. Equivalently, every object is a finite direct sum of subobjects of finite tensor words in visible seeds and their conjugates. For objects X, Y in this category, define

$$\mathrm{Hom}_r(X,Y) := \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X,Y),$$

after transporting all localizations to one reference collar by the zero-obstruction transport functor. Different reference collars give canonically unitarily equivalent Hom spaces by Theorem 5.22. The projector P_α detects a seed in the original one-collar algebra; it is not a declaration that every irreducible generated by tensor products is already a central block of that finite algebra. If X is an irreducible summand of a tensor word T, its categorical support is an orthogonal idempotent $e_X \in \mathrm{End}_{\widehat{K}_{\Sigma,r}}(T)$, represented physically, when that concatenated-collar realization is retained, in the finite algebra for that particular concatenated collar.

Theorem 5.26 (FixedCutoffBosonicSectorCategory)**.** *At every fixed regulator cutoff r, on the ordinary or central-defect zero-obstruction branch and in the bosonic internal-gauge sector of the $3+1$-dimensional EFT regime, the construction of Definition 5.25 produces a semisimple rigid symmetric C^*-tensor category*

$$\mathsf{Sect}_r^{\mathrm{bos}}.$$

Its visible seed objects are the transportable edge-center collar charges (P_α, W_α). Its simple objects are all irreducible summands of finite tensor words in those seeds and their conjugates; they need not be visible blocks of the original one-collar algebra, and there may be infinitely many simple isomorphism classes even though every object and Hom space is finite-dimensional. The tensor product and duals are the representation tensor product and conjugate; the finite tensor-realization clause of Definition 7.2 identifies them physically with collar concatenation and orientation reversal on a certified tail. The *-operation and C^*-norm are the operator adjoint and norm on the finite-dimensional carriers, and the symmetry is the bosonic spacelike-exchange symmetry of the $3+1$-dimensional EFT branch. The inclusion has the canonical faithful symmetric strong monoidal forgetful functor

$$F_r : \mathsf{Sect}_r^{\mathrm{bos}} \longrightarrow \mathsf{Hilb}_{\mathrm{fd}}.$$

Proof. By edge-center completion, every seed charge visible on one reference collar is represented by a minimal central summand P_α together with an irreducible boundary carrier W_α of $\widehat{K}_{\Sigma,r}$. Since $\widehat{K}_{\Sigma,r}$ is compact and acts unitarily on finite-dimensional collar data, its finite-dimensional representations are completely reducible. The replete additive Karoubi tensor subcategory generated by the W_α's and their conjugates is therefore semisimple, with finite-dimensional objects and Hom spaces. Complete reducibility does not imply that its simple skeleton is finite, and no such finiteness is used.

Transport between collars is the path-composite construction of Theorem 5.22. For every closed loop ℓ, the common choice U_r^{str} acts as the identity on each seed block. It therefore acts as the identity on direct sums, as $\mathbf{1} \otimes \mathbf{1}$ on tensor products, as the conjugate identity on duals, and by restriction as the identity on every invariant subobject. Thus every object in the generated category has path-independent transport using the same representative, and those transport identifications are monoidal. In particular, transport of P_α, W_α, and their intertwiners is independent of the overlap path. Hence the Hom spaces defined after moving all localizations to a reference collar do not depend on the chosen path or reference collar, up to canonical transported unitary identification.

For two objects X and Y, take the diagonal tensor product $X \otimes Y$ of $\widehat{K}_{\Sigma,r}$-modules. By definition the generated Karoubi category contains this tensor word and every irreducible summand selected by an idempotent in its intertwiner algebra, so the product stays inside the category without requiring those summands to occur in the center of the original one-collar algebra. The trivial one-dimensional module is the tensor unit. Associativity is implemented by the standard finite-dimensional tensor associator, whose pentagon identity is the equality of the two rebracketings of a fourfold tensor product. When the finite tensor-realization receipt is present, these tensor words, idempotents, and rebracketings are represented by concatenated collars and union-collar gluing.

Duals are the conjugate representations W_α^*; the finite tensor-realization receipt identifies this with reversing collar orientation and swapping the half-collar boundary actions. Evaluation and coevaluation are the standard $\widehat{K}_{\Sigma,r}$-invariant pairings. Their zig-zag identities are the usual finite-dimensional duality identities, equivalently cap/cup cancellation for collar gluing.

The *-structure is the operator adjoint on the lifted finite-dimensional carriers. Each $\mathrm{Hom}_r(X,Y)$ is a closed subspace of operators between finite-dimensional Hilbert spaces, adjoints are again $\widehat{K}_{\Sigma,r}$-intertwiners, composition is operator composition, and the norm is the operator norm. Thus $\|f^*f\| = \|f\|^2$, direct sums and subobjects are implemented by orthogonal projections, and the category is a C^*-category.

The canonical unitary flip $c_{X,Y} : X \otimes_r Y \to Y \otimes_r X$ is an intertwiner. Its naturality, hexagon identities, and $c_{Y,X}c_{X,Y} = \mathrm{id}$ are the standard representation identities, so the braiding is symmetric. On a finite tensor-realization branch this flip is represented physically by spacelike exchange of the codimension-two collar supports; double exchange is isotopic to the identity on the bosonic $3+1$-dimensional branch. Fermionic signs, spinorial matter, and chirality are not part of this bosonic internal-gauge category; they belong to the later super-Tannakian or matter-sector lift. Forgetting the group action and retaining the carrier Hilbert space and the same linear maps is faithful, *-preserving, and symmetric strong monoidal, which gives the displayed F_r. □

Corollary 5.27 (Fixed-cutoff category from sector construction)**.** *The fixed-cutoff collar-sector packages are bosonic symmetric C^*-tensor categories with canonical objectwise finite-dimensional forgetful fibers by Theorem 5.26. Refinement compatibility is a separate question: Theorem 7.3 supplies it only on a cofinal tail carrying the explicit compact-gauge refinement receipt of Definition 7.2. The realized low-energy branch is then supplied by Theorem 7.35.*

Proposition 5.28 (Higher-gauge interacting compatibility)**.** *In a fixed-cutoff crossed-module lattice realization of Theorem 5.17, one may add local collar terms*

$$K^{2g}_{\mathrm{collar}} = \sum_{v\in\Sigma_0} \beta_v(1 - A_v) + \sum_{e\in\Sigma_1} \beta_e(1 - B_e) + \sum_{c\in\mathcal{C}_\Sigma} \beta_c(1 - C_c),$$

where A_v, B_e, and C_c are the local gauge and fake-flat projectors of the chosen higher-gauge realization. These terms are bounded-support and commute in the standard higher-gauge projector realization [42], so the full fixed-cutoff generator is quasi-local and inside the Axiom 3.7 class.

Proof sketch. The additional terms are local projector constraints supported on finitely many collar cells, so they preserve bounded support and quasi-locality. In the standard higher-gauge projector realization they commute, so the same finite-constraint MaxEnt/Lieb–Robinson bookkeeping used on the ordinary branch continues to apply. □

Remark 5.29 (Status of the regulator package and the precise Markov boundary). *Proposition 5.8, Theorem 5.9, and Theorems 5.17–5.20 together with Proposition 5.28 show that edge-center completion is a theorem of overlap consistency plus the finite regulator presentation on the ordinary, central-defect, and genuinely noncentral higher-gauge branches. No preferred quantum-link realization is assumed on the ordinary or central branch, and the genuinely noncentral branch is handled by the compact crossed-module replacement instead of by one more ordinary-group lemma. The fixed-cutoff topological UV package is closed on all three branches. The continuum modular/geometric lift on the support-visible geometric subnet is Theorem 6.13; the compact-gauge realized branch is handled by Theorem 7.35. The distinct state-control issue is that exact splice and modular-additivity identities require either literal exact Markovity or a controlled family that converges to the exact Markov set on one fixed finite-dimensional collar; and, wherever those identities use the preselected edge factors $\mathfrak{b}_L^\alpha, \mathfrak{b}_R^\alpha$, they additionally carry the Markov-split alignment hypothesis of Definition 5.10, which exact Markovity does not imply (Remark 5.11).*

Definition 5.30 (Exact Markov set and collar-distance modulus). *Fix the finite-dimensional collar Hilbert space of Theorem 5.9. Let*

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : \ I(A:D \mid B)_\sigma = 0\}$$

be the set of exact Markov states on that fixed collar. By HJPW, $\sigma \in \mathfrak{M}_{A:B:D}$ iff σ factorizes blockwise over some *state-dependent direct-sum/tensor decomposition of $\mathcal{H}_B$. The subset factorizing over the fixed edge-center decomposition of B is the EC-aligned class $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ of Definition 5.10; the inclusion $\mathfrak{M}^{\mathrm{EC}}_{A:B:D} \subsetneq \mathfrak{M}_{A:B:D}$ is strict in general (Remark 5.11).*

For a general state ρ_{ABD}, define its distance to the exact Markov set by

$$d_{\mathrm{M}}(\rho) := \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1,$$

and the fixed-collar exact-Markov modulus by

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup \{d_{\mathrm{M}}(\rho) : I(A:D \mid B)_\rho \le \varepsilon\}.$$

The modulus $\delta^{\mathrm{M}}_{A:B:D}$ measures distance to the full exact Markov set $\mathfrak{M}_{A:B:D}$. No analogous vanishing modulus exists for the EC-aligned class: by Remark 5.11 there are states with $I(A:D \mid B) = 0$ at fixed nonzero trace distance from $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$, so replacement by an EC-aligned normal form is available only under the Markov-split alignment hypothesis on the comparison family, not from small conditional mutual information alone.

Proposition 5.31 (Controlled exact Markov-collar limit). *On a fixed finite-dimensional collar, the exact Markov set $\mathfrak{M}_{A:B:D}$ is compact and*

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad \text{as } \varepsilon \downarrow 0.$$

Consequently, if $\rho^{(n)}_{ABD}$ is any sequence of states on that same collar with

$$I(A:D \mid B)_{\rho^{(n)}} \le \varepsilon_n, \qquad \varepsilon_n \to 0,$$

then there exist exact Markov states $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ such that

$$\|\rho^{(n)} - \sigma^{(n)}\|_1 \leq \delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) \longrightarrow 0.$$

Thus approximate recoverability converges to the exact Markov set on the fixed collar. Convergence to the EC-aligned normal form used by the compact-gauge branch requires, in addition, that the comparison states $\sigma^{(n)}$ can be chosen in $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$; this is the Markov-split alignment hypothesis on the comparison family (Definition 5.10) and is not supplied by small conditional mutual information alone (Remark 5.11).

Proof. The full state space on a fixed finite-dimensional Hilbert space is compact in trace norm. Conditional mutual information is continuous there because von Neumann entropy is continuous in finite dimension. Hence $\mathfrak{M}_{A:B:D}$ is closed, so it is compact as a closed subset of a compact space.

Suppose $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \not\to 0$. Then there exist $\eta > 0$, a sequence $\varepsilon_n \downarrow 0$, and states $\rho^{(n)}$ with $I(A : D \mid B)_{\rho^{(n)}} \leq \varepsilon_n$ but $d_{\mathrm{M}}(\rho^{(n)}) \geq \eta$ for all n. By compactness, pass to a trace-norm convergent subsequence $\rho^{(n_k)} \to \rho^*$. Continuity of conditional mutual information gives

$$I(A : D \mid B)_{\rho^*} = \lim_{k\to\infty} I(A : D \mid B)_{\rho^{(n_k)}} = 0,$$

so $\rho^* \in \mathfrak{M}_{A:B:D}$. But then

$$d_{\mathrm{M}}(\rho^{(n_k)}) \leq \|\rho^{(n_k)} - \rho^*\|_1 \longrightarrow 0,$$

contradicting $d_{\mathrm{M}}(\rho^{(n_k)}) \geq \eta$. Thus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$. The final statement follows by choosing $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ within $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) + 1/n$ of $\rho^{(n)}$. □

Theorem 5.32 (Finite-collar Markov replacement stability)**.** *Fix one finite-dimensional collar model $A : B : D$ and a faithful floor $\lambda_* > 0$ for the collar state. Let*

$$\mathcal{S}_{\lambda_*} := \{\rho_{ABD} : \rho_{ABD} \geq \lambda_* \mathbf{1}\}$$

inside the affine state space on that fixed collar, and let

$$\mathfrak{M}^{\lambda_*}_{A:B:D} := \mathfrak{M}_{A:B:D} \cap \mathcal{S}_{\lambda_*}.$$

Assume this faithful exact-Markov class is nonempty. There are constants $C_{A:B:D,\lambda_*} > 0$ and $\theta_{A:B:D,\lambda_*} > 0$, depending on the chosen collar model and floor but not on a refinement family, such that every $\rho \in \mathcal{S}_{\lambda_*}$ satisfies

$$d^{\lambda_*}_{\mathrm{M}}(\rho) := \inf_{\sigma \in \mathfrak{M}^{\lambda_*}_{A:B:D}} \|\rho - \sigma\|_1 \leq C_{A:B:D,\lambda_*}\, I(A : D \mid B)_\rho^{\theta_{A:B:D,\lambda_*}}.$$

Equivalently, on that fixed faithful collar class,

$$\delta^{\mathrm{M},\lambda_*}_{A:B:D}(\varepsilon) \leq C_{A:B:D,\lambda_*}\, \varepsilon^{\theta_{A:B:D,\lambda_*}}.$$

The constants are collar-local; this is not a dimension-free stability theorem for arbitrary tripartite systems.

Proof. On $\mathcal{S}_{\lambda_*}$ the von Neumann entropy terms are real analytic functions of the matrix entries, hence

$$F(\rho) := I(A : D \mid B)_\rho$$

is a nonnegative real analytic function. By equality in strong subadditivity, or equivalently the HJPW structure theorem, the zero set of F is precisely $\mathfrak{M}^{\lambda_*}_{A:B:D}$. The compact-set Lojasiewicz inequality for a real analytic nonnegative function gives constants $c > 0$ and $q > 0$ on this fixed compact collar class such that

$$F(\rho) \geq c\, d_{\mathrm{M}}^{\lambda_*}(\rho)^q.$$

Taking $C = c^{-1/q}$ and $\theta = 1/q$ gives the displayed estimate. $\square$

Proposition 5.33 (Approximate collar recovery)**.** *Let A-B-D be a collar tripartition satisfying*

$$I(A : D|B)_\rho \leq \varepsilon,$$

and let $\mathcal{R}_{B\to BD}$ be a recovery map such that

$$F(\rho_{ABD}, (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})) \geq e^{-\varepsilon/2}.$$

Define the recovered comparison state

$$\rho^{\mathrm{rec}}_{ABD} := (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB}), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Then

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \leq r_{\mathrm{FR}}(\varepsilon).$$

Consequently, for every CPTP map $\Lambda_{BD\to B'D'}$ and every bounded observable X supported on $A \cup B'$,

$$|\mathrm{Tr}[X(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})]| \leq \|X\|_\infty \,\|(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})\|_1 \leq \|X\|_\infty\, r_{\mathrm{FR}}(\varepsilon).$$

Proof. The fidelity lower bound and the Fuchs–van de Graaf inequality give

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \leq 2\sqrt{1 - F(\rho_{ABD}, \rho^{\mathrm{rec}}_{ABD})^2} \leq 2\sqrt{1 - e^{-\varepsilon}} = r_{\mathrm{FR}}(\varepsilon).$$

The estimate $r_{\mathrm{FR}}(\varepsilon) \leq 2\sqrt{\varepsilon}$ follows from $1 - e^{-x} \leq x$. Contractivity of trace norm under CPTP maps gives the transported bound, and the observable estimate is the duality between trace norm and operator norm [29]. The bound is constructive and dimension-free. What it controls is closeness to a recovered comparison state; no claim is made that $\rho^{\mathrm{rec}}_{ABD}$ is itself exact Markov. $\square$

Proposition 5.34 (Exact splice theorem and controlled collar approximation)**.** *Under Theorem 5.9, let $\sigma_{ABD} \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ be EC-aligned exact Markov (Definition 5.10), with normal form*

$$\sigma_{ABD} = \bigoplus_\alpha p_\alpha\, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \sigma^{(\alpha)}_{b^\alpha_R D}.$$

Let $\tau^{(\alpha)}_{b^\alpha_R D'}$ be any normalized family of states compatible with the same right-boundary sectors, and define

$$\sigma'_{ABD'} = \bigoplus_\alpha p_\alpha\, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \tau^{(\alpha)}_{b^\alpha_R D'}.$$

Define the common left algebra

$$\mathcal{A}_{Ab_L} := \bigoplus_\alpha \mathcal{B}(\mathcal{H}_{Ab^\alpha_L}) \otimes \mathbf{1}_{b^\alpha_R},$$

canonically represented on both direct-sum Hilbert spaces. Then:

1. *for every* $X \in \mathcal{A}_{Ab_L}$*,*

$$\mathrm{Tr}(X\sigma'_{ABD'}) = \mathrm{Tr}(X\sigma_{ABD});$$

2. *if* ρ_{ABD} *is any state on the same fixed collar and* $\sigma_{\mathrm{al}} \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ *is an EC-aligned exact Markov state with*

$$\|\rho_{ABD} - \sigma_{\mathrm{al}}\|_1 \le \delta_{\mathrm{al}},$$

then the corresponding exact splice σ'_{al} *satisfies*

$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_{\mathrm{al}})\right| \le \|X\|_\infty\, \delta_{\mathrm{al}}$$

for all $X \in \mathcal{A}_{Ab_L}$*.*

The splice construction itself requires the EC-aligned normal form, so item 2 takes closeness to $\mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ *as its input. Small conditional mutual information supplies closeness only to the larger set* $\mathfrak{M}_{A:B:D}$ *via* $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$*; upgrading that to* $\delta_{\mathrm{al}} \to 0$ *is the Markov-split alignment hypothesis on the controlled family (Definition 5.10, Remark 5.11). Hence the exact splice identity used by the compact-gauge branch is justified either at EC-aligned exact Markovity or along a controlled collar family admitting EC-aligned replacements with* $\delta_{\mathrm{al}}(\varepsilon_\delta) \to 0$*.*

Proof. For item 1, blockwise factorization gives

$$\mathrm{Tr}(X\sigma'_{ABD'}) = \sum_\alpha p_\alpha \mathrm{Tr}\Big(X_\alpha \sigma^{(\alpha)}_{Ab^\alpha_L}\Big)\,\mathrm{Tr}\Big(\tau^{(\alpha)}_{b^\alpha_R D'}\Big)\,.$$

Here X_α denotes the α-block of X in $\mathcal{A}_{Ab_L}$. Each $\tau^{(\alpha)}$ is normalized, so the second factor is 1. The same computation for σ_{ABD} gives the same value because the original right factor is likewise normalized.

For item 2, σ_{al} is EC-aligned by hypothesis, so item 1 applies to it:

$$\mathrm{Tr}(X\sigma'_{\mathrm{al}}) = \mathrm{Tr}(X\sigma_{\mathrm{al}}).$$

Therefore

$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_{\mathrm{al}})\right| = \left|\mathrm{Tr}[X(\rho_{ABD} - \sigma_{\mathrm{al}})]\right| \le \|X\|_\infty\, \|\rho_{ABD} - \sigma_{\mathrm{al}}\|_1 \le \|X\|_\infty\, \delta_{\mathrm{al}}.$$

□

Proposition 5.35 (Transport of modular-additivity errors from the exact Markov reference)**.** *Let* $K_X(\omega) := -\log \omega_X$ *for a faithful reduced state on region* X*, and define the modular defect*

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Fix a collar and let ρ *be a faithful state with* $I(A:D\mid B)_\rho \le \varepsilon$*. Choose* $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ *with*

$$\|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Assume that the four marginals ρ_X *and* $(\sigma_\varepsilon)_X$ *for* $X \in \{ABD, AB, BD, B\}$ *all satisfy*

$$\rho_X \ge \lambda_* \mathbf{1}, \qquad (\sigma_\varepsilon)_X \ge \lambda_* \mathbf{1}$$

on their supports for some $\lambda_* > 0$*. Then*

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

For any $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$, the operator $\Delta K(\sigma_\varepsilon)$ is blockwise constant for the state's own *HJPW decomposition; it lies in the collar center $Z(\mathcal{A}_{\mathrm{EC}}(B))$ exactly when σ_ε can be chosen EC-aligned, $\sigma_\varepsilon \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$, which is the Markov-split alignment hypothesis on the comparison family (Definition 5.10). Under that hypothesis every matrix element of the modular defect of ρ is within $4\lambda_*^{-1}\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ of the EC-central exact Markov value.*

Proof. By monotonicity of trace distance under partial trace,

$$\|\rho_X - (\sigma_\varepsilon)_X\|_1 \le \|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$

for each $X \in \{ABD, AB, BD, B\}$. On the interval $[\lambda_*, 1]$, the function $\log x$ has derivative bounded by λ_*^{-1}. By the standard integral representation of the operator logarithm, this implies the operator-norm Lipschitz bound

$$\|K_X(\rho) - K_X(\sigma_\varepsilon)\|_\infty = \|\log(\sigma_\varepsilon)_X - \log\rho_X\|_\infty \le \lambda_*^{-1}\,\|\rho_X - (\sigma_\varepsilon)_X\|_\infty \le \lambda_*^{-1}\,\delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Summing the four region contributions gives

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\,\delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

For $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$, the exact HJPW block factorization implies that $\Delta K(\sigma_\varepsilon)$ is blockwise constant on the blocks of the state's own HJPW decomposition; when $\sigma_\varepsilon \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ those blocks are the edge-center blocks and $\Delta K(\sigma_\varepsilon) \in Z(\mathcal{A}_{\mathrm{EC}}(B))$, so the same bound controls all matrix elements relative to the EC-central exact Markov modular-additivity value. ☐

Theorem 5.36 (Finite-stage modular-defect propagation). *Consider a fixed finite branch calculation that uses N collar or strip modular-additivity identities, followed by finitely many sums, products, commutators with bounded operators, bounded functional-calculus operations on a fixed spectral interval, and bounded-time modular transports. Replace the j-th exact identity by its finite-stage controlled form with errors*

$$r_{\mathrm{FR}}(\varepsilon_j), \qquad \delta^{\mathrm{M}}_j(\varepsilon_j), \qquad \eta^{\mathrm{reg}}_j,$$

where η^{reg}_j denotes the regularized support-visible transport remainder when no common full-algebra floor is used. Then every bounded downstream modular observable O produced by that calculation obeys

$$|\langle O\rangle_{\mathrm{finite}} - \langle O\rangle_{\mathrm{exact}}| \le \mathcal{P}_O\Big(\{r_{\mathrm{FR}}(\varepsilon_j)\}_{j=1}^N, \{\delta^{\mathrm{M}}_j(\varepsilon_j)\}_{j=1}^N, \{\eta^{\mathrm{reg}}_j\}_{j=1}^N\Big),$$

for a polynomial-continuity modulus $\mathcal{P}_O$ whose coefficients depend only on the fixed collar models, the bounded-time interval, the operator norms, and the declared faithful floors or regularization schedules. In particular $\mathcal{P}_O \to 0$ as all listed errors tend to zero.

Proof. Induct over the expression tree defining O. Sums and products are controlled by triangle inequalities and submultiplicativity on the bounded operator class. Commutators satisfy

$$\|[A, B] - [A', B']\| \le \|A - A'\|\,\|B\| + \|A'\|\,\|B - B'\| + \|B - B'\|\,\|A - A'\|,$$

with all norms bounded on the fixed calculation. Bounded functional calculus on a fixed spectral interval is uniformly continuous, and it is Lipschitz for the logarithmic comparisons when the floor λ_* is present. For a bounded-time modular transport generated by K and K', Duhamel's formula gives

$$\|e^{itK}Xe^{-itK} - e^{itK'}Xe^{-itK'}\| \le 2|t|\,\|X\|\,\|K - K'\| + O(\|K - K'\|^2)$$

uniformly for t in the chosen compact interval. When $K - K'$ is not an unregularized full-algebra bounded operator, Theorem 6.18 supplies the support-visible matrix-element replacement and its remainder η_j^{reg}. Composing these finitely many continuity estimates gives the displayed polynomial modulus. $\square$

Theorem 5.37 (Collar-locality of dimension-dependent constants)**.** *All constants used in exact-Markov replacement, logarithmic modular comparison, regularized modular transport, and the finite-stage propagation of modular defects are functions only of the declared fixed local collar model, its support-visible dimension, the chosen faithful floor or regularization schedule, the bounded observable class, and the bounded modular-time interval. No estimate used in the BW/Lorentz/null-modular/Einstein branch requires a dimension-free trace-norm stability theorem from small conditional mutual information to exact Markov normal form for arbitrary tripartite quantum systems.*

Proof. The Fawzi–Renner term r_{FR} is the only one-shot constructive recoverability estimate used here, and it compares the state to a recovered comparison state instead of the exact Markov set. The exact-Markov replacement modulus δ^{M} is defined after pullback to one fixed finite collar model; Theorem 5.32 gives a rate only on a fixed faithful collar class. Logarithmic comparison uses the local floor λ_*, while the BW branch avoids a refinement-uniform full-algebra floor by using the regularized support-visible transport theorem. The propagation theorem above then composes only those collar-local estimates. $\square$

Remark 5.38 (What is and is not carried forward)**.** *There are therefore two distinct quantitative controls, and this paper keeps them separate:*

1. *the constructive one-shot recoverability bound $r_{\mathrm{FR}}(\varepsilon) = O(\varepsilon^{1/2})$, which compares the physical state to a recovered comparison state and is enough for bounded-observable statements at one regulator stage;*
2. *the fixed-collar exact-Markov modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$, which compares the physical state to the exact Markov set and is the correct quantity for justifying exact splice or modular-additivity identities in a controlled limit, with an additional factor $4\lambda_*^{-1}$ when logarithms are involved.*

Later spatial and null collar theorems therefore use exact identities only in two regimes: literal exact Markovity, or a controlled refinement or scaling family in which the relevant $\delta^{\mathrm{M}}(\varepsilon_\delta)$ tends to zero after transport to one fixed finite-dimensional collar model. The manuscript does not *claim a universal dimension-free one-shot trace-norm estimate from small conditional mutual information directly to an exact Markov state.*

Remark 5.39 (Floor transfer to the exact-Markov reference)**.** *On one fixed finite-dimensional collar model, Proposition 5.35 does not require a second independent faithfulness input on the exact-Markov comparison family. Suppose the transported physical marginals satisfy*

$$\rho_X \geq \bar{\lambda}_{m,\delta}\mathbf{1}$$

for all sufficiently late stages for each $X \in \{ABD, AB, BD, B\}$, and choose exact-Markov replacements $\sigma_{n,m,\delta}$ with

$$\|\rho_X - (\sigma_{n,m,\delta})_X\|_1 \leq \delta^{\mathrm{M}}_{m,\delta}(\varepsilon_{n,m,\delta}) \longrightarrow 0.$$

Then $\|\rho_X - (\sigma_{n,m,\delta})_X\|_\infty \leq \|\rho_X - (\sigma_{n,m,\delta})_X\|_1$, so for all sufficiently large n,

$$(\sigma_{n,m,\delta})_X \geq \tfrac{1}{2}\bar{\lambda}_{m,\delta}\mathbf{1}.$$

Thus a lower spectral bound, when available on a fixed collar model, is inherited by the exact-Markov reference once the exact-Markov modulus tends to zero. The support-visible theorem below does not require such a bound uniformly across the full refining matrix algebra.

Proposition 5.40 (Generalized-entropy split)**.** *If the reduced cap state takes the block form*

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha} \right),$$

then

$$S(\rho_C) = S_{\mathrm{bulk}}(C) + \mathrm{Tr}(\rho_C L_C),$$

where

$$S_{\mathrm{bulk}}(C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^{(\alpha)}_{\mathrm{bulk},C}), \qquad L_C = \sum_\alpha (\log d_\alpha) P_\alpha.$$

Proof. The entropy of a direct sum is

$$S\left(\bigoplus_\alpha p_\alpha \sigma_\alpha\right) = H(p_\alpha) + \sum_\alpha p_\alpha S(\sigma_\alpha).$$

Apply this identity with

$$\sigma_\alpha = \rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha}.$$

Since

$$S(\sigma_\alpha) = S(\rho^{(\alpha)}_{\mathrm{bulk},C}) + \log d_\alpha,$$

the first statement follows. □

Definition 5.41 (Effective Newton constant in the refinement dictionary)**.** *In the refinement-scaling regime, if*

$$\mathrm{Tr}(\rho_C L_C) \approx N_\Sigma \bar{\ell}(t), \qquad A(\partial C) \approx N_\Sigma a_{\mathrm{cell}},$$

then matching $\mathrm{Tr}(\rho_C L_C)$ *to the area term* $A(\partial C)/(4G_{\mathrm{geom}})$ *defines the geometric coupling*

$$G_{\mathrm{geom}} := \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

This is a continuum dictionary identification. The area scale entering a_{cell} *is supplied by the separate scale-readback fixed point, not by solving this equation for* G*.*

Proposition 5.42 (Shared edge-entropy identity on the realized product-group branch)**.** *On the realized product-group branch*

$$G_{\mathrm{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6},$$

let

$$R = R_3 \boxtimes R_2 \boxtimes q$$

be a lifted product presentation of one cut sector on the quotient branch, and let the R*-sector contribution of the collar edge-center entropy operator be*

$$(L_C)\,|_R = \log d_R.$$

Then

$$(L_C)\,|_R = L_C^{(3)} + L_C^{(2)}, \qquad L_C^{(3)} := \log d_{R_3}, \qquad L_C^{(2)} := \log d_{R_2},$$

because every irreducible U(1) *representation is one-dimensional and contributes* $\log 1 = 0$*. Consequently, on the product heat-kernel branch*

$$\bar{\ell}_{\text{shared}} = \langle L_C \rangle = \bar{\ell}_{\text{SU}(3)}(t_{3,\text{run}}) + \bar{\ell}_{\text{SU}(2)}(t_{2,\text{run}}).$$

If the same branch satisfies the D10 pixel law

$$\bar{\ell}_{\text{SU}(2)}(t_{2,\text{run}}) \ + \ \bar{\ell}_{\text{SU}(3)}(t_{3,\text{run}}) = P/4,$$

then

$$\bar{\ell}_{\text{shared}} = P/4.$$

Proof. For the lifted product presentation $R = R_3 \boxtimes R_2 \boxtimes q$ of a sector on the quotient branch,

$$d_R = d_{R_3} d_{R_2} d_q.$$

Every irreducible U(1) representation is one-dimensional, so $d_q = 1$. Therefore

$$\log d_R = \log d_{R_3} + \log d_{R_2} + \log d_q = \log d_{R_3} + \log d_{R_2}.$$

This proves the sector identity

$$(L_C)\,|_R = L_C^{(3)} + L_C^{(2)}.$$

On the product heat-kernel branch, expectation values add, so

$$\bar{\ell}_{\text{shared}} = \langle L_C \rangle = \bar{\ell}_{\text{SU}(3)}(t_{3,\text{run}}) + \bar{\ell}_{\text{SU}(2)}(t_{2,\text{run}}).$$

If the same branch satisfies the D10 pixel law, the right-hand side equals $P/4$. □

Proposition 5.43 (Local scale-certificate package on the gravity row)**.** *On the declared local extension surface, let the selected no-G scale certificate supply*

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \qquad B_\star = \frac{3\pi}{\ell_\star^2}$$

on the observation-located scale branch. Equivalently, $\ell_\star^2 = 3\pi/B_\star$*. The local cell has the two readings*

$$a_{\text{cell}} = P\ell_\star^2, \qquad \bar{\ell}_{\text{shared}} = \frac{P}{4}.$$

Then the edge area law gives

$$G_{\text{geom}} := \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}} = \ell_\star^2, \qquad G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

More generally, if $\widehat{L}(P)$*,* $\widehat{T}(P)$*,* $\widehat{E}(P)$*, and* $\widehat{\Theta}(P)$ *denote dimensionless branch outputs on that same local branch, then*

$$L_{\text{loc}} = \sqrt{a_{\text{cell}}}\,\widehat{L}(P), \qquad t_{\text{loc}} = \frac{\sqrt{a_{\text{cell}}}}{c_\star}\,\widehat{T}(P),$$

$$E_{\text{loc}} = \frac{\hbar c_\star}{\sqrt{a_{\text{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\text{loc}} = \frac{\hbar c_\star}{k_B\sqrt{a_{\text{cell}}}}\,\widehat{\Theta}(P).$$

Thus, at fixed P*, local lengths scale with* $a_{\text{cell}}^{1/2}$ *alone, local times use that same* $a_{\text{cell}}^{1/2}$ *scale divided by the structural Lorentz output* $c_\star$*, and local mass/energy and temperature rows use the inverse* $a_{\text{cell}}^{1/2}$ *scale dressed only by the familiar-unit display conventions* $\hbar$ *and* k_B*.*

Proof. The OPH scale-readback map supplies $\ell_\star^2$ before the Newton area-law readout is evaluated. By Proposition 5.42,

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4(P/4)} = \frac{a_{\text{cell}}}{P}.$$

Using the local cell-area reading $a_{\text{cell}} = P\ell_\star^2$, this becomes

$$G_{\text{geom}} = \ell_\star^2.$$

The SI display relation is

$$\ell_\star^2 = \frac{\hbar G_{\text{SI}}}{c^3},$$

so

$$G_{\text{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

The following formulas are dimensional bookkeeping on the same declared local branch. Because P is dimensionless and a_{cell} is the only local microscopic area datum, every local length readout is $\sqrt{a_{\text{cell}}}$ times a dimensionless branch quantity. The structural Lorentz output c converts that same local ruler to seconds, while the familiar-unit display constants $\hbar$ and k_B convert the inverse local ruler to energy/mass rows and to Kelvin rows. □

Proposition 5.34 is the algebraic version of interior invariance under compatible exterior substitutions. Proposition 5.35 is the precise bridge from small collar conditional mutual information to the exact modular identities later used by the spatial and null branches: at finite stage one carries a remainder, and exact identities appear only when that remainder is zero or tends to zero in the controlled collar limit. Proposition 5.40 isolates the exact entropy split, while Definition 5.41 supplies the continuum identification that turns the edge center into a gravitational coupling. Proposition 5.42 identifies the shared edge entropy on the realized product-group branch with the same $\mathrm{SU}(2) + \mathrm{SU}(3)$ quantity used by the D10 pixel law, and Proposition 5.43 makes the scale-readout cancellation and familiar-unit display explicit.

6 Modular Geometry, Relativity, and Einstein Dynamics

6.1 Geometric modular flow

Before stating the continuum modular theorem, fix the algebraic target. At fixed cutoff, the full operational patch algebra contains geometric overlap data, declared record summaries, and compare/write/verify pointer layers. The BW question should be asked only on the overlap-generated geometric subnet, not on the entire operational algebra.

Definition 6.1 (Operational total algebra, geometric subnet, and auxiliary/record sector)**.** *At fixed cutoff n and patch P, let $\mathcal{A}_n^{\text{tot}}(P)$ denote the operational patch algebra consisting of the physical patch algebra together with the declared overlap-readable summaries used by the compare/write/verify surface. Let $B_{IJ} \subset P$ range over overlap collars incident on P, let $\Pi_\alpha^{(IJ)}$ denote the overlap sector projectors, and let $Q_a^{(IJ)}$ denote the boundary observables carrying the geometric cut data. Define the geometric subnet by*

$$\mathcal{A}_n^{\text{geo}}(P) := W^*\Big\langle \bigcup_{B_{IJ}\subset P} \iota_{IJ\to P}\big(\{\Pi_\alpha^{(IJ)}\}_\alpha \cup \{Q_a^{(IJ)}\}_a\big)\Big\rangle_{\text{repair,isotony}} \Big/ \mathcal{N}_n(P),$$

where $\mathcal{N}_n(P)$ is the maximal repair-invariant overlap-trivial kernel, namely the maximal part whose restriction to every overlap algebra inside P is scalar. The complementary operational sector generated by record summaries, pointer registers, and other interface-inert observables is denoted $\mathcal{A}_n^{\mathrm{aux/rec}}(P)$.

Theorem 6.2 (Geometric subnet and operational primeness)**.** *For each fixed-cutoff patch P, the operational algebra admits a canonical quotient*

$$\pi_{n,P}^{\mathrm{geo}} : \mathcal{A}_n^{\mathrm{tot}}(P) \twoheadrightarrow \mathcal{A}_n^{\mathrm{geo}}(P)$$

with the following properties:

(i) *$\mathcal{A}_n^{\mathrm{geo}}(P)$ is generated by the overlap sector projectors and geometric boundary observables, and is closed under repair and isotony;*

(ii) *the record-summary block, pointer algebra, and interface-inert auxiliary observables lie in the complementary operational sector $\mathcal{A}_n^{\mathrm{aux/rec}}(P)$ and do not enlarge $\mathcal{A}_n^{\mathrm{geo}}(P)$;*

(iii) *if $N \subset \mathcal{A}_n^{\mathrm{geo}}(P)$ is a repair-invariant subalgebra whose restriction to every overlap algebra inside P is scalar, then $N = \mathbb{C}\,\mathbf{1}$.*

Proof. Items (i) and (ii) are the content of Definition 6.1. For item (iii), if such an N were nontrivial, then its pullback under $\pi_{n,P}^{\mathrm{geo}}$ would define a nontrivial repair-invariant overlap-trivial component of $\mathcal{A}_n^{\mathrm{tot}}(P)$, contradicting the maximality of $\mathcal{N}_n(P)$. Thus no nontrivial overlap-trivial factor survives inside $\mathcal{A}_n^{\mathrm{geo}}(P)$. □

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a literal statement about the finite regulator matrices. The target object is a support-visible realized scaling-limit geometric cap pair

$$(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$$

for each round cap $C \subset S^2$. Axiom 3.7 controls the state-side realized branch across refinement through one common finite-dimensional MaxEnt family, and Theorem 6.2 removes the overlap-trivial auxiliary/record layers from the BW target. The theorem below uses support-visible regularized modular transport and the finite cap-normal BW certificate of Theorem 6.6. The analytic modular-transport branch is closed under the stated regularization and multiresolution hypotheses; geometric BW identification additionally requires the cap-normal, support-order, cross-ratio, mixed-GNS, and geometric 2π-KMS certificate.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C, with $\delta \downarrow 0$ and $A_\delta \cup B_\delta \cup D_\delta$ covering a neighborhood of the cut. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1-e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\,\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound used to compare modular Hamiltonians. Every exact collar identity below is therefore to be read in one of two ways:

1. literal exactness when the reference collar state is exact Markov; or
2. a controlled collar family for which

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with the finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η^{M}_δ carried explicitly.

On the finite-range conditional-mixing branch, Theorem 5.3 supplies $\varepsilon_\delta \le c|\partial C|_{\mathrm{UV}} e^{-\delta/\xi}$. Vanishing of this quantity uses the full rate condition of Corollary 5.4; the ratio $\delta/\ell_{\mathrm{UV}} \to \infty$ without boundary-count control is not used below.

For each cap C, let $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

6.1.1 Finite cap-normal certificate for the support-visible BW limit

Definition 6.3 (Oriented cap normal and BW frame). *Identify $\Omega \in S^2$ with the future null ray $q(\Omega) = (1, \Omega) \in \mathbb{R}^{3,1}$, with $\eta(x, y) = -x^0y^0 + \mathbf{x}\cdot\mathbf{y}$. For a round cap*

$$C = C(u, \theta) = \{\Omega : u \cdot \Omega \ge \cos\theta\}, \qquad 0 < \theta < \pi,$$

define its oriented spacelike normal by

$$n_C = (\cot\theta, \csc\theta\, u).$$

Then $\eta(n_C, n_C) = 1$ and

$$\eta(n_C, q(\Omega)) = \frac{u\cdot\Omega - \cos\theta}{\sin\theta},$$

so $C = \{\Omega : \eta(n_C, q(\Omega)) \ge 0\}$. The sign reversal $n_C \mapsto -n_C$ selects the complementary cap. A nondegenerate cap class fixes

$$\theta_0 \le \theta \le \pi - \theta_0, \qquad 0 < \theta_0 < \frac{\pi}{2},$$

so the cap-normal map lands in a compact subset of de Sitter space $\{n : \eta(n, n) = 1\}$.

A BW-framed cap is

$$\widehat{C} = (C, n_C, p_C^-, p_C^+),$$

where $p_C^\pm \in \partial C$ are distinct ordered boundary points. Choose any orientation-preserving Möbius map $h_{\widehat{C}}$ with

$$h_{\widehat{C}}(C) = \mathbb{H}, \qquad h_{\widehat{C}}(p_C^-) = 0, \qquad h_{\widehat{C}}(p_C^+) = \infty.$$

The independently normalized geometric cap flow is

$$\lambda_{\widehat{C}}(s)(z) = h_{\widehat{C}}^{-1}(e^{-s}h_{\widehat{C}}(z)).$$

This parameter s is geometric; it is not the modular parameter t.

Definition 6.4 (Finite cap-normal BW certificate). *A cofinal regulator branch carries* $\mathsf{FiniteCapBWCertificate}$ *when it supplies the following primitive data and vanishing envelopes on the support-visible prime geometric subnet:*

C1. *cap and point meshes with $h_r^{\mathrm{cap}} \to 0$, $h_r^{\mathrm{pt}} \to 0$, unit cap-normal residuals tending to zero, nondegenerate cap radii, boundary-incidence control, and refinement-compatible cap normals;*

C2. *ordered BW frame points p_C^-, p_C^+ on every anchor cap, with frame separation and orientation bounded away from degeneration;*

C3. *finite cap algebras satisfying isotony, support-order separation, and a support-visible automorphism quotient whose cap-support action is faithful;*

C4. *the exact modular-compatible reference tower and the cap-family-uniform compact-time regularized modular bound of Theorem 6.21;*

C5. *mixed-GNS Cauchy residuals and support-covariance residuals tending to zero uniformly on compact modular-time intervals, with both t and $-t$ controlled;*

C6. *approximate identity, inverse, and group-law residuals for the finite support flow, compact-time equicontinuity, anchor-cap/frame preservation, held-out complex cross-ratio convergence on separated quartets, and an orientation witness excluding anti-Möbius limits;*

C7. *an independently parameterized geometric 2π-KMS strip residual tending to zero, with uniform strip bounds and refinement-compatible matrix elements;*

C8. *nontriviality plus either a wrong-β lower gap on a predeclared interval or the finite type-I generator-distance/noncentrality bound*

$$\inf_{Z \in Z(M_r(C))_{\mathrm{sa}}} \|K_{r,C}^{(a_r)} - 2\pi B_{r,C} - Z\|_{2,\omega_r} \to 0, \qquad \inf_{Z \in Z(M_r(C))_{\mathrm{sa}}} \|B_{r,C} - Z\|_{2,\omega_r} \geq v_* > 0.$$

The certificate is a branch hypothesis. This paper does not prove

$$\mathsf{BareFiniteOPHConsensus} \Longrightarrow \mathsf{FiniteCapBWCertificate}.$$

That production implication belongs to the Einstein branch-entry program.

Proposition 6.5 (Why the certificate clauses are necessary)**.** *The certificate cannot be compressed to cap normals, finite cap IDs, ordinary KMS language, or pointwise weak matrix convergence.*

Proof. First, a round cap and its oriented normal determine the cap but not a cap-preserving one-parameter subgroup. After conjugating the cap to the disk, its orientation-preserving stabilizer is $\mathrm{PSU}(1,1)$, which contains elliptic, hyperbolic, and parabolic one-parameter subgroups. The ordered boundary frame is the extra datum selecting the BW axis and direction.

Second, a finite cap-ID set cannot carry a nontrivial continuous boost as permutations. Every continuous homomorphism $\mathbb{R} \to \mathrm{Sym}(\mathcal{C}_r)$ is constant because $\mathbb{R}$ is connected and $\mathrm{Sym}(\mathcal{C}_r)$ is discrete. A finite simulator may sample and project a continuous cap flow, but the projection must carry mesh error.

Third, modular KMS alone does not pick 2π. If σ_t^ω is the modular group, then $\alpha_s^{(\kappa)} = \sigma_{s/\kappa}^\omega$ makes ω a κ-KMS state for every $\kappa > 0$. The value 2π is fixed only after the geometric parameter s has been normalized independently and the state is shown to be 2π-KMS for that geometric flow.

Fourth, pointwise weak matrix convergence does not preserve automorphisms: on $\ell^2(\mathbb{Z})$, the bilateral shifts S^n are unitary but converge weakly to 0. The quadratic mixed-GNS Cauchy residual and inverse-time control prevent this collapse. ☐

Theorem 6.6 (Finite cap-net convergence to the support-visible BW automorphism)**.** *Let* $(\mathfrak{B}_r)_{r\in I}$ *be a cofinal family of finite cap-normal BW systems satisfying* FiniteCapBWCertificate*. Then every nondegenerate BW-framed cap* $\widehat{C}$ *has a unique support-visible prime geometric scaling-limit cap pair*

$$(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$$

and

$$\boxed{\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.}$$

Moreover, for every compact modular-time interval and every fixed support-visible test triple, the finite transported regularized modular matrix elements converge to the BW matrix elements:

$$\sup_{|t|\le T} \left|\omega_r(B_r^* \tau_t^{r,C_r}(A_r)D_r) - \omega_\infty(B^*\alpha_{\lambda_{\widehat{C}}(2\pi t)}(A)D)\right| \to 0.$$

Cap inclusion, round-cap structure, and complex cross-ratios of separated quartets are preserved. If the limiting cap algebra is type I, the inner-generator specialization is

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}},$$

and in a factor $Z_C = c_C\mathbf{1}$*. No density-matrix generator identity is asserted in the generic non-type-I case.*

Proof. The mixed-GNS Cauchy residual and the exact reference tower produce a unique strongly continuous unitary implementation of the limiting modular automorphism group on the support-visible GNS direct limit. Support covariance and support-order separation make the modular image of every test-cap algebra read as a unique cap-support flow, while the approximate identity, inverse, and group-law residuals pass to the limit.

Cap-normal compactness gives round-cap limits, and support-order reflection preserves cap inclusion. Held-out complex cross-ratio convergence on dense point meshes, with conditioned anchors and an orientation witness, forces every support-flow cluster limit to lie in $\mathrm{PSL}(2,\mathbb{C})$. Since the flow preserves the anchor cap and the ordered boundary frame, framed-cap rigidity gives

$$f_t^C = \lambda_{\widehat{C}}(\kappa_C t)$$

for a unique $\kappa_C > 0$.

The geometric strip residual, normal-family bound, and refinement-compatible geometric matrix elements pass the 2π-KMS condition to the limiting independently normalized geometric flow. KMS uniqueness for a faithful normal state gives

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

The wrong-β gap excludes any fixed normalization separated from 2π, while nontriviality excludes the only exact multiple-temperature exception. Since every convergent subnet has the same BW limit, compactness upgrades subnet convergence to convergence of the cofinal branch. The type-I generator statement follows because two inner implementations of the same automorphism differ by a central self-adjoint term. $\square$

Corollary 6.7 (Geometric modular action on certified caps)**.** *Let*

$$(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$$

be a scaling-limit cap pair emitted by Theorem 6.6. Then

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit cap algebra is type I, this may be represented as

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}}.$$

In the generic non-type-I case, the automorphism identity is the complete statement.

Remark 6.8 (BW is a conditional scaling-limit branch)**.** *The Lorentz/BW statement is not "finite cells imply Lorentz invariance." The branch theorem is the conditional implication*

$$\begin{gathered}\mathsf{FiniteCapBWCertificate}\\ \Longrightarrow \sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.\end{gathered}$$

The certificate includes support-visible cap-pair extraction, cap-family-uniform regularized modular transport, support-readable covariance, framed-cap/cross-ratio rigidity, independently normalized geometric 2π-KMS convergence, and wrong-normalization controls. Finite type-I regulators supply collar control and regularized support-visible matrix elements; they do not make the full finite operational algebra, pointer/record sectors, or off-support directions Lorentz covariant, and bare finite consensus does not by itself produce this certificate.

Proposition 6.9 (Support-visible cap-pair extraction on the local GNS support quotient)**.** *Fix a round cap $C \subset S^2$. Under Axioms 3.5–3.10, Theorem 6.2, and the derived fixed-cutoff regulator/collar/consensus package, consider the transported geometric cap-local test family on $\mathcal{A}_n^{\mathrm{geo}}(C_n)$, its projectively compatible transported marginals, the asymptotic transport-equivalence certificate, and a cutoff schedule satisfying Theorem 6.18. Along a cofinal refinement subnet:*

(i) *the transported cap marginals admit local weak-$*$ limits on every support-visible geometric cap subalgebra;*

(ii) *the compatible local limits glue to a state on $\mathcal{A}_\infty^{\mathrm{geo,sv}}(C)$;*

(iii) *the GNS representation gives a cap pair*

$$(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$$

faithful on the local support quotient, and the regularized support-visible modular matrix elements converge to its modular automorphism group.

Proof. For each fixed local collar model, the finite-stage state spaces are weak-$*$ compact. The transported marginal family is projectively compatible up to the refinement-equivalence certificate supplied by the consensus package, so the usual diagonal subnet argument gives compatible local weak-$*$ limits on the cap-local test family. Theorem 6.18 controls the regularized modular matrix elements on every bounded support-visible collar observable under the stated cutoff schedule. Passing to the GNS representation of the glued state quotients the null directions and leaves a faithful representation on the local support. This gives the displayed support-visible cap pair and its modular automorphism group. ☐

Lemma 6.10 (Support-readable modular covariance)**.** *Let $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ be the cap pair of Proposition 6.9. Assume the extracted support-visible prime geometric subnet satisfies outer regularity / minimal support: for every cap-local observable O, the intersection of all connected regions P with $O \in \mathcal{A}_\infty^{\mathrm{geo,sv}}(P)$ is again a connected region, denoted* $\mathrm{supp}(O)$*. For every connected cap-local region $R \subset C$, define*

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}_\infty^{\mathrm{geo,sv}}(R)} \mathrm{supp}\Big(\sigma_t^{\omega_\infty^{\mathrm{geo},C}}(O)\Big).$$

Then the modular automorphism group induces a one-parameter support map on cap-local regions:

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}}(\mathcal{A}_\infty^{\mathrm{geo,sv}}(R)) = \mathcal{A}_\infty^{\mathrm{geo,sv}}\Big(f_t^C(R)\Big).$$

Proof. The modular group is an automorphism group of the cap algebra. By outer regularity / minimal support, the support of each cap-local observable is readable from the net labeling after GNS quotienting of null directions. Thus the displayed definition of $f_t^C(R)$ is intrinsic to the support-visible net. The inclusion "$\subseteq$" follows by construction. Applying the same argument to the inverse automorphism $\sigma_{-t}^{\omega_\infty^{\mathrm{geo},C}}$ gives the reverse inclusion, hence equality. $\square$

Theorem 6.11 (Projective Markov replacement compatibility)**.** *Let ρ_n be a cofinal refinement family, and let $R_{n\to m}$ denote restriction to a fixed local collar model m. Suppose that for each fixed m*

$$I(A_m : D_m \mid B_m)_{R_{n\to m}\rho_n} \le \varepsilon_{n,m}, \qquad \delta_m^{\mathrm{M}}(\varepsilon_{n,m}) \to 0 \quad (n \to \infty),$$

and suppose the finite-stage restriction maps commute with the physical quotient and normal-form maps up to the declared refinement-equivalence certificate. Then, after passing to a cofinal subnet, one can choose exact Markov replacements $\sigma_{n,m} \in \mathfrak{M}_m$ such that for every fixed $k \le m$,

$$\|R_{m\to k}\sigma_{n,m} - \sigma_{n,k}\|_1 \to 0.$$

Hence the exact-Markov comparison states define a projective support-visible Markov comparison class in the scaling limit.

Proof. For each fixed m, Proposition 5.31 gives exact Markov replacements within $\delta_m^{\mathrm{M}}(\varepsilon_{n,m}) + o(1)$ of $R_{n\to m}\rho_n$. The exact Markov sets $\mathfrak{M}_m$ are compact on fixed collars, so a diagonal subnet has limits on every fixed m. Restriction maps are continuous and, by hypothesis, commute with the physical quotient and normal form up to the same refinement-equivalence certificate used by the cap-local test family. Therefore the restriction of the m-limit to a smaller fixed collar k agrees with the k-limit. Pulling the chosen approximants along the diagonal subnet gives the displayed asymptotic compatibility. $\square$

Proposition 6.12 (Framed-cap rigidity from surviving cut data)**.** *Let $\widehat{C} = (C, n_C, p_C^-, p_C^+)$ be a BW-framed cap, and let the cap pair be the support-visible extracted pair of Proposition 6.9. Assume the modular support map of Lemma 6.10 acts by orientation-preserving conformal maps on the support-visible cap geometry and preserves the anchor cap and its ordered boundary frame. Then any continuous one-parameter subgroup preserving this framed data is $\lambda_{\widehat{C}}(\kappa t)$ for a unique $\kappa > 0$.*

Proof. Choose the Möbius map $h_{\widehat{C}}$ of Definition 6.3. The orientation-preserving conformal maps of the upper half-plane preserving 0 and ∞ individually are exactly $z \mapsto az$, $a > 0$. For a continuous one-parameter group, $a(t+s) = a(t)a(s)$, so $a(t) = e^{-\kappa t}$ after the declared frame orientation fixes the sign. Transporting this subgroup back to C gives $\lambda_{\widehat{C}}(\kappa t)$. Uniqueness of κ follows from nontriviality of the dilation subgroup. The cap normal alone is not enough: it fixes the cap side, not the ordered boundary fixed points. $\square$

Theorem 6.13 (Finite cap-net support-visible BW scaling theorem)**.** *Assume Axioms 3.5–3.10, Theorem 6.2, and the derived fixed-cutoff regulator/collar/consensus package established above. Assume also that the cofinal branch carries the finite cap-normal BW certificate of Definition 6.4: cap-normal density and nondegeneracy, BW framing, support-order faithfulness, cap-family-uniform mixed-GNS modular convergence, support covariance, held-out oriented cross-ratio convergence, independently normalized geometric 2π-KMS convergence, and wrong-normalization separation. For each OPH-realized observer-supporting refinement branch and each nondegenerate BW-framed cap $\widehat{C} = (C, n_C, p_C^-, p_C^+)$, the support-visible prime geometric cap net admits a weak-$*$/GNS scaling-limit cap pair*

$$(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C}).$$

Then:

(i) *At each finite regulator stage, the cap algebra is type I, the reduced cap state has a density matrix $\rho_C^{(\delta)}$, and its modular Hamiltonian*

$$K_C^{(\delta)} := -\log \rho_C^{(\delta)}$$

has nonadditive part confined to the shrinking collar up to the carried errors measured by $r_{\mathrm{FR}}(\varepsilon_\delta)$, the fixed-collar replacement modulus δ^{M}, and the regularized support-visible modular transport bound of Theorems 6.18 and 6.19. In particular, on each fixed collar model the physical collar state differs from its constructive recovered comparison state by $r_{\mathrm{FR}}(\varepsilon_\delta)$ in trace norm, and the support-visible modular matrix elements converge under the displayed cutoff schedule independently of the exact-Markov replacement sequence.

(ii) *For every bounded support-visible cap-local observable O, the regularized modular matrix elements converge in the local weak-$*$/GNS support-quotient topology supplied by Proposition 6.9. The convergence is controlled by the explicit error budget*

$$r_{\mathrm{FR}}(\varepsilon_\delta), \qquad \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta), \qquad \frac{\Delta_\delta}{a_\delta}, \qquad d_{m,\delta}a_\delta, \qquad \Delta_\delta|\log a_\delta|,$$

which vanishes under the stated Markov-replacement and cutoff schedule.

(iii) *On the support-visible scaling-limit cap pair, the modular automorphism group is geometric:*

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

Equivalently, the modular parameter t and the geometric cap-dilation parameter s are related by

$$s = 2\pi t.$$

(iv) *No separate cap-isotropy/SO(2)-equivariance selector, finite-cell Lorentz-invariance premise, or unregularized full-algebra common floor is used in this theorem. The conformal support-map regularity, framed-cap/cross-ratio rigidity, and geometric 2π-KMS normalization are explicit certificate clauses, not hidden finite-regulator Lorentz premises. The target algebra is the support-visible prime geometric subnet.*

(v) *If the scaling-limit cap algebra happens to be type I, item* (iii) *may be written as the operator identity*

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}}.$$

If the cap algebra is a factor, $Z_C = c_C\mathbf{1}$. In the generic continuum case, where the scaling-limit cap algebra has left the regulator class and is expected in QFT examples to be non-type-I, the correct theorem statement is item (iii) *itself, and the geometric modular action is generally outer instead of inner.*

Consequently, on a branch carrying the multiresolution regulator certificate of Assumption 6.20, the transported regularized modular action on every fixed support-visible geometric local algebra converges uniformly on compact modular-time intervals to the geometric cap-dilation action. Without that certificate, the theorem identifies only the scaling-limit cap-pair modular automorphism. It does not assert compact-time convergence of finite-regulator modular groups beyond the stated certificate, a refinement-uniform unregularized spectral floor for every off-support direction of the full finite matrix algebra, or Lorentz covariance for record/pointer or interface-inert auxiliary registers outside the extracted geometric subnet.

Proof. Step 1: fixed-cutoff collar control. On the finite type-I regulator net, Propositions 5.34, 5.35, and 5.40 localize the modular defect to the shrinking collar and quantify the discrepancy between the physical collar state, its constructive recovered comparison state, and its exact-Markov reference by the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}.

Step 2: support-visible regularization. Proposition 6.17 shows that a refinement-uniform common floor on the full finite matrix algebra is unavailable in general. Theorem 6.18 supplies the needed replacement: for $K_a(\rho) = -\log(\rho + a\mathbf{1})$ and bounded support-visible collar observables, the modular matrix-element difference is bounded by

$$\|O\| \left(\frac{4\Delta_n}{a} + d_{m,\delta}a + 4\Delta_n |\log a| \right).$$

Choosing $a_n \downarrow 0$ with $\Delta_n/a_n \to 0$, $d_{m,\delta}a_n \to 0$, and $\Delta_n|\log a_n| \to 0$ makes the right-hand side vanish on every fixed local collar model. Theorems 6.11 and 6.19 then make the exact-Markov comparison family projectively compatible and replacement-independent on support-visible observables. Thus the theorem concerns the support-visible scaling limit instead of off-support full-algebra directions.

Step 3: scaling-limit cap-pair extraction. Theorem 6.2 fixes the target algebra as the overlap-generated prime geometric subnet, with overlap-trivial auxiliary/record factors removed. Proposition 6.9 supplies the support-visible weak-$*$/GNS cap pair from the transported cap-local test family, the projectively compatible marginal family, the asymptotic transport-equivalence certificate, and the regularized modular transport cutoff schedule.

Step 4: certificate-level geometric identification on the prime subnet. Lemma 6.10 makes the modular automorphism group induce a support map on cap-local regions of the extracted prime geometric subnet. The finite cap-normal certificate supplies support-order faithfulness, cap-frame preservation, held-out complex cross-ratio convergence, orientation preservation, and compact-time equicontinuity. By Theorem 6.6, the limiting support flow is the framed cap flow. Thus

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}.$$

with the certificate-selected frame understood in the notation λ_C.

Step 5: normalization. The certificate does not use the false shortcut that modular KMS alone selects 2π. Instead, its independently normalized geometric strip residual passes the 2π-KMS condition to $s \mapsto \alpha_{\lambda_{\widehat{C}}(s)}$, and KMS uniqueness gives

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

The wrong-β gap excludes every fixed normalization separated from 2π, and nontriviality excludes the only exact multiple-temperature exception.

Step 6: operator versus automorphism form. If the limit cap algebra is type I, one can represent the modular automorphism group by a modular Hamiltonian $K_C = -\log \rho_C$ and a geometric generator B_C. Equality of the implemented automorphisms gives $K_C = 2\pi B_C + Z_C$ for a central self-adjoint Z_C. If the limit algebra leaves the regulator class, the modular automorphism group is well defined while the inner operator representative need not exist inside $\mathcal{A}_\infty^{\mathrm{geo,sv}}(C)$. In that case the automorphism identity is the full theorem statement. This is exactly the observer-facing content needed for Lorentz kinematics, the null half-sided modular bridge, and the local Einstein branch. □

Remark 6.14 (Finite modular-flow core versus BW geometry)**.** *At finite regulator and on finite-dimensional faithful cap algebras, the modular clock itself has a purely algebraic core: a positive density matrix ρ has a modular Hamiltonian $H = -\log \rho$, the maps*

$$\sigma_z(A) = e^{-izH} A e^{izH}$$

form a complex-parameter automorphism family, real $z = t$ gives the one-parameter *-automorphism flow, the state is invariant under the flow, and the $z = -i$ boundary value is the modular operator appearing in the KMS identity. The OPHProofChain finite Lean audit formalizes this matrix-algebra core, including the KMS boundary condition and uniqueness within Hamiltonian-implemented flows up to the additive scalar in H.

This finite result is not the BW theorem. It proves that the finite density-matrix modular object really is a real-time modular flow. The OPH physics in Theorem 6.13 is the additional support-visible scaling-limit identification of that modular flow with the geometric cap-dilation subgroup $\alpha_{\lambda_C(2\pi t)}$.

Definition 6.15 (BW-branch observer-relative time reading)**.** *On the branch satisfying the hypotheses of Theorem 6.13, OPH uses the modular automorphism parameter t of the extracted cap pair $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$ as that cap observer's relative time parameter. This is a declared BW-branch reading of the geometric modular flow derived in Theorem 6.13; it is not an additional proof that arbitrary operational clocks, global time, or the full problem of time have been derived from the axioms.*

Remark 6.16 (BW-side UV scaffold after the geometric-subnet split)**.** *Theorem 6.2 fixes the target algebra: the continuum Lorentz claim is asked only on the support-visible extracted prime geometric subnet, not on the full operational algebra. Record/pointer/interface-inert observables are outside that geometric subnet, and off-support directions that disappear in the limiting GNS support are not observer-facing data.*

The transported geometric cap-local system consists of the cap-local test family on $\mathcal{A}_n^{\mathrm{geo}}(C_n)$, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. On each fixed local collar model, the regularized estimate of Theorem 6.18 replaces the unavailable unregularized full-algebra floor. Proposition 6.9 gives the local GNS support quotient cap pair, Lemma 6.10 reads modular flow as a support map on cap-local regions, and Proposition 6.12 identifies the residual cap-preserving conformal freedom with the standard hyperbolic subgroup. This is the theorem-side content of the BW lift used below.

Proposition 6.17 (Recoverability is not modular geometry: common-floor collapse countermodel)**.** *Exact or asymptotically exact Markov recovery at finite cutoff does not imply the eventual common floor used in the unregularized BW/geometric cap-pair extraction. In $M_2(\mathbb{C})$, the faithful states*

$$\rho_n = \begin{pmatrix} e^{-n} & 0 \\ 0 & 1-e^{-n} \end{pmatrix}$$

have $\lambda_{\min}(\rho_n) \to 0$. Tensoring this family with any fixed finite exact-Markov collar factor gives a full-rank exact-Markov collar family, but no refinement-uniform lower spectral floor survives. On an off-diagonal matrix unit the modular generator carries a logarithmic gap of order n, so unregularized modular transport can fail even though every finite stage is faithful and Markov. Thus the full-algebra common-floor route is the wrong target; Theorem 6.18 supplies the support-visible replacement used by Theorem 6.13.

Theorem 6.18 (Regularized support-visible modular transport)**.** *Fix a local collar model of finite dimension $d_{m,\delta}$. Let ρ_n be the transported physical collar marginal, $\widehat{\rho}_n$ the exact-Markov comparison marginal, and $\Delta_n = \|\rho_n - \widehat{\rho}_n\|_1$. For $a > 0$, set $K_a(\rho) = -\log(\rho + a\mathbf{1})$. For every bounded support-visible collar observable O,*

$$\left|\operatorname{Tr} \rho_n O\big(K_a(\rho_n) - K_a(\widehat{\rho}_n)\big)\right| \le \|O\| \left(\frac{4\Delta_n}{a} + d_{m,\delta}a + 4\Delta_n |\log a| \right).$$

If $a_n \downarrow 0$, $\Delta_n/a_n \to 0$, $d_{m,\delta}a_n \to 0$, and $\Delta_n|\log a_n| \to 0$, then the regularized support-visible modular matrix elements converge on that fixed collar model.

Proof. The integral representation of the operator logarithm gives the displayed Lipschitz control above the spectral cutoff a. Splitting the comparison into the visible support above a, the a-tail, and the trace-distance error gives the three terms in the bound, which vanish under the stated cutoff schedule. □

Theorem 6.19 (Support-visible modular convergence from controlled Markov collars)**.** *Let $(A_\delta, B_\delta, D_\delta)$ be a shrinking collar family around a cap cut. Suppose that after pullback to every fixed local collar model,*

$$r_{\mathrm{FR}}(\varepsilon_\delta) \to 0, \qquad \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0,$$

and suppose the regularized cutoff schedule in Theorem 6.18 is satisfied. Then for every bounded support-visible geometric collar observable O,

$$\lim_{\delta \downarrow 0} \operatorname{Tr} \rho_\delta O\big(K_{a_\delta}(\rho_\delta) - K_{a_\delta}(\sigma_\delta)\big) = 0$$

for any projectively compatible exact-Markov replacement family σ_δ. Consequently, the scaling-limit modular automorphism on the extracted geometric cap pair is independent of the particular exact-Markov replacement sequence used to audit the finite-stage calculation.

Proof. Theorem 6.18 gives the displayed matrix element bound with $\Delta_\delta = \|\rho_\delta - \sigma_\delta\|_1$. The controlled Markov replacement hypotheses make $\Delta_\delta \to 0$ on each fixed collar model, and the chosen a_δ schedule sends

$$\Delta_\delta/a_\delta, \qquad d_{m,\delta}a_\delta, \qquad \Delta_\delta|\log a_\delta|$$

to zero. The projective compatibility theorem ensures that different compatible replacement choices determine the same local weak-$*$ limits after GNS quotienting of null directions. Therefore the support-visible cap modular automorphism emitted in the scaling limit is a property of the physical cap pair, not of an arbitrary replacement choice. □

Assumption 6.20 (Multiresolution regulator certificate)**.** *For every round cap C, the support-visible geometric regulator is represented on a cofinal branch by finite algebras $M_r(C)$, faithful reference states $\widehat{\omega}_r^C$, refinement embeddings ι_{rs}, and faithful state-preserving conditional expectations E_{sr} satisfying*

$$\iota_{st}\iota_{rs} = \iota_{rt}, \qquad E_{tr} = E_{sr}E_{ts}, \qquad \widehat{\omega}_s^C \circ \iota_{rs} = \widehat{\omega}_r^C.$$

The tower is the explicit multiresolution edge-center construction of Theorem `2.6d` *in the main paper: in bare shell coordinates, the fine algebra is the coarse algebra tensored with faithful detail factors, while the OPH patch presentation is obtained by a finite-depth local circuit.*

For a physical regulator state at stage s, let $\rho^{\mathrm{phys}}_{s\downarrow r,C}$ denote its restriction to the embedded fixed local algebra $M_r(C)$, let $\widehat{\rho}_{r,C}$ be the reference density matrix there, and put

$$\epsilon_{s,r,C} := \|\rho^{\mathrm{phys}}_{s\downarrow r,C} - \widehat{\rho}_{r,C}\|_1.$$

The branch certificate requires $\epsilon_{s,r,C} \to 0$ for every fixed support-visible local algebra, or one stronger uniform envelope on the declared local test tower.

Theorem 6.21 (Strong local scaling of regularized modular groups)**.** *Under Assumption 6.20, the reference modular groups are exactly refinement compatible:*

$$\sigma_t^{\widehat{\omega}_s^C}(\iota_{rs}A) = \iota_{rs}(\sigma_t^{\widehat{\omega}_r^C}(A)).$$

For a regularizer $a_s > 0$, define on the transported fixed local algebra

$$K^{(a_s)}_{s\downarrow r,C} := -\log(\rho^{\mathrm{phys}}_{s\downarrow r,C} + a_s\mathbf{1}).$$

If $\lambda_{r,C} = \lambda_{\min}(\widehat{\rho}_{r,C})$, then for every $A \in M_r(C)$ and $T < \infty$,

$$\sup_{|t|\le T} \left\| e^{-itK^{(a_s)}_{s\downarrow r,C}} A e^{itK^{(a_s)}_{s\downarrow r,C}} - \sigma_t^{\widehat{\omega}_r^C}(A) \right\| \le 2T\|A\| \left[\frac{\epsilon_{s,r,C}}{a_s} + \log\left(1 + \frac{a_s}{\lambda_{r,C}}\right)\right].$$

Consequently, $a_s \downarrow 0$ and $\epsilon_{s,r,C}/a_s \to 0$ give uniform-on-compact-time convergence on every fixed support-visible local algebra. Pointwise projective convergence on a countable local test tower always admits a cofinal diagonal subsequence and one cutoff schedule with this property.

Proof. In multiresolution coordinates the reference density at stage s is $\widehat{\rho}_{r,C} \otimes \tau_{s\setminus r}$ and the embedded observable is $A \otimes \mathbf{1}$, which proves exact modular compatibility. For positive matrices above the floor a_s,

$$\|\log X - \log Y\| \le a_s^{-1}\|X - Y\|.$$

Comparing first with $-\log(\widehat{\rho}_{r,C} + a_s\mathbf{1})$ and then with $-\log\widehat{\rho}_{r,C}$ gives the bracketed generator bound. Duhamel's formula gives the displayed compact-time automorphism estimate. For the diagonal schedule, choose stages s_k with $\max_{r\le r_k} \epsilon_{s_k,r,C} \le k^{-4}$ and set $a_{s_k} = k^{-1}$. □

Remark 6.22 (Cap-family uniformity for the finite BW certificate)**.** *For Theorem 6.6, the compact-time estimate above is required uniformly over the declared finite nondegenerate cap family and the fixed separating support-visible test tower. The operator-norm estimate then implies the quadratic mixed-GNS Cauchy residual because $\|XD\|_{2,\omega} \le \|X\|\|D\|_{2,\omega}$. Thus the reference-tower theorems supply the analytic engine; the finite cap-normal certificate adds the geometric support, framing, cross-ratio, normalization, and wrong-scale controls.*

Remark 6.23 (Support-visible closure boundary)**.** *The finite cap-normal certificate implication to the support-visible BW cap automorphism is closed by Theorem 6.13. Production of that certificate from bare finite OPH consensus is excluded by Theorem 6.32; on towers carrying the computable incidence, mesh, cross-ratio, and normalization receipts, the certificate is produced from repair normal forms by Theorem 6.31 (§6.1.2), and those receipts are the irreducible branch content of the parent Einstein branch-entry gate. The theorem does not need, and does not claim, a full-algebra unregularized common floor on directions that collapse out of the limiting observer support. The observer-facing content is the automorphism statement of Theorem 6.13, which is exactly what the Lorentz, null-modular, and local Einstein branches use.*

Remark 6.24 (Exact BW claim boundary)**.** *Theorem 6.13 has two layers. The cap-pair theorem identifies the modular automorphism of the extracted support-visible continuum pair with $\alpha_{\lambda_C(2\pi t)}$. The finite-to-continuum group statement is Theorem 6.21; it applies only to transported regularized modular actions on fixed local algebras. Neither layer asserts an unregularized full-algebra spectral floor, convergence on record/pointer or interface-inert sectors, or finite-cell Lorentz covariance. A black-box AQFT BW route would be a separate certificate requiring conformal-net properties such as isotony, additivity, locality, duality, standardness, positive energy, and suitable modular inclusions.*

6.1.2 Producing the certificate from repair normal forms: incidence receipts, the producer chain, and the dimension-selection boundary

The certificate of Definition 6.4 was introduced above as a hypothesis. This subsection upgrades its status. We construct, for a concrete cofinal family of OPH repair systems, a quotient-intrinsic and refinement-natural readout chain

$$\text{repaired finite quotient normal form} \longrightarrow \text{support-visible incidence complex} \longrightarrow \text{screen/cap data} \longrightarrow \mathsf{FiniteCap}$$

in which no chart, group action, modular flow, or scale is placed in the codomain by declaration. Every geometric object below is computed from the normal form itself, and the residual inputs are reduced to finitely many *computable receipts*: decidable predicates on finite repair stages. The subsection closes with an underdetermination theorem and explicit finite countermodels showing that repair confluence alone selects none of these receipts, so the receipts are exactly the irreducible geometric branch content.

Support-visible incidence data.

Definition 6.25 (Support-visible incidence complex)**.** *Fix a finite transactional quotient repair branch as in Definition 4.4, with patch set $\mathcal{P}_r$ at stage r, quotient algebra $\mathcal{A}_r = \bigvee_{p\in\mathcal{P}_r} \mathcal{A}_{r,p}$, and repaired normal form $W_r = \mathrm{Rep}_\lambda(x_r)$ with MaxEnt reference state ω_{W_r}. For $p \in \mathcal{P}_r$, let $s_p \in \mathcal{A}_{r,p}$ be the central support projection of the restriction $\omega_{W_r}|_{\mathcal{A}_{r,p}}$. The* support-visible incidence complex *$K_r = K(W_r)$ is the abstract simplicial complex with vertex set the quotient-visible patch classes $[p]$ and*

$$\sigma = \{[p_0], \ldots, [p_k]\} \in K_r \quad :\Longleftrightarrow \quad \omega_{W_r}(s_{p_0} s_{p_1} \cdots s_{p_k}) \neq 0,$$

where the product is taken after transport into any common refining patch algebra; the transactional gluing clauses of node D1 make the nonvanishing condition independent of the transport route. Write $K_r^{(2)}$ for the 2-skeleton.

Lemma 6.26 (Quotient, gauge, schedule, and refinement invariance of the incidence complex)**.** *K_r depends only on the quotient normal-form class of x_r:*

1. *(Schedule.) Any two admissible repair schedules from x_r yield the same K_r.*

2. *(Gauge.) The gauge groupoid of node D1 acts on support projections by conjugation, $s_p \mapsto u\, s_p\, u^*$, and leaves every incidence value $\omega_{W_r}(s_{p_0} \cdots s_{p_k}) \neq 0$ unchanged; hence K_r is gauge-invariant.*

3. *(Refinement.) If $\pi_{r'r}$ is a refinement projection compatible with finite-stage normal forms, and the stated Petz support/CPTP control of node D1 holds, then $[p'] \mapsto [\pi_{r'r} p']$ extends to a simplicial map $K_{r'} \to K_r$, and these maps compose along the refinement order.*

Proof. (1) Theorem 4.8 makes the normal form W_r, hence ω_{W_r} and each s_p, independent of the schedule. (2) A gauge transformation acts by a quotient automorphism $\alpha_u = u(\cdot)u^*$ intertwining $\omega_{W_r} \circ \alpha_u^{-1}$ with the transported normal form; supports transform covariantly, $s_p \mapsto u s_p u^*$, and $\omega(u s_{p_0} \cdots s_{p_k} u^*) = \omega'(s'_{p_0} \cdots s'_{p_k})$, so simplex membership is unchanged. (3) Petz support control states that the refinement channel $E_{r'r}$ maps the support of $\omega_{W_{r'}}|_{\mathcal{A}_{r',p'}}$ into the support of $\omega_{W_r}|_{\mathcal{A}_{r,\pi p'}}$; a nonvanishing joint support value at stage r' therefore pushes forward to a nonvanishing joint support value of the images, which is the simplicial-map property. Functoriality follows from compatibility of the projections with normal forms. □

Computable receipts.

Definition 6.27 (Spherical incidence receipt)**.** *Stage r carries the* spherical incidence receipt SphInc_r *when the 2-skeleton $K_r^{(2)}$ is a closed combinatorial surface (every edge lies in exactly two 2-simplices, every vertex link is a single cycle), is connected, satisfies $\chi(K_r^{(2)}) = 2$, and admits a coherent orientation (a choice of cyclic order on each 2-simplex, compatible across shared edges); the orientation is part of the receipt datum. All clauses are decidable by finite search on K_r.*

Definition 6.28 (Cap, mesh, and cross-ratio receipts)**.** *On a refinement tower carrying SphInc_r cofinally:*

1. *(*Disk receipt.*) A stage-r* combinatorial cap *is a subcomplex $C \subset K_r^{(2)}$ that is a combinatorial disk whose boundary ∂C is an incidence cycle; the receipt datum is the finite list Cap_r of such disks selected by the record layer, closed under complements.*

2. *(*Mesh receipt.*) $\mathsf{Mesh}_r(\theta_r)$: every vertex star is contained in some $C \in \mathrm{Cap}_r$, and every $C \in \mathrm{Cap}_r$ contains at least one and at most $N(\theta_r)$ vertices, with $\theta_r \downarrow 0$ along the tower (cofinal nondegenerate cap mesh).*

3. *(*Modular cross-ratio receipt.*) For a combinatorial cap C with boundary cycle $(v_1, \ldots, v_m)$, the cap modular clock of the record layer assigns to each ordered boundary quadruple (v_a, v_b, v_c, v_d) the modular transport time $t_r(v_a, v_b; v_c, v_d)$ needed to flow v_a past v_b relative to the anchors v_c, v_d, and the receipt $\mathsf{CR}_r(\varepsilon_r)$ asserts that the induced finite cross-ratios*

$$\mathrm{cr}_r(v_a, v_b; v_c, v_d) := \exp(2\pi\, t_r(v_a, v_b; v_c, v_d))$$

   form a Cauchy family along refinement with modulus $\varepsilon_r \downarrow 0$, compatible with the simplicial refinement maps of Lemma 6.26.

4. *(*Normalization receipt.*) $\mathsf{KMS}_r(2\pi; \delta_r)$: the independently normalized geometric KMS comparison of Definition 6.4 holds at inverse modular temperature 2π with residual $\delta_r \downarrow 0$, together with the wrong-normalization separation clause.*

The producer chain.

Theorem 6.29 (Topology production: spherical incidence receipts produce S^2)**.** *If stage r carries SphInc_r, then the geometric realization $|K_r^{(2)}|$ is PL-homeomorphic to S^2, uniquely up to PL homeomorphism, and the receipt orientation induces the orientation of S^2. If moreover $r' \geq r$ both carry the receipt and the refinement map of Lemma 6.26 is surjective on 2-simplices and orientation-coherent, then its realization $|K_{r'}^{(2)}| \to |K_r^{(2)}|$ is a degree-one map of 2-spheres.*

Proof. A connected closed combinatorial surface is classified up to PL homeomorphism by orientability and Euler characteristic; orientable with $\chi = 2$ gives S^2. Uniqueness of the PL structure on surfaces is Radó's theorem. A simplicial map between oriented closed surfaces that is surjective and orientation-coherent on top simplices has a well-defined simplicial degree, which the coherence clause fixes to one. □

Theorem 6.30 (Conformal structure and cap production)**.** *Let $(W_r)_{r\in I}$ be a refinement tower of repaired normal forms carrying, cofinally, the receipts SphInc_r, the disk and mesh receipts with $\theta_r \downarrow 0$, and $\mathsf{CR}_r(\varepsilon_r)$ with $\varepsilon_r \downarrow 0$. Then:*

1. *The inverse limit of the boundary-cycle vertex sets, completed in the cross-ratio uniformity, is a topological S^2 (by Theorem 6.29 and mesh density), carrying a unique conformal structure for which the limit cross-ratios of CR_r are the Möbius cross-ratios; with this structure every limit of receipt caps is a round cap, and the assignment is refinement-natural.*

2. *The limit cap family therefore lands in $\mathrm{Cap}^{\mathrm{or}}_{\mathrm{round}}(S^2)$ with nondegenerate radii controlled by the mesh modulus θ_r, producing the cap-normal map of Proposition 6.37* as output*: the cap normals $n_C \in dS_3^{\mathrm{cap}}$ are computed from the produced round caps, with cap-normal residuals bounded by an explicit function $\kappa(\varepsilon_r, \theta_r) \to 0$.*

3. *The produced data are quotient-, gauge-, and schedule-invariant and refinement-natural, by Lemma 6.26 and receipt compatibility.*

Proof. (1) Mesh density with $\theta_r \downarrow 0$ makes the boundary-vertex sets cofinally dense in the inverse-limit surface. On a topological S^2, a separating dense family of boundary circles with coherent cross-ratio data determines a unique atlas of circular charts: fix three limit vertices as Möbius gauge; the Cauchy cross-ratio data then embed every other limit vertex into $\mathbb{C} \cup \{\infty\}$ uniquely, since a point of the Riemann sphere is determined by its cross-ratios against a fixed triple, and consistency across overlapping quadruples is exactly the receipt compatibility clause. Uniform Cauchy moduli ε_r give well-definedness of the limit embedding and its continuity; bijectivity onto S^2 follows from density and compactness. The conformal structure is the pullback of the standard structure; uniqueness holds because Möbius cross-ratios separate conformal structures on S^2 (two structures with identical cross-ratio functionals on a dense family differ by a Möbius map, which is the declared gauge freedom). (2) In the produced conformal chart, a receipt cap has boundary an incidence cycle whose limit quadruple cross-ratios are those of a circle (real cross-ratios for concyclic quadruples, a closed condition implied in the limit by the Cauchy receipt applied to the disk boundary); hence the limit boundary is a round circle and the disk a round cap. The normal n_C is then the finite formula of Proposition 6.37; the residual bound $\kappa(\varepsilon_r, \theta_r)$ follows by continuity of $C \mapsto n_C$ on nondegenerate caps, quantitatively since cot and csc are Lipschitz on mesh-bounded ranges $\alpha \in [\theta_r, \pi - \theta_r]$. (3) is Lemma 6.26 together with the receipt compatibility requirements, which are themselves stated on normal-form data only. □

Theorem 6.31 (Producer theorem: receipts produce the finite cap-normal BW certificate)**.** *Let (W_r) be a cofinal tower as in Theorem 6.30, carrying in addition $\mathsf{KMS}_r(2\pi;\delta_r)$, the mixed-GNS state Cauchy clause of Axiom 3.7, and the D1 gauge/transport package. Then the tower carries* FiniteCapBWCertificate*, with clause-by-clause error envelopes:*

1. *cap-normal density, nondegeneracy, and residuals $\leq \kappa(\varepsilon_r,\theta_r)$ from Theorem 6.30;*
2. *BW framing and support covariance from the produced cap normals plus gauge invariance (Lemma 6.26);*
3. *support-order faithfulness from injectivity of the produced incidence order embedding: distinct receipt caps have distinct produced normals once $\kappa(\varepsilon_r,\theta_r)$ is below the mesh separation θ_r-gap;*
4. *cap-family-uniform mixed-GNS modular convergence from the Axiom 3.7 Cauchy clause evaluated on the produced (finite, mesh-bounded) cap family;*
5. *held-out oriented cross-ratio convergence: this is literally $\mathsf{CR}_r(\varepsilon_r)$ evaluated on held-out quadruples;*
6. *independently normalized geometric 2π-KMS convergence and wrong-normalization separation: this is $\mathsf{KMS}_r(2\pi;\delta_r)$.*

Consequently Theorem 6.13 applies on this tower with all hypotheses produced from repair normal forms and receipts, and the failure modes are localized: if a receipt fails at cofinally many stages, the corresponding certificate clause fails with the same witness, and the produced object degrades exactly as quantified above (topology loss for SphInc*, conformal-class loss for* CR*, scale loss for* KMS*).*

Proof. Each clause is either produced by Theorems 6.29–6.30 (clauses 1–3, with the quantitative separation argument for clause 3: two caps whose normals coincide up to κ have boundary circles within a mesh gap, contradicting the disk-receipt separation once $\kappa <$ gap), restated from an axiom-level Cauchy clause on the produced finite family (clause 4), or is itself a receipt (clauses 5–6). The failure localization is immediate because each clause consumed exactly one receipt. ☐

The dimension-selection boundary.

Theorem 6.32 (Underdetermination: confluence selects no topology, dimension, framing, or normalization)**.** *There exist finite transactional quotient repair systems $\mathfrak{B}^{(S^2)}$, $\mathfrak{B}^{(T^2)}$, $\mathfrak{B}^{(S^3)}$, and $\mathfrak{B}^{(\vee)}$ such that:*

1. *each satisfies every node-D1 hypothesis (repair completeness, transactional acceptance with validation-complete read sets, termination, confluence, unique schedule-independent quotient normal form, boundary/sector preservation);*
2. *their repair transition systems are isomorphic as abstract rewrite systems (same state count, same conflict components, same repair-path category), so no confluence-level invariant distinguishes them;*
3. *their support-visible incidence complexes are, respectively, a triangulated S^2, a triangulated T^2, the boundary complex $\partial\Delta^4$ (a combinatorial S^3, locally three-dimensional), and two triangulated 2-spheres wedged at one patch (not a manifold);*
4. *for every $\beta > 0$ there is a variant $\mathfrak{B}^{(S^2)}_\beta$ with the same repair structure whose cap modular comparison holds at inverse temperature β instead of 2π.*

Consequently SphInc*, the disk/mesh receipts, and* $\mathsf{KMS}(2\pi)$ *are not consequences of bare finite OPH consensus, and the implication displayed above (*$\mathsf{BareFiniteOPHConsensus} \not\Rightarrow \mathsf{FiniteCapBWCertificate}$*) is strict at the level of topology, dimension, and normalization.*

Proof. Fix any finite abstract patch set P and equip each patch with the same two-level record algebra $\mathbb{C}^2$, with overlap constraints requiring equality of the shared bit on adjacent patches and a single soft conflict seeded on one fixed overlap; repair resolves the conflict by the recovery-derived law in one transactional step. Termination, confluence, and schedule independence are immediate (one conflict component; Newman's lemma degenerates), and all D1 clauses hold by construction. The construction is functorial in the adjacency structure: taking P with adjacency given by (i) the icosahedral triangulation of S^2, (ii) a 7×7 triangulated torus, (iii) $\partial\Delta^4$, and (iv) two icosahedra identified at a vertex yields the four systems; their rewrite systems are isomorphic because the conflict component and repair path structure never see the adjacency beyond the seeded overlap, while the MaxEnt supports reproduce exactly the declared adjacency as joint-support nonvanishing, so the incidence complexes are as listed. For (4), rescale the declared cap modular clock by $\beta/2\pi$: the repair layer is unchanged, while the independently normalized KMS comparison certifies β. Each system is finite and explicit; machine receipts are provided in the repository (`code/geometry/`). $\square$

Corollary 6.33 (Status of the screen dimension claim)**.** *On towers carrying the receipts of Definitions 6.27–6.28, the screen S^2, its orientation, its round-cap incidence structure, and its conformal class are* derived *from repair normal-form data by Theorems 6.29–6.31. The selection of those receipts is* underdetermined *by repair confluence (Theorem 6.32) and is therefore carried corpus-wide as the explicit branch content of the Einstein entry: every downstream $3+1$-dimensional claim in this manuscript is conditional on* SphInc *plus the mesh, cross-ratio, and 2π-normalization receipts, and on nothing weaker. The identity* $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ *is downstream of the produced S^2 and is not itself a dimension selection.*

6.1.3 Cap-normal reconstruction and the canonical H^3 chart

This subsection makes explicit the geometric step from the support-visible round two-sphere to the Lorentz-equivariant cap space and the canonical hyperbolic space of observer rest frames. We use time-first coordinates

$$V := \mathbb{R} \oplus \mathbb{R}^3, \qquad x = (x^0, \mathbf{x}),$$

with Lorentz form

$$\eta(x, y) := -x^0 y^0 + \mathbf{x} \cdot \mathbf{y}.$$

Write $G := \mathrm{SO}^+(\eta)$, the proper orthochronous Lorentz group denoted $\mathrm{SO}^+(3,1)$ below.

Definition 6.34 (Future null cone and celestial projectivization)**.** *The future null cone is*

$$\mathcal{N}^+ := \left\{ q \in V \setminus \{0\} : \eta(q,q) = 0,\ q^0 > 0 \right\}.$$

Its positive projectivization is

$$\mathbb{P}\mathcal{N}^+ := \mathcal{N}^+ / \mathbb{R}_{>0}.$$

Fix $u_0 := (1, \mathbf{0})$. For $\Omega \in S^2 \subset \mathbb{R}^3$, define

$$q(\Omega) := (1, \Omega).$$

Lemma 6.35 (Celestial null section)**.** *The map*

$$S^2 \longrightarrow \mathbb{P}\mathcal{N}^+, \qquad \Omega \longmapsto [q(\Omega)]$$

is a bijection. Moreover,

$$\eta(q(\Omega), q(\Omega)) = 0, \qquad -\eta(u_0, q(\Omega)) = 1.$$

Thus $q(\Omega)$ is the unique representative of its future null ray normalized against u_0.

Proof. Since $\|\Omega\| = 1$,

$$\eta(q(\Omega), q(\Omega)) = -1 + \|\Omega\|^2 = 0,$$

and $q(\Omega)^0 = 1 > 0$. Also $-\eta(u_0, q(\Omega)) = 1$. Conversely, let $q = (q^0, \mathbf{q}) \in \mathcal{N}^+$. Nullness and future orientation give

$$q^0 = \|\mathbf{q}\| > 0.$$

After multiplying q by $(q^0)^{-1}$, one obtains

$$(1, \mathbf{q}/q^0) = q(\Omega), \qquad \Omega := \mathbf{q}/q^0 \in S^2.$$

The normalization $q^0 = 1$ is unique on each positive null ray. □

Definition 6.36 (Oriented round cap)**.** *For $\mathbf{c} \in S^2$ and $0 < \alpha < \pi$, define the oriented closed round cap*

$$C(\mathbf{c}, \alpha) := \left\{\Omega \in S^2 : \mathbf{c} \cdot \Omega \geq \cos\alpha\right\}.$$

Its boundary circle is

$$\partial C(\mathbf{c}, \alpha) = \left\{\Omega \in S^2 : \mathbf{c} \cdot \Omega = \cos\alpha\right\}.$$

The endpoints $\alpha = 0, \pi$ are excluded as degenerate point/full-sphere limits.

Proposition 6.37 (De Sitter normal representation of oriented round caps)**.** *Let $C = C(\mathbf{c}, \alpha)$ be a nondegenerate oriented round cap. Define*

$$\boxed{n_C := (\cot\alpha,\ \csc\alpha\,\mathbf{c})\,.}$$

Then:

1. *n_C is unit spacelike:*

$$\eta(n_C, n_C) = 1.$$

2. *For every $\Omega \in S^2$,*

$$\boxed{\eta(n_C, q(\Omega)) = \frac{\mathbf{c} \cdot \Omega - \cos\alpha}{\sin\alpha}.}$$

3. *Consequently,*

$$\eta(n_C, q(\Omega)) \begin{cases} > 0, & \Omega \in \operatorname{int} C, \\ = 0, & \Omega \in \partial C, \\ < 0, & \Omega \notin C. \end{cases}$$

In particular,

$$\eta(n_C, q(\Omega)) = 0 \quad \Longleftrightarrow \quad \Omega \in \partial C.$$

4. *The oriented cap is recovered from its normal by*

$$C(n_C) = \left\{\Omega \in S^2 : \eta(n_C, q(\Omega)) \geq 0\right\}.$$

5. *The assignment $C \mapsto n_C$ is a bijection*

$$\mathrm{Cap}^{\mathrm{or}}_{\mathrm{round}}(S^2) \cong dS_3^{\mathrm{cap}}, \qquad dS_3^{\mathrm{cap}} := \{n \in V : \eta(n,n) = 1\}.$$

6. *Reversing the cap orientation sends $n_C \mapsto -n_C$. In particular, the complementary oriented cap has normal $n_{C^c} = -n_C$.*

Proof. Using $\|\mathbf{c}\| = 1$,

$$\eta(n_C, n_C) = -\cot^2\alpha + \csc^2\alpha\,\|\mathbf{c}\|^2 = -\cot^2\alpha + \csc^2\alpha = 1.$$

For $\Omega \in S^2$,

$$\eta(n_C, q(\Omega)) = -\cot\alpha + \csc\alpha\,\mathbf{c}\cdot\Omega = \frac{\mathbf{c}\cdot\Omega - \cos\alpha}{\sin\alpha}.$$

Since $0 < \alpha < \pi$, $\sin\alpha > 0$, so the sign is exactly the sign of $\mathbf{c}\cdot\Omega - \cos\alpha$. The incidence and sign statements follow.

It remains to prove surjectivity and uniqueness. Let $n = (n^0, \mathbf{n}) \in dS_3^{\mathrm{cap}}$. Then

$$-\left(n^0\right)^2 + \|\mathbf{n}\|^2 = 1, \qquad \|\mathbf{n}\| = \sqrt{1 + (n^0)^2} > 0.$$

Set $\mathbf{c} := \mathbf{n}/\|\mathbf{n}\|$. The function $\cot : (0,\pi) \to \mathbb{R}$ is a bijection, so there is a unique $\alpha \in (0,\pi)$ satisfying $\cot\alpha = n^0$. Then

$$\csc\alpha = \sqrt{1 + \cot^2\alpha} = \sqrt{1 + (n^0)^2} = \|\mathbf{n}\|,$$

and hence $n = n_{C(\mathbf{c},\alpha)}$. Finally,

$$\eta(-n_C, q(\Omega)) \geq 0 \quad \Longleftrightarrow \quad \eta(n_C, q(\Omega)) \leq 0,$$

which reverses the selected side of the boundary circle. $\square$

Remark 6.38 (Cap space is de Sitter, not hyperbolic space)**.** *The parameter space of oriented round caps is the spacelike unit hyperboloid*

$$dS_3^{\mathrm{cap}} \cong G/\mathrm{SO}^+(2,1).$$

It is not H^3. Cap normals are spacelike vectors. Observer-frame points in H^3 are future unit timelike vectors. The two spaces are related by the cap/half-space duality proved below, not by identifying their points.

Theorem 6.39 (Projective Lorentz action and cap-normal equivariance)**.** *For every $\Lambda \in G$ and every $\Omega \in S^2$, define*

$$\omega_\Lambda(\Omega) := \left(\Lambda q(\Omega)\right)^0.$$

Then $\omega_\Lambda(\Omega) > 0$, and there is a unique $g_\Lambda(\Omega) \in S^2$ such that

$$\boxed{\Lambda q(\Omega) = \omega_\Lambda(\Omega)\, q(g_\Lambda(\Omega))\,.}$$

Explicitly,

$$\boxed{g_\Lambda(\Omega) = \frac{\big(\Lambda q(\Omega)\big)^{\text{spatial}}}{\big(\Lambda q(\Omega)\big)^0}.}$$

The assignment $\Lambda \mapsto g_\Lambda$ is an effective action of G on S^2. If γ_{S^2} is the round metric, then

$$\boxed{g_\Lambda^* \gamma_{S^2} = \omega_\Lambda^{-2} \gamma_{S^2}.}$$

Hence the action is conformal. Moreover,

$$G \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{Conf}^+(S^2),$$

and for every oriented round cap C,

$$\boxed{g_\Lambda(C) = C(\Lambda n_C), \qquad n_{g_\Lambda C} = \Lambda n_C.}$$

Proof. Because Λ is proper orthochronous, it preserves $\mathcal{N}^+$. Thus $\Lambda q(\Omega) \in \mathcal{N}^+$, and $\omega_\Lambda(\Omega) > 0$. A future null vector $Q = (Q^0, \mathbf{Q})$ obeys $\|\mathbf{Q}\| = Q^0$, so $\mathbf{Q}/Q^0 \in S^2$. This proves the displayed formula for g_Λ, and uniqueness follows from the time-component normalization $q(\Omega)^0 = 1$.

To prove conformality, let $v \in T_\Omega S^2$. Since $dq_\Omega(v) = (0, v)$, differentiating

$$\Lambda q(\Omega) = \omega_\Lambda(\Omega)\, q(g_\Lambda(\Omega))$$

gives

$$\Lambda dq_\Omega(v) = d\omega_\Lambda(v)\, q(g_\Lambda \Omega) + \omega_\Lambda(\Omega)\, dq_{g_\Lambda \Omega}(dg_\Lambda v).$$

The identities $\eta(q, q) = 0$ and $\eta(q, dq(w)) = 0$ remove the first term and the cross term when taking Lorentz norms. Hence

$$\eta(dq_\Omega(v), dq_\Omega(v)) = \omega_\Lambda(\Omega)^2 \eta(dq_{g_\Lambda \Omega}(dg_\Lambda v), dq_{g_\Lambda \Omega}(dg_\Lambda v))\,.$$

Since $\eta(dq_\Omega(v), dq_\Omega(v)) = \|v\|^2$, polarization gives

$$g_\Lambda^* \gamma_{S^2} = \omega_\Lambda^{-2} \gamma_{S^2}.$$

For the global group identification, identify V with Hermitian 2×2 matrices by

$$\mathsf{X}(x) = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}.$$

Then

$$\det \mathsf{X}(x) = (x^0)^2 - \|\mathbf{x}\|^2 = -\eta(x, x).$$

For $A \in \mathrm{SL}(2, \mathbb{C})$, the map $\mathsf{X} \mapsto A\mathsf{X}A^\dagger$ preserves Hermiticity, determinant, orientation, and the future cone. It defines a homomorphism $\mathrm{SL}(2, \mathbb{C}) \to G$ with kernel $\{\pm I\}$, and the image is the connected Lorentz group. Future null rays are rank-one positive Hermitian rays, i.e. $\mathbb{CP}^1 \cong S^2$, and the induced projective action is the Möbius action. Thus

$$G \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{Conf}^+(S^2).$$

Finally, let $C = C(n_C)$. Since

$$q(g_\Lambda \Omega) = \omega_\Lambda(\Omega)^{-1} \Lambda q(\Omega),$$

Lorentz invariance gives

$$\eta(\Lambda n_C, q(g_\Lambda \Omega)) = \omega_\Lambda(\Omega)^{-1} \eta(n_C, q(\Omega)).$$

The factor $\omega_\Lambda(\Omega)^{-1}$ is positive, so the interior, boundary, and exterior signs are preserved. Therefore

$$g_\Lambda(C(n_C)) = C(\Lambda n_C).$$

Because Λ preserves η, Λn_C is the uniquely normalized oriented normal of the transformed cap, hence

$$n_{g_\Lambda C} = \Lambda n_C.$$

□

Corollary 6.40 (Lorentz kinematics on the screen)**.** *Under the hypotheses of Theorem 6.13,*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{SO}^+(3, 1).$$

The cap modular flow is represented by the corresponding one-parameter Lorentz boost/dilation subgroup. More precisely, if $\lambda_C(t)C = C$ is the cap-preserving conformal modular flow and $\Lambda_C(t) \in G$ is its Lorentz representative, then

$$\boxed{\Lambda_C(t) n_C = n_C.}$$

Thus the cap modular subgroup lies in

$$\mathrm{Stab}_G(n_C) \cong \mathrm{SO}^+(2, 1).$$

Proof. The group identification and cap-normal action are Theorem 6.39. Theorem 6.13 identifies the realized scaling-limit cap modular automorphism with the standard cap-preserving conformal dilation and fixes its 2π normalization. Since $\lambda_C(t)$ preserves the oriented cap, equivariance and uniqueness of the unit normal give

$$n_C = n_{\lambda_C(t)C} = \Lambda_C(t) n_C.$$

□

Definition 6.41 (Future observer-frame hyperboloid)**.** *Define*

$$H^3 := \left\{ u \in V : \eta(u, u) = -1,\ u^0 > 0 \right\}.$$

A point $u \in H^3$ is a future unit timelike observer-frame direction. Its linear instantaneous rest space is

$$E_u := u^\perp = \{ v \in V : \eta(u, v) = 0 \}.$$

Proposition 6.42 (Canonical observer-frame symmetric space)**.** *The proper orthochronous Lorentz group acts transitively on H^3. The stabilizer of $u_0 = (1, \mathbf{0})$ is*

$$K := \mathrm{Stab}_G(u_0) \cong \mathrm{SO}(3).$$

Consequently,

$$\boxed{H^3 \cong G/K \cong \mathrm{SO}^+(3, 1)/\mathrm{SO}(3).}$$

The subgroup K is a maximal compact subgroup of G, and every maximal compact subgroup of G is conjugate to K. Thus G/K is canonical up to G-equivariant isometry. Furthermore:

1. *$T_uH^3 = u^\perp = E_u$, and $\eta|_{E_u}$ is positive definite and three-dimensional.*

2. *The induced metric $g_{H^3} := \eta|_{TH^3}$ is G-invariant. Every G-invariant Riemannian metric on G/K is a positive scalar multiple of this metric.*

3. *For each $u \in H^3$, the normalized celestial section*

$$\mathcal{S}_u := \Big\{ q \in \mathcal{N}^+ : -\eta(u,q) = 1 \Big\}$$

is naturally the unit two-sphere in E_u. Explicitly,

$$q = u + s, \qquad s \in E_u, \qquad \eta(s,s) = 1.$$

4. *The construction is Lorentz-natural:*

$$\Lambda \mathcal{S}_u = \mathcal{S}_{\Lambda u}.$$

5. *The ideal boundary of H^3 is the projective future null cone:*

$$\boxed{\partial_\infty H^3 \cong \mathbb{P}\mathcal{N}^+ \cong S^2.}$$

Proof. Every future unit timelike vector can be completed to an oriented, time-oriented Lorentz frame, so G acts transitively on H^3. A Lorentz transformation fixing u_0 acts orthogonally on

$$u_0^\perp \cong \mathbb{R}^3,$$

and proper orientation gives precisely SO(3). Hence

$$H^3 \cong G/\mathrm{SO}(3).$$

If $L \subseteq G$ is compact, average u_0 over L using Haar measure. The future timelike cone is convex, so the average is a future timelike vector fixed by L; after normalization, L is conjugate into a point stabilizer. Thus K is maximal compact and all maximal compact subgroups are conjugate.

Differentiating $\eta(u,u) = -1$ gives $T_uH^3 = u^\perp$. The orthogonal complement of a timelike vector is positive definite and has dimension three. Since G preserves η, the induced metric is G-invariant. At u_0, the isotropy representation of $K \cong \mathrm{SO}(3)$ on $T_{u_0}H^3 \cong \mathbb{R}^3$ is the standard irreducible representation, so every K-invariant positive inner product is a positive scalar multiple of the Euclidean one. Homogeneity gives the metric uniqueness.

Take $q \in \mathcal{S}_u$ and set $s := q - u$. Then

$$\eta(u,s) = \eta(u,q) - \eta(u,u) = -1 + 1 = 0,$$

and

$$\eta(s,s) = \eta(q-u, q-u) = 0 - 2(-1) - 1 = 1.$$

Thus s lies on the unit sphere in E_u. Conversely, if $s \in E_u$ and $\eta(s,s) = 1$, then $q = u + s$ is future null and satisfies $-\eta(u,q) = 1$. Lorentz naturality follows from preservation of η.

Finally, for $s \in E_u$ with $\eta(s,s) = 1$, the curve

$$\gamma_{u,s}(t) = \cosh t\, u + \sinh t\, s$$

is a unit-speed geodesic ray in H^3. Projectively,

$$[\gamma_{u,s}(t)] \longrightarrow [u+s] \qquad (t \to +\infty),$$

and $\eta(u+s, u+s) = 0$. Every future null ray has such a normalized decomposition relative to u. Hence the ideal boundary is $\mathbb{P}\mathcal{N}^+ \cong S^2$. □

Proposition 6.43 (No cap-to-observer-point map from cap data alone)**.** *There is no G-equivariant map*

$$F : dS_3^{\mathrm{cap}} \longrightarrow H^3.$$

Equivalently, an oriented round cap by itself does not canonically select a point of the observer-frame hyperboloid.

Proof. Fix $n_0 \in dS_3^{\mathrm{cap}}$. Its stabilizer is

$$H := \mathrm{Stab}_G(n_0) \cong \mathrm{SO}^+(2,1),$$

which is noncompact. Suppose an equivariant map F existed. For every $h \in H$,

$$F(n_0) = F(hn_0) = hF(n_0).$$

Thus the full noncompact group H would fix $F(n_0) \in H^3$. But the stabilizer of every point of H^3 is conjugate to $\mathrm{SO}(3)$, hence compact. This is impossible. □

Remark 6.44 (Meaning of the no-go statement)**.** *The canonical output of one oriented cap is therefore not an observer point. It is the oriented geodesic plane and hyperbolic half-space constructed in the next proposition. Selecting a particular observer frame $u \in H^3$ requires additional observer, tetrad, clock, or record data.*

Proposition 6.45 (Round-cap/hyperbolic-half-space duality)**.** *Let $n \in dS_3^{\mathrm{cap}}$. Define*

$$\Pi_n := \Big\{u \in H^3 : \eta(n,u) = 0\Big\}, \qquad \mathcal{H}_n^+ := \Big\{u \in H^3 : \eta(n,u) \geq 0\Big\}.$$

Then:

1. *Π_n is a totally geodesic copy of H^2.*
2. *Its ideal boundary is the round circle represented by n:*

$$\partial_\infty \Pi_n = \Big\{[q] \in \mathbb{P}\mathcal{N}^+ : \eta(n,q) = 0\Big\} = \partial C(n).$$

3. *The ideal boundary of the oriented half-space is the oriented cap:*

$$\partial_\infty \mathcal{H}_n^+ = C(n) = \Big\{[q] \in \mathbb{P}\mathcal{N}^+ : \eta(n,q) \geq 0\Big\}.$$

4. *The construction is Lorentz-equivariant:*

$$\Lambda\Pi_n = \Pi_{\Lambda n}, \qquad \Lambda\mathcal{H}_n^+ = \mathcal{H}_{\Lambda n}^+.$$

5. *The signed distance from $u \in H^3$ to Π_n, positive on $\mathcal{H}_n^+$, is*

$$\boxed{s_n(u) = \operatorname{arsinh}\big(\eta(n,u)\big).}$$

Proof. Because n is unit spacelike, $n^\perp$ has signature $(-++)$. Its future unit timelike hyperboloid is a copy of H^2, and

$$\Pi_n = H^3 \cap n^\perp.$$

Intersections of the hyperboloid with linear subspaces through the origin are totally geodesic, proving the first claim. The ideal boundary of $H^3 \cap n^\perp$ consists of future null rays contained in $n^\perp$, which are exactly those satisfying $\eta(n,q) = 0$. Under Lemma 6.35, this is the circle $\partial C(n)$; the sign condition gives the corresponding oriented ideal half-space boundary.

Equivariance follows from $\eta(\Lambda n, \Lambda u) = \eta(n,u)$. For the distance formula, let

$$a := \eta(n,u), \qquad p := \frac{u - an}{\sqrt{1+a^2}}.$$

Then $\eta(p,n) = 0$, $\eta(p,p) = -1$, and p lies on the future sheet, so $p \in \Pi_n$. With $s := \operatorname{arsinh}(a)$,

$$u = \cosh s\, p + \sinh s\, n,$$

which is the unit-speed geodesic normal to Π_n through p. Hence s is the signed distance. $\square$

Corollary 6.46 (Three-dimensional observer-facing spatial chart)**.** *Under the hypotheses of Corollary 6.40, the canonical observer-frame/rest-space chart on the Lorentz branch is*

$$\boxed{H^3 \cong \mathrm{SO}^+(3,1)/\mathrm{SO}(3).}$$

Its dimension is exactly

$$\boxed{\dim H^3 = \dim \mathrm{SO}^+(3,1) - \dim \mathrm{SO}(3) = 6 - 3 = 3.}$$

The associated invariant metric is fixed only up to one positive curvature radius. With radius $R_H > 0$,

$$H^3_{R_H} = \left\{ X \in V : \eta(X,X) = -R_H^2,\ X^0 > 0 \right\},$$

and

$$d_H(X,Y) = R_H \operatorname{arcosh}\left(-\frac{\eta(X,Y)}{R_H^2} \right).$$

The group/cap theorem fixes the hyperbolic homogeneous-space geometry and its dimension; it does not fix a physical numerical value of R_H.

Proof. The homogeneous-space statement and dimension follow from Proposition 6.42. The standard isotropy representation of SO(3) is irreducible, so the invariant metric is unique up to one overall positive multiplier. Writing that multiplier as R_H^2 gives the radius-R_H hyperboloid and the displayed distance formula. $\square$

Remark 6.47 (Rest-frame chart versus populated bulk)**.** *The theorem proves the exact three-dimensional Lorentz/H^3 kinematic chart on the support-visible round-cap BW branch. It makes four distinctions explicit:*

1. *An oriented cap is represented by a spacelike $n_C \in dS_3^{\mathrm{cap}}$.*
2. *An observer rest frame is represented by a future timelike $u \in H^3$.*
3. *The individual observer's linear instantaneous rest space is $u^\perp = T_u H^3$.*
4. *A cap determines a geodesic plane and half-space in H^3, not a preferred observer point.*

The theorem does not, by itself, select a preferred u, populate the chart with records or objects, construct a chart-blind neutral bulk, fix R_H in physical units, produce a stress tensor, or establish the Einstein equation. Populated observer-facing bulk records require the conditional cap-response localization certificate of Corollary 6.64. Object-family interpretation, chart-blind neutral-bulk reconstruction, physical scale, stress, and Einstein dynamics remain separate certificates. The conditional event-manifold continuation of these gates, an event base with a Lorentzian atlas over which this H^3 is strictly the frame fiber, is constructed in §6.3.

6.1.4 Record-populated H^3 from modular cap-response localization

This subsection attaches records to the observer-facing H^3 chart without promoting the chart itself into a neutral bulk. Fix $R_H > 0$ and

$$H^3_{R_H} := \{X \in \mathbb{R}^{1,3} : \eta(X,X) = -R_H^2,\ X^0 > 0\}.$$

For $X, Y \in H^3_{R_H}$, write

$$\bar{d}(X,Y) := \frac{1}{R_H} d_H(X,Y) = \operatorname{arcosh}\left(-\frac{\eta(X,Y)}{R_H^2}\right).$$

For an oriented round cap C, let n_C be its unit spacelike normal from Proposition 6.37. Define

$$h_C(X) := \frac{\eta(X, n_C)}{R_H}, \qquad q_C(X) := [h_C(X)]_+, \qquad \chi_C(X) := \mathbf{1}_{\{h_C(X) \geq 0\}}.$$

Lemma 6.48 (Cap/half-space response dictionary)**.** *The map $C \mapsto n_C$ identifies oriented round caps with unit spacelike normals. The cap C determines the hyperbolic plane*

$$\Pi_C := \{X \in H^3_{R_H} : \eta(X, n_C) = 0\}$$

and oriented half-space

$$\mathcal{H}_C^+ := \{X \in H^3_{R_H} : \eta(X, n_C) \geq 0\}.$$

The signed distance $s_C(X)$ from X to Π_C, positive on $\mathcal{H}_C^+$, obeys

$$h_C(X) = \sinh\left(\frac{s_C(X)}{R_H}\right), \qquad s_C(X) = R_H \operatorname{arsinh} h_C(X).$$

Proof. The first statement is Proposition 6.37. If $Y \in \Pi_C$ is the foot of the perpendicular from X, the normal geodesic has form

$$\gamma(s) = \cosh(s/R_H) Y + R_H \sinh(s/R_H) n_C.$$

Pairing with n_C gives $\eta(\gamma(s), n_C)/R_H = \sinh(s/R_H)$. □

Theorem 6.49 (Round-cap half-space functions separate H^3)**.** *The families $\{h_C\}$, $\{q_C\}$, and $\{\chi_C\}$, over oriented round caps, separate points of $H^3_{R_H}$. Equivalently, for every $X \neq Y$ there is an oriented round cap C with $h_C(X) < 0 < h_C(Y)$.*

Proof. Let M be the geodesic midpoint between X and Y, and let $v \in T_M H^3_{R_H}$ be the unit tangent pointing from X to Y. Then

$$X = \cosh(\delta/2R_H) M - R_H \sinh(\delta/2R_H) v, \qquad Y = \cosh(\delta/2R_H) M + R_H \sinh(\delta/2R_H) v,$$

where $\delta = d_H(X,Y) > 0$. The vector v is unit spacelike and therefore is n_C for a unique oriented cap C. Thus

$$h_C(X) = -\sinh(\delta/2R_H) < 0, \qquad h_C(Y) = +\sinh(\delta/2R_H) > 0.$$

The nonnegative and binary responses then distinguish the same ordered pair. $\square$

Remark 6.50 (Separation versus inverse stability)**.** *Qualitative point separation proves exact injectivity for the full response family. It does not provide a uniform linear inverse bound; a continuous injective map such as $x \mapsto x^3$ has no Lipschitz inverse at zero. Finite noisy localization therefore requires a quantitative observability constant $\alpha > 0$, supplied by a conditioned cap frame or an independently verified response frame.*

Definition 6.51 (Centered tight cap frame)**.** *Let $u_1, \dots, u_m \in S^2$ satisfy*

$$\sum_{j=1}^m u_j = 0, \qquad \sum_{j=1}^m u_j u_j^\mathsf{T} = \frac{m}{3} I_3.$$

Set $\theta_0 = \arccos(1/\sqrt{3})$ and

$$n_j = \left(\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}\, u_j\right), \qquad h_j(X) = \frac{\eta(X, n_j)}{R_H}.$$

Theorem 6.52 (Explicit finite-frame reconstruction)**.** *For the frame of Definition 6.51, every $X = (X^0, \mathbf{X}) \in H^3_{R_H}$ is recovered from $h(X) = (h_1(X), \dots, h_m(X))$ by*

$$X^0 = -\frac{\sqrt{2}\, R_H}{m} \sum_{j=1}^m h_j(X), \qquad \mathbf{X} = \frac{\sqrt{6}\, R_H}{m} \sum_{j=1}^m h_j(X) u_j.$$

Moreover

$$|h(X) - h(Y)|_2^2 = \frac{m}{2R_H^2} |X - Y|_{\mathrm{E}}^2$$

and hence

$$|h(X) - h(Y)|_2 \geq \sqrt{\frac{m}{2}}\, \bar{d}(X,Y).$$

For the twelve icosahedral directions, $\alpha_h = \sqrt{6}$.

Proof. Since

$$h_j(X) = \frac{1}{R_H}\left(-\frac{X^0}{\sqrt{2}} + \sqrt{\frac{3}{2}}\, \mathbf{X} \cdot u_j\right),$$

summing and using the centered tight-frame identities gives the two displayed reconstruction formulas. Applying the same identities to $\Delta X = X - Y$ gives the Euclidean norm identity. Finally $|X - Y|_{\mathrm{E}} \geq R_H \bar{d}(X,Y)$ on the hyperboloid, because the Euclidean norm dominates the spacelike Minkowski norm and $2\sinh(r/2) \geq r$. $\square$

Proposition 6.53 (General cap-frame condition)**.** *For unit spacelike cap normals $n_1, \dots, n_m$, form the matrix B with rows $(-n_j^0, n_j^1, n_j^2, n_j^3)$. If $W \succ 0$ and B has rank 4, then*

$$X_{\mathrm{E}} = R_H (B^\mathsf{T} W B)^{-1} B^\mathsf{T} W h(X), \qquad \alpha_{\text{frame}} = \sigma_{\min}(W^{1/2} B),$$

and

$$|h(X) - h(Y)|_W \geq \alpha_{\text{frame}} \bar{d}(X,Y).$$

Proof. The response vector is $h(X) = BX_{\mathrm{E}}/R_H$. The left inverse gives the reconstruction formula, and the smallest singular value of $W^{1/2}B$ gives the lower bound after using $|X-Y|_{\mathrm{E}} \geq R_H \bar{d}(X,Y)$. $\square$

Lemma 6.54 (Paired hinge response)**.** *For $\Phi(z) = ([z]_+, [-z]_+)$,*

$$\frac{1}{\sqrt{2}}|z-w| \leq |\Phi(z) - \Phi(w)|_2 \leq |z-w|.$$

Thus cap/complement nonnegative responses inherit a lower constant $\alpha_Q = \alpha_h/\sqrt{2}$. For the twelve-direction signed frame, $\alpha_Q = \sqrt{3}$.

Proof. If z, w have the same sign, the norm is $|z-w|$. If they have opposite signs, it is $\sqrt{z^2+w^2}$, which lies between $(|z|+|w|)/\sqrt{2}$ and $|z|+|w|$. $\square$

Definition 6.55 (Record-conditioned modular cap response)**.** *Let O be an observer and let $P_{i,O}$ be the central, or declared approximately central, projector for quotient-visible record token i, with $\omega_O(P_{i,O}) > 0$. Define*

$$\omega_{i,O}(A) := \frac{\omega_O(P_{i,O}AP_{i,O})}{\omega_O(P_{i,O})}.$$

For a cap C, let $\widehat{M}_{C,0,O}$ be the declared cap-response probe observable and set

$$\widehat{M}_{C,t,O} := \sigma_t^{C,O}(\widehat{M}_{C,0,O}), \qquad R_i(C,t,O) := \omega_{i,O}(\widehat{M}_{C,t,O}).$$

On the geometric branch, the modular-time convention is the same one used in Theorem 6.13: $\sigma_t^{C,O} = \alpha_{\lambda_C(2\pi t)}$.

Definition 6.56 (Point-local response factorization)**.** *For a probe sample $j = (C,t,O)$, the point-local branch asserts that there is a source $X_i(t) \in \Omega \subset H^3_{R_H}$ such that*

$$R_{i,j} = b_j + a_{i,j}\psi_j(h_j(X_i(t))) + \xi_{i,j}, \qquad h_j(X) = \frac{\eta(X,n_j)}{R_H}.$$

Here b_j is background, $a_{i,j} > 0$ is a calibrated gain or declared source amplitude, ψ_j is a frozen scalar sensor kernel, and $\xi_{i,j}$ collects measurement, record-centrality, calibration, modular-transport, finite-record, point-model, and numerical error. The raw half-space kernel is $\psi_j(z) = [z]_+$; a signed calibrated channel may use $\psi_j(z) = z$. This factorization is a branch hypothesis or finite source-factor receipt, not a consequence of fitting a point.

Lemma 6.57 (Calibration error)**.** *Suppose $R_j = b_j + a_j f_j + \xi_j$, $y_j = (R_j - \widehat{b}_j)/\widehat{a}_j$, $a_j \geq a_{\min} > 0$, $|\widehat{a}_j - a_j| \leq \delta_{a,j} < a_{\min}$, $|\widehat{b}_j - b_j| \leq \delta_{b,j}$, $|\xi_j| \leq \delta_{R,j}$, and $|f_j| \leq M_j$. Then*

$$|y_j - f_j| \leq \frac{\delta_{R,j} + \delta_{b,j} + \delta_{a,j}M_j}{a_{\min} - \delta_{a,j}}.$$

Proof. Subtract f_j from y_j. The numerator is $(b_j - \widehat{b}_j) + \xi_j + (a_j - \widehat{a}_j)f_j$, and the denominator is at least $a_{\min} - \delta_{a,j}$. $\square$

Definition 6.58 (Quantitative response observability)**.** *For a compact source region $\Omega \subset H^3_{R_H}$, let $F : \Omega \to \mathbb{R}^{|J|}$ be the calibrated response map*

$$F_j(X) := \psi_j(h_j(X)).$$

With $|z|_W = (z^\top W z)^{1/2}$, $W \succ 0$, the map is quantitatively observable on Ω when

$$\alpha \bar{d}(X,Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X,Y)$$

for all $X, Y \in \Omega$, with $0 < \alpha \le L < \infty$.

Theorem 6.59 (Exact residual identifiability)**.** *If F satisfies the lower bound in Definition 6.58 and exact data are $y = F(X_\star)$, then $X_\star$ is the unique global minimizer on Ω of*

$$\mathcal{J}(X) := |F(X) - y|_W^2.$$

Proof. $\mathcal{J}(X_\star) = 0$. If another point also has zero residual, then $F(X) = F(X_\star)$, so $\alpha\bar{d}(X, X_\star) = 0$, hence $X = X_\star$. □

Theorem 6.60 (Finite net on a bounded hyperbolic source region)**.** *Let $\Omega = B_H(o, D)$ be a closed hyperbolic ball. For every $\varepsilon > 0$ there is a finite ε-net $\mathcal{N}_\varepsilon \subset \Omega$, measured in $\bar{d}$, with*

$$|\mathcal{N}_\varepsilon| \le \frac{V_{R_H}(D + R_H\varepsilon/2)}{V_{R_H}(R_H\varepsilon/2)},$$

where

$$V_{R_H}(r) = \pi R_H^3 \left[\sinh(2r/R_H) - 2r/R_H\right].$$

There is no finite global ε-net for all of noncompact H^3.

Proof. Choose a maximal subset of Ω whose distinct points are more than ε apart in normalized distance. Maximality gives coverage. The balls of physical radius $R_H\varepsilon/2$ around net points are disjoint and lie in $B_H(o, D + R_H\varepsilon/2)$, giving the volume bound. Noncompact H^3 has infinite volume, so a finite global net cannot exist. □

Theorem 6.61 (Finite-net bounded-noise localization)**.** *Let $\Omega \subset H^3_{R_H}$ be compact and let F satisfy Definition 6.58. Let measured calibrated data be $y = F(X_\star) + e$, with $|e|_W \le \sigma$. If $\mathcal{N}_\varepsilon$ is an ε-net of Ω and $\widehat{X}_\varepsilon \in \mathcal{N}_\varepsilon$ satisfies*

$$|F(\widehat{X}_\varepsilon) - y|_W \le \min_{Z \in \mathcal{N}_\varepsilon} |F(Z) - y|_W + \tau,$$

then

$$d_H(\widehat{X}_\varepsilon, X_\star) \le R_H \left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma + \frac{1}{\alpha}\tau\right].$$

Proof. Choose $X_\varepsilon \in \mathcal{N}_\varepsilon$ with $\bar{d}(X_\varepsilon, X_\star) \le \varepsilon$. Then $|F(X_\varepsilon) - F(X_\star)|_W \le L\varepsilon$. Near-optimality and $|e|_W \le \sigma$ give

$$|F(\widehat{X}_\varepsilon) - F(X_\star)|_W \le L\varepsilon + 2\sigma + \tau.$$

The lower observability bound converts this response error into the displayed hyperbolic distance bound. □

Theorem 6.62 (Certified noisy finite uniqueness)**.** *For $Z \in \mathcal{N}_\varepsilon$, let $r(Z) = |F(Z) - y|_W$. Suppose certified interval arithmetic gives $\underline{r}(Z) \le r(Z) \le \overline{r}(Z)$. If a candidate $\widehat{X}_\varepsilon$ satisfies*

$$\overline{r}(\widehat{X}_\varepsilon) < \min_{Z \ne \widehat{X}_\varepsilon} \underline{r}(Z),$$

then it is the unique residual minimizer for every realization consistent with the intervals. Define

$$\Delta_{\mathrm{loc}} := \min_{Z \ne \widehat{X}_\varepsilon} \underline{r}(Z) - \overline{r}(\widehat{X}_\varepsilon).$$

A unique finite point is licensed exactly when $\Delta_{\mathrm{loc}} > 0$; when $\Delta_{\mathrm{loc}} \le 0$, the required finite output is AMBIGUOUS *together with the admissible localization set.*

Proof. Every competitor has residual at least its lower interval bound, while the candidate has residual at most its upper interval bound. A strict positive gap therefore rules out ties and competitors. Without such a gap, a finite procedure cannot choose a unique output without adding an external tie-breaker. □

Theorem 6.63 (Lorentz-chart naturality of localization)**.** *For $A \in \mathrm{SO}^+(3,1)$, transform points and cap normals by $X \mapsto AX$ and $n_j \mapsto An_j$. Then*

$$\eta(AX, An_j) = \eta(X, n_j).$$

Consequently the calibrated feature vector, residual objective, observability constants, and localization/gap verdicts are unchanged; if $\widehat{X}$ is the reconstructed point in one chart, $A\widehat{X}$ is the reconstructed point in the transformed chart.

Proof. Lorentz transformations preserve η, so all response coordinates and all residual inequalities are invariant under the common chart transform. □

Corollary 6.64 (Conditional record-populated H^3 localization)**.** *Fix a clock slice t and a quotient-visible record token i. Assume:*

1. *a valid observer-facing $H^3_{R_H}$ chart and cap-normal map are available;*
2. *$X_i(t) \in \Omega \subset H^3_{R_H}$ for a declared compact source region;*
3. *calibrated modular cap responses satisfy the point-local factorization of Definition 6.56;*
4. *the finite response map has constants $0 < \alpha \le L$ on Ω;*
5. *the total response/model error obeys $|e_i|_W \le \sigma_i$;*
6. *$\mathcal{N}_\varepsilon$ is an ε-net of Ω;*
7. *$\widehat{X}_i(t)$ is a residual minimizer up to tolerance τ_i.*

Then the localized record support is the ball

$$S_i(t) = B_H\left(\widehat{X}_i(t), R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i + \frac{1}{\alpha}\tau_i\right]\right).$$

Exact continuous data identify $X_i(t)$ uniquely. Finite noisy data give the displayed certified localization ball; a unique finite point additionally requires $\Delta_{\mathrm{loc},i}(t) > 0$. If that gap fails, the theorem output is AMBIGUOUS*.*

Proof. Exact uniqueness is Theorem 6.59. The finite-net error estimate is Theorem 6.61. Noisy finite uniqueness is Theorem 6.62, and chart naturality is Theorem 6.63. □

Remark 6.65 (Record-population claim boundary)**.** *The H^3 chart remains a recovered geometric result on the support-visible Lorentz branch. Record population is a conditional continuation theorem: calibrated quotient-visible cap responses localize a record token when they satisfy the point-local response factorization and quantitative cap-frame hypotheses. Exact data give a unique point. Finite noisy data give a certified localization ball; a unique finite point additionally requires a positive residual gap. This theorem does not derive point locality, unlabeled source mixture resolution, extended-source collapse to a point, particle species, physical stress, neutral third-person bulk, or Einstein branch entry. The localized records constructed here are the germ inputs of the conditional event-manifold packet of §6.3, where they populate an event base rather than the frame fiber.*

6.2 The null modular bridge

The fixed-cutoff strip analysis begins from the transferred cut-center data and the inherited-strip Markov structure. At fixed regulator scale one obtains exact or controlled four-term strip additivity together with endpoint-Lipschitz control of the renormalized half-line family; on the same scaling-limit geometric-cap branch used in Theorem 6.13, the null half-line blow-up net then inherits its geometric dilation and therefore half-sided modular inclusion. Standard Borchers–Wiesbrock theory [13] applies only after that derived half-sided modular pair is in hand, yielding positivity and unitary implementation of null translations. The downstream bounded-interval transport is discharged by Lemma 6.135, and the tensor upgrade carries the standard null-invisible metric ambiguity. The half-line generator itself is identified below with the effective local null-stress charge.

The present subsection therefore establishes the following chain:

1. null-strip center transfer and the inherited left/right split hypothesis;
2. exact strip additivity on the exact inherited Markov model, and a carried defect operator on controlled exact-Markov replacements on one fixed strip model;
3. endpoint-Lipschitz matrix elements for the renormalized half-line family and the resulting weak tail generator;
4. derived half-sided modular inclusion on the null half-line blow-up net from the declared scaling-limit geometric-cap branch, and the resulting Borchers positive translation generator.

The only explicit downstream boundary retained below is the bounded-interval transport/projective branch together with the later tensor upgrade; the half-line generator/charge identification is proved inside the bridge itself. Lemma 6.72 states the positive null-translation generator explicitly: the Borchers unitary group is constructed on its Stone domain, positivity is theorem-level instead of implicit, and the half-line modular Hamiltonians are tied to it by explicit affine covariance and quadratic-form identities.

Proposition 6.66 (Null-strip center transfer and inherited split)**.** *Fix a null generator Ω and a regulated tripartition*

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts

$$\Gamma_- := \{v = v_2\}, \qquad \Gamma_+ := \{v = v_3\}.$$

Assume the derived fixed-cutoff regulator/collar package established above, and assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data of Proposition 5.8 in the following precise sense:

(i) each cut $\Gamma_\pm$ carries a compact derived boundary action $\widehat{K}_\pm$ on a reference cut Hilbert space, with $\widehat{K}_\pm = K_\pm$ on the ordinary branch;

(ii) in a compatible type-I regulator presentation, the adjacent regions carry inverse transport across each cut, so for irreducible $\widehat{K}_\pm$-modules $W_{\alpha_\pm}$ one has decompositions

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W^*_{\alpha_+} \otimes \mathcal{H}_{i_+^{\alpha_+}};$$

(iii) the gauge-invariant strip algebra is the commutant of the transported boundary action on $\tilde{\mathcal{H}}_J$, so that the pair (α_-, α_+) is a central cut label;

(iv) for each (α_-, α_+), the multiplicity space $\mathcal{H}_{j^{\alpha_-,\alpha_+}}$ admits a chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

such that, blockwise, $\mathcal{A}(I_- \cup J)$ acts only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acts only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$.

Then the strip algebra has the inherited central decomposition

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+},$$

and, under item (iv),

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}).$$

The glued tripartition carries the block decomposition

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Thus the two null cuts transfer the same center data as the spatial collar branch, while the left/right split of the strip multiplicity spaces is exactly the extra decomposition-inheritance hypothesis and is not forced by center transfer alone.

Proof. By complete reducibility of the finite-dimensional unitary actions $\widehat{K}_\pm$, the displayed decompositions of $\tilde{\mathcal{H}}_{I_-}$, $\tilde{\mathcal{H}}_J$, and $\tilde{\mathcal{H}}_{I_+}$ exist. Under item (iii), the strip algebra is the commutant of the transported $\widehat{K}_- \times \widehat{K}_+$ action on

$$\tilde{\mathcal{H}}_J = \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+}.$$

Write an operator $X \in \mathcal{B}(\tilde{\mathcal{H}}_J)$ in matrix form relative to that direct sum. The block

$$X_{(\alpha_-,\alpha_+),(\beta_-,\beta_+)}$$

intertwines

$$W^*_{\alpha_-} \otimes W_{\alpha_+} \longrightarrow W^*_{\beta_-} \otimes W_{\beta_+}$$

for the product action of $\widehat{K}_- \times \widehat{K}_+$. By Schur's lemma, such an intertwiner is zero unless $(\alpha_-, \alpha_+) = (\beta_-, \beta_+)$, and when the sector pair agrees it is scalar on the representation factors. Therefore the commutant is exactly

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}),$$

and the center is generated by the corresponding block projectors P_{α_-,α_+}.

For the glued tripartition, tensor the three regulator Hilbert spaces and decompose:

$$\tilde{\mathcal{H}}_{I_-} \otimes \tilde{\mathcal{H}}_J \otimes \tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_-,\beta_-,\alpha_+,\beta_+} (W_{\alpha_-} \otimes W^*_{\beta_-}) \otimes \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\beta_-,\beta_+}} \otimes (W_{\beta_+} \otimes W^*_{\alpha_+}) \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Taking $\widehat{K}_- \times \widehat{K}_+$-invariants and applying Schur's lemma to each cut gives

$$(W_{\alpha_-} \otimes W^*_{\beta_-})^{\widehat{K}_-} \cong \begin{cases} \mathbb{C}, & \alpha_- = \beta_-, \\ 0, & \alpha_- \neq \beta_-, \end{cases} \qquad (W_{\beta_+} \otimes W^*_{\alpha_+})^{\widehat{K}_+} \cong \begin{cases} \mathbb{C}, & \beta_+ = \alpha_+, \\ 0, & \beta_+ \neq \alpha_+, \end{cases}$$

so only matching cut sectors survive and one obtains

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

If item (iv) holds, substitute the chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

to obtain the displayed inherited split and the stated action of the left and right union algebras. Nothing in the Schur-lemma argument itself forces item (iv); it is exactly the extra structural condition required to reproduce the same blockwise tensor pattern as Theorem 5.9. ☐

Corollary 6.67 (Exact or controlled four-term null modular relation on an inherited strip model)**.** *Fix one finite-dimensional regulated strip model satisfying Proposition 6.66, and define*

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \; I(I_- : I_+ \mid J)_\sigma = 0\},$$

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \leq \varepsilon \right\}.$$

For any state η on this strip model, write

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the reference state ω is exact Markov and EC-aligned on this inherited decomposition (the inherited-split form of Definition 5.10, with sector pairs $\alpha = (\alpha_-, \alpha_+)$ and factors j_L, j_R), then

$$\Delta K_J(\omega) \in Z(\mathcal{A}(J)),$$

and in the canonical blockwise identification of Proposition 6.66 one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, without fixing that normalization there exists a central operator

$$K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$$

such that

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \leq \varepsilon$, then on this fixed strip model one may choose an exact Markov state $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$, assumed EC-aligned on the inherited split wherever the central identification below is used (the strip form of the Markov-split alignment hypothesis), with

$$\|\omega - \widetilde{\omega}_J\|_1 \leq \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

where

$$K_{\partial,J}(\widetilde{\omega}_J) := \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J)).$$

For every bounded observable O in $\mathcal{A}(I_-\cup J)$ or $\mathcal{A}(J\cup I_+)$,

$$\left|\omega(O) - \widetilde{\omega}_J(O)\right| \le \|O\|_\infty\, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

If the marginals entering ΔK_J for ω and $\widetilde{\omega}_J$ are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, the Fawzi–Renner recovery map supplies a constructive comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Hence the exact four-term strip relation is available only at EC-aligned exact Markovity, or along a controlled strip family on one fixed inherited strip model admitting EC-aligned replacements for which

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0.$$

In the controlled case, $\mathfrak{D}_J$ is carried explicitly at each finite stage.

Proof. Assume first that ω is exact Markov and EC-aligned on the inherited split of Proposition 6.66. The HJPW structure theorem gives blockwise factorization over a state-dependent decomposition of $\mathcal{H}_J$; the alignment hypothesis identifies that decomposition with the inherited split, so that on each sector pair $\alpha = (\alpha_-, \alpha_+)$,

$$\omega_{I_-JI_+}|_\alpha = p_\alpha\, \omega^{(\alpha)}_{I_-\,j_L} \otimes \omega^{(\alpha)}_{j_R\,I_+}.$$

Accordingly,

$$K_{I_-\cup J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_-\,j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{I_-\,j_L} \otimes K^{(\alpha)}_{j_R I_+}.$$

Tracing over I_+, I_-, or both gives the marginal block formulas

$$K_{I_-\cup J}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_-\,j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{I_-\,j_L} \otimes K^{(\alpha)}_{j_R},$$

$$K_{J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R I_+},$$

$$K_J(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R}.$$

Subtracting the last three expressions from the first gives zero on every block. Thus, in the canonical blockwise identification supplied by Proposition 6.66,

$$\Delta K_J(\omega) = 0.$$

If one chooses to keep the blockwise endpoint-label bookkeeping explicit instead of absorbing those constants into the canonical identification, the remainder is a direct sum of block constants and is therefore central. This yields the stated central form with $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$.

For the controlled statement, apply Proposition 5.31 on this fixed finite-dimensional strip model, relabeling A, B, D there as I_-, J, I_+. This yields an exact Markov replacement $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with the displayed trace-norm bound. The identity

$$\Delta K_J(\omega) = \Delta K_J(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J)$$

is tautological, so the displayed four-term formula follows from the exact-Markov case applied to $\widetilde{\omega}_J$. The observable estimate is immediate from

$$\big|\omega(O) - \widetilde{\omega}_J(O)\big| \leq \|O\|_\infty \, \|\omega - \widetilde{\omega}_J\|_1.$$

Finally, Proposition 5.35, again relabeled $A, B, D \mapsto I_-, J, I_+$, yields the operator-norm bound on $\mathfrak{D}_J$ whenever the relevant marginals have the stated lower spectral bound. The Fawzi–Renner estimate is the same one-shot recoverability bound used elsewhere and is independent of the exact-Markov replacement modulus. □

Definition 6.68 (Renormalized null modular functional)**.** *For a null interval I on generator Ω, let $K_\partial(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 6.67 on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define*

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega).$$

For a null half-line $H_a := (a, \infty)$, abbreviate

$$\widetilde{K}_a(\Omega) := \widetilde{K}[H_a, \Omega].$$

Proposition 6.69 (Endpoint-Lipschitz null modular families and weak tail generator)**.** *Under the derived quasi-local propagation and bounded-interval endpoint-Lipschitz control internal to Axiom 3.7, the renormalized interval family obeys the following matrix-element bounds on every fixed local-energy-bounded domain. For bounded intervals*

$$I = (a, b), \qquad I' = (a', b'),$$

contained in one compact endpoint window,

$$\big|\langle \psi, (\widetilde{K}[I', \Omega] - \widetilde{K}[I, \Omega])\phi \rangle\big| \leq C_{\psi,\phi,\Omega} \left(|a' - a| + |b' - b|\right).$$

In particular the matrix elements of $\widetilde{K}[I, \Omega]$ are jointly Lipschitz in the two endpoints and therefore have finite variation there.

For null half-lines $H_a = (a, \infty)$, one likewise has

$$\big|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle\big| \leq C_{\psi,\phi,\Omega} \, |a' - a|$$

for a, a' in every compact window. Hence, for fixed ψ, ϕ,

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is locally Lipschitz and therefore absolutely continuous. Its distributional derivative defines a locally L^∞ sesquilinear-form density

$$\langle \psi, q(a, \Omega)\phi \rangle := -\frac{1}{2\pi} \, \partial_a f_{\psi,\phi}(a),$$

and for $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle \psi, q(v, \Omega)\phi\rangle \, dv.$$

The object $q(a, \Omega)$ is the weak tail generator instead of a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after the derived half-sided modular inclusion of Corollary 6.71 and Lemma 6.72.

Proof. The bounded-interval endpoint-control statement derived from Axiom 3.7 applies precisely to the renormalized family obtained after removal of the central endpoint term. Therefore, for intervals contained in one fixed compact endpoint window,

$$|\langle \psi, (\widetilde{K}[I', \Omega] - \widetilde{K}[I, \Omega])\phi\rangle| \leq C_{\psi,\phi,\Omega} \, |I'\Delta I|,$$

where $I'\Delta I$ is the symmetric difference of the two intervals. If

$$I = (a, b), \qquad I' = (a', b'),$$

then

$$|I'\Delta I| \leq |a' - a| + |b' - b|,$$

which gives the displayed joint endpoint-Lipschitz bound.

For half-lines $H_a = (a, \infty)$ and $H_{a'} = (a', \infty)$, the symmetric difference is the bounded interval between the two endpoints, so

$$|H_{a'}\Delta H_a| = |a' - a|.$$

Because the central endpoint-label term has been removed in the definition of $\widetilde{K}_a$, the same endpoint-control estimate yields

$$|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle| \leq C_{\psi,\phi,\Omega} \, |a' - a|.$$

Hence, on every compact a-interval, the function

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

is Lipschitz. A Lipschitz function on a compact interval is absolutely continuous, so it has an almost-everywhere derivative in L^∞ and obeys the fundamental theorem of calculus:

$$f_{\psi,\phi}(b) - f_{\psi,\phi}(a) = \int_a^b \partial_v f_{\psi,\phi}(v) \, dv.$$

Define

$$\langle \psi, q(v, \Omega)\phi\rangle := -\frac{1}{2\pi} \, \partial_v f_{\psi,\phi}(v)$$

distributionally. Substituting this definition into the previous identity gives

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle \psi, q(v, \Omega)\phi\rangle \, dv.$$

This is exactly the claimed weak-tail formula. $\square$

Theorem 6.70 (Downstream density-upgrade template)**.** *Fix a null generator Ω and the renormalized half-line family $\widetilde{K}_a(\Omega)$ of Proposition 6.69. For fixed vectors ψ, ϕ in a common dense domain, let*

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle, \qquad q_{\psi,\phi}(a) := \langle \psi, q(a,\Omega)\phi \rangle = -\frac{1}{2\pi}\,\partial_a f_{\psi,\phi}(a).$$

Assume in addition that $q_{\psi,\phi}$ is weakly differentiable in a, that

$$\lim_{a\to+\infty} f_{\psi,\phi}(a) = 0, \qquad \lim_{a\to+\infty} q_{\psi,\phi}(a) = 0,$$

and define distributionally

$$\langle \psi, p(a,\Omega)\phi \rangle := -\partial_a q_{\psi,\phi}(a) = \frac{1}{2\pi}\,\partial_a^2 f_{\psi,\phi}(a).$$

Then

$$q_{\psi,\phi}(a) = \int_a^\infty \langle \psi, p(v,\Omega)\phi \rangle\, dv,$$

and

$$\langle \psi, \widetilde{K}_a(\Omega)\phi \rangle = 2\pi \int_a^\infty (v-a)\,\langle \psi, p(v,\Omega)\phi \rangle\, dv.$$

This statement is not part of the present fixed-cutoff bridge; it records the exact additional input/output package carried into the downstream density-upgrade task.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v = 0$. For $a \geq 0$, write

$$H_a := (a,\infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

Thus $\mathcal{M}_a(\Omega)$ is the half-line blow-up algebra generated by bounded null interval algebras inside H_a. By isotony of the null interval net,

$$a \leq b \quad \Longrightarrow \quad \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 6.71 (Derived half-sided modular inclusion on null half-lines)**.** *On the scaling-limit geometric-cap branch of Theorem 6.13, the blow-up modular action near a smooth entangling cut acts on the null coordinate by*

$$v \mapsto e^{-2\pi t} v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \qquad (t \leq 0),$$

so the inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$$

is half-sided modular. Equivalently, after the harmless convention change $t \mapsto -t$, one recovers the standard positive-time half-sided-inclusion form. The half-sided-inclusion step is therefore derived from the null-interval structure, isotony, and the scaling-limit geometric action instead of imported as an additional modular-QFT ingredient.

Proof. By Theorem 6.13, on every controlled collar family whose carried remainder vanishes in the refinement limit, the cap modular flow converges to the exact geometric cap dilation with 2π normalization. Blow up the cap near a smooth cut and restrict to the chosen null generator. In that tangent limit, the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c, d) \subset H_a$, the blow-up action therefore sends

$$\mathcal{A}((c, d), \Omega) \longmapsto \mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d), \Omega).$$

Taking the von Neumann closure of the algebra generated by all such intervals gives

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

If $t \leq 0$, then $e^{-2\pi t}a \geq a$, hence

$$H_{e^{-2\pi t}a} \subseteq H_a.$$

Isotony of the null interval net therefore implies

$$\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

This is exactly the half-sided modular-inclusion property for the pair $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$. Reparametrizing the modular parameter by $t \mapsto -t$ gives the more common positive-time convention if desired. $\square$

Lemma 6.72 (Positive null-translation generator)**.** *For the derived half-sided modular inclusion*

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \qquad (a > 0)$$

of Corollary 6.71, let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define

$$K_0(\Omega) := -\log \Delta_0(\Omega).$$

Then Borchers–Wiesbrock yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\, \mathcal{M}_b(\Omega)\, U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

Also,

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \qquad \Delta_0(\Omega)^{it} P_\Omega \Delta_0(\Omega)^{-it} = e^{-2\pi t} P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω,

$$[K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega.$$

For each $a \geq 0$, let $\Delta_a(\Omega)$ be the modular operator of the translated standard pair $(\mathcal{M}_a(\Omega), \omega)$ and define

$$K_a(\Omega) := -\log \Delta_a(\Omega).$$

Then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

hence

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. *Equivalently,*

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi\rangle = -2\pi(b-a)\,\langle \psi, P_\Omega \phi\rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$.

In the canonical blockwise normalization of Corollary 6.67, where the half-line endpoint term has been absorbed into the central part, the weak endpoint derivative of Proposition 6.69 agrees with the Borchers generator:

$$\langle \psi, P_\Omega \phi\rangle = -\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

for the same domain vectors. The half-sided-inclusion step proves only this affine half-line pair $(K_0(\Omega), P_\Omega)$*; bounded-interval modular-Hamiltonian formulas are downstream consequences of the bounded-interval kernel discharged in Lemma 6.135.*

Proof. Corollary 6.71 gives the required half-sided modular inclusion. The standard Borchers–Wiesbrock theorem for a standard half-sided inclusion [13] then yields a unique strongly continuous unitary group $U_\Omega(a)$ with $U_\Omega(a)\omega = \omega$, the translation law for the nested half-line net, and a positive self-adjoint generator P_Ω; Stone's theorem gives the displayed domain formula.

The affine covariance relation

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a)$$

is the Borchers relation. Differentiating at $a = 0$ gives

$$\Delta_0(\Omega)^{it} P_\Omega \Delta_0(\Omega)^{-it} = e^{-2\pi t} P_\Omega.$$

Since $\Delta_0(\Omega)^{it} = e^{-itK_0(\Omega)}$, differentiating at $t = 0$ on the common analytic core yields

$$-i[K_0(\Omega), P_\Omega] = -2\pi P_\Omega,$$

hence

$$[K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and

$$\mathcal{M}_a(\Omega) = U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^*,$$

the modular operator of the translated standard pair is

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*.$$

Differentiating this identity in a on $D(K_0(\Omega)) \cap D(P_\Omega)$ gives

$$\frac{d}{da}K_a(\Omega) = i[P_\Omega, K_a(\Omega)] = -2\pi P_\Omega,$$

because the commutator relation just proved is invariant under conjugation by $U_\Omega(a)$. Integrating from 0 to a yields the displayed quadratic-form identity, and taking matrix elements gives the equivalent form. Proposition 6.69 had produced the weak endpoint derivative of the renormalized half-line family; in the canonical normalization of Corollary 6.67 the central term has been removed, so that weak derivative is exactly the Borchers generator. The final sentence is the explicit claim boundary: bounded intervals require the additional projective interval action not supplied by half-sided inclusion alone. $\square$

Theorem 6.73 (The half-line generator is the local null-stress charge)**.** *Fix a null generator Ω and work in the canonical blockwise normalization of Corollary 6.67. Let $\widetilde{K}_a(\Omega)$ denote the renormalized modular Hamiltonian of the half-line H_a, and let P_Ω be the positive Borchers generator of Lemma 6.72. Then on*

$$D_\Omega := D(K_0(\Omega)) \cap D(P_\Omega)$$

one has

$$\langle \psi, P_\Omega \phi\rangle = -\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle \qquad (\psi, \phi \in D_\Omega),$$

and this operator is exactly the local null-stress charge of the effective spacetime description on that same half-line family.

Equivalently, if the effective spacetime description writes the same renormalized family as

$$\widetilde{K}_a^{\mathrm{eff}}(\Omega) = 2\pi\, Q_a^{\mathrm{mod}}(\Omega) + K_\partial(a, \Omega),$$

with $Q_a^{\mathrm{mod}}(\Omega)$ the standard local null modular-energy term and $K_\partial(a,\Omega)$ central and endpoint-supported, then

$$Q_{kk}(\Omega) := -\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega)$$

satisfies

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on D_Ω.

Hence the renormalized half-line null generator is not a separate EFT-side input: it is the same operator that the effective spacetime description calls the local null-stress charge of the null half-line.

Proof. Lemma 6.72 proves the first displayed identity: in the canonical normalization of Corollary 6.67, the weak endpoint derivative of the renormalized half-line family is exactly the Borchers generator. Passing to the effective spacetime description does not change the operator family; it rewrites the same $\widetilde{K}_a(\Omega)$ in local continuum variables appropriate to the local Lorentzian regime. In that description, the local null-stress charge is precisely the right endpoint derivative

$$-\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega)$$

of the same half-line modular family. Because $\widetilde{K}_a^{\mathrm{eff}}(\Omega)$ and $\widetilde{K}_a(\Omega)$ represent the same renormalized modular Hamiltonians, their right derivatives agree on D_Ω. Therefore the OPH generator and the effective null-stress charge coincide:

$$P_\Omega = Q_{kk}(\Omega).$$

The endpoint-supported central term does not alter the noncentral generator; in the canonical blockwise normalization it is absorbed into the central part fixed in Corollary 6.67. This proves the identification. $\square$

Imported input and downstream boundary. The only continuum input used above is the standard local modular-Hamiltonian form on the effective description of that same half-line family. It is used only to name, in continuum language, the operator fixed by the OPH half-line derivative. There is no separate "null-stress identification" assumption. The downstream boundary is only:

(i) transport of the half-line statement to bounded null intervals, carried by the affine/projective covariance and endpoint control of Lemma 6.135; and

(ii) reconstruction of a full local symmetric tensor from the directional charges, which is determined only up to the null-invisible metric term ϕg_{ab}.

So the gap closed here is the generator/charge identification itself: inside the null bridge and on the declared half-line family, P_Ω is exactly the local null-stress charge.

Lemma 6.74 (Null data determine the stress tensor modulo a metric term)**.** *Let X_{ab} be a symmetric tensor. If*

$$X_{ab}k^a k^b = 0$$

for every null vector k at a point, then

$$X_{ab} = \phi g_{ab}$$

for some scalar ϕ.

Proof. Write

$$X_{ab} = Y_{ab} + \phi g_{ab}$$

with Y_{ab} traceless. Since $g_{ab}k^a k^b = 0$ on null vectors, the hypothesis implies $Y_{ab}k^a k^b = 0$ for all null k. In local inertial coordinates let $k = (1, \hat{n})$ with $|\hat{n}| = 1$. Then

$$Y_{ab}k^a k^b = Y_{00} + 2\hat{n}^i Y_{0i} + \hat{n}^i \hat{n}^j Y_{ij}.$$

Oddness under $\hat{n} \mapsto -\hat{n}$ forces $Y_{0i} = 0$. Averaging that relation over S^2 gives

$$Y_{00} + \frac{1}{3}\delta^{ij}Y_{ij} = 0.$$

Tracelessness of Y_{ab} gives

$$-Y_{00} + \delta^{ij}Y_{ij} = 0,$$

so both Y_{00} and the spatial trace $\delta^{ij}Y_{ij}$ vanish. The condition is therefore

$$\hat{n}^i \hat{n}^j Y_{ij}^{\mathrm{TF}} = 0 \qquad \text{for all } \hat{n} \in S^2,$$

where Y_{ij}^{TF} is the traceless spatial part. The $\ell = 2$ spherical harmonics are linearly independent, so this implies $Y_{ij}^{\mathrm{TF}} = 0$. Hence $Y_{ab} = 0$ and therefore $X_{ab} = \phi g_{ab}$. □

Remark 6.75 (Claim boundary in the null bridge)**.** *The c.12-local bridge ends with explicit theorem-level objects: the positive self-adjoint null-translation generator P_Ω on its Stone domain, its Borchers affine covariance relation, the half-line modular identities above, and the exact half-line generator/charge identification of Theorem 6.73. The downstream boundary is transport to bounded null intervals through Lemma 6.135 and the tensor upgrade modulo the null-invisible metric term. The density-upgrade template of Theorem 6.70 is recorded separately and does not carry the operator-identification burden.*

6.2.1 Null-net standardness, derived translations, and the four-translation assembly

The bridge above consumes four inputs whose status this subsubsection settles: (i) cyclicity/separation of the common vector for the half-line and interval algebras actually used, which the Borchers–Wiesbrock step requires and which no previous statement proved on the realized branch; (ii) the inference from locality of the global finite-range MaxEnt generator to endpoint-Lipschitz control of *reduced* interval modular Hamiltonians, which is invalid in general; (iii) the affine/projective bounded-interval covariance used by Lemma 6.135; and (iv) the compatibility of the per-direction Borchers generators with one four-dimensional translation representation. Each item below is classified as a finite theorem, a scaling-limit theorem, or an explicit receipt; no Lorentz or translation representation is assumed anywhere in this subsubsection before it is derived.

The common null GNS net.

Definition 6.76 (Common null interval net)**.** *Fix the produced support-visible cap pair of Theorem 6.13, on the producer branch of Theorem 6.31, a produced round cap C_0, and a boundary point $\xi \in \partial C_0$. Let $\{H_a\}_{a\in\mathbb{R}}$, $H_a = (a,\infty)$, be the null half-line blow-up family at ξ used by Corollary 6.71, and let $(\mathcal{H},\Omega)$ be the GNS space of the produced scaling-limit state. For half-lines set $\mathcal{N}(H_a)$ equal to the weak closure of the blow-up strip algebras of Proposition 6.66; for a bounded open interval $I = (a,b)$ set*

$$\mathcal{N}(I) := \mathcal{N}(H_a) \wedge \mathcal{N}(H_b)',$$

and for the left half-lines $\check{H}_b = (-\infty,b)$ set $\check{\mathcal{N}}(\check{H}_b) := \mathcal{N}(H_b)' \wedge \mathcal{N}(\mathbb{R})$, where $\mathcal{N}(\mathbb{R})$ is the full blow-up algebra. Isotony holds by construction, and locality holds in the form $\mathcal{N}(H_b) \subseteq \mathcal{N}(H_a)' \vee \mathcal{N}(I)$ inherited from support-local disjoint commutation of node D1 at every finite stage. The nontriviality receipt $\mathsf{NTI}_r(d_{\min})$ *asserts that at stage r every strip difference algebra over a nonempty interval has a relative commutant of dimension at least $d_{\min} \geq 2$ inside the ambient strip; on towers where NTI holds cofinally, all $\mathcal{N}(I) \neq \mathbb{C}\,1$.*

Theorem 6.77 (Stagewise standardness and its limit)**.** *At every finite stage r, the MaxEnt reference state is faithful on the finite quotient algebra, so its GNS vector Ω_r is separating for every strip subalgebra. Assume additionally:*

1. *the cap-family-uniform mixed-GNS Cauchy clause of Definition 6.4 on the blow-up family;*
2. *the support-visible branch (no support collapse: Proposition 6.17 marks the excluded boundary);*
3. *the* half-line cyclicity receipt $\mathsf{Cyc}_r(\delta_r)$*: for every half-line H of either orientation and every vector in the stage-r reference frame of the tower, the subspace $\mathcal{N}_r(H)\Omega_r$ approximates it to accuracy δ_r, with $\delta_r \downarrow 0$ cofinally.*

Then in the limit net of Definition 6.76:

1. *Ω is cyclic for $\mathcal{N}(H)$ for every half-line H of either orientation;*
2. *Ω is separating for every $\mathcal{N}(H)$ and every $\mathcal{N}(I)$;*
3. *every inclusion $\mathcal{N}(H_a) \subseteq \mathcal{N}(H_b)$ $(a \geq b)$ is an inclusion of algebras standard with respect to the same Ω.*

Proof. Finite stage: a faithful state on a finite-dimensional von Neumann algebra has a separating GNS vector, and separation passes to subalgebras. Limit cyclicity: fix $\psi \in \mathcal{H}$ and $\varepsilon > 0$; the mixed-GNS Cauchy clause supplies a stage r and a reference-frame vector ψ_r with $\|\psi - \psi_r\| < \varepsilon/3$ together with an approximation of stage-r strip elements by limit elements with defect $< \varepsilon/3$ on Ω; the receipt $\mathsf{Cyc}_r(\delta_r)$ with $\delta_r < \varepsilon/3$ then gives $n \in \mathcal{N}(H)$ with $\|n\Omega - \psi\| < \varepsilon$. Separation: the opposite-orientation half-line $\check{H}$ obeys $\check{\mathcal{N}}(\check{H}) \subseteq \mathcal{N}(H)'$ by locality, and Ω is cyclic for $\check{\mathcal{N}}(\check{H})$ by item (1); a vector cyclic for a subalgebra of the commutant is separating for the algebra. Intervals inherit separation from $\mathcal{N}(I) \subseteq \mathcal{N}(H_a)$. Standardness of the inclusions is the conjunction of (1) and (2) for both members. $\square$

Remark 6.78 (Reeh–Schlieder upgrade: interval cyclicity is derived, not assumed)**.** *Cyclicity for* bounded *intervals is deliberately not a receipt. After Theorem 6.79 below produces the positive translation generator, the standard Reeh–Schlieder/Borchers argument applies: given weak additivity $\bigvee_a U(a)\mathcal{N}(I)U(a)^* = \mathcal{N}(\mathbb{R})$ (a finite-stage receipt on the same tower), positivity of P makes $a \mapsto U(a)$ boundary values of an operator-valued analytic function, and the usual edge-of-the-wedge argument propagates cyclicity from half-lines to every $\mathcal{N}(I)$, I open nonempty. Thus interval standardness is a theorem downstream of one half-line receipt, not an independent assumption.*

The derived half-sided inclusion.

Theorem 6.79 (Produced geometric dilation and derived half-sided inclusion)**.** *On the producer branch of Theorem 6.31, the modular automorphism group σ_t^Ω of $(\mathcal{N}(H_0), \Omega)$ acts on the blow-up net by the produced cap flow:*

$$\sigma_t^\Omega(\mathcal{N}(H_a)) = \mathcal{N}(H_{e^{-2\pi t}a}), \qquad a \geq 0.$$

In particular $\Delta^{it}\mathcal{N}(H_1)\Delta^{-it} \subseteq \mathcal{N}(H_1)$ for $t \leq 0$, so $(\mathcal{N}(H_1) \subseteq \mathcal{N}(H_0), \Omega)$ is a half-sided modular inclusion of algebras standard by Theorem 6.77. No dilation, translation, or Lorentz unitary is assumed: the geometric action is the support-flow identification produced by Theorems 6.31 and 6.13, transported through the blow-up exactly as in Corollary 6.71. The Borchers–Wiesbrock theorem [13, 14] then yields a unique strongly continuous unitary group $U(a) = e^{iaP}$ with

1. *$P \geq 0$ self-adjoint on its Stone domain, $U(a)\Omega = \Omega$;*
2. *$U(a)\mathcal{N}(H_b)U(a)^* = \mathcal{N}(H_{a+b})$ for $a \geq 0$, and for all $a \in \mathbb{R}$ on the algebras where both sides are defined;*
3. *$\Delta^{it}U(a)\Delta^{-it} = U(e^{-2\pi t}a)$ and $JU(a)J = U(-a)$.*

This re-derives Corollary 6.71 and Lemma 6.72 with the declared scaling-limit branch replaced by the produced one, and with the standardness hypotheses of the Borchers–Wiesbrock input proved rather than labeled.

Proof. The produced cap flow of Theorem 6.13 is, on the blow-up family, the one-parameter dilation of the half-line endpoint with the certified 2π normalization; this is precisely the computation of Corollary 6.71, now with the certificate produced by Theorem 6.31 instead of declared. Monotonicity $e^{-2\pi t} \geq 1$ for $t \leq 0$ gives the compression property. Theorem 6.77 supplies the standard-pair hypotheses. The conclusion package, including the Stone domain and the affine commutation relations, is the corrected Borchers–Wiesbrock theorem [13, 14]. $\square$

Regional modular locality: theorem and counterexample boundary.

Theorem 6.80 (Markov modular locality on the central-interface branch)**.** *Work on the declared central-interface branch of Axiom 3.7, on which Theorem 5.15 derives exact collar Markovianity of the MaxEnt reference states across every collar cut. Let I be a strip interval whose two endpoint cuts both carry the exact Markov property. Then the reduced modular Hamiltonian $K_I = -\log \rho_I$ decomposes as*

$$K_I = \sum_{\text{cells } c \subseteq I} k_c \; + \; k^L_{\partial I} + k^R_{\partial I},$$

where each k_c is the retained local MaxEnt term of cell c and the two boundary blocks act on the Hayden–Jozsa–Petz–Winter interface factors of the corresponding cuts [30], with $\|k^{L/R}_{\partial I}\|$ bounded by the uniform interface constants of the declared collar family, independently of $|I|$. Consequently the renormalized half-line family of Corollary 6.67 has endpoint-Lipschitz matrix elements on the finite-rank domain of Proposition 6.69, with Lipschitz constant the uniform interface bound. This replaces the earlier informal inference from locality of the global *generator, which is invalid without the Markov clause by Proposition 6.81.*

Proof. Vanishing conditional mutual information across a cut $I(A : C \mid B)_\rho = 0$ gives, by the structure theorem [30], a decomposition of the cut collar $\mathcal{H}_B = \bigoplus_k \mathcal{H}_{b^L_k} \otimes \mathcal{H}_{b^R_k}$ with $\rho_{ABC} = \bigoplus_k p_k \, \rho_{Ab^L_k} \otimes \rho_{b^R_k C}$. Applying this at both endpoint cuts of I and restricting to I yields $\rho_I = \bigoplus_{k,l} p_{kl} \, \rho^L_k \otimes \rho^{\text{bulk}}_I \otimes \rho^R_l$ with the bulk factor the MaxEnt Gibbs state of the retained cell terms in I; taking $-\log$ gives the displayed sum, the direct-sum labels being carried by the central interface projections, whose blocks are exactly $k^{L/R}_{\partial I}$. The interface blocks are functions of the boundary charges retained by the central-interface clause, so their norms carry the declared uniform constants. Endpoint variation of K_{I_a} then changes only one boundary block and the cells in the swept collar, giving the Lipschitz bound cellwise. ☐

Proposition 6.81 (Finite witness: Gibbs locality does not imply modular locality)**.** *There is an explicit four-site spin chain with translation-invariant nearest-neighbour Hamiltonian H and inverse temperature $\beta = 1$ such that, for I the first three sites, the interaction expansion of $-\log \rho_I$ contains a nonzero term coupling the two endpoint sites of I whose norm exceeds every bound obtained from the naive locality inference; the witness values are machine-verified by exact diagonalization with certified tolerances in the repository receipts (`code/geometry/`). Hence exact or controlled Markovianity (or the central-interface clause that derives it) is a genuine additional hypothesis, and the boundary between Theorem 6.80 and its failure is exactly the failure of collar Markovianity, quantified by the collar conditional mutual information.*

Proof. Take $H = \sum_{i=1}^{3} (Z_i Z_{i+1} + g X_i) + g X_4$ with $g = 1$. At second order in β, the cluster expansion of $-\log \rho_I$ acquires the commutator correction $\frac{\beta^2}{2} \, \mathbb{E}_{\text{tr}}[[H_{I\partial}, H_{\partial I^c}]]$-type terms which do not cancel for noncommuting nearest-neighbour couplings and which, after the partial trace over site 4, generate a weight-two Pauli term supported on sites $\{1, 3\}$; its coefficient is nonzero at $g = 1$, $\beta = 1$, as certified numerically to interval-arithmetic accuracy in the repository receipt. Since the collar CMI of this state across the cut $3 \,|\, 4$ is strictly positive (also certified), the example sits strictly outside the Markov branch, as claimed. ☐

Bounded intervals and the four-translation assembly.

Theorem 6.82 (Möbius covariance and the projective bounded-interval action)**.** *Assume Theorem 6.79 for the half-line family at ξ and for the opposite-orientation family (equivalently, the* *reflected inclusion obtained from $J\mathcal{N}(H_0)J$), with common Ω, on a tower carrying* NTI *cofinally. Then the two derived translation groups and the modular dilations generate a strongly continuous (anti-)unitary positive-energy representation of* $\mathrm{PSL}(2,\mathbb{R})$ *acting on the compactified null line [13, 15], and*

$$\mathcal{N}(gI) = U_g\,\mathcal{N}(I)\,U_g^*$$

for every Möbius g and bounded or half-infinite I, with isotony and locality preserved. The bounded-interval modular kernels are the projective transports of the half-line kernels; if the finite-stage kernel residuals obey $\epsilon_r = o(\ell_r^4)$ along the declared scaling, the transported kernels satisfy the $o(\ell^4)$ requirement consumed by Lemma 6.135. The projective bounded-interval covariance previously used is therefore derived on this branch, with the kernel-residual scaling the only quantitative receipt.

Proof. The generating result for two half-sided modular inclusions with common standard vector is the Wiesbrock/Guido–Longo–Wiesbrock theorem [13, 15]: the modular group of $\mathcal{N}(H_0)$ and the two derived translation groups satisfy the $\mathrm{PSL}(2,\mathbb{R})$ commutation relations proved in Theorem 6.79(3) and its reflection; integrability to the simply connected cover and descent to $\mathrm{PSL}(2,\mathbb{R})$ follow from the affine relations plus $U(a)\Omega = \Omega$, and positivity of the rotation generator is equivalent to positivity of the two translation generators. Covariant interval algebras are then defined by transport; consistency on overlaps of group elements mapping I to itself follows because the stabilizer of an interval is generated by its own modular dilations, which fix $\mathcal{N}(I)$ by Takesaki modular theory [26]. NTI guarantees the transported algebras are nontrivial and that intersections realize $\mathcal{N}(I) = \mathcal{N}(H_a) \wedge \mathcal{N}(H_b)'$. The kernel-residual propagation is linear along the transport, giving the stated $o(\ell^4)$ transfer. □

Definition 6.83 (Modular-intersection receipt)**.** *For two produced caps $C, \tilde{C}$ with transverse boundary circles, write $\mathsf{MI}_r(C,\tilde{C};\varepsilon)$ for the finite-stage statement that the stage-r modular data of the pair $(\mathcal{M}_r(C), \mathcal{M}_r(\tilde{C}), \Omega_r)$ satisfy the Wiesbrock modular-intersection relations [16] up to residual ε: the compressions of each modular group into the intersection algebra agree with the intersection's own modular group, and the mixed commutation relations of the two modular groups close on a common dense core. The receipt is decidable at every finite stage and carries a convergence envelope $\varepsilon_r \downarrow 0$.*

Theorem 6.84 (Four-translation assembly with future-cone spectrum)**.** *Assume the producer branch with BW framing (so the produced $G = \mathrm{SO}^+(3,1)$ action on caps is unitarily implemented with invariant Ω), Theorem 6.79 applied at every produced null direction $\Omega \in S^2$ (yielding positive generators P_Ω), and MI_r cofinally on transverse produced cap pairs. Then:*

1. *(*Covariance; theorem, no receipt.*) For every $\Lambda \in G$,*

   $$U(\Lambda)\,P_\Omega\,U(\Lambda)^* = \omega_\Lambda(\Omega)\,P_{g_\Lambda(\Omega)},$$

   with $\omega_\Lambda, g_\Lambda$ as in Theorem 6.39: the blow-up construction is natural, so the generator transforms with the null-vector weight.

2. *(*Additivity on the MI branch.*) Generators of distinct null directions strongly commute on a common core, and whenever $\sum_i c_i\, q(\Omega_i) = 0$ in V one has $\sum_i c_i P_{\Omega_i} = 0$ on that core. Hence there is a unique four-parameter strongly continuous unitary group $x \mapsto U(x)$, $x \in V$, with self-adjoint generators P^μ satisfying $P(q(\Omega)) = P_\Omega$.*

3. *(*Spectrum.*) $\eta(q,P) \geq 0$ for every future null q, so the joint spectrum of P^μ lies in the closed future cone.*

4. *(*Poincaré closure.*)* $U(\Lambda)U(x)U(\Lambda)^* = U(\Lambda x)$, *so* $(U(\Lambda), U(x))$ *is a positive-energy unitary representation of the proper orthochronous Poincaré group [12].*

A family of per-direction generators without the MI *relations does not satisfy item (2); the receipt is exactly the acceptance boundary between one commuting four-translation representation and unrelated one-generator-per-direction constructions.*

Proof. (1) Conjugating the half-sided inclusion at direction Ω by $U(\Lambda)$ yields the half-sided inclusion at $g_\Lambda(\Omega)$ with the affine parameter rescaled by $\omega_\Lambda(\Omega)$, by Theorem 6.39 applied to the blow-up family; uniqueness of the Borchers generator in Theorem 6.79 then forces the displayed weight. (2) The MI relations in the limit give, for each transverse pair, the commutation of the associated translation unitaries through the modular-intersection theorem of [16]; strong commutativity of the generators follows on the joint Stone core. For the linearity relation, fix a dependent family $\sum_i c_i q(\Omega_i) = 0$; the covariance (1) and commutativity reduce the claim to the stabilizer direction, where the two sides generate the same one-parameter group by the uniqueness clause of the Borchers–Wiesbrock theorem applied to the common compressed inclusion; the finite-stage residual is controlled by ε_r and vanishes in the limit. Uniqueness and existence of $U(x)$ then follow by choosing four null directions in general position and checking independence of the choice via the same relation. (3) is positivity of each P_Ω plus density of $\{q(\Omega)\}$ in the generating rays of the dual cone. (4) combines (1) and (2); the group law on the semidirect product holds on the core and extends by continuity. □

Remark 6.85 (Re-audit ledger for the earlier conditional closures)**.** *Relative to the earlier conditional packets: the left/right null-strip tensor split of Proposition 6.66 is a fixed-cutoff branch input; exact-or-controlled Markovianity is a theorem on the central-interface branch (Theorem 6.80) with an explicit counterexample boundary (Proposition 6.81); the standardness hypotheses consumed by every Borchers–Wiesbrock invocation are proved (Theorem 6.77), with one half-line cyclicity receipt and the Reeh–Schlieder upgrade (Remark 6.78); the half-sided inclusion itself is derived on the produced branch (Theorem 6.79); bounded-interval projective covariance is derived (Theorem 6.82) with the kernel-residual scaling as the only quantitative receipt; and the four-translation assembly is a theorem on the* MI *branch (Theorem 6.84). Finite theorems: stagewise standardness, receipts decidability, the witness proposition. Scaling-limit theorems: limit standardness, derived inclusion, Möbius covariance, assembly. Branch receipts:* Cyc*,* NTI*, weak additivity,* MI*, kernel-residual scaling, and the fixed-cutoff strip split.*

6.3 From repaired records to a conditional Lorentzian event manifold

Corollaries 6.40 and 6.46 are statements about Lorentz-group and observer-frame *kinematics*: H^3 is the hyperboloid of future unit timelike frames, not a spatial slice of physical events, and Remark 6.47 lists what the chart theorem does not do. The Einstein package below (Theorems 6.99–6.107) nevertheless consumes a locally Lorentzian $d = 4$ regime with local inertial coordinates, causal diamonds, geodesic balls, a metric-compatible connection, and curvature. This subsection constructs that consumer interface: a quotient-intrinsic event space $\mathcal{E}$, a four-dimensional atlas of signature $(-+++)$, tetrads, metric, connection, and glued observer clocks, from the outputs of the producer chain (Theorems 6.31, 6.13), the record localization packet (Corollary 6.64), and the derived translations (Theorems 6.79, 6.84). The construction cleanly separates the event-manifold *base* from the H^3 *fiber* of unit timelike frames at each event, keeps every additional input as an explicitly named receipt, and closes with countermodels showing that $\dim H^3 = 3$ by itself promotes nothing.

Events as coincidence classes of record germs.

Definition 6.86 (Record germs and intrinsic coincidence)**.** *Fix a refinement tower on the producer branch. A stage-r* located record datum *is a quotient-visible record token i together with its certified localization output $S_i(t) = B_H(\widehat{X}_i(t), \rho_i(t)) \subset H^3_{R_H}$ from Corollary 6.64 in some observer chart, a modular clock coordinate τ_i read from the derived translation/clock line of Theorem 6.79 with the certified 2π normalization, and a clock error bar ς_i. Its* certified box *is $[\tau_i \pm \varsigma_i] \times S_i(t)$ transported between observer charts by the D1 gauge/transport package and Theorem 6.63. A* record germ *is a refinement-compatible family $(i_r)_{r \geq r_0}$ of located record data whose transported certified boxes are nested up to transport defect and whose radii $\rho_{i_r} + \varsigma_{i_r} \to 0$ cofinally; a germ is a Cauchy filter of certified boxes. Two germs are* coincident *when, after common refinement, their transported certified boxes intersect at cofinally many stages. Coincidence uses only normal-form data: certified balls, clock reads, and transport maps.*

Lemma 6.87 (Quotient, gauge, schedule, refinement, and chart invariance)**.** *The germ set, the coincidence relation, and the box uniformity are invariant under: repair schedule (Theorem 4.8 fixes the normal form); gauge (support covariance of node D1 and Lemma 6.26); refinement (Petz support/CPTP transport of certified boxes along the tower maps); and Lorentz chart change (Theorem 6.63 for the spatial factor, the covariance clause of Theorem 6.84(1) for the clock factor). In particular coincidence is an observer-independent relation.*

Proof. Each datum entering Definition 6.86 was individually proved invariant in the quoted statements; intersection and nesting of transported boxes are expressed through η-invariant quantities (hyperbolic distances, affine clock differences), so the derived relation inherits every invariance. □

Definition 6.88 (Event space and its unconditional regularity class)**.** *The* event space *$\mathcal{E}$ is the set of coincidence classes of record germs, equipped with the uniformity generated by the certified boxes. Unconditionally, $\mathcal{E}$ is a second-countable, completely regular uniform space (the tower is countable and each stage carries finitely many tokens); it need be neither Hausdorff nor a manifold. This is the precise regularity class obtained without receipts; everything stronger below is conditional and cited as such.*

Definition 6.89 (Event-manifold receipts)**.** *On a declared bounded region of interest:*

(E1) Population density*: every certified box of every stage contains a germ, and the covering radii shrink with a declared modulus $\rho_r \downarrow 0$ (record-dense branch).*

(E2) Separation gaps*: distinct germs achieve certified disjoint boxes at some common stage, with the residual-gap certificates $\Delta_{\mathrm{loc}} > 0$ of Theorem 6.62 uniform on the region.*

(E3) Rank-four frame*: every point of the region is covered by a calibrated response frame of rank four (a rank-three conditioned spatial cap frame as in Proposition 6.53, augmented by one derived modular clock line), with uniform observability constants $0 < \alpha \leq L$.*

(E4) Overlap cocycle*: on chart overlaps the frame transition data agree with a Poincaré transformation (element of $\mathrm{SO}^+(3,1) \ltimes V$) up to residuals $\varepsilon_r \downarrow 0$, and the transitions satisfy the cocycle law up to the same residuals; decidable at every finite stage.*

Theorem 6.90 (Conditional event four-manifold)**.** *Assume* (E1)–(E3) *on a region. Then the corresponding subspace $\mathcal{E}_U \subseteq \mathcal{E}$ is Hausdorff, second countable, and locally compact, and every rank-four frame chart*

$$\varphi : \ \mathcal{E}_U \supseteq W \longrightarrow \mathbb{R}^4, \qquad \varphi(e) = (\tau(e), X_{\mathrm{E}}(e))$$

(clock read, then frame reconstruction as in Proposition 6.53) is a bi-Lipschitz homeomorphism onto an open subset of $\mathbb{R}^4$ with constants L/α; overlapping charts have bi-Lipschitz transition maps. Hence $\mathcal{E}_U$ is a topological four-manifold with a locally bi-Lipschitz atlas. Under the additional second-order response modulus ($\mathsf{E4}'$) *(uniform $C^{1,1}$ control of the calibrated responses), the atlas is $C^{1,1}$. Without* ($\mathsf{E2}$) *the Hausdorff property can fail and only Definition 6.88 survives (Proposition 6.96(iii)); without* ($\mathsf{E1}$) *the dimension can drop (Proposition 6.96(i)).*

Proof. Hausdorff. Distinct classes contain non-coincident germs, so ($\mathsf{E2}$) supplies a stage with certified disjoint boxes; boxes generate the uniformity, so the corresponding box neighborhoods separate the classes. *Second countability* is Definition 6.88. *Charts.* On a box neighborhood W, ($\mathsf{E3}$) gives for germ representatives the two-sided bound

$$\alpha\,\bar{d}(e,e') \;\le\; |\varphi(e)-\varphi(e')| \;\le\; L\,\bar{d}(e,e')$$

with $\bar{d}$ the box (product hyperbolic-plus-clock) distance: the spatial component is Proposition 6.53, the clock component is the affine clock line with its calibration constants, and the constants combine in quadrature. Thus φ is injective and bi-Lipschitz onto its image. *Open image.* $\mathcal{E}_U$ is complete in the box uniformity (germs are Cauchy filters of boxes, and every Cauchy filter of certified boxes is itself a germ by Definition 6.86), so $\varphi(W)$ is complete, hence closed in the target box; ($\mathsf{E1}$) makes $\varphi(W)$ ρ_r-dense in that box for every r; closed and dense gives the whole open box. Local compactness follows. Transitions of bi-Lipschitz charts are bi-Lipschitz; the $C^{1,1}$ upgrade is the chain rule for the second-order modulus ($\mathsf{E4}'$), Rademacher differentiability supplying the almost-everywhere derivatives with Lipschitz representatives. The failure statements are the countermodels below. $\square$

Causal structure, signature, and time orientation.

Definition 6.91 (Intrinsic precedence and diamonds)**.** *For germs x, y, declare $x \preceq y$ when some located representative of y has a validated transactional read-set ancestry (node D1) containing a located representative of x, cofinally in refinement. The relation is schedule-independent (ancestry of the normal form is reachability-invariant by node D1) and gauge/refinement invariant as in Lemma 6.87. Causal diamonds are $D(p,q) := \{e \in \mathcal{E} : \; p \preceq e \preceq q\}$.*

Theorem 6.92 (Local cone structure, signature $(-{+}{+}{+})$, and time orientation)**.** *Assume* ($\mathsf{E1}$)–($\mathsf{E4}$) *and the* MI*/assembly branch of Theorem 6.84. Then, near every event p in the region:*

1. *the chart φ transports $\preceq$ to the cone order of η up to $o(1)$ envelopes: $e \in D(p,q)$ iff $\varphi(e)-\varphi(p)$ and $\varphi(q)-\varphi(e)$ lie in the closed future cone up to the receipt residuals;*
2. *the local cone at p is linearly isomorphic to the standard cone of η: its sphere of null generators is the produced celestial S^2 of Theorem 6.30 with its produced conformal structure, and a proper convex cone in $\mathbb{R}^4$ whose projectivized boundary is conformally that S^2 is the cone of a quadratic form of Lorentz signature; hence each tangent space carries signature $(-{+}{+}{+})$, with exactly one time dimension and three spatial dimensions;*
3. *the transactional direction of repair ancestry selects one component of the timelike double cone at every event, consistently across overlaps ((*$\mathsf{E4}$*) transitions are orthochronous because they preserve $\preceq$); this is the time orientation;*
4. *causal compatibility on overlaps: the transition maps preserve cones and cone orders exactly in the limit, up to the ε_r residuals at finite stage.*

Proof. (1) The derived translations move certified boxes along null and timelike directions of the chart with unit affine weight (Theorem 6.79(2)); a located record in the read-set ancestry of another is connected to it by a finite chain of transactionally validated collar transports, each of which the blow-up chart represents inside the closed future cone up to the calibration residuals of the frame; composing the chain and passing to the germ limit gives the cone order with $o(1)$ defect. Conversely, a chart pair in the open cone interior admits, by (E1), a chain of populated boxes realizing ancestry. (2) The null directions at p are exactly the blow-up directions of the produced caps through p's boxes, so the projectivized cone boundary is the produced S^2; if a proper convex cone $K \subset \mathbb{R}^4$ has projectivized boundary a quadric conformally equivalent to the round $S^2 \subset \mathbb{RP}^3$, then $K = \{x : \; q(x) \leq 0, \; x^0 \geq 0\}$ for a quadratic form q vanishing on that quadric; a real quadratic form in four variables whose zero set is a nondegenerate real quadric projectively equivalent to a sphere has signature $(1, 3)$ up to overall sign, which is Lorentz signature; normalizing the time direction inside K gives $(-+++)$. (3) Repair transactions are directed (commits are not invertible moves on the quotient by strict descent), so ancestry orients each timelike chain; schedule independence makes the orientation well defined on germs, and (E4) transitions preserve $\preceq$, hence map future cones to future cones. (4) is (E4) plus (2), Poincaré maps being exactly the affine maps preserving η-cones. ☐

Proposition 6.93 (H^3 is the frame fiber over events, not a spatial slice)**.** *On the branch of Theorem 6.92, define the frame bundle*

$$F(\mathcal{E}) := \{(e, u) : \; e \in \mathcal{E}_U, \; u \in T_e\mathcal{E} \text{ future unit timelike}\}.$$

Each fiber is the hyperboloid $H^3 \cong \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ of Corollary 6.46, and the record-localization chart of Corollary 6.64 is a chart on the fiber *data of located observers, not on the event base. Moreover, in the homogeneous local model there is no Lorentz-equivariant section $\mathcal{E} \to F(\mathcal{E})$: the stabilizer of an event in the local model is the full Lorentz group, which fixes no point of H^3 (Proposition 6.42); selecting u requires observer, tetrad, clock, or record data, exactly as in Remark 6.44. Identifying velocity space H^3 with a spatial slice of $\mathcal{E}$ is therefore excluded structurally, independently of the dimension coincidence* $\dim H^3 = 3$.

Proof. The fiber statement is Definition 6.41 applied in $(T_e\mathcal{E}, g_e)$, which carries Lorentz signature by Theorem 6.92(2). For the no-section statement, an equivariant section would assign to the model event a point of H^3 fixed by its stabilizer $\mathrm{SO}^+(3,1)$, whose point stabilizers on H^3 are the compact $\mathrm{SO}(3)$ subgroups; a noncompact group cannot fix a point with compact stabilizer, as in Proposition 6.43. ☐

Tetrads, metric, connection, clocks.

Theorem 6.94 (Tetrads, Lorentzian metric, and curvature readout)**.** *Assume* (E1)*–*(E4)*. Then:*

1. *each rank-four frame chart defines a tetrad e^a_μ (the frame differentials of the three conditioned cap responses and the clock line), and the transition functions of overlapping tetrads are the* (E4) *Poincaré data, satisfying the cocycle law in the limit;*

2. *$g := \eta_{ab}\, e^a \otimes e^b$ is a well-defined Lorentzian metric on $\mathcal{E}_U$, independent of the chart by (1); its conformal class is produced (Theorem 6.30 on the celestial factor, Theorem 6.84 fixing the relative time/space normalization through the affine clock weights), while the single overall physical scale is the separate scale certificate (the R_H/cell-ruler receipts of Definition 5.41); the Weyl representative is therefore produced and only the global conformal factor's absolute normalization is a receipt;*

3. *on the* (E4′) *branch the atlas is* $C^{1,1}$*, the Levi-Civita connection of* g *exists with locally bounded Christoffel symbols, geodesics and geodesic balls exist locally, and the curvature is defined almost everywhere as a locally bounded measurable readout; on the declared smooth scaling branch the readout upgrades to the smooth curvature tensor consumed by Lemma 6.101 and Theorems 6.105–6.107. This is the precise interface through which the Einstein package's "locally Lorentzian* $d = 4$ *scaling regime" is supplied rather than assumed.*

Proof. (1) is Theorem 6.90 plus (E4); tetrad transitions are the derivatives of the chart transitions, which are affine Poincaré in the limit, so the cocycle law is inherited. (2) The η-contraction of Poincaré-related tetrads is chart-independent; conformal-class production is quoted; the scale statement repeats Corollary 6.46's boundary (no physical R_H from group data) in the produced setting. (3) On a $C^{1,1}$ atlas the metric is Lipschitz, so Christoffel symbols exist a.e. and are locally bounded (Rademacher), local geodesic existence follows from Picard–Lindelöf with measurable right-hand side in the Carathéodory sense, and the Riemann tensor is a.e. defined and locally bounded; smoothness on the declared branch is a declared upgrade, exactly as consumed downstream. ☐

Theorem 6.95 (Glued modular clocks and scheduler-independent histories)**.** *Assume* (E1)–(E4)*. Along every maximal* $\preceq$*-chain of germs carrying frame data (an observer history), the derived modular clock of Theorem 6.79, in the certified* 2π *normalization, defines a monotone proper-time parameterization; on overlaps of two histories the transition law is the Lorentz reparameterization* $d\tau'/d\tau = -\eta(u, u')$ *determined by the two frame vectors, and it is scheduler-independent because every ingredient is a normal-form invariant (Lemma 6.87). Global statements are explicitly assumption-gated: existence of a global time function is equivalent to stable causality of* $(\mathcal{E}_U, g)$*, and global hyperbolicity is carried as the* record-Cauchy receipt *(every inextendible* $\preceq$*-chain in the region meets the declared record family), which this manuscript does not derive. No global causality claim is made beyond these named assumptions.*

Proof. Monotonicity along a history follows from Theorem 6.92(3): ancestry orients the chain, and the derived clock increases along the transactional direction by positivity of the generator (Theorem 6.79(1)). The overlap law is the chain rule for the two affine clock lines evaluated on the same germ chain, with the Lorentz factor $-\eta(u, u')$ produced by the frame transition of (E4). Scheduler independence is Lemma 6.87. The global equivalences are the standard Lorentzian statements, quoted here only to name the receipts. ☐

Countermodels and claim boundary.

Proposition 6.96 (Countermodels: the receipts are jointly irreducible)**.** *Each of the following explicit systems satisfies the full producer branch (Theorems 6.31, 6.13), hence carries the exact Lorentz/*H^3 *kinematics of Corollaries 6.40 and 6.46 verbatim, and fails exactly one event receipt with the stated consequence; machine receipts are in the repository (*`code/geometry/`*).*

1. *(*Population deficit; fails (E1)*.) Records confined to a single observer worldline stack:* $\mathcal{E}$ *is a one-dimensional chain (or a* $(1{+}2)$*-dimensional sheet for a planar stack), while* $\dim H^3 = 3$ *throughout. The event dimension is a population output, not a kinematic one.*

2. *(*Label inflation; fails the intended reading of (E3)*.) Doubling every record with a persistent internal label that survives all separation tests yields* $\mathcal{E} \cong M^4 \times \{0, 1\}$*, and a continuum label yields a five-dimensional event space, with identical screen and frame kinematics; the rank-four frame receipt is what excludes surviving extra coordinates.*

3. *(*Reconciliation branching; fails $(\mathsf{E2})$.*) Two germ families sharing every certified box up to stage $r_\star$ and separating afterwards with permanently failed residual gaps ($\Delta_{\rm loc} \le 0$, output* AMBIGUOUS *by Theorem 6.62) produce the standard non-Hausdorff doubled-germ quotient: a line with doubled origin pattern inside $\mathcal{E}$. The finite-stage witness of the failure is exactly the* AMBIGUOUS *verdict, so non-Hausdorff formation is detectable, and Definition 6.88 is the unconditional regularity class.*

In particular no argument of the form "$\dim H^3 = 3$ therefore spacetime is 3+1-dimensional" is valid: dimension, Hausdorffness, and manifoldness of the event base are controlled by $(\mathsf{E1})$–$(\mathsf{E4})$ *and by nothing weaker.*

Proof. (i) Take the producer-branch tower and restrict the record layer to tokens whose localization balls follow one timelike chain (one located observer); every construction upstream of records is untouched, while Definition 6.88 yields the chain. (ii) Tensor the record layer with a two-point (respectively interval-valued) internal register transported trivially; boxes never separate the register, so coincidence classes split accordingly. (iii) Seed a reconciliation defect at stage $r_\star$ as in the obstruction gates of node D1 so that two token families carry intersecting boxes before $r_\star$ and disjoint uncertified boxes after; the residual-gap certificate then fails permanently by construction. Each construction preserves the screen-side producer data because it modifies only the record layer downstream of the certificate. □

Remark 6.97 (Claim boundary for the event-manifold packet)**.** *Proved here, conditionally and quotient-intrinsically: event classes and their invariances (Lemma 6.87); the unconditional regularity class (Definition 6.88); the conditional four-manifold atlas (Theorem 6.90); local cones, signature $(-{+}{+}{+})$, time orientation, and overlap causal compatibility (Theorem 6.92); the strict base/fiber separation with a no-section no-go (Proposition 6.93); tetrads, produced conformal metric, connection/curvature readout with explicit regularity (Theorem 6.94); glued scheduler-independent clocks (Theorem 6.95); and the countermodels (Proposition 6.96). Carried as named receipts:* $(\mathsf{E1})$–$(\mathsf{E4})$ *(with* $(\mathsf{E4}')$ *and the smooth upgrade), the* MI*/assembly branch, the absolute conformal scale, stable causality, and the record-Cauchy receipt. On branches without these receipts, the corpus-wide claim is exactly Lorentz-group and observer-frame kinematics, as stated in Remark 6.47.*

6.4 Generalized entropy and the Einstein equation

From this point through Theorem 6.107 we specialize to the physical $d = 4$ scaling branch. The claim boundary is explicit. The only standard geometric input used below, beyond D3 and D4, is the fixed-volume area-variation identity for a small geodesic ball. The small-ball modular kernel used below is *not* imported from a separate EFT law: it is the local expression of the geometric cap generator B_C from Theorem 6.13, evaluated using the half-line null-stress identification of Theorem 6.73 together with the bounded-interval kernel of Lemma 6.135.

For a reference state ω and a small cap C,

$$\delta S_C = \delta\langle K_C\rangle.$$

By Theorem 6.13,

$$\delta S_C = 2\pi\, \delta\langle B_C\rangle.$$

By Proposition 5.40,

$$S_{\rm gen}(C) = \mathrm{Tr}(\rho L_C) + S_{\rm bulk}(C).$$

Together with Definition 5.41, this identifies the area term as the coarse-grained form of the edge entropy density instead of an independent postulate. After this edge-center split, the first-law piece entering the small-ball argument is the bulk contribution:

$$\delta S_{\text{bulk}}(C) = 2\pi\, \delta\langle B_C \rangle.$$

Definition 6.98 (Admissible fixed-cap MaxEnt variation)**.** *Fix a cap C and a realized reference state ω_C on the cap-reduced realized MaxEnt branch. An admissible fixed-cap MaxEnt variation is a first-order variation*

$$\delta\rho_C = \frac{d}{ds}\Big|_{s=0} \rho_C(s)$$

along a C^1 curve $s \mapsto \rho_C(s)$ such that:

1. *every $\rho_C(s)$ lies in the same cap-label-preserving block class of Proposition 5.40, so the same edge-center operator L_C and sector projectors P_α apply;*
2. *the cap C, its boundary sector, charge labels, size/volume, and the optional conserved charges Q_b are held fixed;*
3. *the support-visible cap geometry varies only through the declared geometric tangent readout and does not change hidden carrier representatives, port labels, worker schedule, or gauge presentation;*
4. *the Axiom 3.7 constraint values are fixed to first order, except for the declared stress-energy perturbation channel:*

   $$\frac{d}{ds}\Big|_{s=0} \mathrm{Tr}(\rho_C(s) O_a(x)) = 0, \qquad \frac{d}{ds}\Big|_{s=0} \mathrm{Tr}(\rho_C(s) Q_b) = 0$$

   for every retained local constraint $O_a(x)$ supported in the fixed cap neighborhood and every retained conserved charge Q_b; and
5. *in the associated small-ball/diamond readout, the carried modular/stress remainder satisfies the Einstein-branch scaling condition*

   $$\delta\langle E^{(\eta)}_{C,\ell} \rangle = o(\ell^4).$$

Shape or null-cut deformations are denoted separately by δ_{shape} or ∂_λ; they are not part of this fixed-cap variation class.

Theorem 6.99 (Derived fixed-cap generalized-entropy stationarity)**.** *Let C be a fixed cap and let ω_C be a realized reference state on the cap-reduced realized MaxEnt branch of Axiom 3.7. For every admissible fixed-cap MaxEnt variation δ of Definition 6.98,*

$$\delta S_{\text{gen}}(C) = 0.$$

Equivalently, the realized reference state is stationary for generalized entropy on the allowed fixed-cap variation class selected by the realized MaxEnt family.

Proof. Axiom 3.7 selects the realized branch by entropy maximization on the finite-dimensional constraint surface cut out by the retained local constraints and optional conserved charges. Definition 6.98 restricts to first-order tangent directions that stay on that same fixed-cap constraint surface. Hence the first variation of the cap entropy vanishes at the realized reference state:

$$\delta S(\rho_C)\big|_{\omega_C} = 0.$$

For the same cap-label-preserving block class, Proposition 5.40 gives

$$S(\rho_C) = S_{\text{bulk}}(C) + \mathrm{Tr}(\rho_C L_C) = S_{\text{gen}}(C).$$

Differentiating this identity along the admissible curve yields

$$\delta S_{\text{gen}}(C) = \delta S(\rho_C)\big|_{\omega_C} = 0.$$

□

Remark 6.100 (Claim boundary of the fixed-cap generalized-entropy stationarity theorem)**.** *Theorem 6.99 internalizes only the fixed-cap, cap-label-preserving, constraint-preserving first-order variation class selected by Axiom 3.7. It does not claim generalized-entropy stationarity for arbitrary shape deformations, arbitrary null-cut deformations, or arbitrary off-branch perturbations outside that realized MaxEnt family.*

Lemma 6.101 (Internal d=4 small-ball bridge)**.** *Assume the hypotheses of Theorem 6.73, together with the bounded-interval kernel of Lemma 6.135. Let p be the cap center and let u^a be the future-directed unit tangent of the local diamond rest frame at p. In local inertial coordinates (t, x^i) adapted to u^a, let D_ℓ be the causal diamond whose $t = 0$ slice is the Euclidean ball*

$$B_\ell = \{t = 0,\ r < \ell\}, \qquad r^2 = \delta_{ij}x^i x^j.$$

Then

$$\delta S_{\text{bulk}}(C) = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle\, d^3x + \delta\langle E^{(\eta)}_{C,\ell}\rangle.$$

If $\delta\langle T_{00}\rangle$ is also approximately constant across B_ℓ and

$$\delta\langle E^{(\eta)}_{C,\ell}\rangle = o(\ell^4)$$

along the same scaling family, then in $d = 4$,

$$\delta S_{\text{bulk}}(C) = \frac{8\pi^2\ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5 \partial T) + o(\ell^4).$$

Proof. Work in the local Lorentzian scaling regime around p. On the geometric-subnet branch of Theorem 6.13, the cap modular flow is the geometric flow of the diamond-preserving conformal Killing field. In the tangent diamond D_ℓ, that field is

$$\xi_{D_\ell} = \frac{1}{2\ell}\Big((\ell^2 - r^2 - t^2)\,\partial_t - 2t\, x^i \partial_i\Big),$$

so on the $t = 0$ slice one has

$$\xi_{D_\ell} \cdot n = \frac{\ell^2 - r^2}{2\ell},$$

with $n^a = u^a$ the slice normal at the center.

The same kernel is the bounded-interval kernel of Lemma 6.135. Along each null generator of the diamond, Theorem 6.73 identifies the half-line generator constructed above with the local null-stress charge, and Lemma 6.135 transports that identification to the affine-covariant interval weight that vanishes at the two diamond endpoints. On the $t = 0$ slice that weight is exactly $(\ell^2 - r^2)/(2\ell)$. Therefore the null-stress bridge reconstructs the bulk modular charge of the geometric generator as

$$\delta\langle B_C\rangle = \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle\, d^3x + \frac{1}{2\pi}\, \delta\langle E^{(\eta)}_{C,\ell}\rangle.$$

Multiplying by the first-law factor 2π gives the displayed formula for $\delta S_{\mathrm{bulk}}(C)$.

If $\delta\langle T_{00}\rangle$ is approximately constant across B_ℓ, then

$$\int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle\, d^3x = \frac{4\pi\ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5 \partial T).$$

The additional hypothesis $\delta\langle E^{(\eta)}_{C,\ell}\rangle = o(\ell^4)$ then yields

$$\delta S_{\mathrm{bulk}}(C) = 2\pi \cdot \frac{4\pi\ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5\partial T) + o(\ell^4),$$

which is exactly the stated coefficient. No separate EFT small-ball first law has been used: the kernel comes from the geometric cap generator together with the D4 null-stress bridge. □

Theorem 6.102 (Finite-cutoff Einstein remainder bound)**.** *On the Lorentz/null-modular/Einstein branch, before taking the controlled collar and small-ball limits, the small-ball rest-frame relation has the form*

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle + \mathcal{E}_{\ell,\delta},$$

where, on one fixed faithful collar model and for a bounded support-visible observable class,

$$|\mathcal{E}_{\ell,\delta}| \le C_1 r_{\mathrm{FR}}(\varepsilon_\delta) + C_2 \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) + C_3 \eta^{\mathrm{reg}}_\delta + C_4 \ell\, \|\partial T\| + o_\delta(1) + o_\ell(1).$$

Here $\eta^{\mathrm{reg}}_\delta$ is the regularized support-visible modular transport remainder, and the constants depend only on the fixed collar model, the bounded observable class, and the local small-ball chart. If the controlled collar limit removes the first three terms and the small-ball limit removes the derivative and geometric remainders, the exact rest-frame relation of Theorem 6.106 follows.

Proof. Lemma 6.101 expresses the bulk entropy variation as the geometric cap-kernel stress term plus the carried modular/collar operator $E^{(\eta)}_{C,\ell}$ and the ordinary long-wavelength derivative remainder. The finite-stage modular-defect propagation theorem bounds expectations of $E^{(\eta)}_{C,\ell}$ by the Fawzi–Renner observable error, the fixed-collar Markov replacement modulus, and the support-visible regularized modular-transport remainder. Inserting that estimate into fixed-cap generalized-entropy stationarity and dividing by the small-ball coefficient gives the displayed $\mathcal{E}_{\ell,\delta}$. □

Proposition 6.103 (Conditional modular-remainder stress continuation)**.** *The finite-cutoff remainder in Theorem 6.102 is not, by itself, a dark-sector stress tensor. On a declared quotient-invariant collar-localization continuation branch, suppose the carried collar modular identity*

$$-\langle K_{ABD} - K_{AB} - K_{BD} + K_B\rangle = I(A : D|B)$$

is source-localized to the same small-ball operator $E^{(\delta)}_{B_\ell,r}$ that enters fixed-cap generalized-entropy stationarity, with residual ϵ_{loc}. Suppose also that the source is allocated to positive quotient-visible *causal packets and lifted covariantly with rest frame u^a. Then the continuation branch defines an effective repair stress whose rest-energy density is*

$$T_A^{ab} u_a u_b = \frac{15\hbar c}{8\pi^2 \ell^4} R_{r,\ell} + O(\epsilon_{\mathrm{loc}}) + \text{small-ball and regulator errors},$$

where $R_{r,\ell}$ is in nats and ℓ is the proper geodesic small-ball radius. The stress is separately conserved only on the zero first-moment repair branch; otherwise conservation is joint with the explicit recipient stress.

Remark 6.104 (Scalar-to-tensor boundary)**.** *The scalar collar quantity $I(A : D|B)$ cannot, alone, determine a symmetric rank-two stress tensor or a universal local energy expectation. The stress continuation also needs the packet factorization, causal packet directions, allocation rule, rest frame, source-localization map, and transport law. Until those receipts pass, finite collar CMI is a recoverability and diagnostic quantity instead of a physical dark/anomaly density source.*

Theorem 6.105 (Fixed-volume small-ball area variation)**.** *For a small geodesic ball B_ℓ centered at p with local rest-frame velocity u^a, the fixed-volume first variation of the boundary area is*

$$\delta A\big|_{V,\Lambda} = -\frac{4\pi\ell^4}{15}\,\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] + o(\ell^4).$$

Therefore the area part of the generalized entropy satisfies

$$\delta S_{\mathrm{area}} = \frac{\delta A}{4G} = -\frac{1}{4G}\frac{4\pi\ell^4}{15}\,\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] + o(\ell^4).$$

Proof. This is the standard fixed-volume small-geodesic-ball identity used in Jacobson-style entanglement-equilibrium derivations. In 3+1 dimensions the coefficient is $\Omega_2\ell^4/(4^2-1) = 4\pi\ell^4/15$. The sign is fixed by the fact that positive curvature decreases the area of a fixed-volume small ball. The metric-term ambiguity left by null data is represented by Λg_{ab}. □

Theorem 6.106 (Jacobson-type rest-frame relation from derived fixed-cap generalized-entropy stationarity)**.** *Assume Axioms 3.5–3.10, Assumption 3.15, and the hypotheses of Theorem 6.73, Lemma 6.101, Theorem 6.99, and Theorem 6.105. Then, for every sufficiently small cap centered at p and every admissible fixed-cap MaxEnt variation δ about a realized reference state ω in the locally Lorentzian $d = 4$ scaling regime, if u^a is the future-directed unit tangent of the local diamond rest frame at p, one has*

$$\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] = 8\pi G\, u^a u^b\, \delta\langle T_{ab}\rangle$$

at p. In the adapted rest frame $u^a = (1, 0, 0, 0)$, this is

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

Proof. Theorem 6.99 gives the fixed-cap generalized-entropy stationarity condition

$$0 = \delta S_{\mathrm{gen}}(C) = \delta S_{\mathrm{bulk}}(C) + \frac{\delta A}{4G}.$$

By Lemma 6.101,

$$\delta S_{\mathrm{bulk}}(C) = \frac{8\pi^2\ell^4}{15}\, u^a u^b\, \delta\langle T_{ab}\rangle + O(\ell^5 \partial T) + o(\ell^4),$$

where in the adapted rest frame the contraction is $\delta\langle T_{00}\rangle$.

By Theorem 6.105, the fixed-volume small-ball area variation gives

$$\delta A\big|_{V,\Lambda} = -\frac{4\pi\ell^4}{15}\,\delta\Big[(G_{ab}+\Lambda g_{ab})u^a u^b\Big].$$

Therefore

$$0 = \frac{8\pi^2\ell^4}{15}\,u^a u^b\,\delta\langle T_{ab}\rangle - \frac{\pi\ell^4}{15G}\,\delta\Big[(G_{ab}+\Lambda g_{ab})u^a u^b\Big] + O(\ell^5\partial T) + o(\ell^4).$$

Dividing by ℓ^4 and taking the small-ball limit removes the $O(\ell^5\partial T)+o(\ell^4)$ terms and yields

$$\delta\Big[(G_{ab}+\Lambda g_{ab})u^a u^b\Big] = 8\pi G\,u^a u^b\,\delta\langle T_{ab}\rangle.$$

This is precisely the desired rest-frame scalar relation. □

Theorem 6.107 (Timelike scalar data upgrade to tensor Einstein)**.** *If the first-variation relation of Theorem 6.106 holds for all local observer four-velocities and all reference states in a connected scaling branch, then*

$$G_{ab}+\Lambda g_{ab} = 8\pi G\,\langle T_{ab}\rangle$$

throughout that branch. The only local ambiguity is the metric term isolated by Lemma 6.74, and that ambiguity is exactly the same branch constant Λ*.*

Proof. Define

$$Y_{ab} := G_{ab}+\Lambda g_{ab} - 8\pi G\,\langle T_{ab}\rangle.$$

At any point p, Theorem 6.106 gives

$$u^a u^b\,\delta Y_{ab} = 0$$

for the unit future-directed four-velocity u^a of the local diamond rest frame. By the Lorentz branch of Theorem 6.13 and overlap consistency across observers through the same point, these local rest frames exhaust all timelike directions. Hence

$$u^a u^b\,\delta Y_{ab} = 0 \qquad \text{for every unit timelike } u^a.$$

Choose local inertial coordinates at p and write

$$u^a = \gamma(1, v^i), \qquad |\vec v| < 1, \qquad \gamma = (1-|\vec v|^2)^{-1/2}.$$

Then

$$0 = u^a u^b\,\delta Y_{ab} = \gamma^2\Big(\delta Y_{00} + 2v^i\delta Y_{0i} + v^i v^j\delta Y_{ij}\Big)$$

for every $|\vec v|<1$. The quadratic polynomial in v^i therefore vanishes on an open ball, so all of its coefficients vanish:

$$\delta Y_{00} = 0, \qquad \delta Y_{0i} = 0, \qquad \delta Y_{ij} = 0.$$

Thus

$$\delta Y_{ab} = 0$$

as a full tensor.

Along any path of admissible fixed-cap MaxEnt variations inside the connected scaling branch, Y_{ab} is therefore constant. The null bridge isolated the only purely local freedom as a metric term via Lemma 6.74; fixing that term on one maximally symmetric reference state determines the same branch constant Λ. Equivalently,

$$Y_{ab} = 0,$$

so

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$$

throughout the branch. □

Lemma 6.108 (Constancy of the metric-term residue)**.** *Let U be a connected local chart on the Lorentzian scaling branch. Suppose the pointwise null-data/Jacobson step gives*

$$F_{ab} = \kappa T_{ab} + \lambda(x) g_{ab}$$

on U, where F_{ab} is the geometric side, T_{ab} is the matter stress tensor, $\nabla^a F_{ab} = 0$, $\nabla^a T_{ab} = 0$, and $\nabla_c g_{ab} = 0$. Then $\lambda(x)$ is constant on U. Thus the local metric-term ambiguity is one branch constant Λ, not an arbitrary scalar field, once the Bianchi identity, local stress conservation, and chart connectivity are present.

Proof. Taking the divergence of the displayed equation gives

$$0 = \nabla^a F_{ab} = \kappa \nabla^a T_{ab} + \nabla^a(\lambda g_{ab}) = \nabla_b \lambda.$$

Hence the gradient of λ vanishes. On a connected chart, a scalar with zero gradient is constant. Without connectivity the same pointwise null-data argument can assign different constants on different components, so the connectivity hypothesis is load-bearing. □

Remark 6.109 (Formal algebra core of the Einstein branch)**.** *The OPHProofChain finite-algebra audit contains a sorry-free Lean formalization of the algebraic part of this subsection. The formalized core includes: the rest-frame arithmetic in Theorem 6.106; the upgrade from all future unit timelike directions to the full tensor equation in Theorem 6.107; the null-cone/Jacobson residue $F = \kappa T + \lambda g$; and Lemma 6.108 in a discrete chart model with an explicit connectivity hypothesis. This formalization does not derive the branch hypotheses themselves. Fixed-cap generalized-entropy stationarity, the internal small-ball bridge, the fixed-volume area-variation identity, bounded-interval transport, and the support-visible BW/geometric scaling branch remain exactly the named inputs listed in the dependency table. It therefore sharpens the status of the Einstein result: the algebra after those branch inputs is theorem-grade, while bare finite consensus alone is not Einstein-complete.*

6.5 Edge-entropy/area law identifies the Einstein/Newton coupling

The Jacobson-type chain of Theorems 6.99, 6.106, and 6.107 produces an Einstein equation $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$ on every connected locally Lorentzian $d = 4$ scaling branch. The next theorem identifies the coefficient G in that equation with the OPH edge-entropy/area-law ratio $a_{\text{cell}}/(4\bar{\ell}_{\text{shared}})$ of Definition 5.41, and tracks the same G through the standard linearization into the Newton coupling of the weak-field, slow-motion, classical-source limit. The identification is structural: the same G propagates literally through the algebra from the edge-entropy/area-law dictionary into the Einstein equation and into the Newton-Poisson limit, with no separate Newton coefficient introduced or matched against a measured constant at any step.

Theorem 6.110 (Edge-entropy/area law identifies the Einstein/Newton coupling)**.** *Assume Axioms 3.5–3.10, Assumption 3.15, the hypotheses of Theorem 6.13 (BW scaling on the prime geometric subnet), Theorem 6.73 (half-line generator/null-stress identification), Lemma 6.74 (null-data uniqueness modulo a metric term), Theorem 6.99 (derived fixed-cap generalized-entropy stationarity), Lemma 6.101 (internal $d = 4$ small-ball bridge), Theorem 6.106 (Jacobson-type rest-frame relation), and Theorem 6.107 (internal tensor upgrade). Work in the refinement-scaling regime* *of Definition 5.41, in which* $\mathrm{Tr}(\rho_C L_C) \approx N_\Sigma\, \bar{\ell}(t)$ *and* $A(\partial C) \approx N_\Sigma\, a_{\mathrm{cell}}$*. Define the OPH edge-entropy/area-law coupling*

$$G_{\mathrm{geom}} := \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

Then on every connected locally Lorentzian $d = 4$ *scaling branch:*

(i) Einstein branch. *The coefficient* G *appearing in*

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab} \rangle$$

of Theorem 6.107 is exactly G_{geom}*.*

(ii) Realized product-group local-pixel form. *If, in addition, the realized product-group hypotheses of Proposition 5.42 hold and the local pixel branch satisfies the D10 pixel law, so that* $\bar{\ell}_{\mathrm{shared}} = P/4$ *and* $a_{\mathrm{cell}} = P\, \ell_\star^2$ *(Proposition 5.43), then*

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}} = \ell_\star^2,$$

and the Einstein equation on that branch reads $G_{ab} + \Lambda g_{ab} = 8\pi\, \ell_\star^2\, \langle T_{ab} \rangle$*.*

(iii) Newton-Poisson weak-field limit. *On a static, weak-field background* $g_{ab} = \eta_{ab} + h_{ab}$ *with* $|h_{ab}| \ll 1$*, in the slow-motion (*$|v| \ll 1$ *in geometric units) classical-source reduction* $T_{00} \to \rho$*,* $|T_{ij}| \ll T_{00}$*, the linearization of (i) in the harmonic gauge gives*

$$\nabla^2 \Phi = 4\pi\, G_{\mathrm{geom}}\, \rho, \qquad \ddot{\mathbf{x}} = -\nabla \Phi,$$

where $\Phi := -\frac{1}{2} h_{00}$ *is the Newtonian potential and the geodesic equation in the same limit yields the displayed acceleration. The Newton coupling in this limit is the same* $G_{\mathrm{geom}} = a_{\mathrm{cell}}/(4\bar{\ell}_{\mathrm{shared}})$ *as in (i).*

In particular, the OPH edge-entropy/area-law ratio $a_{\mathrm{cell}}/(4\bar{\ell}_{\mathrm{shared}})$ *is the structural Einstein and Newton coupling on the realized branch.*

Proof. **(i)** The first variation of the generalized entropy on a fixed cap C is, by Proposition 5.40 together with the area-term identification of Definition 5.41,

$$\delta S_{\mathrm{gen}}(C) = \delta S_{\mathrm{bulk}}(C) + \delta\Big(\frac{A(\partial C)}{4G_{\mathrm{geom}}}\Big) = \delta S_{\mathrm{bulk}}(C) + \frac{\delta A}{4G_{\mathrm{geom}}}.$$

The coefficient $1/(4G_{\mathrm{geom}}) = \bar{\ell}(t)/a_{\mathrm{cell}}$ in front of δA is fixed by Definition 5.41: it is the refinement-scaling matching of $\mathrm{Tr}(\rho_C L_C)$ to the area term, and uses no input outside the OPH edge-center entropy operator and the cell-area scaling. Theorem 6.99 gives $\delta S_{\mathrm{gen}}(C) = 0$ on the admissible fixed-cap MaxEnt class, so the same coefficient enters the small-ball balance

$$0 = \delta S_{\mathrm{bulk}}(C) + \frac{\delta A}{4G_{\mathrm{geom}}}.$$

Lemma 6.101 expresses $\delta S_{\mathrm{bulk}}(C)$ in terms of $u^a u^b\, \delta \langle T_{ab} \rangle$ using only the geometric cap generator and the D4 null-stress bridge, with no extra small-ball first law and no measured constant. The standard fixed-volume area-variation identity for a small geodesic ball in $d = 4$ gives

$$\delta A\big|_{V,\Lambda} = -\frac{4\pi \ell^4}{15}\, \delta\Big[(G_{ab} + \Lambda g_{ab}) u^a u^b\Big],$$

and the proof of Theorem 6.106 divides through by $\ell^4/(15)$ to obtain the rest-frame scalar relation. The factor of G on the right-hand side of Theorem 6.106 is exactly the same G_{geom} introduced through Definition 5.41, because nothing else has been substituted between $\delta S_{\mathrm{gen}}(C) = 0$ and the rest-frame relation. Theorem 6.107 then promotes the rest-frame scalar relation to the tensor equation $G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle$ using the Lorentz branch of Theorem 6.13, the rest-frame coverage of timelike directions, and the null-data uniqueness of Lemma 6.74. Hence the coefficient G in Theorem 6.107 is G_{geom}.

(ii) On the realized product-group branch, Proposition 5.42 together with the D10 pixel law gives $\bar{\ell}_{\mathrm{shared}} = P/4$, and Proposition 5.43 gives $a_{\mathrm{cell}} = P\,\ell_\star^2$. Therefore

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}} = \frac{P\,\ell_\star^2}{4(P/4)} = \ell_\star^2.$$

Substituting into part (i) yields the displayed local-pixel Einstein equation.

(iii) Reduce (i) in the Newton-Poisson regime, using signature $(-,+,+,+)$ and geometric units $c = 1$. On laboratory and solar-system scales the curvature radius r of the Newtonian source is much smaller than the cosmological horizon $r_{\mathrm{dS}} = \sqrt{3/\Lambda}$ (Corollary 6.142), so all Λ-dependent corrections to Φ are bounded by $\Lambda\, r^2$ and are dropped at this scale; the Newton-Poisson reduction of (i) therefore coincides with the reduction of $G_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle$. Write $g_{ab} = \eta_{ab} + h_{ab}$ with $|h_{ab}| \ll 1$, $h := \eta^{ab} h_{ab}$, and $\bar{h}_{ab} := h_{ab} - \frac{1}{2}\eta_{ab} h$. In the harmonic gauge $\partial^a \bar{h}_{ab} = 0$ the linearization of $G_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle$ about flat space is the standard linearized-Einstein wave equation

$$\Box\, \bar{h}_{ab} = -16\pi G_{\mathrm{geom}}\, T_{ab}, \qquad \Box := \eta^{ab}\partial_a\partial_b = -\partial_t^2 + \nabla^2.$$

For a static, classical, slow-motion source with $T_{00} = \rho$ and $|T_{0i}|, |T_{ij}| \ll T_{00}$, time derivatives drop and the dominant component is the time-time component:

$$\nabla^2 \bar{h}_{00} = -16\pi G_{\mathrm{geom}}\, \rho.$$

On the static weak-field branch with the standard Newtonian-limit metric $g_{00} = -(1 + 2\Phi)$, $g_{ij} = (1 - 2\Phi)\delta_{ij}$, one has $h_{00} = -2\Phi$, $h_{ij} = -2\Phi\, \delta_{ij}$, and

$$h = \eta^{ab} h_{ab} = -h_{00} + \delta^{ij} h_{ij} = 2\Phi - 6\Phi = -4\Phi, \qquad \bar{h}_{00} = h_{00} - \tfrac{1}{2}\eta_{00} h = -2\Phi - \tfrac{1}{2}(-1)(-4\Phi) = -4\Phi.$$

Substituting $\bar{h}_{00} = -4\Phi$ into the displayed wave equation gives

$$\nabla^2(-4\Phi) = -16\pi G_{\mathrm{geom}}\, \rho \quad \Longleftrightarrow \quad \nabla^2 \Phi = 4\pi G_{\mathrm{geom}}\, \rho,$$

which is the Newton-Poisson equation with coupling G_{geom}. The geodesic equation $\ddot{x}^a + \Gamma^a_{bc}\dot{x}^b\dot{x}^c = 0$ in the slow-motion limit reduces to $\ddot{x}^i = -\Gamma^i_{00} = -\partial^i \Phi$, i.e. $\ddot{\mathbf{x}} = -\nabla\Phi$. All factors of G in this reduction are the same G_{geom} as in part (i): no measured Newton constant, no Planck area, and no measured cosmological constant has been substituted. ☐

Remark 6.111 (Dependencies and forbidden inputs for Theorem 6.110)**.** ***Dependencies.*** *The proof composes Axioms 3.5–3.10 (screen net, overlap consistency, MaxEnt, MAR, recoverable generalized entropy), Assumption 3.15, Theorem 6.13 (BW scaling theorem on the prime geometric subnet), Theorem 6.73 (half-line generator/null-stress identification), Lemma 6.74 (null-data uniqueness modulo a metric term), Proposition 5.40 (generalized-entropy split), Definition 5.41 (refinement-scaling area-term identification), Theorem 6.99 (derived fixed-cap generalized-entropy stationarity), Lemma 6.101 (internal $d = 4$ small-ball bridge), Theorem 6.106 (Jacobson-type rest-frame relation), and Theorem 6.107 (internal tensor upgrade). Part (ii) additionally invokes Proposition 5.42 (shared edge-entropy identity on the realized product-group branch with the D10 pixel law) and Proposition 5.43 (local scale-certificate package on the gravity row, supplying $a_{\mathrm{cell}} = P\,\ell_\star^2$*

from the no-G scale certificate of Theorem 3.3). Part (iii) uses only the standard linearization of part (i) on a flat background; it imports no further OPH structure beyond the validity of the small-ball/scaling regime that supports Theorems 6.106 and 6.107.

Forbidden inputs. *The proof of Theorem 6.110 does not use:*

- *the measured Newton constant G_N in any unit system;*
- *the measured Planck area $\hbar G_N/c^3$ or any other G_N-derived length scale;*
- *the measured cosmological constant $\Lambda_{\rm obs}$ (the cosmological constant appearing in (i) is the same branch constant isolated by Lemma 6.74 and fixed by the global capacity closure of Proposition 6.131, not a measured input);*
- *a gravity-calibrated clock scale, in particular the cesium frequency $\nu_{\rm Cs}$ tied to a chosen value of G_N or to a chosen Planck unit.*

The constant $G_{\rm geom} = a_{\rm cell}/(4\bar{\ell}_{\rm shared})$ is supplied entirely by the OPH edge-entropy/area-law data: $a_{\rm cell}$ and $\bar{\ell}_{\rm shared}$ come from Axioms 3.5–3.10, Proposition 5.42, and Proposition 5.43, with the cell-area scale $\ell_\star^2$ supplied by the separate no-G scale certificate of Theorem 3.3. The SI display $G_{\rm SI} = c^3\ell_\star^2/\hbar$ of Proposition 5.43 is a downstream unit-translation step and is not used in the proof of (i)-(iii).

Claim boundary. *Theorem 6.110 asserts only the identification of the coefficient G in (i)-(iii) with $G_{\rm geom}$. It does not extend to non-scaling regimes, to fixed-cap variations outside the admissible MaxEnt class of Definition 6.98, or to gravitational regimes that violate the small-ball/scaling hypotheses of Lemma 6.101. The finite-cutoff remainder of Theorem 6.102 controls the carried error before those limits are taken.*

Refinement-scaling dictionary scope. *The structural content of (i)-(iii) is derived: the same $G_{\rm geom}$ propagates literally through Theorems 6.99, 6.106, and 6.107 into the Einstein equation and through the standard linearization into the Newton-Poisson limit. Theorem 6.110 as stated takes the matching $\mathrm{Tr}(\rho_C L_C) = A(\partial C)/(4G_{\rm geom})$ of Definition 5.41 as a starting form, so the factor of 4 is inherited at the level of this theorem. Corollary 6.112 below tightens this by running the same Jacobson chain with only the linear-in-area scaling $\mathrm{Tr}(\rho_C L_C) \propto A(\partial C)$, itself derived inside OPH from the Markov-collar/refinement-stability structure of the screen net (Theorem 5.9, Proposition 5.40, together with the discrete-to-continuum scaling $\mathrm{Tr}(\rho_C L_C) \approx N_\Sigma \bar{\ell}(t)$ and $A(\partial C) \approx N_\Sigma a_{\rm cell}$ of Definition 5.41), and identifying the factor of 4 as the ratio of the resulting OPH structural Einstein-equation coefficient to the constant 8π entering the standard Newton-Poisson convention $\nabla^2\Phi = 4\pi G_N \rho$. The convention used by Corollary 6.112 is the standard naming of the Newton coupling.*

Corollary 6.112 (Factor of 4 in $G_{\rm geom}$ from the OPH structural Einstein-equation coefficient and the Newton-Poisson convention)**.** *Assume Axioms 3.5–3.10, Assumption 3.15, and the hypotheses of Theorems 6.13, 6.73, 6.99, Lemma 6.74, and Lemma 6.101. Take as the only refinement-scaling input the linear-in-area scaling*

$$\mathrm{Tr}(\rho_C L_C) \approx \frac{\bar{\ell}(t)}{a_{\rm cell}} A(\partial C),$$

which follows from $\mathrm{Tr}(\rho_C L_C) \approx N_\Sigma \bar{\ell}(t)$ and $A(\partial C) \approx N_\Sigma a_{\rm cell}$ (the two cell-decomposition identifications of Definition 5.41, with no $1/(4G_{\rm geom})$ form invoked). Then on every connected locally Lorentzian $d = 4$ scaling branch:

(a) OPH structural Einstein-equation coefficient. *The Jacobson chain produces*

$$G_{ab} + \Lambda g_{ab} = \kappa_{\mathrm{OPH}}\, \langle T_{ab} \rangle, \qquad \kappa_{\mathrm{OPH}} = \frac{2\pi\, a_{\mathrm{cell}}}{\bar{\ell}(t)},$$

where κ_{OPH} is determined by the structural OPH inputs alone, with no naming convention chosen for the gravitational coupling.

(b) Newton coupling from the Newton-Poisson convention. *Repeating the slow-motion, weak-field, classical-source linearization of (a) as in part (iii) of Theorem 6.110, with the general coefficient κ in place of $8\pi G_{\mathrm{geom}}$, yields*

$$\nabla^2 \Phi = \frac{\kappa_{\mathrm{OPH}}}{2}\, \rho.$$

Define the Newton coupling G_N by the standard physics convention $\nabla^2\Phi = 4\pi G_N\, \rho$. Then $\kappa_{\mathrm{OPH}} = 8\pi G_N$.

(c) Factor of 4. *Combining (a) and (b),*

$$G_N = \frac{\kappa_{\mathrm{OPH}}}{8\pi} = \frac{2\pi\, a_{\mathrm{cell}}/\bar{\ell}(t)}{8\pi} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)} = G_{\mathrm{geom}}.$$

Hence the factor of 4 in $G_{\mathrm{geom}} = a_{\mathrm{cell}}/(4\bar{\ell}(t))$ is the ratio of the OPH structural coefficient $\kappa_{\mathrm{OPH}} = 2\pi\, a_{\mathrm{cell}}/\bar{\ell}(t)$ to the Newton-Poisson constant 8π entering the convention $\nabla^2\Phi = 4\pi G_N \rho$. Equivalently, the dictionary identification $\mathrm{Tr}(\rho_C L_C) = A(\partial C)/(4G_{\mathrm{geom}})$ of Definition 5.41 is recoverable from the structural OPH inputs and the Newton-Poisson convention via (c).

Proof. **(a)** Theorem 6.99 and Proposition 5.40 give

$$0 = \delta S_{\mathrm{gen}}(C) = \delta S_{\mathrm{bulk}}(C) + \delta\big(\mathrm{Tr}(\rho_C L_C)\big) = \delta S_{\mathrm{bulk}}(C) + \frac{\bar{\ell}(t)}{a_{\mathrm{cell}}}\, \delta A,$$

where the last equality uses the linear-in-area scaling. By Lemma 6.101,

$$\delta S_{\mathrm{bulk}}(C) = \frac{8\pi^2 \ell^4}{15}\, u^a u^b\, \delta\langle T_{ab}\rangle + O(\ell^5 \partial T) + o(\ell^4),$$

and the standard fixed-volume area-variation identity for a small geodesic ball in $d = 4$ gives

$$\delta A\big|_{V,\Lambda} = -\frac{4\pi \ell^4}{15}\, \delta\big[(G_{ab} + \Lambda g_{ab}) u^a u^b\big].$$

Substituting,

$$0 = \frac{8\pi^2 \ell^4}{15}\, u^a u^b\, \delta\langle T_{ab}\rangle - \frac{\bar{\ell}(t)}{a_{\mathrm{cell}}}\, \frac{4\pi \ell^4}{15}\, \delta\big[(G_{ab} + \Lambda g_{ab}) u^a u^b\big] + O(\ell^5 \partial T) + o(\ell^4).$$

Dividing by ℓ^4 and taking the small-ball limit removes the remainder terms and yields

$$\delta\big[(G_{ab} + \Lambda g_{ab}) u^a u^b\big] = \frac{2\pi\, a_{\mathrm{cell}}}{\bar{\ell}(t)}\, u^a u^b\, \delta\langle T_{ab}\rangle.$$

The tensor upgrade of Theorem 6.107 (using the Lorentz branch of Theorem 6.13 and the null-data uniqueness of Lemma 6.74) promotes this rest-frame relation to $G_{ab} + \Lambda g_{ab} = (2\pi\, a_{\mathrm{cell}}/\bar{\ell}(t))\, \langle T_{ab}\rangle$, which is the displayed equation with $\kappa_{\rm OPH} = 2\pi\, a_{\rm cell}/\bar\ell(t)$. This step uses the structural linear-in-area scaling before a $1/(4G_{\rm geom})$ naming convention is introduced.

(b) Linearize $G_{ab} + \Lambda g_{ab} = \kappa\,\langle T_{ab}\rangle$ on a flat background as in part (iii) of Theorem 6.110, with the coefficient κ kept general. Signature $(-,+,+,+)$, geometric units $c = 1$. The harmonic-gauge linearized Einstein equation, with Λ-dependent corrections at scales $r \ll r_{\rm dS} = \sqrt{3/\Lambda}$ bounded by Λr^2 and dropped exactly as in part (iii), is

$$\Box\,\bar h_{ab} = -2\kappa\,T_{ab}.$$

For a static, classical, slow-motion source with $T_{00} = \rho$ and $|T_{0i}|, |T_{ij}| \ll T_{00}$, this reduces to

$$\nabla^2 \bar h_{00} = -2\kappa\,\rho.$$

The standard Newtonian-limit metric $g_{00} = -(1 + 2\Phi)$, $g_{ij} = (1 - 2\Phi)\delta_{ij}$ gives $\bar h_{00} = -4\Phi$, so

$$\nabla^2(-4\Phi) = -2\kappa\,\rho \quad\Longleftrightarrow\quad \nabla^2\Phi = \frac{\kappa}{2}\,\rho.$$

Comparing with the Newton-Poisson convention $\nabla^2\Phi = 4\pi G_N\,\rho$ gives $\kappa = 8\pi G_N$. Substituting $\kappa_{\rm OPH}$ from (a) yields $\kappa_{\rm OPH} = 8\pi G_N$, as displayed.

(c) Solve (b) for G_N and substitute (a):

$$G_N = \frac{\kappa_{\rm OPH}}{8\pi} = \frac{2\pi\, a_{\rm cell}/\bar\ell(t)}{8\pi} = \frac{a_{\rm cell}}{4\bar\ell(t)}.$$

This is exactly $G_{\rm geom}$ of Definition 5.41; the factor of 4 is the ratio of the OPH structural coefficient $\kappa_{\rm OPH}$ to the Newton-Poisson constant 8π. Equivalently, the dictionary form $\mathrm{Tr}(\rho_C L_C) = A(\partial C)/(4G_{\rm geom})$ of Definition 5.41 follows by substituting $G_{\rm geom} = a_{\rm cell}/(4\bar\ell(t))$ into the structural linear-in-area scaling $\mathrm{Tr}(\rho_C L_C) = (\bar\ell(t)/a_{\rm cell})\,A(\partial C)$. $\square$

Remark 6.113 (Forbidden inputs preserved by Corollary 6.112)**.** *The forbidden-input list of Remark 6.111 carries through Corollary 6.112 unchanged: the proof uses no measured Newton constant G_N (the symbol G_N appears only as the constant defined by the convention $\nabla^2\Phi = 4\pi G_N\rho$, with its OPH value derived in (c)), no measured Planck area, no measured cosmological constant, and no gravity-calibrated clock scale. The only convention imported beyond the OPH inputs is the standard physics naming of the Newton coupling via the Newton-Poisson equation.*

6.6 Einstein branch closure: local stress, entropy bridge, uniform small-diamond limit, and the composed branch-entry theorem

The Einstein package above consumes four inputs whose construction this subsection owns: (i) a local, symmetric, normalized, conserved stress tensor whose null charges are the modular generators, rather than an effective-field stress tensor named after the fact; (ii) a generalized-entropy first law with the edge/center term carried explicitly and a stationarity proof compatible with a *changing* stress channel; (iii) a single uniform scaling family replacing the per-radius diagonal choice of Lemma 6.136, and a base condition converting first-variation constancy into an absolute tensor equation; and (iv) the composition of the entire chain from repaired normal forms to the semiclassical Einstein equation into one typed theorem with a partitioned input list, no-hidden-geometry audit, and an explicit nonemptiness status for the realized branch. Throughout, every statement is classified as exact finite, scaling-limit, semiclassical expectation-value, conditional physical identification, or numerical/empirical.

6.6.1 The local conserved stress tensor from modular charges

Theorem 6.114 (Null tomography for symmetric tensors)**.** *Let T be a symmetric bilinear form on $(\mathbb{R}^4, \eta)$. The restriction of $v \mapsto T(v,v)$ to the null cone determines T up to a multiple of η: if $T(k,k) = T'(k,k)$ for every null k, then $T' - T = \phi\,\eta$ for some $\phi \in \mathbb{R}$. Conversely, a function $k \mapsto \tau(k)$ on the future null cone arises as $k \mapsto T(k,k)$ for a (then η-ambiguous) symmetric T if and only if it is homogeneous of degree two and satisfies the finite linear compatibility relations*

$$\sum_i c_i\, k_i \otimes k_i = 0 \implies \sum_i c_i\, \tau(k_i) = 0,$$

over all finite null families. Nine linearly independent null directions suffice to reconstruct the η-trace-free part of T by solving the corresponding linear system, and the reconstruction is Lipschitz in the charge data on any direction family with nonvanishing ninth singular value.

Proof. The symmetric square vectors $k \otimes k$, k null, span the η-orthogonal complement of nothing less than the full ten-dimensional symmetric space except for the relation $\eta^{ab}(k \otimes k)_{ab} = 0$: concretely, polarization of $T(k,k)$ over null $k = (1, \Omega)$, $\Omega \in S^2$, yields all components T_{00}, $T_{0i} + T_{i0}$, and the symmetric spatial moments $T_{ij}\Omega^i\Omega^j$ against the $\ell \le 2$ spherical harmonics, which determine T_{ij} up to its trace; the only undetermined combination is $T_{00} + \delta^{ij}T_{ij}$ paired against $\ell = 0$, i.e. the η-trace. A symmetric form vanishing on the whole null cone is therefore proportional to η (the standard null-invisibility statement, used in the tensor upgrade of Theorem 6.106). The converse and the nine-direction count follow because $\dim \operatorname{span}\{k \otimes k : k \text{ null}\} = 9$; Lipschitz stability is the finite-dimensional least-squares bound with constant $1/\sigma_9$ of the direction design matrix. $\square$

Definition 6.115 (Local charge domain and smeared null charges)**.** *Work on an event region of Theorem 6.90 carrying receipts* E1–E4*, the derived translations of Theorem 6.79, and the assembly branch of Theorem 6.84. The test space $\mathcal{D}$ consists of Lipschitz compactly supported functions on the region. For a null direction Ω and $f \in \mathcal{D}$ supported on a chart, define the smeared null charge as the weak refinement limit*

$$T[f;\Omega] \;:=\; \lim_r \sum_{I \in \mathcal{P}_r} f(a_I)\, \langle \widetilde{K}_I(\Omega) \rangle',$$

where $\mathcal{P}_r$ are the stage-r bounded null intervals of Theorem 6.82 along the Ω-generator through the chart, $\widetilde{K}_I$ their renormalized modular Hamiltonians, and $\langle\cdot\rangle'$ the endpoint derivative of Proposition 6.69 in the canonical normalization of Corollary 6.67. Existence of the limit on $\mathcal{D}$ with refinement-uniform errors is the kernel-residual receipt of Theorem 6.82 (scaling-limit statement).

Theorem 6.116 (Construction of the local conserved stress tensor)**.** *On the domain of Definition 6.115, with the* MI*/assembly branch and the collar-uniform recovery constants (the collar CMI interface of node D2), the family $\{T[f;\Omega]\}_{\Omega \in S^2}$ satisfies the compatibility relations of Theorem 6.114 for every f, and therefore defines a unique semiclassical expectation tensor field $\langle T_{ab} \rangle \in L^\infty_{\mathrm{loc}}$ modulo $\phi\, g_{ab}$, with:*

1. *(*Symmetry and locality.*) $\langle T_{ab} \rangle$ is symmetric by construction, and $T[f;\Omega]$ depends only on the record data in the causal support of f (support-local commutation of node D1 through the blow-up).*

2. *(*Covariance and common normalization.*) Under the produced Poincaré action, $T[f;\Omega]$ transforms with the null weight of Theorem 6.84(1); the linearity clause 6.84(2) is exactly the statement that the directional charges carry one common normalization, so the tomography system is consistent across directions rather than per-direction rescaled.*

3. *(*Generator identification.*) For every Ω, $P_\Omega = \int T_{kk}$ as quadratic forms on the common Stone core: this is Theorem 6.73 upgraded from named identification to construction, because the right-hand side is Definition 6.115 rather than an assumed effective density; endpoint/collar errors are uniform under refinement by the kernel-residual receipt.*

4. *(*Ward identity.*) In the weak (distributional) sense on the $C^{1,1}$ atlas, $\nabla^a \langle T_{ab}\rangle = 0$: for b a translation direction, invariance of the charges under the derived translations ($U(x)$-conjugation moves f by x and leaves the limit unchanged) gives vanishing coordinate divergence in each chart, and the* E4 *Poincaré transitions preserve this statement, so the covariant divergence vanishes almost everywhere.*

5. *(*Ambiguity classification.*) The full ambiguity is $\phi\, g_{ab}$ (null-invisible; fixed downstream by the base condition of Theorem 6.125) plus improvement terms $\partial^c\partial^d\chi_{[ca][db]}$ with χ supported like f, which do not change any charge on closed faces; no other ambiguity survives the tomography relations.*

Statement levels: (1), (2), (5) scaling-limit theorems on the named branch; (3) scaling-limit theorem given the kernel-residual receipt; (4) semiclassical expectation-value statement.

Proof. Compatibility: a dependent null family $\sum c_i k_i \otimes k_i = 0$ pulls back under the blow-up to a dependent family of translation generators; by Theorem 6.84(2), $\sum c_i P_{\Omega_i} = 0$, and the same relation holds for the endpoint-derivative densities after smearing because each $T[f;\Omega]$ is the f-weighted disintegration of P_Ω along its generator (Definition 6.115 and Lemma 6.72); the finite-stage defect is bounded by the MI residual ε_r and vanishes in the limit. Theorem 6.114 then yields the tensor with the stated ambiguity, measurably in the base point by the Lipschitz reconstruction bound applied chartwise. Symmetry is built into the tomography. Locality is support-local commutation transported through the blow-up: intervals outside the causal support of f contribute commuting differences whose endpoint derivatives vanish in the canonical normalization. Covariance is Theorem 6.84(1) applied to each interval charge, plus naturality of the partition family under the Möbius transport of Theorem 6.82. The generator identification integrates the disintegration against $f \to 1$ on the generator through the chart, with the monotone refinement errors controlled by the kernel-residual receipt; this is exactly the content of Theorem 6.73 with the density constructed. For the Ward identity, fix a chart and a coordinate direction x^μ; then $T[f(\cdot - x);\Omega] = \langle U(x)^*(\cdots)U(x)\rangle$-transported charges equal $T[f;\Omega]$ up to the state-variation term, which vanishes because Ω-invariance $U(x)\Omega = \Omega$ holds for the reference state and the semiclassical expectation is taken in that state; dividing by $|x|$ and letting $x \to 0$ gives $\partial^\mu T[\partial_\mu f;\Omega] = 0$, i.e. the weak coordinate conservation, for every Ω, hence for the reconstructed tensor. Chart transitions are Poincaré by E4, so the statement globalizes to the a.e. covariant divergence on the $C^{1,1}$ atlas. The ambiguity classification is Theorem 6.114 plus the standard improvement-term observation that double-divergence additions with antisymmetrized index blocks change no face charge. □

Definition 6.117 (Universal-coupling receipt)**.** UC*: the tensor of Theorem 6.116 (constructed from the* same *modular data as the entropy first law below) is the sole source entering the geometric branch: no record sector carries a second, differently normalized coupling to the produced metric. This is a physical-identification receipt, not a theorem; it is the equivalence-principle step, and it is falsifiable on any branch carrying a sector whose modular charges decouple from its entropy response.*

Proposition 6.118 (Countermodel: positive generators without a local stress tensor)**.** *There are directional charge families in which every P_Ω exists, is positive, and is translation-covariant* per direction*, but no symmetric tensor field reproduces them: any family violating one linear compatibility relation of Theorem 6.114 (equivalently, any branch without the* MI *linearity clause) admits no rank-two source, and the least-squares tomography residual is bounded below by the violation size times* σ_9^{-1}*. Machine receipts construct such a family explicitly and verify both the irreducible residual and the exact reconstruction of consistent families (*`code/geometry/`*). Scalar collar CMI data alone likewise admit no promotion to a rank-two source: a scalar function of cuts determines only the* $\ell = 0$ *moment of Theorem 6.114 and leaves the nine-dimensional trace-free part undetermined.*

Proof. Take any consistent family and add to one direction's charge a bump violating one dependent-family relation; Theorem 6.114 shows solvability fails, and the quantitative residual bound is the least-squares lower bound. The scalar statement is the rank count in the same theorem. □

6.6.2 The generalized-entropy first law with the edge term, and stationarity with a changing stress channel

Work on the central-interface branch of Axiom 3.7 (Theorem 5.15), where the collar algebra of a cap C has center generated by the boundary-charge functions, the reference state decomposes as $\rho_C = \bigoplus_\alpha p_\alpha \, \rho_\alpha$ over central sectors α, and the type-I generator form $K_C = 2\pi B_C + Z_C$ holds with Z_C central (the form used by Theorem 6.13 and the compact first-law bookkeeping). Define

$$S_{\text{bulk}}(C) := \sum_\alpha p_\alpha S(\rho_\alpha), \qquad S_{\text{edge}}(C) := H(\{p_\alpha\}) + \sum_\alpha p_\alpha \log d_\alpha,$$

with d_α the declared edge-sector dimension weights, so that $S(\rho_C) = S_{\text{bulk}} + H(p)$ and S_{edge} is the edge/center functional whose refinement limit supplies the area term in Definition 5.41.

Theorem 6.119 (Bulk/edge/central first law, exact finite form)**.** *For any differentiable variation* $\rho(\epsilon)$ *of the reference state with* $\operatorname{Tr} \delta\rho = 0$*, writing* $\delta X := \frac{d}{d\epsilon} X|_0$*:*

1. *(*First law.*)* $\delta S(\rho_C) = \langle K_C \, \delta\rho \rangle = 2\pi \, \delta\langle B_C \rangle + \delta\langle Z_C \rangle$*, exactly, at fixed operators* B_C, Z_C*.*

2. *(*Edge term identification and corrected bookkeeping.*) On variations preserving the central-interface class,* Z_C *acts sectorwise as* $(-\log p_\alpha + z_\alpha)\mathbf{1}$ *with* z_α *fixed by the declared normalization, and* $\delta\langle Z_C \rangle = \delta H(p) + \sum_\alpha z_\alpha \, \delta p_\alpha = \delta S_{\text{edge}}$ *exactly when* $z_\alpha = \log d_\alpha$*. Substituting into (1), the corrected bookkeeping is the exact split*

$$\boxed{\delta S(\rho_C) = 2\pi \, \delta\langle B_C \rangle + \delta S_{\text{edge}}}$$

   with no dropped term: total entropy variation equals modular (Clausius) flux plus edge/area response, and the same δS_{edge} *appears exactly once, as* $\delta A/(4G)$*, through Definition 5.41. The earlier step* $\delta S_{\text{bulk}} = 2\pi\delta\langle B_C \rangle$ *is* not *an identity: since* $\delta S = \delta S_{\text{bulk}} + \delta H(p)$*, one has* $\delta S_{\text{bulk}} = 2\pi\delta\langle B_C \rangle + \sum_\alpha z_\alpha \, \delta p_\alpha$*, so the naive form is valid exactly on the fixed-edge-sector-weight class (*$\delta p_\alpha = 0$*) and otherwise carries the explicit central flux* $\sum_\alpha z_\alpha \delta p_\alpha = \delta S_{\text{edge}} - \delta H(p)$*. Both the exact split and the quantitative naive-step defect are certified numerically in the repository receipts.*

3. *(*Type-I boundary.*) Item (2) is a finite/type-I statement; on the generic automorphism-level branch of Theorem 6.13 the algebraic replacement is the relative-entropy form* $\delta S(\rho \| \rho_{\text{ref}}) = 0$ *at first order together with sector-weight variation, and the identity holds for the declared central-sector weights without invoking a global density matrix. This is the precise algebraic substitute for the density-matrix step, stated at expectation-value level.*

Proof. (1) is $\delta S = -\operatorname{Tr}(\delta\rho \log \rho)$ (the $\operatorname{Tr}\delta\rho = 0$ term removes the normalization), followed by the split $-\log\rho = K_C = 2\pi B_C + Z_C$. (2) On the central-interface class, $\rho = \oplus_\alpha p_\alpha \rho_\alpha$ gives $-\log\rho = \oplus_\alpha(-\log p_\alpha + (-\log\rho_\alpha))$; the central part is $-\log p_\alpha + z_\alpha$ after moving the sectorwise constant z_α between Z and $2\pi B$ per the declared normalization. The central term pairs with $\delta\rho$ through the sector traces: $\delta\langle Z\rangle = \sum_\alpha(-\log p_\alpha + z_\alpha)\delta p_\alpha = \delta H(p) + \sum_\alpha z_\alpha \delta p_\alpha$ using $\sum_\alpha \delta p_\alpha = 0$; choosing $z_\alpha = \log d_\alpha$ matches δS_{edge} term by term, and substituting into (1) gives the boxed split. The naive-step statement follows from $S = S_{\text{bulk}} + H(p)$: $\delta S_{\text{bulk}} = \delta S - \delta H(p) = 2\pi\delta\langle B\rangle + \delta\langle Z\rangle - \delta H(p) = 2\pi\delta\langle B\rangle + \sum_\alpha z_\alpha \delta p_\alpha$. (3) restates (2) through Araki relative entropy: first-order stationarity of $S(\rho\|\rho_{\text{ref}})$ at $\rho = \rho_{\text{ref}}$ is the same identity with $\langle K\,\delta\rho\rangle$ read as the relative-modular pairing, and sector weights are declared data of the central-interface branch. ☐

Theorem 6.120 (MaxEnt stationarity with a changing stress channel)**.** *Let $\rho(t)$ be the MaxEnt family maximizing S subject to the modular stress constraint $\langle T_{ff}\rangle = t$ (the realized cap-label-preserving family of Theorem 6.99), with Lagrange multiplier $\lambda(t)$. Then, exactly and without assuming the constraint surface is fixed:*

$$\frac{dS}{dt} = \lambda(t),$$

and on the branch where the constraint is the certified modular charge with the 2π-KMS normalization, $\lambda = 2\pi$ at the reference point. Consequently the generalized-entropy stationarity of Theorem 6.99 extends to the coupled *variation class (fiber variations at fixed constraint value* plus *first-order changes of the constraint value along the MaxEnt family): fiber variations satisfy $\delta S \le 0$ with equality at the MaxEnt point, along-family variations satisfy $\delta S = \lambda\,\delta t$ exactly, and combining with the exact split of Theorem 6.119 yields the Clausius balance consumed downstream,*

$$\delta S_{\text{edge}} = \lambda\,\delta t - 2\pi\,\delta\langle B_C\rangle, \qquad \lambda = 2\pi,$$

for every coupled variation: the edge/area response equals the constraint-channel flux minus the bulk modular flux. This resolves the fixed-surface/changing-channel mismatch: the term missing from the fixed-constraint argument when the stress channel moves is exactly $\lambda\,\delta t$, supplied by the envelope identity rather than asserted. The variation class is nonempty and distinguishes state, constraint-value, and edge-sector directions; shape/null-cut and metric variations are not *covered by this theorem and retain their separate status.*

Proof. The MaxEnt state at constraint value t is $\rho(t) = e^{-\lambda(t)T_{ff}-\mu(t)}/Z$-type within the declared family; differentiating $S(\rho(t)) = \lambda t + \mu + \log Z$-form along the family and using the constraint derivative gives the exact envelope identity $dS/dt = \lambda$ (the standard Legendre/envelope computation, exact at finite dimension: $dS/dt = -\operatorname{Tr}(\dot\rho\log\rho) = \lambda\operatorname{Tr}(\dot\rho T_{ff}) + \mu\operatorname{Tr}\dot\rho = \lambda$). The KMS normalization identifies $\lambda = 2\pi$ at the reference point by Theorem 6.13. For the coupled class, decompose any first-order variation into a component along the MaxEnt family (constraint value moves by δt) and a component in the fixed-t fiber; the fiber component obeys $\delta S \le 0$ with equality on the family (MaxEnt optimality, first order zero), and the along-family component contributes $\delta S = \lambda\delta t$ by the envelope identity, with the constraint charge identified as the modular charge through Theorem 6.116(3). Substituting $\delta S = 2\pi\delta\langle B\rangle + \delta S_{\text{edge}}$ from the boxed split gives the displayed Clausius balance. Nonemptiness of the class is witnessed by the family itself and by the sector-weight variations of Theorem 6.119(2). ☐

Proposition 6.121 (Countermodels: MaxEnt, small CMI, or a central split alone do not give the gravitational law)**.** *(i) For any coefficient $\gamma \ne 1/(4G)$ in a trial functional $S_\gamma := S_{\text{out}} + \gamma A$,* *the coupled-class stationarity of Theorem 6.120 fails at first order by exactly* $(\gamma - \frac{1}{4G})\,\delta A \neq 0$ *on any variation moving the edge sector: the coefficient is fixed by the edge/center dictionary of Definition 5.41 and by nothing weaker. (ii) A blockwise state with exactly zero collar CMI and a central split, but edge weights* $z_\alpha \neq \log d_\alpha$*, breaks the edge identification: the defect* $\delta\langle Z\rangle - \delta S_{\mathrm{edge}} = \sum_\alpha (z_\alpha - \log d_\alpha)\delta p_\alpha$ *is nonzero for generic sector-weight variations, so the area dictionary misprices the edge response by a computable amount; on the declared normalization the naive step* $\delta S_{\mathrm{bulk}} = 2\pi\delta\langle B\rangle$ *fails off the fixed-weight class by exactly* $\sum_\alpha \log d_\alpha\,\delta p_\alpha$*. Machine receipts exhibit both defects numerically with certified magnitudes (`code/geometry/`). Hence the Axiom-4 leading-term input and the declared edge normalization are genuine branch content, consistent with the audit boundary that generalized entropy is an independent axiom on this corpus.*

Proof. (i) is immediate from the balance identity: replacing $1/(4G)$ by γ leaves the uncancelled multiple of δA. (ii) is the computation in Theorem 6.119(2) with the mismatched normalization, plus the naive-step defect formula proved there; the machine receipt instantiates both with explicit blockwise states. ☐

6.6.3 Uniform small-diamond limit and the absolute Einstein equation

Theorem 6.122 (One uniform scaling family)**.** *Let* (r, ℓ_r, δ_r)*, with* $\ell_r \downarrow 0$*, be one declared cofinal scaling family on which Theorem 5.3 holds uniformly over the event region and cap family:*

$$\varepsilon_r := c|\partial C_r|_{\mathrm{UV}} e^{-\delta_r/\xi_r}.$$

Assume that the fixed-collar replacements used on this family have uniform constants $C_{\mathrm{M}} < \infty$*,* $\theta > 0$ *with*

$$\delta_r^{\mathrm{M}} \leq C_{\mathrm{M}}\varepsilon_r^{\theta},$$

and that, for some

$$s > \max\{8, 4/\theta\}, \qquad \omega_r \longrightarrow +\infty,$$

the same family satisfies the rate receipt

$$\frac{\delta_r}{\xi_r} - \log(c|\partial C_r|_{\mathrm{UV}}) \geq s\log(\ell_0/\ell_r) + \omega_r.$$

Assume also the kernel-residual receipt $\epsilon_r^{\mathrm{ker}} = o(\ell_r^4)$ *of Theorem 6.82. Then, uniformly over the declared region and cap family,*

$$r_{\mathrm{FR}}(\varepsilon_r) = o(\ell_r^4), \qquad \delta_r^{\mathrm{M}} = o(\ell_r^4), \qquad \epsilon_r^{\mathrm{ker}} = o(\ell_r^4).$$

These estimates hold on one common domain of finite-rank test vectors. No per-radius diagonal choice is used.

Proof. The rate receipt gives

$$\varepsilon_r \leq (\ell_r/\ell_0)^s e^{-\omega_r}.$$

Proposition 5.33 therefore gives

$$r_{\mathrm{FR}}(\varepsilon_r) \leq 2\sqrt{\varepsilon_r} \leq 2(\ell_r/\ell_0)^{s/2}e^{-\omega_r/2} = o(\ell_r^4)$$

because $s/2 > 4$. The uniform fixed-collar modulus gives

$$\delta_r^{\mathrm{M}} \leq C_{\mathrm{M}}(\ell_r/\ell_0)^{s\theta}e^{-\theta\omega_r} = o(\ell_r^4)$$

because $s\theta > 4$. The kernel estimate is a premise. The domain statement holds because the finite-rank cores of Proposition 6.69 are nested along the declared family and their union is a common core by the mixed-GNS Cauchy clause. Uniformity is the region-uniformity of the hypotheses. ☐

Theorem 6.123 (Variation coverage)**.** *On a connected event region carrying* E1–E4*, the coupled variation class of Theorem 6.120, transported by the produced Lorentz action, covers all local timelike unit directions at all events of the region with realized reference states, and the covered set is connected. Consequently the polarization argument of the tensor upgrade (Theorem 6.106 via Lemma 6.137) applies eventwise on the region without importing an external all-observer postulate.*

Proof. E1 populates every chart box with events carrying frame data; Lemma 6.137 produces, from cap pairs, modular data whose rest-frame directions realize a neighborhood of any given timelike direction; the produced G-action (unitarily implemented on the producer branch) acts transitively on the fiber H^3 of directions (Proposition 6.42) and continuously in the region, so the orbit of the realized set is all of the direction bundle over the region; connectedness follows since G and the region are connected and E4 transitions are continuous. □

Definition 6.124 (Vacuum-reference receipt)**.** VR*: the region carries a realized MaxEnt reference state whose constructed stress tensor (Theorem 6.116) is maximally symmetric, $\langle T_{ab}\rangle_{\rm ref} = -\rho_{\rm vac}\, g_{ab}$, and whose declared normalization sets the first-variation integration tensor to zero at the reference point, $Y_{ab}[\omega_{\rm ref}] = 0$.* VR *is a physical base-condition receipt: it is* not *implied by $\delta Y_{ab} = 0$, and Proposition 6.126 shows it is irreducible.*

Theorem 6.125 (Absolute semiclassical Einstein equation on the closed branch)**.** *Assume the composed hypotheses: producer branch (Theorem 6.31), null-net packet (Theorems 6.77–6.84), event region with* E1–E4 *and the smooth scaling upgrade, constructed stress (Theorem 6.116) with* UC*, the repaired first law and coupled stationarity (Theorems 6.119, 6.120), the uniform family (Theorem 6.122), coverage (Theorem 6.123), and* VR*. Then on each connected component of the region, as a semiclassical expectation-value equation for the nonlinear fields,*

$$\boxed{G_{ab} + \Lambda\, g_{ab} = 8\pi G\, \langle T_{ab}\rangle}$$

with $G = a_{\rm cell}/(4\bar{\ell}_{\rm shared})$ the structural coupling of Definition 5.41 (forbidden-input audit unchanged: no measured G, Planck area, measured Λ, or gravity-calibrated clock enters), and Λ one constant per component, fixed relative to the VR *reference by $\Lambda = 8\pi G\rho_{\rm vac} + \Lambda_{\rm ref}$ with $\Lambda_{\rm ref}$ the reference-curvature constant; its global numerical closure is the separate D6 capacity gate. Without* VR*, the conclusion degrades exactly to: $G_{ab}+8\pi G\langle T_{ab}\rangle$-balance up to one undetermined covariantly constant tensor $Y_{ab} = c\, g_{ab}$ per component (constancy, not absoluteness).*

Proof. The chain Theorem 6.99 → Lemma 6.101 → Theorem 6.105 → Theorem 6.106 runs with every previously assumed input now supplied: the locally Lorentzian $d = 4$ regime and the geodesic-ball identity are consumed from the event packet on its smooth branch (the fixed-volume area expansion is a metric identity requiring only the constructed C^∞-branch metric and no field equation); the small-ball kernel is the constructed density of Theorem 6.116(3) through Lemma 6.101; the remainders are uniform by Theorem 6.122; the first-variation identity with the moving stress channel is Theorem 6.120; and coverage is Theorem 6.123. The output of the tensor upgrade is $\delta Y_{ab} = 0$ for $Y_{ab} := G_{ab} + \Lambda' g_{ab} - 8\pi G\langle T_{ab}\rangle$ along the covered class; coverage plus polarization make Y_{ab} direction-independent, the Ward identity of Theorem 6.116(4) and the contracted Bianchi identity (valid on the smooth branch) force $\nabla^a Y_{ab} = 0$, and Y proportional to g with $\nabla(Y - cg) = 0$ forces $Y_{ab} = c\, g_{ab}$ with c locally constant, hence constant per connected component. VR evaluates c at the reference state: $Y[\omega_{\rm ref}] = 0$ fixes c in terms of $\rho_{\rm vac}$ and the reference curvature, absorbing it into the displayed Λ. The coupling propagates literally from Definition 5.41 through the edge/area dictionary as in the existing structural-coupling theorem; no step introduces a measured constant. The degraded conclusion without VR is the same argument stopped before the evaluation step. □

Proposition 6.126 (Failure examples: uniformity, coverage, connectedness, baseline)**.** *Each hypothesis of Theorem 6.125 is load-bearing: (i) with per-radius stage choices in place of Theorem 6.122, the diagonal family can satisfy $\eta \le \ell^5$ stagewise while no single cofinal family does (choose stages with incompatible domains: the receipt fails and the limit order is undefined); (ii) with population confined as in Proposition 6.96(i), coverage fails and the polarization argument determines only the uu-component along one worldline; (iii) on a disconnected region, Λ genuinely differs between components: two receipt-carrying components with different reference curvatures satisfy all variational identities with different constants; (iv) without* VR*, de Sitter and flat reference branches satisfy the* same *first-variation identities with different $Y_{ab} = c\, g_{ab}$: $\delta Y_{ab} = 0$ never fixes c. Machine receipts instantiate (iv) and the tomography/coefficient countermodels numerically (`code/geometry/`).*

Proof. (i) is the observation in the theorem's proof that the union-core exists only along a declared family; an adversarial stage assignment with disjoint cores has no common domain. (ii) is Proposition 6.96(i) fed through Theorem 6.123. (iii) constancy is proved per component only; the example is two disjoint receipt regions. (iv) both backgrounds are maximally symmetric, so all covered variations produce identical $\delta Y = 0$ data while c differs by Λ_{dS}; this is the standard integration-constant freedom, here pinned only by VR. ☐

6.6.4 The composed branch-entry theorem and the realized-branch status

Theorem 6.127 (Composed Einstein branch-entry theorem)**.** *Let $(\mathfrak{B}_r)$ be a refinement tower of finite transactional OPH repair systems satisfying the node-D1 hypotheses and Axioms 3.5–3.10, with inputs partitioned as:*

- ***(Ax)** foundational axioms: the MaxEnt/recovery package with the central-interface clause; the generalized-entropy leading term and edge normalization $z_\alpha = \log d_\alpha$ (the Axiom-4 boundary);*
- ***(R_{fin})** decidable finite-stage receipts:* SphInc*, disk/mesh,* CR*,* KMS(2π)*;* Cyc*,* NTI*, weak additivity,* MI*;* E1–E4*;*
- ***(R_{scale})** scaling receipts: kernel residuals $o(\ell^4)$ on one declared family; uniform strong conditional Gibbs mixing constants, boundary counts, fixed-collar replacement modulus, and the sharp collar rate margin of Theorem 6.122; the $C^{1,1}$/smooth upgrade;*
- ***(R_{phys})** physical-identification receipts:* UC*,* VR*, the absolute conformal scale, stable causality/record-Cauchy clauses.*

Then the composition

$$D1 \to D3h \to D3b\text{–}f \to D4 \to D4b \to T_{ab} \to S_{\mathrm{gen}}\text{-}law \to uniform\ small\text{-}diamond \to \big(G_{ab}+\Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle\big)$$

holds with each arrow the cited theorem of this manuscript, on common domains with composable error envelopes, at semiclassical expectation-value level, with G structural and Λ per-component. Every target object is produced by an arrow; none is assumed except through the listed inputs. The branch is nonempty *exactly when some realized tower carries (R_{fin}) cofinally with (R_{scale}) moduli, a machine-checkable condition at every finite stage.*

Proof. Composition and bookkeeping: D1 is Theorem 4.8 and its package; D3h is Theorems 6.29–6.31; D3b–f are Theorem 6.13 with Corollaries 6.40, 6.46 (now consuming the produced certificate); D4 is Theorems 6.77–6.84; D4b is Theorems 6.90–6.95; the stress arrow is Theorem 6.116;

the entropy arrow is Theorems 6.119 and 6.120; the final arrows are Theorems 6.122, 6.123, and 6.125. Domains compose because each packet's output object is the next packet's declared input on the same tower, and every error envelope is stated on the common declared scaling family; the statement-level classification is inherited from the packets. Nonemptiness is by definition of the receipts, whose finite-stage clauses are decidable (Definitions 6.27, 6.28, 6.83, 6.89). ☐

Proposition 6.128 (No hidden geometry in the inputs)**.** *The input class (Ax)+(R_{fin}-form) hides none of the target objects: within the class of towers satisfying (Ax) and the D1 hypotheses, there are members realizing (i) non-spherical and non-manifold screen topologies and arbitrary modular normalizations (Theorem 6.32); (ii) event sets of dimension* 1*,* 5*, or non-Hausdorff type (Proposition 6.96); (iii) positive directional generators with no rank-two local source (Proposition 6.118); (iv) identical first-variation data with distinct absolute baselines (Proposition 6.126(iv)). Hence no target (S^2,* 3+1*, Lorentzian metric, local stress, area law, or Einstein relation) is a consequence of the input* form*; each enters exactly through the named receipt that excludes the corresponding countermodel, and the receipt list of Theorem 6.127 is thereby minimal in the sense that deleting any receipt family readmits its countermodel.*

Proof. Each item cites its countermodel theorem; minimality is the observation that each countermodel satisfies every input except the named receipt. ☐

Remark 6.129 (Realized-branch nonemptiness: machine status)**.** *The nonemptiness clause is empirical and machine-checkable. The repository evaluation (`code/geometry/realized_branch_receipts.py`) records the following state.*

(i) Topology and mesh. *The verified rooted-tree packet-net domain fails* SphInc *($\chi = 1$, no* 2*-simplices), as any tree must. A designed cellulated-sphere mesh passes* SphInc *but is declared geometry, not a consensus product. The cyclic cap-net repair run (`code/geometry/cyclic_cap_net_run.py`) is a genuine transactional repair tower with seeded conflicts and runtime-verified termination, conflict-free normal forms, and schedule independence across randomized schedules on a three-stage refinement tower of* 20/80/320 *records. Its repaired normal-form output passes* SphInc *at every stage with decreasing mesh modulus and simplicial refinement naturality. The overlap net of that run is chosen spherical, which is explicit branch selection in the sense of Corollary 6.33 and never a consensus output; the run certifies joint realizability of the node-D1 clauses with the* SphInc*/mesh/naturality families on one repair tower.*

(ii) Boundary-collar modular data. *The free-fermion modular-clock instrumentation (`code/geometry/modular_clock_instrumentation.py`) computes arc entanglement Hamiltonians $h = \log((1 - C_I)/C_I)$ at* 120*-digit precision on Gaussian MaxEnt collar states (critical hopping, anti-periodic twist, half filling, unit Fermi velocity) for collar rings $N_r = 16, 32, 64$; the arc spectrum reaches 10^{-30} of its endpoints, beyond double precision. The modular velocity profile, extracted by the odd-range chiral resummation $\beta(j) = -2\sum_{r\,\mathrm{odd}} r(-1)^{(r-1)/2} h_{i,i+r}$, matches the* 2π*-normalized conformal Bisognano–Wichmann profile with median interior residual 4.4×10^{-3}, 1.1×10^{-3}, and 1.1×10^{-4} at the three stages, and the $\times 1.2$ wrong-normalization control is separated by more than a factor five at every stage. The nearest-neighbour truncation of the same data misses the profile by roughly nine percent at every stage and serves as the wrong-extraction control. Modular transport cross-ratios $\exp(2\pi \sum_j 1/\beta_j)$ over fixed angular quadruples are Cauchy along refinement and match the Möbius cross-ratios with relative errors 1.1×10^{-2}, 2.3×10^{-3} at the two finer stages. The boundary-collar cross-ratio and* 2π*-KMS clauses are witnessed on this tower.*

(iii) Null-net standardness. *The null-net instrumentation (`code/geometry/null_net_receipts.py`) witnesses, on the same states: nontrivial relative commutants and weak additivity (combinato-*

rial, every stage); strictly positive separating moduli (occupation spectra strictly inside $(0,1)$*); mixed-GNS Cauchy behaviour of fixed smeared observables; the one-particle half-sided-inclusion compression condition, as a chiral-packet leakage asymmetry of at least* 9.6 *with one consistent direction across stages, where a real packet shows exactly zero asymmetry by time-reversal symmetry and serves as the null control; and the modular Lie-closure receipt, in which the one-particle commutator* $i[h_A, h_B]$ *matches the conformal-algebra prediction* $-\int(\beta_A\beta'_B - \beta'_A\beta_B)P$ *after even-range chiral resummation of the lattice momentum density, with shape residuals* 0.2% *and* 0.7% *against unresummed controls of* 3.6% *and* 6.0%*. The decay rate of that residual is uncertified and is an open clause.*

(iv) Events on the screen. *The realized-event instrumentation (`code/geometry/realized_event_receipts.py`) runs the event pipeline on records generated by the sequential transactional repair dynamics, with the clock in the certified* 2π*-normalized modular unit calibrated to the lattice propagation speed and chart noise calibrated to the measured cross-ratio residuals. Screen population with shrinking radii (*E1*, seeding scale), certified pairwise separation (*E2*), and the Möbius chart cocycle at noise level (*E4*) hold, and the intrinsic read-after-write ancestry order is separated by a quadratic cone of Lorentzian signature* $(1,2)$ *whose timelike eigenvector aligns with the modular clock to* 0.9995*. With screen-only records the event set is a* $(1{+}2)$*-dimensional sheet and is measured as such (chart PCA rank three): countermodel (i) of Proposition 6.96 operating as designed, since event dimension is a population output.*

(v) Bulk depth. *The bulk-depth instrumentation (`code/geometry/bulk_depth_receipts.py`) places records at the cells of all tower stages with the produced depth dictionary* $\rho = -\log\tan(\alpha/2)$*, the Möbius rapidity of the record's cap. Coarse cells are deep records and fine cells near-boundary records, the scale/radius duality realized by the refinement tower, and the hyperbolic metric equalizes lattice propagation speeds across stages (measured max/min ratio* 1.16*). Two dynamics are on file. Under the corpus-consistent inverse-system coupling (per-stage patches with one-step interface patches between consecutive stages, the node-D1 refinement-projection structure), the record set witnesses* E3 *rank four with positive observability, bulk chart PCA dimension four, and an intrinsic ancestry cone of Lorentzian signature* $(1,3)$ *at all five seeds, with the modular clock timelike to* 0.999–1.000 *and classification* 84–86%*. Under the strong-coupling variant (one global patch namespace, so a coarse commit writes every scale in one tick), the measured signature is* $(2,2)$*; depth degenerates to a second causal direction. That variant violates the inverse-system structure and is retained as a countermodel: the receipt detects pathological depth dynamics. Two caveats are explicit: the timelike cone margin is small (normalized eigenvalue* -0.028 *to* -0.009 *across seeds, stable in sign), and margin convergence with commit density and tower depth is uncertified work in progress.*

State of the gate. *Every structural receipt family of Theorem 6.127 carries a realized witness at finite or one-particle level: topology/mesh (with explicit branch selection), boundary-collar modular, null-net standardness, screen-event, and the bulk-depth* $(1,3)$ *cone. The open obligations are the cone-margin convergence study, the* Cyc *limit clause, the Lie-closure rate, second-quantized* MI*, cap-interior modular data, and the physical-identification receipts. The realized geometric branch is therefore not certified nonempty, and the composed theorem is conditional. Public summaries must state exactly this: derivation complete on the receipt branch; all structural receipt families realized at finite or one-particle level, including a seed-stable Lorentzian* $(1,3)$ *ancestry cone on the corpus-consistent multi-scale dynamics; margin-convergence, limit-clause, cap-interior, and physical-receipt work in progress.*

6.7 Cosmological-constant / screen-capacity closure

Lemma 6.130 (Vacuum-Energy Blindness)**.** *For any null vector k and any vacuum-energy contribution $T^{\mathrm{vac}}_{ab} = -\rho_{\mathrm{vac}} g_{ab}$,*

$$T^{\mathrm{vac}}_{kk} = 0.$$

Proposition 6.131 (Structural Separation of Λ)**.** *Local null-modular data fix T_{ab} only up to ϕg_{ab}. Therefore Λ is not determined by local overlap consistency alone and requires a global capacity closure; that zero-input closure is the cosmic record-closure readback fixed point of Definition 6.144.*

Corollary 6.132 (Cosmological capacity relation)**.** *If the cosmic record-capacity fixed point is identified with the de Sitter static-patch entropy on the observed branch,*

$$N_{\mathrm{CRC}} = S_{\mathrm{dS}},$$

and the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

then

$$\Lambda_{\mathrm{CRC}} = \frac{3\pi}{G N_{\mathrm{CRC}}}.$$

Equivalently, for the geometric scale $G_{\mathrm{geom}} = \ell_\star^2$, the capacity branch fixes the dimensionless product

$$\Lambda_{\mathrm{CRC}} \ell_\star^2 = \frac{3\pi}{N_{\mathrm{CRC}}}.$$

The SI value of Λ_{CRC} or of the scale product $\Lambda_{\mathrm{CRC}} N_{\mathrm{CRC}}$ requires the selected scale certificate.

Corollary 6.133 (Global screen-capacity closure of the Einstein branch)**.** *Assume the hypotheses of Theorem 6.107 and Corollary 6.132. Then on the same connected scaling branch,*

$$G_{ab} + \frac{3\pi}{G N_{\mathrm{CRC}}} g_{ab} = 8\pi G \langle T_{ab} \rangle.$$

Thus the local Einstein recovery closes globally at the cosmic record-capacity fixed point N_{CRC}.

Proof. Theorem 6.107 gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$$

on the connected scaling branch. Corollary 6.132 identifies

$$\Lambda_{\mathrm{CRC}} = \frac{3\pi}{G N_{\mathrm{CRC}}}$$

at the cosmic record-capacity fixed point. Substituting that value of Λ yields the stated equation. □

6.8 No-smuggling dependency discharge for the Einstein bridge

Theorem 6.134 (Einstein bridge dependency discharge (E0))**.** *Write OPH5 for Axioms 3.5–3.11: the S^2 screen net, overlap consistency, local MaxEnt/refinement, recoverable generalized entropy, and MAR. On any recovered-core, support-visible scaling branch of a cofinal OPH regulator family satisfying quotient confluence and the regularity hypotheses stated in Sections 2–5, OPH5 supplies the branch inputs used by the Einstein bridge:*

- *(i) the geometry readout factors through quotient normal forms via the screen fold $\chi_{S,r} \circ n_r \circ \pi_r$;*
- *(ii) round-cap pairs carry the Lorentz/H3 readout because $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ and the observer rest space is $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$;*
- *(iii) the support-visible BW theorem on the extracted prime geometric subnet fixes the 2π-normalized cap modular flow;*
- *(iv) the null modular bridge identifies the positive half-line translation generator with the local null-stress charge on the same branch;*
- *(v) the bounded-interval kernel is supplied by Lemma 6.135;*
- *(vi) fixed-cap generalized-entropy stationarity is the MaxEnt/refinement stationarity theorem for admissible cap-label-preserving variations;*
- *(vii) the small-ball area term is the standard fixed-volume geometric identity after the geometry readout has been discharged;*
- *(viii) the carried collar and derivative remainders are $o(\ell^4)$ by Lemma 6.136; and*
- *(ix) all timelike directions are covered by cap-pair geometry as in Lemma 6.137.*

Thus the finite consensus reduct supplies quotient normal forms, while OPH5 on the declared recovered-core, support-visible scaling branch supplies the geometric, modular, stress, entropy, and coverage inputs needed by the Einstein bridge. Bare finite consensus without those OPH5 branch closures does not imply the Einstein equation.

Proof. The quotient reduct supplies schedule-independent normal forms; the screen-net and support-visible extraction of Theorem 6.2 make the observer-readable cap data quotient-safe. The cap automorphism and KMS/BW normalization are Theorem 6.13, and the Lorentz/H3 statement is the standard conformal action of round cuts on S^2. Corollary 6.71, Lemma 6.72, and Theorem 6.73 supply the half-sided modular pair, positive null generator, and half-line generator/charge identity. Axioms 3.7 and 3.10 together with Theorem 6.99 supply the fixed-cap generalized-entropy stationarity statement on the realized cap-label-preserving MaxEnt family. Lemmas 6.135, 6.136, and 6.137 discharge the remaining downstream bridge inputs. The last sentence records the scope: the theorem is an OPH5 recovered-core branch theorem, not a claim about arbitrary terminating finite rewrite systems. □

Lemma 6.135 (Bounded-interval kernel from null projective covariance (E0.5))**.** *On the null blow-up of the support-visible geometric branch, assume the half-line generator/charge identity, endpoint-Lipschitz renormalized interval control, and affine/projective covariance of the interval-preserving null action. Then for a null interval (a,b) the local interval modular generator has the universal kernel*

$$K^{\mathrm{null}}_{(a,b)} = 2\pi \int_a^b \frac{(v-a)(b-v)}{b-a}\, T_{kk}(v)\, dv + R_{(a,b)},$$

where $R_{(a,b)}$ is the controlled endpoint/collar remainder carried by the same refinement family. In the small-ball limit this is the ball modular kernel used in Lemma 6.101.

Proof. The half-line bridge fixes the affine null translation generator as a local stress charge. Projective covariance transports this generator from half-lines to finite intervals and fixes the unique quadratic interval weight that vanishes at both endpoints and has the correct affine limit: $(v-a)(b-v)/(b-a)$. Endpoint-Lipschitz interval control prevents endpoint counterterms from changing the leading local charge; the remaining collar/endpoint contribution is exactly the carried $R_{(a,b)}$. □

Lemma 6.136 (Diagonal remainder lemma (E0.6))**.** *Let ℓ be the small-ball radius and $\eta_\delta \to 0$ the carried collar/Markov/recovery error at collar scale δ on the cofinal refinement branch. Assume one declared family $\delta = \delta(\ell)$ satisfies the quantitative rate receipt $\eta_{\delta(\ell)} = o(\ell^4)$, uniformly on the region and cap family. Consequently the interval endpoint, collar, and long-wavelength derivative remainders do not contribute to the Einstein coefficient in the $\ell \to 0$ bridge limit.*

Proof. The rate premise controls the carried collar contribution uniformly by $o(\ell^4)$. The ordinary small-ball derivative remainders are of higher order in ℓ, so their sum is $o(\ell^4)$. Mere pointwise convergence $\eta_\delta \to 0$ would permit a different stage for every radius and would not prove this statement; Theorem 6.122 supplies one sufficient common-family receipt. □

Lemma 6.137 (Cap-pair timelike coverage (E0.7))**.** *On the Lorentz/H3 branch of Theorem 6.13, overlapping observer cap pairs supply every local timelike direction needed for the scalar-to-tensor upgrade.*

Proof. By Proposition 6.42, every observer frame is a future unit timelike vector $u \in H^3$ with normalized sky

$$\mathcal{S}_u = \{u + s : s \in u^\perp,\ \eta(s,s) = 1\}.$$

Let v be any future timelike vector and put

$$r := \sqrt{-\eta(v,v)}, \qquad u := v/r \in H^3.$$

For any unit $s \in u^\perp$, define

$$q_\pm := u \pm s.$$

Then

$$\eta(q_\pm, q_\pm) = \eta(u,u) + \eta(s,s) = -1 + 1 = 0, \qquad q_\pm^0 > 0,$$

in a frame where $u = u_0$, and hence in every proper orthochronous frame. Moreover,

$$\boxed{v = \frac{r}{2}\,(q_+ + q_-).}$$

Thus normalized future null directions in observer skies cover every future timelike direction once the observer-frame chart is supplied. The cap-pair data select the corresponding local causal diamonds; the overlap atlas therefore provides the all-directions hypothesis used in Theorem 6.107. This argument does not say that two unnormalized projective rays alone select a unique observer frame; the observer normalization or equivalent causal-diamond scale data is also required. □

Proposition 6.138 (Recovered-core stress closure (E2))**.** *If the recovered stress tensor is reconstructed from the full compatible family of null and timelike modular charges on the recovered core, then it is covariantly conserved on that branch:*

$$\nabla^a \langle T_{ab} \rangle = 0.$$

Proof. The charge family is defined by overlap-compatible local modular generators. For an infinitesimal recovered-core diamond, opposite faces inherit the same overlap charge with opposite orientation, so the net first-order flux defect vanishes in the refinement limit. After the cap-pair tensor upgrade this local charge conservation is exactly the covariant divergence-free condition. □

Corollary 6.139 (OPH physical Einstein equation on the discharged branch (E1))**.** *Under Theorem 6.134, the recovered-core scaling branch satisfies*

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle$$

modulo the local null-data ambiguity in Λg_{ab}. With D6 cosmic record-capacity closure,

$$\Lambda_{\text{CRC}} \ell_\star^2 = \frac{3\pi}{N_{\text{CRC}}}, \qquad G_{\text{geom}} = \ell_\star^2.$$

Proof. Theorem 6.134 supplies the geometry readout, BW normalization, null-stress charge, bounded-interval kernel, fixed-cap stationarity, small-ball area term, remainder control, and timelike coverage. Theorems 6.106 and 6.107 then give the tensor Einstein equation modulo the expected metric ambiguity. Corollary 6.133 and the scale convention $G_{\text{geom}} = \ell_\star^2$ give the displayed D6 capacity relation. □

Theorem 6.140 (OPH Einstein-branch closure theorem)**.** *For a cofinal OPH5 regulator family on a recovered-core, support-visible scaling branch satisfying quotient confluence, support-visible branch regularity, and the hypotheses discharged in Theorem 6.134, the scaling-limit Einstein branch satisfies*

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$$

modulo the expected local null-data ambiguity in Λg_{ab}. With D6 capacity closure,

$$\Lambda = \frac{3\pi}{G N_{\text{CRC}}}.$$

Proof. Theorem 6.134 turns the OPH5 recovered-core branch into the exact hypothesis package required by the Jacobson-type bridge: quotient-safe geometry readout, 2π-normalized BW cap flow, local null stress charge, bounded-interval ball kernel with $o(\ell^4)$ remainder, fixed-cap generalized-entropy stationarity, the fixed-volume small-ball area identity, and all timelike directions. Lemma 6.101 gives the bulk entropy variation coefficient $8\pi^2\ell^4/15$, while Theorem 6.105 gives the area variation coefficient $-\pi\ell^4/(15G)$. Theorem 6.99 forces their sum to vanish, yielding Theorem 6.106. The scalar-to-tensor upgrade in Theorem 6.107 gives the tensor equation, and Corollary 6.133 fixes the remaining metric term on the D6 capacity branch. □

Remark 6.141 (Einstein-branch falsifiers)**.** *The OPH Einstein branch fails if quotient-normal-form confluence fails; geometry readout depends on hidden carrier coordinates, port labels, worker schedule, or repair schedule; support-visible BW modular flow does not become the 2π-normalized cap-preserving conformal flow; the null-stress bridge fails to identify the half-line generator with local stress charge; bounded-interval transport fails to give the ball kernel with $o(\ell^4)$ remainder; fixed-cap MaxEnt stationarity fails for the declared admissible variations; the small-ball area coefficient or area-law normalization fails; timelike-direction completeness fails; or D6 capacity closure is nonunique or target-leaking.*

Corollary 6.142 (De Sitter static-patch parameter display on the D6 branch)**.** *Assume the hypotheses of Corollary 6.132 and a supplied geometric scale $G_{\text{geom}} = \ell_\star^2$. Then the same cosmic record-capacity fixed point gives the coherent static-patch display*

$$S_{\text{dS}} = N_{\text{CRC}}, \qquad A_{\text{dS}} = 4GN_{\text{CRC}}, \qquad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\text{dS}}}{c}.$$

Together with Corollary 6.133, this packages the D6 branch as the global closure of the same Einstein branch plus the de Sitter entropy relation. The radius and timescale are displayed after the selected scale certificate is supplied.

Proof. Corollary 6.132 fixes $\Lambda G_{\text{geom}} = 3\pi/N_{\text{CRC}}$. Once G_{geom} is supplied by the selected scale certificate, the de Sitter entropy-area relation gives

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = N_{\text{CRC}},$$

and the standard static-patch formulas give the stated r_{dS} and t_Λ. □

Remark 6.143 (Capacity normalization)**.** *The entropy capacity and the bare horizon area ratio differ by the factor π. In Planck units,*

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P}\right)^2 = \frac{3}{\Lambda \ell_P^2}, \qquad N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

For $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$, this gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$ and $N_{\text{scr}} \simeq 3.31 \times 10^{122}$. The D6 branch uses the second quantity because N_{scr} is defined as the de Sitter entropy capacity.

Definition 6.144 (Cosmic record-closure capacity)**.** *For a candidate entropy capacity N, let $\mathfrak{U}_N$ be the OPH universe candidate with supplied active boundary capacity N, let* nf *be the quotient normal-form map on the declared terminating/confluent repair surface, let* Obs *select the stable self-reading observer sector, and let* Cap_{read} *return the capacity reconstructed by that sector. The capacity readback map is*

$$F(N) = \text{Cap}_{\text{read}}(\text{Obs}(\text{nf}(\mathfrak{U}_N))).$$

The cosmic record-closure capacity is the fixed point

$$N_{\text{CRC}} = F(N_{\text{CRC}}), \qquad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The screen-normalized count representation is as follows. Let Ω_N^{sc} denote the terminal OPH normal forms at capacity N that are closed under repair, support at least one stable observer/checkpoint subfederation, carry the recovered local package, and whose own horizon record surface reads back capacity N. Define

$$\Pi(N) := \frac{|\Omega_N^{\text{sc}}|}{\dim \mathcal{H}_{\partial,N}} = |\Omega_N^{\text{sc}}| e^{-N}, \qquad \log \dim \mathcal{H}_{\partial,N} = N.$$

This is a horizon record-capacity statement. It does not select the dimension of a primitive observer algebra. Combined with the pixel area relation $P_\star = a_{\text{cell}}/\ell_\star^2$, the capacity gives the equal-area chart count

$$K_{\text{cell}} = \frac{A_{\text{screen}}}{a_{\text{cell}}} = \frac{4N_{\text{CRC}}}{P_\star},$$

which is about 8.12×10^{122} *on the observed branch. The value* $P_\star/4 \simeq 0.408$ *nats is an edge/cut density, not* $\log \dim \mathcal{H}_{\rm cell}$ *for an autonomous cell factor. A fixed elementary observer carrier would require additional hypotheses about homogeneous cellulation, isomorphic cell algebras, equivariant overlaps, refinement preservation, and selection of a one-cell or fixed-block carrier. The corresponding input-free selector target is*

$$N_\star = \mathrm{MAR}\,\arg\max_N \left[\log|\Omega_N^{\rm sc}| - N\right].$$

Equivalently, with $\ell(N) = \log|\Omega_N^{\rm sc}| - N$*, the OPH-derived stationarity map*

$$T_\eta(N) = N + \eta\ell'(N)$$

has a unique stable fixed point under the derivative-sign certificate stated on the synthesis surface. Informally, $N_{\rm CRC}$ *is the single capacity where the universe reads back its own boundary without deficit or slack.*

Remark 6.145 (Capacity fixed-point reading)**.** *Definition 6.144 gives the precise D6 screen-capacity fixed point: the readback map* F *is defined on the OPH finite normal-form grammar, and the full screen dimension supplies the non-tunable normalization* e^{-N} *in the count representation. In differentiable form, the pressure certificate is the unique fixed-point condition* $T_\eta(N) = N$*, equivalently*

$$\frac{d}{dN}\left[\log|\Omega_N^{\rm sc}| - N\right] = 0.$$

On the observed branch the read-off is

$$N_{\rm scr} \simeq 3.31 \times 10^{122}.$$

The observed branch identifies this fixed point with the de Sitter entropy capacity used by Theorem 6.151.

Remark 6.146 (Observed-age benchmark boundary)**.** *The branch quantity* t_Λ *is the de Sitter timescale fixed by the same D6 closure. The observed cosmic age* t_0 *is a downstream FLRW comparison quantity outside the D6 theorem outputs, after one chooses a cosmological model above the local/global D5→D6 stack. On the flat* Λ*CDM benchmark,*

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_\Lambda}}\sinh^{-1}\left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\right).$$

Definition 6.147 (Clocked FLRW continuation boundary)**.** *A clocked FLRW continuation above the D6 gravity branch is a tuple*

$$(g_{ab}, \chi, u^a, h_{ab}, a(\tau), \kappa, T_{ab}, \mathsf{Top})$$

where $u_a = -N\nabla_a\chi$*,* $u^a u_a = -1$*,* $h_{ab} = g_{ab} + u_a u_b$*, the u-orthogonal slices are homogeneous and isotropic, and*

$$K(\tau) = \frac{\kappa}{a(\tau)^2}, \qquad {}^{(3)}R = 6K(\tau), \qquad \kappa \in \{-1, 0, +1\}.$$

The topology policy Top *records whether globally inequivalent flat quotients such as* $\mathbb{R}^3$ *and* T^3 *are distinguished or identified. The observer-facing* H^3 *record atlas used elsewhere is a Lorentz/rest-space chart; it is not by itself an FLRW curvature measurement.*

Lemma 6.148 (FLRW curvature as visible scalar holonomy)**.** *On a clocked homogeneous-isotropic spatial slice of Definition 6.147, let $\nabla^{(3)}$ be the spatial Levi–Civita connection. The small-loop holonomy in the u-v plane satisfies*

$$\mathrm{Hol}_{\partial\square_{uv}}(\nabla^{(3)}) = \exp\Big(KA_\square J_{uv} + O(A_\square^{3/2})\Big)\,,$$

where $A_\square$ is the loop area and J_{uv} is the infinitesimal rotation generator in that two-plane. On a visibly separated OPH refinement system, the refinement-limit scalar spatial holonomy vanishes if and only if $K = 0$. Thus the flat FLRW branch is the zero-visible spatial holonomy branch.

Proof. The displayed formula is the standard infinitesimal holonomy expansion for a connection with constant sectional curvature. If $K = 0$, every area-normalized small-loop curvature readout vanishes. Conversely, if $K \neq 0$, the area-normalized holonomy converges to KJ_{uv} in each visible two-plane. Visible separation forbids quotienting this nonzero family away as a mere gauge representative difference. Homogeneity and isotropy leave no independent scalar spatial-curvature obstruction beyond K, so vanishing refinement-limit scalar holonomy forces $K = 0$. □

Remark 6.149 (Flatness is not a D6 output)**.** *Lemma 6.148 only names the OPH meaning of an FLRW flat branch. A selection statement needs an additional cosmological continuation hypothesis or theorem. Capacity closure at D6 does not select $\kappa = 0$: de Sitter admits $\kappa = 0, +1, -1$ FLRW presentations with the same local expansion relation $H^2 + \kappa/a^2$. Curvature damping likewise gives only $\dot{K} = -2HK$ and hence bounds $|K|$ or $|\Omega_K|$ along a fixed sector; it does not change κ. The allowed branch labels are* DirectTheorem*,* ConditionalCMH*,* ExplicitAssumption*, and* OpenTheorem*, with* OpenTheorem *as the default.*

Theorem 6.150 (Conditional cosmological minimal holonomy)**.** *Fix a clocked cosmological boundary datum $B_{\cos,r}$ containing the clock, congruence, source and stress packets, screen checkpoints, orientation data, and topology policy, but not κ when flatness is to be derived. Let $\widetilde{C}_{b,r}^{\mathrm{FLRW}}$ be the raw FLRW fiber over that boundary. Suppose a refinement-natural functional J_{curv} on the physical quotient measures only quotient-visible spatial Levi–Civita curvature obstruction and satisfies $J_{\mathrm{curv}} = 0$ exactly on $\kappa = 0$. If a flat extension exists and the zero set is a single physical class after the topology policy, then the selected fiber*

$$C_{b,r}^{\mathrm{sel}} = \underset{C\in\widetilde{C}_{b,r}^{\mathrm{FLRW}}}{\operatorname{argmin}}\; J_{\mathrm{curv}}(C)$$

has $\kappa = 0$, uniquely modulo that topology policy.

Proof. The hypotheses put every candidate in one clocked boundary fiber, so J_{curv} compares extensions without changing source, stress, clock, or screen data. Since a flat extension exists, the infimum of J_{curv} is zero. The equivalence $J_{\mathrm{curv}} = 0 \iff \kappa = 0$ forces every minimizer into the flat sector, and the topology policy turns the zero set into one physical class. □

Theorem 6.151 (Cosmological-constant / screen-capacity closure stack)**.** *Assume the hypotheses of Theorem 6.107, the cosmic record-capacity fixed point*

$$N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}}),$$

its observed-branch de Sitter entropy readout

$$N_{\mathrm{CRC}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

the selected scale certificate $G_{\mathrm{geom}} = \ell_\star^2$, *and the standard de Sitter static-patch formulas*

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Then:

(i) *the local null-modular data determine the Einstein branch only modulo a metric term* Λg_{ab}*;*

(ii) *the same branch closes globally as*

$$G_{ab} + \frac{3\pi}{GN_{\mathrm{CRC}}}\, g_{ab} = 8\pi G \, \langle T_{ab} \rangle;$$

(iii) *the same D6 closure, together with the selected scale certificate, gives the static-patch display*

$$S_{\mathrm{dS}} = N_{\mathrm{CRC}}, \qquad A_{\mathrm{dS}} = 4GN_{\mathrm{CRC}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c};$$

(iv) *the observed cosmic age* t_0 *is not an additional theorem output of this stack and is a downstream FLRW benchmark.*

Thus the cosmological-constant package is one local/global theorem stack: the local null-modular branch leaves exactly the null-invisible metric freedom, the cosmic record-capacity fixed point closes that same Einstein branch globally, and the SI static-patch display uses the selected scale certificate.

Proof. Item (i) is Proposition 6.131, whose local ambiguity statement rests on the null-data blindness isolated by Lemma 6.130 together with the D5 Einstein branch. Item (ii) is Corollary 6.133. Item (iii) is Corollary 6.142. Item (iv) is Remark 6.146. □

Theorem 6.151 packages the local/global D5→D6 handoff. The D6 hypotheses are exactly the local Einstein branch, the cosmic record-capacity fixed point $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$, its observed-branch de Sitter entropy readout $N_{\mathrm{CRC}} = S_{\mathrm{dS}}$, the standard de Sitter entropy relation, the selected scale certificate, and the standard static-patch radius/time formulas; none of these are hidden inside the local null-modular reconstruction. The local theorem stack does not derive the global capacity from bare null data, and the two dimensionless closures do not derive the SI scale by themselves; the OPH-derived readback closure fixes the capacity, and the selected scale certificate supplies the SI display. The vacuum-energy problem is reorganized by separating local null data from the global capacity fixed point.

7 Gauge Reconstruction and Standard Model Structure

7.1 Compact gauge reconstruction

At any fixed UV cutoff, edge-center completion equips collars with finitely many sector labels, boundary charge carriers, and finite-dimensional intertwiner spaces. These are fixed-cutoff collar data, not refinement-limit assumptions. The local MaxEnt / collar-mixing package established earlier controls only fixed-cutoff recoverability, modular-support localization, and carried error terms on the realized branch. It is logically separate from whether transportable edge sectors survive refinement, and it supplies no theorem that the refinement-limit sector category is trivial or nontrivial.

On the ordinary or central-defect branch, path-independent movement of collar charges is supplied by the overlap-gluing theorem. Theorem 5.22 constructs transport from paths in the overlap recharting groupoid and proves the exact combined criterion: on the ordinary branch the represented loop holonomy must be trivial; on the central branch $[z]_\Sigma = 0$ must first strictify the triangle defect and at least one allowed strictification must have trivial sector holonomy. On the genuinely noncentral branch, $o_\Sigma^{(2)} = 0$ must first strictify the higher associator, after which at least one allowed strict G_Σ-valued representative must have trivial represented holonomy. A nonzero $o_\Sigma^{(2)}$ remains a higher-gauge sector, while an orbit for which every strict representative has residual loop holonomy is not an ordinary path-independent DR sector. The full orbit q_Σ need not determine a unique ordinary H^1 class and is not itself the strictification test.

For the ordinary or central-defect bosonic zero-obstruction branch, the fixed-cutoff categories $\mathsf{Sect}_r^{\mathrm{bos}}$ and their forgetful fibers are theorem-produced. The refinement/fiber ladder is constructed below only on a cofinal tail carrying the explicit compact-gauge refinement receipt of Definition 7.2. On such a tail, the resulting directed colimit

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r^{\mathrm{bos}}$$

is the category on which Doplicher–Roberts / Tannaka reconstruction is applied. Without that receipt this colimit is a conditional object, not an output of the finite-state refinement maps. The theorem below is neutral about whether Sect_∞ is trivial or nontrivial: if only the tensor unit persists, the reconstructed compact group is the trivial group.

7.1.1 Classification is not realization

Proposition 7.1 (Obstruction neutrality of the Standard Model selection step)**.** *The ordinary trivial-holonomy condition, the combined central condition $[z]_\Sigma = 0$ plus a trivial-holonomy strictification, and the combined noncentral condition $o_\Sigma^{(2)} = 0$ plus an allowed strict representative with trivial represented G_Σ-holonomy are transportability conditions. After one common stagewise strict representative is chosen, they permit an ordinary transportable bosonic sector category generated only by sectors trivial under that choice; they do not select*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

On a zero-obstruction branch carrying the compact-gauge refinement receipt, DR/Tannaka reconstruction returns

$$G = \mathrm{Aut}_\otimes(\mathcal{F})$$

for the constructed sector category and fiber functor. If the category is trivial, G is trivial; in general G is whatever compact group that tensor-fiber data reconstructs. The Standard Model quotient enters only after Axiom 3.11 is applied to a nonempty realized one-Higgs chiral sector package.

Proof. Theorem 5.22 identifies associator strictifiability together with vanishing residual represented loop holonomy with strict path-independent transport, and with no more structure than that. Theorems 5.26, 7.3, and 7.6 use that transport condition together with the explicit refinement receipt to build the ordinary bosonic refinement-limit sector category and its finite-dimensional fiber functor. Theorem 7.7 then reconstructs G from $(\mathsf{Sect}_\infty, \mathcal{F})$. None of those steps names a weak doublet, color triplet, Higgs doublet, hypercharge lattice, generation count, or finite $\mathbb{Z}_6$ kernel. Those data are supplied by the MAR-realized branch through Theorem 7.36. □

Definition 7.2 (Compact-gauge refinement receipt)**.** *Let R_0 be a cofinal tail of the separated refinement set. A compact-gauge refinement receipt on R_0 consists of the following data and checks for every $r \preceq s$ in R_0.*

1. Finite extendability. *The finite-state restriction $\rho_{sr} : Q_s \to Q_r$ is supplied with a surjectivity certificate and the restrictions compose. Thus every coarse record has at least one fine extension, and only on this certified branch does commutative pullback give an injective unital map*
$$\rho_{sr}^* : C(Q_r) \hookrightarrow C(Q_s), \qquad \rho_{sr}^* f = f \circ \rho_{sr}.$$
2. Noncommutative collar embeddings. *For each retained collar and its refinement, write*
$$\mathcal{A}_{\mathrm{EC},r}(B) = \bigoplus_{\alpha \in A_r} M_{d_{\alpha,r}}(\mathbb{C}), \qquad \mathcal{A}_{\mathrm{EC},s}(B_s) = \bigoplus_{\beta \in A_s} M_{d_{\beta,s}}(\mathbb{C}).$$
The receipt supplies nonnegative multiplicities $m^{rs}_{\beta\alpha}$ and unitaries
$$V^{rs}_\beta : \bigoplus_{\alpha \in A_r} \left(\mathbb{C}^{d_{\alpha,r}} \otimes \mathbb{C}^{m^{rs}_{\beta\alpha}}\right) \xrightarrow{\simeq} \mathbb{C}^{d_{\beta,s}}, \qquad d_{\beta,s} = \sum_\alpha m^{rs}_{\beta\alpha} d_{\alpha,r},$$
and defines, rather than infers,
$$\left(\jmath^B_{rs}(a)\right)_\beta = V^{rs}_\beta \left[\bigoplus_\alpha (a_\alpha \otimes \mathbf{1}_{m^{rs}_{\beta\alpha}})\right] (V^{rs}_\beta)^*.$$
Every coarse α must occur for some β, which is exactly the injectivity check. For the center-tracking branch used below, every β has a unique coarse shadow $\mathrm{sh}_{sr}(\beta)$, meaning $m^{rs}_{\beta\alpha} > 0$ for exactly one α. Then and only then for this block form does $\jmath^B_{rs}$ carry the whole coarse center into the fine center, and
$$\jmath^B_{rs}(P_{\alpha,r}) = \sum_{\beta : \mathrm{sh}_{sr}(\beta) = \alpha} P_{\beta,s}.$$
For $r \preceq s \preceq t$, the receipt verifies
$$m^{rt}_{\gamma\alpha} = \sum_\beta m^{st}_{\gamma\beta} m^{rs}_{\beta\alpha}$$
and coherence of the V's, equivalently $\jmath^B_{rt} = \jmath^{B_s}_{st} \circ \jmath^B_{rs}$ after the declared collar identifications.
3. Charge refinement. *The receipt supplies continuous surjective homomorphisms*
$$\pi^\Sigma_{sr} : \widehat{K}_{\Sigma,s} \twoheadrightarrow \widehat{K}_{\Sigma,r}, \qquad \pi^\Sigma_{rr} = \mathrm{id}, qquad \pi^\Sigma_{tr} = \pi^\Sigma_{sr} \circ \pi^\Sigma_{ts},$$
together with one common stagewise strict representative U^{str}_r at every $r \in R_0$. The maps intertwine overlap transport and these common choices, including the chosen central or crossed-module strictification where applicable and the represented action of every closed overlap loop; the representative is not changed object by object. In particular, closed fine loops with trivial coarse shadow must act trivially rather than create new residual holonomy. For every visible seed carrier $W_{\alpha,r}$, its pullback $(\pi^\Sigma_{sr})^* W_{\alpha,r}$ lies in $\mathsf{Sect}^{\mathrm{bos}}_s$, and its chosen refined collar realization has support $\jmath^B_{rs}(P_{\alpha,r})$. Fine central blocks under that support are realization blocks of the same pulled-back charge; they do not define a categorical direct sum with a larger carrier.

4. Finite tensor realization. *Every finite tensor word in visible seeds and conjugates, and every invariant subobject projection in its intertwiner algebra, has a compatible finite concatenated-collar realization. These realizations respect tensor product, duals, subobjects, overlap transport, and the preceding refinement maps. This is a family of finite realizations, one for each finite word; it does not place an infinite simple skeleton inside one finite collar algebra.*

The finite-set restrictions of Theorem 4.22 do not by themselves provide the matrix multiplicities, center condition, compact-group surjections, or tensor realizations in this receipt.

Theorem 7.3 (RefinementFunctorAndFiberDescent)**.** *Let R be the separated cofinal OPH refinement system used by Theorem 4.22, and suppose a cofinal tail $R_0 \subset R$ carries a compact-gauge refinement receipt in the sense of Definition 7.2. For each $r \in R_0$, let $\mathsf{Sect}_r^{\mathrm{bos}}$ be the fixed-cutoff zero-obstruction bosonic collar-sector category constructed in Theorem 5.26, with strict transport supplied by Theorem 5.22. Then for every $r \preceq s$ in R_0, pullback along π_{sr}^{Σ} defines a fully faithful symmetric strong monoidal *-functor*

$$U_{rs} : \mathsf{Sect}_r^{\mathrm{bos}} \longrightarrow \mathsf{Sect}_s^{\mathrm{bos}}$$

carrying $\mathbf{1}_r$ to $\mathbf{1}_s$, preserving irreducibles and zero-obstruction transportable sector classes, and satisfying coherent composition identifications

$$U_{st} \circ U_{rs} = U_{rt} \qquad (r \preceq s \preceq t).$$

The canonical fixed-cutoff forgetful functors

$$F_r : \mathsf{Sect}_r^{\mathrm{bos}} \longrightarrow \mathsf{Hilb}_{\mathrm{fd}}$$

are objectwise finite-dimensional and obey

$$F_s \circ U_{rs} = F_r$$

with identity monoidal comparisons satisfying the refinement-composition coherence equation.

Proof. Fix $r \preceq s$ in the certified tail. For a representation $X = (H_X, \varrho_X)$, define

$$U_{rs}X := (\pi_{sr}^{\Sigma})^* X = (H_X, \varrho_X \circ \pi_{sr}^{\Sigma}), \qquad U_{rs}(f) := f.$$

The generator-landing and finite-tensor clauses of the receipt put this pullback in $\mathsf{Sect}_s^{\mathrm{bos}}$. Since π_{sr}^{Σ} is surjective, a linear map between two pulled-back carriers intertwines the fine actions exactly when it intertwines the coarse actions. Hence

$$\mathrm{Hom}_{\widehat{K}_{\Sigma,s}}(U_{rs}X, U_{rs}Y) = \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X, Y).$$

Thus U_{rs} is fully faithful, *-preserving, and preserves irreducibility; in particular, a coarse simple does not split into several fine simples on the certified branch.

Pullback preserves the trivial representation, tensor products, conjugates, evaluation and coevaluation, and the canonical bosonic flip. With the standard underlying-Hilbert-space identifications these are symmetric strong monoidal identities. Exact composition of the π's gives $U_{st}U_{rs} = U_{rt}$ on objects and morphisms. Compatibility of the π's with overlap transport and obstruction data, required by the receipt, carries the common U_r^{str} to the common U_s^{str}: the central or crossed-module defect remains strictified and every represented fine-loop action remains trivial, including loops with trivial coarse shadow. No sector-specific change of strict representative is used. Thus zero obstruction persists. The separately supplied $\jmath^B_{rs}$ records where a seed is realized in the refined collar algebra; its central summands are not used to alter the carrier or to define U_{rs}.

Define $F_r(H_X, \varrho_X) = H_X$ and $F_r(f) = f$. This is the ordinary faithful symmetric strong monoidal forgetful functor, so every fiber is finite-dimensional objectwise even when the category has infinitely many simple classes. Pullback changes only the group action. Therefore

$$F_s U_{rs} X = H_X = F_r X, \qquad F_s U_{rs}(f) = f = F_r(f).$$

The comparison $\theta_{rs,X}$ is the identity on H_X. Naturality and monoidality are identity diagrams, and $\theta_{rt} = \theta_{rs} \circ (\theta_{st} U_{rs})$ is literal equality.

Finally, the algebra portion of the receipt is not deduced from the finite-state map. Surjectivity of ρ_{sr} proves only the injectivity of the commutative pullback ρ^*_{sr}. The displayed multiplicity formula directly verifies that $\jmath^B_{rs}$ is a unital *-homomorphism; occurrence of every coarse block proves injectivity, the unique-shadow condition proves the stated center formula, and multiplicity/coherent-unitary composition proves functorial composition. This discharges the non-commutative refinement obligation independently of the representation pullback construction. □

Remark 7.4 (The U(1) charge-one refinement test)**.** *Let $\widehat{K}_{\Sigma,r} = \mathrm{U}(1)$ and suppose the fixed collar sees the charge-one carrier $W_1(z) = z$, retained as a seed by the common stagewise representative U^{str}_r. The category of Definition 5.25 contains*

$$W_n = W_1^{\otimes n} \quad (n \geq 0), \qquad W_{-n} = (W_1^*)^{\otimes n} \quad (n > 0),$$

so it has the infinite simple family indexed by $\mathbb{Z}$. In particular charge two exists as $W_2 = W_1 \otimes W_1$. Its subobject idempotent, equivalently the unique nonzero central projection in its intertwiner algebra, is

$$\mathbf{1}_{W_2} \in \mathrm{End}_{\mathrm{U}(1)}(W_1^{\otimes 2}) \cong \mathbb{C};$$

in a certified physical realization this is the corresponding idempotent in the two-collar concatenated algebra. It is not asserted to be a projector in the original one-collar finite algebra $\mathcal{A}_{\mathrm{EC},r}(B)$. Under the next certified refinement,

$$U_{rs}(W_2) = (\pi^{\Sigma}_{sr})^* W_2 = \left((\pi^{\Sigma}_{sr})^* W_1\right)^{\otimes 2}$$

remains simple by surjectivity, its projector pulls back to the identity, and $F_s(U_{rs} W_2) = \mathbb{C} = F_r(W_2)$ with comparison map $\mathrm{id}_{\mathbb{C}}$. Thus tensor closure is explicit and does not hide an infinite center inside one finite algebra. A finite fusion or charge-truncated alternative would require a different stated category and an explicit associative fusion quotient; it is not used here.

Remark 7.5 (Conditionality of the refinement limit)**.** *Without Definition 7.2, the stagewise tensor-generated categories and their forgetful functors still exist, but the functors U_{rs}, their compatible fibers, the directed colimit, and $\mathrm{Aut}_{\otimes}(\mathcal{F})$ are not outputs of the finite-state consensus refinement maps. All refinement-limit gauge statements below are therefore conditional on a cofinal tail carrying that receipt.*

Theorem 7.6 (Construction of the refinement-stable bosonic sector category)**.** *Assume Axioms 3.5–3.10. On the ordinary or central-defect bosonic zero-obstruction branch, suppose a cofinal tail R_0 carries the compact-gauge refinement receipt of Definition 7.2. Let the fixed-cutoff sector categories $\mathsf{Sect}^{\mathrm{bos}}_r$ be those constructed in Theorem 5.26, and let the refinement functors and stagewise finite-dimensional bosonic fibers be those constructed in Theorem 7.3. Then the directed colimit*

$$\mathsf{Sect}_\infty := \varinjlim_{r \in R_0} \mathsf{Sect}^{\mathrm{bos}}_r$$

inherits a well-defined semisimple rigid symmetric C^*-tensor structure. Its tensor unit is the persistent vacuum sector, its tensor product is induced from collar concatenation, its duals are induced from orientation reversal / charge conjugation, and its braiding is the induced bosonic spacelike-exchange symmetry of the $3+1$*-dimensional branch. The finite-dimensional stagewise fibers descend to a faithful bosonic fiber functor*

$$\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}.$$

The theorem is neutral about nontriviality: if only the tensor unit persists, then Sect_∞ *is the trivial bosonic tensor category.*

Proof. The fixed-cutoff categorical structure is supplied by Theorem 5.26. The fully faithful symmetric monoidal *-refinement functors, their composition coherence, their preservation of zero-obstruction transportable sector classes on the certified cofinal tail, and the compatible finite-dimensional forgetful fibers are constructed from the receipt in Theorem 7.3.

The directed colimit category has objects represented by refinement tails that agree from some stage onward and morphisms represented by eventual intertwiner classes. Faithfulness of the U_{rs} makes the morphism equivalence relation separated: a nonzero finite-stage intertwiner cannot become zero on a cofinal tail. The tensor unit on each fixed-cutoff category is carried to the tensor unit by every U_{rs}, so the vacuum tail defines $\mathbf{1}_\infty$.

For objects represented by tails X_r and Y_r, define

$$[X] \otimes [Y] := [X_r \otimes_r Y_r]$$

at any sufficiently fine common stage. This is independent of the chosen stage because the monoidal structure maps of U_{rs} identify

$$U_{rs}(X_r \otimes_r Y_r) \cong U_{rs}(X_r) \otimes_s U_{rs}(Y_r).$$

The same argument descends the associators, unitors, duality evaluation and coevaluation maps, and the bosonic symmetry. Since each fixed-cutoff category is rigid, symmetric, semisimple, and C^*, and since the refinement functors preserve *, direct sums, and subobjects, the colimit inherits the same structure on persistent tails.

For the fiber functor, choose a representative X_r of a colimit object $[X]$ and define

$$\mathcal{F}([X]) := F_r(X_r)$$

using the literal carrier identifications

$$F_s(U_{rs}X_r) = F_r(X_r)$$

from Theorem 7.3. The composition compatibility of those identities makes this independent of representative. On morphisms, $\mathcal{F}$ is induced by the eventual action of the corresponding finite-stage F_r. Objectwise finite dimensionality, monoidality, and faithfulness descend from the stagewise F_r's. Thus $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$ is a faithful bosonic fiber functor. □

Theorem 7.7 (Compact gauge reconstruction in the bosonic branch)**.** *Assume Axioms 3.5–3.10 and work on the ordinary or central-defect bosonic zero-obstruction branch. Suppose a cofinal tail carries the compact-gauge refinement receipt of Definition 7.2, and let*

$$\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$$

be the faithful bosonic fiber functor constructed in Theorem 7.6 from Theorems 5.22, 5.26, and 7.3. Let $\mathrm{Aut}_\otimes(\mathcal{F})$ *denote the group of unitary symmetric monoidal natural automorphisms of* $\mathcal{F}$*. Then*

$$G := \mathrm{Aut}_\otimes(\mathcal{F})$$

is a compact group and

$$\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$$

as a symmetric C^**-tensor category with fiber functor. In particular* G *is uniquely determined up to isomorphism by the constructed pair* $(\mathsf{Sect}_\infty, \mathcal{F})$*.*

Proof. Theorem 5.22 supplies the strict zero-obstruction transport criterion. Theorem 5.26 constructs the fixed-cutoff bosonic symmetric C^*-tensor categories. Given the explicit compact-gauge refinement receipt, Theorem 7.3 constructs the fully faithful monoidal *-refinement functors and compatible finite-dimensional forgetful fibers, and Theorem 7.6 descends them to $(\mathsf{Sect}_\infty, \mathcal{F})$.

Fix a small skeleton of Sect_∞. For each object X, a unitary symmetric monoidal natural automorphism $\eta \in \mathrm{Aut}_\otimes(\mathcal{F})$ has a unitary component $\eta_X \in U(\mathcal{F}(X))$. Hence

$$\mathrm{Aut}_\otimes(\mathcal{F}) \hookrightarrow \prod_X U(\mathcal{F}(X)), \qquad \eta \mapsto (\eta_X)_X.$$

Naturality imposes the closed relations

$$\mathcal{F}(f)\eta_X = \eta_Y \mathcal{F}(f) \qquad (f : X \to Y),$$

and monoidality imposes the closed relations

$$\eta_{X\otimes Y} = J_{X,Y}(\eta_X \otimes \eta_Y)J_{X,Y}^{-1}, \qquad \eta_{\mathbf{1}} = \mathrm{id}_{\mathbb{C}}.$$

Thus $G = \mathrm{Aut}_\otimes(\mathcal{F})$ is a closed subgroup of a product of compact unitary groups, and is compact.

Doplicher–Roberts / Tannaka reconstruction applies to the rigid symmetric C^*-tensor category with faithful finite-dimensional bosonic fiber functor constructed above, giving

$$\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G).$$

Uniqueness follows because the reconstructed compact group is the tensor automorphism group of the fiber functor. If Sect_∞ is trivial, then G is the trivial compact group; the realized nontrivial branch is supplied by Theorem 7.35, not hidden as a hypothesis in this reconstruction theorem. $\square$

Theorem 7.7 is a Tannakian-level statement: it produces the compact group G and the equivalence $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ from the constructed category and fiber functor, and nothing more. No observable net $O \mapsto \mathcal{A}(O)$, no realization of the persistent sectors as localized transportable endomorphisms, and no field algebra $\mathcal{F}_{\mathrm{net}}$ with fixed-point identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ is constructed or claimed in this paper. The Doplicher–Haag–Roberts/DR field-net level requires the additional net-realization hypotheses (observable net, localized endomorphisms, unitary transporters, locality/duality, finite statistics with conjugates, and sector completeness) stated as the DHR/DR realization hypotheses in the synthesis paper; any field-algebra language on this lane is conditional on that package. Vanishing collar/represented holonomy is the categorical transport input to that package, not a substitute for net-level DHR transportability.

If a fermionic sign object is present, the correct reconstruction statement is super-Tannakian instead of purely Tannakian. This paper does not prove the full fermionic/chiral extension; it works only in the bosonic internal-gauge branch once the fixed-cutoff sector packages and their monoidal refinement transport have been constructed from that receipt.

Theorem 7.8 (Gauge-sector classification–selection factorization). *On the OPH compact-gauge lane, conditional on a cofinal tail carrying the compact-gauge refinement receipt, the realized Standard Model claim factors as*

$$\text{overlap/gluing data} \longrightarrow \text{associator strictification test} \longrightarrow \text{strict-representative holonomy test} \longrightarrow \mathsf{Sect}_\infty \longrightarrow G = \mathrm{Aut}_\otimes(\mathcal{F}) \longrightarrow \mathfrak{S}_{\mathrm{MAR}}.$$

The first arrow computes the central or crossed-module associator defect and strictifies it when possible. The second computes represented holonomy across the allowed strict representatives. The third keeps only sectors for which at least one such representative has trivial loop action. The fourth reconstructs the compact group from the persistent tensor category and fiber functor. These are classification and reconstruction steps. The final arrow is the realization/selection step: MAR acts on realized admissible sector packages, not on obstruction classes alone.

Proof. The first three arrows are Theorem 5.22: nonzero $[z]_\Sigma$ or nonzero $o^{(2)}_\Sigma$ blocks strictification, while a strictifiable associator proceeds only if the orbit contains an allowed strict representative with trivial sector holonomy. The fourth arrow is Theorem 7.7. Proposition 7.1 shows that these arrows are neutral about the Standard Model quotient. The final arrow is Axiom 3.11 applied to the nonempty witness class of Theorem 7.35, with uniqueness summarized in Theorem 7.36. $\square$

7.2 MAR minimality and the Standard Model quotient

Throughout this subsection, χ_{cpl} means the dimension of the minimal coupled carrier supporting the required weak-type and color-type nonabelian charges on a common block. It is not the abstract minimal faithful representation dimension of the final gauge group.

Definition 7.9 (Weak-type and color-type nonabelian roles). *A weak-type nonabelian charge is a pseudoreal nonabelian doublet role. A color-type nonabelian charge is an intrinsically complex nonabelian role, meaning that its nonabelian factor itself acts by an irreducible representation not equivalent to its conjugate. Abelian twisting of a pseudoreal nonabelian irreducible representation does not count as a color-type role.*

The derivation of the Standard Model quotient separates into five steps:

1. existence of a compact reconstructed group;
2. identification of the minimal nonabelian sector content;
3. identification of the minimal coupled carrier;
4. uniqueness of the connected abelian factor once admitted on that carrier;
5. determination of product gauge structure up to finite quotient, then of the global finite quotient from the realized matter spectrum.

Lemma 7.10 (Minimal nonabelian sector content). *Any admissible low-energy sector must contain both a weak-type pseudoreal nonabelian charge role and an intrinsically complex color-type nonabelian charge role.*

Proof. Light chiral matter requires left-handed multiplets and right-handed singlets to carry inequivalent gauge data. A pseudoreal doublet structure is the minimal way to realize the weak sector without immediately producing vectorlike masses. A genuinely complex nonabelian representation is required to distinguish quarks from antiquarks. A single nonabelian simple factor cannot simultaneously supply both a minimal weak-type role and an intrinsically complex color-type role; abelian twisting does not count toward the color-type requirement. □

Lemma 7.11 (Connected 2D pseudoreal image classification)**.** *Let H be a connected compact group with a faithful irreducible 2-dimensional pseudoreal unitary representation. Then the connected derived subgroup of its image is conjugate to* SU(2)*. Equivalently, the nonabelian factor acting in that role is* SU(2) *up to finite central quotient.*

Proof sketch. Irreducibility places the image inside U(2) with scalar commutant. Its connected derived subgroup is therefore a connected compact subgroup of SU(2). The connected compact subgroups of SU(2) are either tori or SU(2) itself. The representation is nonabelian and pseudoreal, so the torus case is excluded. Hence the derived subgroup is conjugate to SU(2), with only a finite central kernel left invisible in the representation. □

Lemma 7.12 (Connected 3D intrinsically complex image classification)**.** *Let H be a connected compact group with a faithful irreducible intrinsically complex 3-dimensional unitary representation. Then the connected derived subgroup of its image is conjugate to* SU(3)*. Equivalently, the nonabelian factor acting in that role is* SU(3) *up to finite central quotient.*

Proof sketch. Because the representation is irreducible on $\mathbb{C}^3$, the semisimple part of the Lie algebra of the image acts irreducibly on $\mathbb{C}^3$. A product of two nontrivial simple factors would force a tensor-product decomposition of dimension at least 4, so only one simple factor can occur. Among compact simple Lie algebras, the only ones with nontrivial irreducible representations of dimension at most 3 are $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$. The 3-dimensional $\mathfrak{su}(2)$ representation is real, not intrinsically complex, whereas the fundamental $\mathfrak{su}(3)$ representation is intrinsically complex. Therefore the connected derived subgroup is conjugate to SU(3), again up to finite central kernel. □

Lemma 7.13 (Minimal coupled carrier under MAR)**.** *Within the connected positive-dimensional Lie admissible class used by MAR, and under the weak-type / intrinsically complex color-type criteria above, the minimal coupled carrier supporting both roles on a common block has the form*

$$V = \mathbb{C}^3 \otimes \mathbb{C}^2, \qquad \chi_{\mathrm{cpl}} = 6.$$

Proof. By Lemma 7.11, the minimal weak-type nonabelian role is the pseudoreal doublet of SU(2), hence has dimension 2. By Lemma 7.12, the minimal intrinsically complex color-type nonabelian role is the triplet of SU(3), hence has dimension 3.

By definition of χ_{cpl}, the weak-type and color-type roles must act nontrivially on one common irreducible nonabelian block. Disconnected direct-sum summands do not qualify. Commuting irreducible actions on such a common block therefore require dimension at least the product of the minimal weak and color dimensions:

$$\chi_{\mathrm{cpl}} \geq 2 \cdot 3 = 6.$$

Equality is achieved by the tensor-product carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$. The block-diagonal representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ is faithful of dimension 5, but it is not coupled and hence is not the MAR minimizer. The explicit color-type criterion excludes connected non-semisimple counterexamples of $U(2)$ type. □

Remark 7.14. *The dimension comparisons used in Lemma 7.13 apply to the positive-dimensional connected Lie image carrying the admissible nonabelian charges, equivalently to the identity component of the reconstructed group on that sector. Finite, disconnected, or connected non-semisimple counterexamples exist. They are outside the admissible class fixed by the explicit weak-type / color-type criteria.*

MAR supplies existence of a connected abelian factor; the next lemma proves only uniqueness and identification on the minimal coupled carrier.

Lemma 7.15 (Uniqueness of the connected abelian factor on the minimal coupled carrier)**.** *Inside* $\mathrm{U}(6)$*, the commutant of* $\mathrm{SU}(3) \times \mathrm{SU}(2)$ *acting on* $\mathbb{C}^3 \otimes \mathbb{C}^2$ *is exactly* $\mathrm{U}(1)$*.*

Proof. The $\mathrm{SU}(3)$ action is irreducible on the first tensor factor and trivial on the second. The $\mathrm{SU}(2)$ action is irreducible on the second tensor factor and trivial on the first. By Schur's lemma, any operator commuting with both actions is scalar on each irreducible factor. Hence the full commutant is a single $\mathrm{U}(1)$. ☐

Theorem 7.16 (Product Gauge Structure up to Finite Quotient)**.** *Assume Axiom 3.11, the hypotheses of Theorem 7.7 on the ordinary or central-defect realized low-energy branch, the weak-type / color-type MAR inputs above, and that the realized connected gauge image acts faithfully on the minimal coupled carrier. Then the realized connected gauge structure has the form*

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ*.*

Proof. Theorem 7.7 yields some compact group G. Lemma 7.10 identifies the minimal admissible sector content, Lemmas 7.11 and 7.12 identify the connected nonabelian factors realizing the minimal weak-type and color-type roles, and Lemma 7.13 fixes the minimal coupled carrier. The tensor-product structure of that carrier implies commuting weak and color actions, so the connected semisimple part of the realized image is locally $\mathrm{SU}(3) \times \mathrm{SU}(2)$. Axiom 3.11 supplies the existence of one connected abelian charge factor acting nontrivially on the coupled carrier, and Lemma 7.15 shows that any such connected abelian factor is necessarily a single $\mathrm{U}(1)$. A compact connected Lie group with Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ is a quotient of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ by a finite central subgroup. Faithfulness rules out an invisible extra torus, so the only ambiguity is the finite central quotient Γ. ☐

Theorem 7.17 (Hypercharge lattice on the realized matter package)**.** *Assume gauge group* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$*, one generation of chiral matter* (Q, u^c, d^c, L, e^c)*, one Higgs doublet* H*, and Yukawa terms*

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly cancellation and Yukawa invariance determine the hypercharge ratios up to an overall $\mathrm{U}(1)_Y$ *normalization:*

$$Y_L = -N_c Y_Q, \quad Y_H = N_c Y_Q, \quad Y_u = -(N_c+1)Y_Q, \quad Y_d = (N_c-1)Y_Q, \quad Y_e = 2N_c Y_Q.$$

Fixing the normalization by $Q = T_3 + Y$ *and* $Q(\nu_L) = 0$ *gives*

$$Y_Q = \frac{1}{2N_c}.$$

For $N_c = 3$ *one recovers the exact Standard Model lattice*

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_e = 1, \quad Y_H = \frac{1}{2}.$$

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \qquad Y_d = -Y_Q + Y_H, \qquad Y_e = -Y_L + Y_H.$$

The mixed anomalies yield

$$N_c Y_Q + Y_L = 0,$$

and

$$2N_c Y_Q + N_c Y_u + N_c Y_d + 2Y_L + Y_e = 0.$$

Solving the first anomaly gives

$$Y_L = -N_c Y_Q.$$

Substituting the Yukawa relations and $Y_L = -N_c Y_Q$ into the mixed gravitational anomaly then yields

$$Y_H = N_c Y_Q, \qquad Y_u = -(N_c + 1)Y_Q, \qquad Y_d = (N_c - 1)Y_Q, \qquad Y_e = 2N_c Y_Q.$$

With these relations in place, the $\mathrm{SU}(N_c)^2\mathrm{U}(1)$ and $\mathrm{U}(1)^3$ anomalies cancel identically. The normalization of Y_Q is fixed by the electric charge operator $Q = T_3 + Y$ together with $Q(\nu_L) = 0$, which implies

$$Y_L = -\frac{1}{2} = -N_c Y_Q, \qquad \text{hence} \qquad Y_Q = \frac{1}{2N_c}.$$

□

Proposition 7.18 (Borel–Weil carrier for the one-Higgs branch)**.** *On the realized one-Higgs electroweak branch, suppose the support-visible weak chart is locally*

$$C_{\mathrm{EW}} \cong \mathbb{CP}^1$$

with the $\mathrm{SU}(2)_L$ *action and Hopf-fiber* $\mathrm{U}(1)$ *weight normalized by the OPH hypercharge lattice. If the scalar carrier is required to be minimal, holomorphic, nontrivial, one-Higgs, and Yukawa-completable, then the canonical local carrier is*

$$H_{\mathrm{OPH}} = H^0(C_{\mathrm{EW}}, \mathcal{O}(1)) \cong \mathbb{C}^2.$$

With the OPH convention fixed by $Q(\phi^0) = T_3(\phi^0) + Y(\phi^0) = 0$ *on the lower component, this carrier has* $Y(H_{\mathrm{OPH}}) = +1/2$*. Projectivization and symmetry breaking are distinct. For*

$$\phi_0 = \frac{v}{\sqrt{2}}\begin{pmatrix}0\\1\end{pmatrix}, \qquad v \neq 0,$$

the projective ray has stabilizer

$$\mathrm{Stab}([\phi_0]) \cong \mathrm{U}(1)_{T_3} \times \mathrm{U}(1)_Y$$

on the lifted electroweak product, whereas the nonzero vacuum vector has stabilizer

$$\mathrm{Stab}(\phi_0) = \mathrm{U}(1)_Q, \qquad Q = T_3 + Y.$$

Both statements are understood modulo the finite central identifications fixed by the physical gauge quotient. Thus $\mathbb{P}(H_{\mathrm{OPH}}) \cong \mathbb{CP}^1$ *classifies carrier rays but cannot by itself identify the unbroken electromagnetic diagonal, because it forgets the scalar hypercharge phase.*

Proof. Borel–Weil gives $H^0(\mathbb{CP}^1, \mathcal{O}(n)) \cong \mathrm{Sym}^n(\mathbb{C}^2)^*$ for $n \geq 0$, up to the dual convention. The $n = 0$ section space is a singlet and cannot be the weak Higgs doublet. The first nontrivial holomorphic section carrier is $n = 1$, giving the two-complex-dimensional fundamental SU(2) representation. Sections are scalar 0-form fields valued in the internal line bundle, so the Lorentz spin-zero statement follows after the OPH Lorentz branch separates this internal weak chart from the external tangent representation. The Hopf weight fixes the abelian charge up to sign and normalization; Theorem 7.17, the $\mathbb{Z}_6$ quotient convention, and $Q(\phi^0) = 0$ select $Y = +1/2$.

It remains to distinguish the two stabilizers. Use the paper's integer normalization $q = 6Y$, so the Higgs has $q_H = 3$. On the cover $\widetilde{G}_{\mathrm{EW}} = \mathrm{SU}(2)_L \times \mathrm{U}(1)_q$, write the action as

$$(g, z)\phi = z^3 g\phi.$$

An element of SU(2) preserves the line $\mathbb{C}\phi_0$ exactly when it is

$$g = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}, \qquad a \in \mathrm{U}(1).$$

The scalar z^3 never changes the ray, so

$$\mathrm{Stab}_{\widetilde{G}_{\mathrm{EW}}}([\phi_0]) = \left\{ \left(\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}, z \right) : a, z \in \mathrm{U}(1) \right\} \cong \mathrm{U}(1)_{T_3} \times \mathrm{U}(1)_Y.$$

In particular, every pure hypercharge phase fixes $[\phi_0]$. Exact invariance of the vector imposes the additional equation $z^3 a^{-1} = 1$, hence

$$\mathrm{Stab}_{\widetilde{G}_{\mathrm{EW}}}(\phi_0) = \left\{ \left(\begin{pmatrix} z^3 & 0 \\ 0 & z^{-3} \end{pmatrix}, z \right) : z \in \mathrm{U}(1) \right\}.$$

Equivalently, in conventional angle variables,

$$e^{i\alpha T_3} e^{i\beta Y} \phi_0 = e^{i(\beta - \alpha)/2} \phi_0.$$

The ray is unchanged for independent α and β, while vector invariance ties them locally by $\beta = \alpha$, leaving the generator $Q = T_3 + Y$. Passing to the physical gauge group divides both stabilizers by the inherited finite central kernel and leaves the connected vector stabilizer $\mathrm{U}(1)_Q$. Higher $\mathcal{O}(n)$ give larger SU(2) multiplets and fail one-Higgs minimality. □

Remark 7.19 (Carrier only, not the Higgs mass)**.** *Proposition 7.18 identifies the representation, charge convention, and symmetry-breaking geometry of the OPH one-Higgs slot. It does not derive the weak scale, the Higgs quartic, m_H, or Coleman–Weinberg dynamics; those remain D10/D11 quantitative and hierarchy/naturality branch statements.*

Corollary 7.20 (Three Colors on the realized MAR branch)**.** *Under the hypotheses of Theorem 7.16, the realized color-type factor acts in the fundamental triplet on the coupled carrier. Hence the realized quark doublet carries exactly*

$$N_c = 3.$$

Proof. Lemma 7.12 identifies the minimal intrinsically complex color-type role in the connected Lie admissible class as the 3-dimensional fundamental of SU(3). Lemma 7.13 then shows that MAR realizes this role on the coupled block $\mathbb{C}^3 \otimes \mathbb{C}^2$. The color multiplicity of the realized weak doublet Q is therefore fixed by the same D8 carrier to be 3. No later admissibility selector is needed to promote oddness to the specific value 3. □

Corollary 7.21 (Three Generations on the realized MAR branch)**.** *Under the hypotheses of Theorem 7.16, with $N_c = 3$ from Corollary 7.20, the CP-capability and weak-sector UV clauses contained in Axiom 3.11 force*

$$N_g = 3.$$

Proof. On the realized one-Higgs chiral branch, intrinsic CKM CP capability requires

$$\frac{(N_g - 1)(N_g - 2)}{2} > 0,$$

hence $N_g \geq 3$. For one Higgs doublet, the one-loop weak-sector coefficient is

$$b_2 = \frac{11}{3} C_A - \frac{2}{3} \sum_f T(R_f) - \frac{1}{3} \sum_s T(R_s) = \frac{22}{3} - \frac{N_g(N_c + 1)}{3} - \frac{1}{6} = \frac{43}{6} - \frac{N_g(N_c + 1)}{3}.$$

Asymptotic freedom therefore implies

$$N_g(N_c + 1) < \frac{43}{2}.$$

With $N_c = 3$, this gives

$$4N_g < \frac{43}{2}, \qquad \text{hence} \qquad N_g \leq 5.$$

These are not extra post-MAR selectors: they are the explicit numerical consequences of the CP-capability and weak-sector UV clauses present in Axiom 3.11 on the same realized branch. Once the first three MAR components are fixed by the D8 color/weak structure, the fourth component N_g is minimized on the allowed set $\{3, 4, 5\}$, so $N_g = 3$. □

Corollary 7.22 (MAR-minimal SM package is unique up to physical equivalence)**.** *On the ordinary or central zero-obstruction bosonic branch, assume the MAR-admissible class is nonempty and contains the explicit realized one-Higgs chiral matter package used in Theorem 7.16 through Corollary 7.21. Then every MAR-minimal representative has the same observer-visible Standard Model package,*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

with the hypercharge lattice of Theorem 7.17. The only residual freedom is physical equivalence in the sense of Definition 3.12.

Proof. Proposition 3.13 supplies MAR minima. Lemmas 7.11–7.15 fix the minimal weak, color, coupled-carrier, and connected abelian roles at the least first three complexity entries. Corollary 7.20 fixes $N_c = 3$, Corollary 7.21 fixes $N_g = 3$, Theorem 7.17 fixes the charge lattice, and Proposition 7.24 fixes the finite kernel. Definition 3.12 removes only relabelings, gauge-center conventions, and inert implementation data. □

Remark 7.23 (Witten parity on the realized color branch)**.** *With $N_c = 3$, each generation contributes $N_c + 1 = 4$ left-handed $\mathrm{SU}(2)$ doublets, so Witten's global anomaly constraint is automatically satisfied generation by generation. In this theorem stack the anomaly is therefore a consistency check on the realized triplet-doublet package, not the step that creates the color count [37].*

Proposition 7.24 (Global quotient from the realized matter spectrum)**.** *If the realized matter spectrum carries the hypercharges of Theorem 7.17 with $N_c = 3$, then the subgroup of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ acting trivially on all realized states is exactly $\mathbb{Z}_6$.*

Proof. Write $q := 6Y \in \mathbb{Z}$ and $\eta := e^{i\pi/3}$, so the U(1) factor acts by η^q on charge-q multiplets. The element

$$g_0 := (\omega_3, -1, \eta)$$

acts trivially on the realized matter and Higgs multiplets:

$$Q: \ \omega_3(-1)\eta = 1, \qquad u^c: \ \omega_3^{-1}\eta^{-4} = 1, \qquad d^c: \ \omega_3^{-1}\eta^2 = 1,$$
$$L: \ (-1)\eta^{-3} = 1, \qquad e^c: \ \eta^6 = 1, \qquad H: \ (-1)\eta^3 = 1.$$

Its powers therefore generate a subgroup of order six acting trivially on all realized states. Conversely, any central element acting trivially on e^c must have U(1) part η^n, and triviality on Q then fixes the SU(3) and SU(2) center factors uniquely. Hence every trivial central element is a power of g_0, so the kernel is exactly $\mathbb{Z}_6$. □

Remark 7.25 (Formal algebra core of hypercharge and the Z_6 quotient)**.** *The algebraic part of Theorem 7.17 and Proposition 7.24 has an OPHProofChain finite Lean audit: Yukawa invariance together with the stated anomaly equations fixes the hypercharge ray, the neutrino neutrality convention fixes the normalized lattice $Y_Q = 1/(2N_c)$, and for $N_c = 3$ the subgroup acting trivially on the realized one-generation matter-plus-Higgs package is exactly the cyclic group generated by $(\omega_3, -1, e^{i\pi/3})$, hence is isomorphic to $\mathbb{Z}_6$. The formalization covers this algebra; it does not replace the MAR-admissibility and realized matter-package hypotheses that select the branch on which the algebra is applied.*

Remark 7.26 (Arithmetic toy model for finite quotient labels)**.** *DULA's congruence-grading construction gives a simple number-theoretic analogue of this bookkeeping [6]. For integers coprime to 6, the number of prime factors congruent to 5 modulo 6, counted with multiplicity and reduced modulo 2, is an additive finite label that determines whether the product is 1 or 5 modulo 6. OPH uses this only as a toy model for the same formal pattern used here: local factor or sector data accumulate in a finite abelian label, and that label determines the global quotient or obstruction class.*

Corollary 7.27 (Standard Model Gauge Group)**.** *Under Theorem 7.16, Theorem 7.17, Corollaries 7.21 and 7.20, and Proposition 7.24,*

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

Proof. Theorem 7.16 yields the product structure up to finite quotient, and Proposition 7.24 fixes that quotient to $\mathbb{Z}_6$ once the realized hypercharge lattice and $N_c = 3$ are in place. □

Corollary 7.28 (Structural electroweak force and charges)**.** *On the realized one-Higgs branch of Theorem 7.17 and Corollary 7.22, the electroweak gauge factor has Lie algebra*

$$\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

The Higgs doublet $H = (1, 2)_{1/2}$ selects a nonzero neutral vacuum vector

$$\phi_0 = \frac{v}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad v \neq 0,$$

with

$$Q\phi_0 = (T_3 + Y)\phi_0 = 0,$$

so its vector stabilizer has unbroken generator

$$Q = T_3 + Y,$$

and the unbroken gauge factor is $\mathrm{U}(1)_Q$*. The charged weak generators give*

$$W^{\pm} = \frac{1}{\sqrt{2}}(W^1 \mp iW^2),$$

while the neutral $\mathrm{SU}(2)_L$ *and* $\mathrm{U}(1)_Y$ *gauge fields span the* Z/A_Q *connection basis on the D10 quantitative branch. Thus the recovered structural package contains the charged* $W^{\pm}$ *connection directions, the neutral broken* Z *direction, the unbroken electromagnetic connection direction* A_Q*, and the exact charge operator* $Q = T_3 + Y$*. These algebraic and connection labels alone assert neither a propagating mode nor a quantum particle. The mixing angle,* v*, and the numerical* W/Z *masses belong to the D10 running/matching surface instead of this recovered-core structural corollary.*

Proof. The D8 result fixes the weak-type role to the SU(2) doublet and the connected abelian role to $\mathrm{U}(1)_Y$. Theorem 7.17 fixes $Y_H = 1/2$ and the matter hypercharge lattice, while Proposition 7.18 identifies the same one-Higgs slot with the minimal nontrivial holomorphic section carrier on the local weak chart. For conventional angles α, β, the action on the chosen nonzero vector is

$$e^{i\alpha T_3} e^{i\beta Y} \phi_0 = e^{i(\beta-\alpha)/2} \phi_0.$$

Thus the vector is fixed exactly by the diagonal relation $\beta = \alpha$, locally, and its connected stabilizer is generated by $Q = T_3 + Y$. By contrast, its projective ray is fixed for independent α and β; that larger projective stabilizer is not the unbroken gauge group. The two noncommuting charged SU(2) generators are broken directions and combine into the $W^{\pm}$ connection components; the orthogonal neutral broken direction defines the Z connection component, while the gauge field of the unbroken vector stabilizer is the electromagnetic connection component A_Q. The global $\mathbb{Z}_6$ quotient fixes the compatible charge lattice used in Proposition 7.24 and only adds the corresponding finite central identification to this stabilizer calculation. None of these algebraic labels alone asserts a kinetic term, propagating mode, or quantum particle. $\square$

Corollary 7.29 (Conditional Maxwell sector on the realized $\mathrm{U}(1)_Q$ branch)**.** *On the realized ordinary zero-obstruction one-Higgs branch, let* A_Q *be the support-visible connection component on the unbroken electromagnetic factor* $\mathrm{U}(1)_Q$ *of Corollary 7.28. Then*

$$F_Q = dA_Q.$$

Assume in addition the low-energy Lorentzian Maxwell action with conserved current J_Q*,*

$$S_{\mathrm{EM}}[A_Q, J_Q] = -\frac{1}{2g_Q^2}\int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q,$$

equivalently

$$S_{\mathrm{EM}}[A_Q] = -\frac{1}{4g_Q^2}\int F_{Q,\mu\nu} F_Q^{\mu\nu}\, d^4x$$

for the pure field term. Its Euler–Lagrange equations are

$$dF_Q = 0, \qquad d * F_Q = g_Q^2 * J_Q.$$

In local coordinates,

$$\partial_{[\lambda} F_{Q,\mu\nu]} = 0, \qquad \partial_\mu F_Q^{\mu\nu} = g_Q^2 J_Q^\nu.$$

After canonical electromagnetic normalization, these are Maxwell's equations. The numerical value of g_Q, equivalently α_{em}, belongs to the D10 Ward-projected electromagnetic readout. The compact-group reconstruction and connection label do not themselves supply this action or its nonzero kinetic coefficient; they are independent hypotheses of this corollary.

Proof. Corollary 7.28 supplies the unbroken electromagnetic factor $\mathrm{U}(1)_Q$ with generator $Q = T_3 + Y$. The compact-gauge curvature theorem supplies a support-visible continuum connection and curvature

$$F = dA + A \wedge A$$

on the compact-gauge scaling chart. Restricting to $\mathrm{U}(1)_Q$ removes the Lie-bracket term because $\mathfrak{u}(1)$ is abelian, hence

$$F_Q = dA_Q.$$

The identity $d^2 = 0$ gives $dF_Q = 0$.

For the sourced equation, vary

$$S_{\mathrm{EM}}[A_Q, J_Q] = -\frac{1}{2g_Q^2} \int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q.$$

Since $\delta F_Q = d(\delta A_Q)$, integration by parts on compactly supported variations gives

$$\delta S_{\mathrm{EM}} = \int \delta A_Q \wedge \left(\frac{1}{g_Q^2} d * F_Q - *J_Q \right)$$

up to the boundary term. Stationarity for arbitrary δA_Q gives

$$d * F_Q = g_Q^2 * J_Q.$$

Applying d to the sourced equation gives $d * J_Q = 0$, the current-conservation compatibility condition. The component equations are the standard homogeneous and inhomogeneous Maxwell equations with the OPH coupling convention. □

Definition 7.30 (Carrier-mode and quantum-particle receipts)**.** *Fix the structural invariant speed $c_\star$. A classical massless carrier-mode receipt for a field q_X consists of a stated background and phase, an explicit quadratic action with positive nonzero physical kinetic coefficient, a gauge fixing or constraint reduction with physical projector Π_X, and a positive reduced Hamiltonian. Separately, the action Hessian restricted to the physical subspace must have*

$$K_X^{\mathrm{phys}}(\omega, \mathbf{k}) = Z_X(\omega^2 - c_\star^2 |\mathbf{k}|^2)\Pi_X, \qquad Z_X > 0,$$

so the free reduced Green function is

$$D_X^{\mathrm{phys}}(\omega, \mathbf{k}) = \frac{i\,\Pi_X / Z_X}{\omega^2 - c_\star^2 |\mathbf{k}|^2 + i0}.$$

This is an action-level classical-mode statement, not a particle-mass statement.

A quantum-particle receipt additionally supplies: a positive-energy vacuum quantization; a physical Hilbert space obtained by constraint reduction or BRST cohomology; a physical two-point function with Källén–Lehmann measure

$$d\rho_X(\mu^2) = Z_X^{\mathrm{pole}} \delta(\mu^2) d\mu^2 + d\rho_X^{\mathrm{cont}}(\mu^2), \qquad Z_X^{\mathrm{pole}} > 0;$$

or an equivalent joint energy–momentum spectrum containing a positive-residue $p^0 > 0$, $p^2 = 0$ mass shell (at fixed $\mathbf{k}$, $\omega = c_\star|\mathbf{k}|$); and the stability/asymptotic-state or LSZ hypotheses appropriate to the phase, including a deconfined asymptotic sector for a colored carrier. Only the combined classical and quantum receipt licenses a massless quantum-particle claim. A compact group, connection label, or classical field equation alone does not pass either missing step.

Theorem 7.31 (Action-level carrier modes on the Maxwell, pure-Yang–Mills, and Einstein branches)**.** *The following statements hold under the displayed additional action, background, field-content, and phase hypotheses.*

(i) *On the source-free ordinary electromagnetic vacuum branch $J_Q = 0$, assume the Maxwell action of Corollary 7.29 with $0 < g_Q^2 < \infty$ and no Higgs, Stueckelberg, medium, or nonlocal quadratic mass operator. Radiation gauge leaves two transverse components and*

$$H_Q^{(2)} = \frac{1}{2g_Q^2}\int (|\mathbf{E}_T|^2 + |\mathbf{B}|^2)d^3x, \qquad K_{Q,T} = g_Q^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T, \qquad \operatorname{rank}\Pi_T = 2.$$

In covariant ξ-gauge the free kernel has inverse

$$D^Q_{\mu\nu}(k) = \frac{-ig_Q^2}{k^2 + i0}\left(\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 + i0}\right),$$

whose longitudinal part decouples from a conserved current. Hence the branch has two classical massless electromagnetic carrier modes.

(ii) *Let a compact group G carry the Lorentzian two-derivative pure Yang–Mills action with $0 < g^2 < \infty$, expanded about the topologically trivial background in a gauge $\bar{A}_\mu = 0$, in a perturbative or deconfined phase with no Higgs condensate. Writing the perturbation as a^a_μ, its quadratic action is*

$$S^{(2)}_{\mathrm{YM}} = -\frac{1}{4g^2}\sum_a \int (\partial_\mu a^a_\nu - \partial_\nu a^a_\mu)(\partial^\mu a^{a,\nu} - \partial^\nu a^{a,\mu})d^4x.$$

Lorenz gauge gives the standard covariant inverse, while Gauss' law and radiation gauge leave two transverse components per generator, with

$$H^{(2)}_{\mathrm{YM}} = \frac{1}{2g^2}\sum_a \int (|\mathbf{E}^a_T|^2 + |\mathbf{B}^a|^2)d^3x \geq 0,$$

with strict positivity on nonzero reduced finite-energy modes of $\mathbf{k} \neq 0$ under the stated boundary conditions. Their kernel and free transverse Green function are

$$K^{ab}_{\mathrm{YM},T} = \delta^{ab}g^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T, \qquad D^{ab}_{\mathrm{YM},T} = \frac{ig^2\delta^{ab}\Pi_T}{\omega^2 - c_\star^2|\mathbf{k}|^2 + i0}.$$

Consequently there are $2\dim G$ transverse classical modes and the corresponding gauge-fixed free $k^2 = 0$ pole. This is a perturbative quadratic statement. It does not give a gauge-invariant asymptotic gluon on the confined QCD branch; the positive gauge-invariant Yang–Mills gap and the colored connection pole concern different spectral questions.

(iii) *In addition to the derived classical Einstein equation, assume the two-derivative Einstein–Hilbert action with $G > 0$ about a Minkowski vacuum with $\Lambda = 0$, without higher-curvature* *kinetic terms, bimetric mixing, or extra scalar/vector fields. De Donder gauge and residual-gauge reduction give the two transverse-traceless components with*

$$S^{(2)}_{\mathrm{TT}} = \frac{1}{64\pi G}\int \Big[(\partial_t h^{\mathrm{TT}}_{ij})^2 - c_\star^2(\nabla h^{\mathrm{TT}}_{ij})^2\Big]\, d^4x,$$

$$H^{(2)}_{\mathrm{TT}} = \frac{1}{64\pi G}\int \Big[(\partial_t h^{\mathrm{TT}}_{ij})^2 + c_\star^2(\nabla h^{\mathrm{TT}}_{ij})^2\Big]\, d^3x, \qquad D^{\mathrm{TT}}_{ij,kl}(k) \propto \frac{i\Pi^{\mathrm{TT}}_{ij,kl}}{k^2+i0}, \qquad \operatorname{rank}\Pi^{\mathrm{TT}} = 2.$$

Thus the pure Einstein linearization has two classical massless spin-two wave modes. On a curved Einstein background the appropriate object is the Lichnerowicz spectrum with stated boundary conditions, not an automatic Poincaré particle pole.

Proof. For item (i), the second variation of the Maxwell action is the standard vector kinetic Hessian. Gauss' law removes the longitudinal canonical pair, leaving two transverse polarizations with positive Hamiltonian and dispersion $\omega^2 = c_\star^2|\mathbf{k}|^2$; inversion after gauge fixing gives the displayed propagator. For item (ii), $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + O(A^2)$, so the quadratic Hessian is one Maxwell Hessian per Lie-algebra generator and the nonabelian interactions begin at cubic order. For item (iii), the second variation of the Einstein–Hilbert action is the Fierz–Pauli kinetic form. De Donder gauge and its residual freedom reduce it to the displayed transverse-traceless action with $+$ and $\times$ polarizations. These calculations establish the classical receipt of Definition 7.30; none constructs its quantum Hilbert-space, physical spectral-measure, or asymptotic-state clauses. ☐

Corollary 7.32 (Conditional particle interpretation)**.** *If a branch in Theorem 7.31 is also supplied with the quantum-particle receipt of Definition 7.30, its positive-residue $\delta(\mu^2)$ contribution defines a massless particle pole with the displayed physical polarizations. Without that receipt, this paper claims only the corresponding classical or perturbative carrier mode. In particular, compact-group reconstruction does not itself create a Maxwell kinetic term or deconfined phase, and the classical Einstein equation does not itself create a graviton state.*

Remark 7.33 (Hard quadratic terms versus the physical spectrum)**.** *The zero hard parameters in the displayed Maxwell, pure-Yang–Mills, and pure-Einstein quadratic operators are branch-specific action statements, not universal exact particle masses. Gauge or diffeomorphism redundancy can coexist with Higgs/Stueckelberg completions, plasma or medium self-energies, confinement, higher-derivative poles, bimetric sectors, or additional massive fields. Excluding those possibilities requires the declared field-content and phase hypotheses.*

Remark 7.34 (Maxwell branch boundary)**.** *Corollary 7.29 concerns the ordinary electromagnetic branch. Topologically singular sectors with magnetic sources obey modified homogeneous equations. On the certified support-visible compact simple nonabelian branch, Theorem A.37 gives a positive gauge-invariant repair gap under its named finite and continuum receipts. On the abelian branch, the Maxwell action and phase hypotheses of Theorem 7.31 give two classical massless transverse modes; a photon particle requires Corollary 7.32.*

Theorem 7.35 (Realized compact-gauge witness and physical UV landing)**.** *On the ordinary or central zero-obstruction bosonic branch, suppose a cofinal tail carries the compact-gauge refinement receipt of Definition 7.2. Then the OPH compact reference architecture admits an OPH-realizable cofinal heat-kernel edge-sector witness*

$$\mathcal{W}_{\mathrm{SM}} = \{Q_i, u^c_i, d^c_i, L_i, e^c_i, H\}_{i=1}^3$$

with

$$Q_i = (3,2)_{1/6}, \qquad u_i^c = (\bar{3},1)_{-2/3}, \qquad d_i^c = (\bar{3},1)_{1/3},$$
$$L_i = (1,2)_{-1/2}, \qquad e_i^c = (1,1)_1, \qquad H = (1,2)_{1/2}.$$

Every witness sector has positive realized support on the compact heat-kernel branch, and the witness is MAR-admissible. Hence the MAR-admissible class used above is nonempty. Every OPH-admissible microscopic UV completion of this realized branch has observer-visible low-energy package, modulo physical equivalence in Definition 3.12,

$$\mathfrak{S}_{\rm SM} = \left(\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \mathcal{R}_{\rm SM}, H, \mathcal{Y}_{\rm SM}\right), \qquad N_c = 3, \qquad N_g = 3,$$

with the hypercharge lattice of Theorem 7.17.

Proof. Use the fixed-cutoff compact-gauge patch-carrier architecture on the zero-obstruction branch. The microphysics carrier may be represented by finite gauge-register, quantum-link, or federated echosahedral regulator charts; only the declared overlap sector algebra and refinement-compatible compact labels are used here. At a sufficiently fine compact cutoff, retain the Peter–Weyl labels listed in $\mathcal{W}_{\rm SM}$. They are finite-dimensional unitary representations of $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$, and with integer normalization $q = 6Y$ Proposition 7.24 shows that the generator $g_0 = (\omega_3, -1, e^{i\pi/3})$ acts trivially on exactly the displayed matter and Higgs multiplets. The witness therefore descends to the quotient $(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$.

The fixed-cutoff edge heat-kernel law on the microphysics surface gives sector weights

$$p_R(t) \propto d_R e^{-tC_2(R)}$$

on the compact branch. For finite t, $d_R > 0$ and $e^{-tC_2(R)} > 0$, so every witness projector has positive support. On the tail certified by the stated receipt, the refinement functors of Theorem 7.3 carry those zero-obstruction labels forward, and Theorem 7.6 places the corresponding objects in Sect_∞.

The witness is loop-coherent by the zero-obstruction branch. Theorem 7.17 supplies anomaly cancellation and one-Higgs Yukawa completeness on the displayed chiral package; Lemmas 7.11–7.13 give the weak-type and color-type roles on the minimal coupled carrier; Corollary 7.20 gives $N_c = 3$; and Corollary 7.21 gives $N_g = 3$ from the CP-capability and weak-sector UV clauses in Axiom 3.11. Thus $\mathcal{W}_{\rm SM}$ is an occupied MAR-admissible package.

Proposition 3.13 supplies minima for the resulting nonempty admissible class. Corollary 7.22 and Proposition 7.24 identify every MAR-minimal observer-visible package with the displayed Standard Model package. Finally, Remark 3.14 and Definition 3.12 remove only microscopic regulator choices, implementation hiding, gauge-center conventions, generation relabeling, charge-conjugation convention, and inert ancillary stabilization. Thus any OPH-admissible UV completion of the realized branch lands on the same observer-visible low-energy package, not on a unique microscopic representative. □

Theorem 7.36 (Realized Standard Model branch)**.** *Assume:*

(1) *the ordinary or central zero-obstruction bosonic branch of Theorem 5.22;*

(2) *a cofinal tail carrying the compact-gauge refinement receipt of Definition 7.2, together with the fixed-cutoff bosonic sector category, refinement/fiber descent, and compact-group reconstruction of Theorems 5.26, 7.3, 7.6, and 7.7;*

(3) *a nonempty realized one-Higgs chiral MAR-admissible class, witnessed by* $\mathcal{W}_{\mathrm{SM}}$ *in Theorem 7.35;*

(4) *the weak-type and intrinsically complex color-type roles on a common coupled carrier;*

(5) *one connected abelian charge factor acting nontrivially on that carrier; and*

(6) *the anomaly-free, one-Higgs Yukawa-complete, CP-capable, and weak-sector UV clauses of Axiom 3.11.*

Then MAR minima exist, and every MAR-minimal observer-visible package in that class is physically equivalent to

$$\left(\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \mathcal{R}_{\mathrm{SM}}, H, \mathcal{Y}_{\mathrm{SM}}\right), \qquad N_c = 3, \qquad N_g = 3,$$

where the matter package is

$$\mathcal{W}_{\mathrm{SM}} = \{Q_i, u_i^c, d_i^c, L_i, e_i^c, H\}_{i=1}^3,$$

with

$$Q_i = (3,2)_{1/6}, \quad u_i^c = (\bar{3},1)_{-2/3}, \quad d_i^c = (\bar{3},1)_{1/3}, \quad L_i = (1,2)_{-1/2}, \quad e_i^c = (1,1)_1, \quad H = (1,2)_{1/2}.$$

Proof. Proposition 3.13 gives existence of MAR minima because the admissible class is nonempty and $C(\mathfrak{S}) \in \mathbb{N}^4$ is well ordered lexicographically. Lemmas 7.11 and 7.12 identify the minimal weak-type and color-type connected nonabelian roles as SU(2) and SU(3). Lemma 7.13 puts those roles on the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$, not on the uncoupled $\mathbb{C}^3 \oplus \mathbb{C}^2$, and Lemma 7.15 identifies the unique connected abelian commutant as U(1). Theorem 7.16 therefore gives product structure up to finite central quotient. Theorem 7.17 fixes the hypercharge lattice, Corollary 7.20 fixes $N_c = 3$, Corollary 7.21 fixes $N_g = 3$ as a MAR-branch consequence instead of as anomaly cancellation alone, and Proposition 7.24 fixes the global kernel. Corollary 7.22 then identifies all minima modulo the physical equivalence relation of Definition 3.12. □

Corollary 7.37 (Product-Group Consequences)**.** *The adjoint representation of the full connected gauge group contains only*

$$(8,1,0) \oplus (1,3,0) \oplus (1,1,0).$$

Equivalently, the adjoint representation of the derived gauge group contains only

$$(8,1,0) \oplus (1,3,0).$$

There are no mixed $(3,2,\pm 5/6)$ *gauge generators. Hence the ordinary simple-GUT gauge-mediated proton-decay channel is absent. This adjoint-content statement does not by itself supply a gauge-field kinetic term, select a phase, or determine a particle mass; those questions are governed by Definition 7.30 and Theorem 7.31.*

Remark 7.38. *The same product-group structure removes the usual grand-unification monopole sector. In the present framework, any retained coupling-unification discussion does not proceed by embedding the gauge group into a simple group that is later broken. It proceeds, if at all, only through the separate D10 calibration branch built from shared edge heat-kernel data together with the printed running/matching/threshold/scheme conventions and extra calibration assumptions. Absence of gauge-mediated proton decay and unification-like running therefore coexist without tension on that branch. Any confinement statement is a separate infrared dynamical issue, not a corollary of the product-group adjoint argument alone.*

7.3 Fundamental scales

The quantitative D10 branch is a forward quantitative-closure sector, not part of the recovered-core theorem package. Its particle-physics scale variable is

$$P \equiv a_{\mathrm{cell}}/\ell_\star^2,$$

with the selected no-G scale certificate, which supplies $\ell_\star^2 = 3\pi/B_\star$ and becomes the usual Planck-area display after $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$. The same quantitative surface also uses the realized discrete gauge data of the structural branch, in particular

$$N_c = 3, \qquad \beta_{\mathrm{EW}} = N_c + 1 = 4.$$

The logical direction is therefore

$$P \longmapsto \big(M_U(P), E_{\mathrm{cell}}(P)\big) \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\mathrm{tr}}(P)\big) \longmapsto \big(t_2(P), t_3(P), v(P)\big) \longmapsto \alpha_i(\mu_*; P).$$

The edge-law input on this lane is split explicitly by claim tier. The fixed-cutoff same-overlap thermal/Casimir theorem is carried on the separate microphysics surface. This compact paper imports that closure through the declared merge boundary: one fixed-cutoff edge-law package below, then the compact-group / Peter–Weyl lift used in the D10 pixel constraint above it. The compact-group lift is a declared D10 branch handoff, while large-N_{edge} and critical-string claims belong to the continuation lane. The finite-cutoff Casimir branch itself is not a missing input. No hardware evidence, private run log, or laboratory module claim is imported into this D10 theorem step. Measured electroweak quantities are not used to infer the internal transmutation parameters. They enter only at the end as compare-only validation targets for the printed D10 running/matching/threshold/scheme package. The forward transmutation certificate on the live D10 lane makes that logical order explicit: the source-only basis reconstructs the same $\alpha_U(P)$, hence the same unified diffusion parameter

$$t_U(P) = 4\pi^2\alpha_U(P),$$

and transmutation exponent

$$t_{\mathrm{tr}}(P) = \frac{2\pi}{(N_c+1)\alpha_U(P)}$$

as the pixel-closure solve itself. Here the paper-side transmutation factor is the same $\beta_{\mathrm{EW}} = N_c + 1$ used below; legacy overloaded β-ratios are compare-only diagnostic readouts and are not part of the theorem contract.

This fixes the hierarchy interpretation used by the compact paper. The cosmic capacity number N_{CRC} belongs to the global horizon-capacity branch. The electroweak hierarchy is local:

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P_\star)}\right].$$

The hierarchy bundle certifies this local branch by the interval

$$I_U = [0.041123336195630494,\ 0.041125336195630496],$$

with a Krawczyk image

$$K(I_U) \subset [0.0411243357185544983,\ 0.0411243366727064662] \subset \mathrm{int}(I_U)$$

and a strictly negative derivative enclosure

$$\mathcal{F}'(\alpha_U; P_\star) \in [-10.995768,\ -10.985284].$$

The public endpoint branch then gives

$$\begin{aligned}
\alpha_U(P_\star) &= 0.041124336195630495,\\
\frac{v}{E_\star} &= 2.0199803239725553 \times 10^{-17},\\
\log_{10}\frac{E_\star}{v} &= 16.69465286086613.
\end{aligned}$$

The source-audit branch gives

$$\begin{aligned}
P_{\mathrm{cand}} &= \texttt{1.630972095694329018179678925611918842701 69} \qquad ,\\
\frac{v(P_{\mathrm{cand}})}{E_\star} &= 2.0198114150099223 \times 10^{-17},
\end{aligned}$$

and is the branch to use when the public Thomson endpoint is excluded upstream. The Higgs naturalness row is read on the same D10/D11 normal-form surface. The observer-visible scalar mass is the branch readout $m_H = H_{\mathrm{OPH}}(P_\star)$. The split into a bare scalar mass plus a cutoff correction is a regulator coordinate description; the selected branch fixes the scalar row through its own equations. The source-to-Higgs coarse-graining square has the exact selected-branch defect bound

$$\epsilon_H = 0, \qquad \epsilon_H \in [0, 0],$$

with measured weak-scale, Higgs/top, W/Z, gravity, Planck-area, and Λ inputs excluded. This is the OPH hierarchy/naturality solution on the selected exact branch and a quantitative-closure statement outside the recovered-core theorem. The adjacent electroweak projection bridge defines

$$\Pi_{\mathrm{EW}}(P, N) = \frac{24\pi}{\alpha_U(P)\log(N/\pi)}.$$

The target $\Pi_{\mathrm{EW}}(P_\star, N_{\mathrm{CRC}}) = 4P_\star$ is equivalent to

$$\mathcal{B}_{\mathrm{EW}}(P, N) := \alpha_U(P)\log(N/\pi) - \frac{6\pi}{P} = 0,$$

or $N_{\mathrm{EW}}(P) = \pi\exp[6\pi/(P\alpha_U(P))]$. On the public endpoint branch this target is 3.5323546226929906511187512962 10^{122}. The rounded 3.31×10^{122} capacity display is diagnostic and fails the exact bridge residual. The factor 12 in the exponent has the screen-sieve theorem as its geometric source: on the triangulated S^2 branch, $q_v = 6 - \deg(v)$ obeys $\sum_v q_v = 12$; convex MaxEnt defect cost spreads this into twelve unit fivefold defects; edge-center collars make them central ports; A_5 symmetry places them on the icosahedral vertex orbit. With reversible write/check orientation, the same branch supplies a 24-slot oriented repair register; the hierarchy exponent uses that 24-fold repair normalization.

The scales fixed directly from P are

$$M_U(P) = \frac{E_P}{e^{2\pi}}\,P^{1/6}, \qquad E_{\mathrm{cell}}(P) = \frac{E_P}{\sqrt{P}}.$$

The factor $e^{-2\pi}$ is the same modular normalization used on the Lorentz/BW side, and $P^{1/6}$ is the cell-area scaling relation carried by the present D10 implementation.

Golden-ratio equilibrium benchmark. The total/bulk/edge hierarchy has one exact self-similar balance point. Writing

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)},$$

the exact self-similar balance condition

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle}$$

gives

$$x = \frac{1}{x-1}, \qquad x^2 - x - 1 = 0.$$

Hence the unique positive equilibrium point is

$$x = \varphi := \frac{1+\sqrt{5}}{2}.$$

Equivalently, the equilibrium-breaking order parameter

$$A_\varphi(x) := x - 1 - \frac{1}{x}$$

vanishes exactly at $x = \varphi$. In that sense φ is the exact self-similar balance point instead of a numerological comparison constant. The synthesis paper gives the outer/inner closure relation that fixes the realized value of P by matching that detuning to the inner electromagnetic observation scale emitted by the same cell. The role of the equilibrium theorem is to explain why the realized value sits close to φ. Exact equilibrium is too symmetric to support durable records, structure, and dynamics, so the realized branch sits at a small equilibrium-breaking detuning away from it. The technical question for the quantitative branch is the size of that detuning together with the reduced-residual/root-control analysis of the printed solve.

The one-dimensional internal variable solved on the D10 branch is the unified coupling α_U. For a trial value of α_U, define

$$v(\alpha_U, P) := E_{\text{cell}}(P)\,\exp\!\left(-\frac{2\pi}{\beta_{\text{EW}}\alpha_U}\right).$$

Run the one-loop D10 couplings from

$$\alpha_1(M_U) = \alpha_2(M_U) = \alpha_3(M_U) = \alpha_U$$

down to a scale μ using

$$\alpha_i^{-1}(\mu; \alpha_U, P) = \alpha_U^{-1} + \frac{b_i}{2\pi}\log\frac{M_U(P)}{\mu},$$

with the printed one-loop coefficients b_i. Let $\mu_* = \mu_*(\alpha_U, P)$ be the fixed point determined by the tree-level Z-mass relation

$$\mu_* = \frac{v(\alpha_U, P)}{2}\sqrt{g_2(\mu_*)^2 + g_Y(\mu_*)^2}, \qquad \alpha_Y = \frac{3}{5}\,\alpha_1.$$

At that fixed point define

$$t_2(\alpha_U, P) = 4\pi^2\alpha_2(\mu_*; \alpha_U, P), \qquad t_3(\alpha_U, P) = 4\pi^2\alpha_3(\mu_*; \alpha_U, P).$$

Let $\bar{\ell}_{\mathrm{SU}(2)}(t)$ and $\bar{\ell}_{\mathrm{SU}(3)}(t)$ denote the nonabelian edge-entropy functions appearing in the D10 pixel constraint. The pixel-closure functional is

$$\mathcal{F}(\alpha_U;P) := \bar{\ell}_{\mathrm{SU}(2)}(t_2(\alpha_U,P)) + \bar{\ell}_{\mathrm{SU}(3)}(t_3(\alpha_U,P)) - \frac{P}{4}.$$

The printed D10 package takes $\alpha_U(P)$ to be the branch value selected by solving

$$\mathcal{F}(\alpha_U;P) = 0.$$

Only after that forward solve are the internal transmutation parameters fixed:

$$t_U(P) := 4\pi^2\alpha_U(P), \qquad t_{\mathrm{tr}}(P) := \frac{2\pi}{(N_c+1)\alpha_U(P)},$$

and with them the downstream scale data

$$t_2(P) := t_2(\alpha_U(P),P), \qquad t_3(P) := t_3(\alpha_U(P),P), \qquad v(P) := v(\alpha_U(P),P).$$

Thus the branch runs from OPH input P to the internal transmutation data $(t_U(P), t_{\mathrm{tr}}(P))$, then to the downstream local scale data $(t_2(P), t_3(P), v(P))$, and only then to the low-energy couplings. It does not use measured $\alpha_i(m_Z)$ to infer those internal t-parameters.

Every quoted D10 electroweak number is downstream of this forward map. In particular, the source-locked running-family anchor $a_0(P) = \alpha_{\mathrm{em}}^{-1}(m_Z^2;P)$, the declared electromagnetic transport family $\alpha_{\mathrm{em}}^{-1}(q^2;P)$ and $\sin^2\theta_W(q^2;P)$, and the frozen public compare-only W/Z validation rows sit on the printed quantitative surface. The Thomson endpoint

$$\alpha_{\mathrm{Th}}^{-1}(P) = \lim_{q^2\to 0} \alpha_{\mathrm{em}}^{-1}(q^2;P)$$

on the Ward-projected $\mathrm{U}(1)_Q$ lane is declared through the source-spectral reduction theorem and remains gated by the populated source spectral measure payload, same-scheme remainder, and interval certificate. When compared with observation, these rows serve as validation of that printed implementation on the quantitative-closure branch recorded in the theorem checklist. The repository carries a numerical witness for the outer/inner fixed-point closure, and the public endpoint value is kept out of the closure solve.

8 Relation to Holography and Existing UV Frameworks

OPH belongs to the holographic family of ideas, but its primitive data are different from those used in asymptotic-boundary constructions. The distinction matters because several of the central claims of this paper, including the treatment of Λ, the handling of factorization, and the emergence of gauge structure, depend on it.

8.1 Static-patch screen versus asymptotic boundary

In asymptotic-boundary holography one starts from a dual theory at infinity and reconstructs the bulk inward. Here one starts from a finite-capacity screen equipped with a net of local patch algebras and reconstructs the effective bulk from overlap consistency. The primitive object is therefore a family of observer patches with nontrivial overlaps, not a single global boundary theory.

This shift has two technical consequences. First, subsystem factorization is treated from the beginning as a gluing problem with centers and sector labels. Second, the physical role of the horizon is local and operational: every observer has direct access only to a patch algebra, and global law is whatever survives reconciliation on overlaps.

8.2 Positive Λ and finite capacity

The D5→D6 cosmological-capacity stack is explicit: the null-modular reconstruction of Section 6 determines the stress tensor only up to a metric term, so local null data do not fix Λ. The global capacity datum is the cosmic record-capacity fixed point

$$N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$$

together with its observed-branch de Sitter entropy readout

$$N_{\mathrm{CRC}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

On that same branch this yields the dimensionless relation

$$\Lambda_{\mathrm{CRC}} G_{\mathrm{geom}} = \frac{3\pi}{N_{\mathrm{CRC}}}.$$

After the selected scale certificate supplies $G_{\mathrm{geom}} = \ell_\star^2$, this is displayed as $\Lambda_{\mathrm{CRC}} = 3\pi/(G_{\mathrm{geom}} N_{\mathrm{CRC}})$ and gives the static-patch package

$$S_{\mathrm{dS}} = N_{\mathrm{CRC}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c},$$

while the observed cosmic age is a downstream FLRW benchmark instead of an additional D6 theorem output. The compact paper is therefore formulated in a de Sitter-first language: the positive cosmological-constant capacity relation is tied to finite capacity instead of a deformation of a negative-Λ starting point. The input-free global closure is the cosmic record-closure readback fixed point

$$N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}}), \qquad F(N) = \mathrm{Cap}_{\mathrm{read}}\big(\mathrm{Obs}(\mathrm{nf}(\mathfrak{U}_N))\big).$$

The screen-normalized self-closure density is its count representation:

$$\Pi(N) = |\Omega_N^{\mathrm{sc}}| e^{-N}.$$

In differentiable form, the OPH-derived stationarity map $T_\eta(N) = N + \eta\, d\log\Pi(N)/dN$ has a unique stable fixed point under the derivative-sign certificate. On the observed branch this fixed point is the de Sitter entropy capacity.

8.3 Factorization, gauge structure, and the string sector

The factorization problem of gauge theory and gravity is often treated as a technical nuisance. Here it is part of the architecture. Edge-center completion turns the collar center into a first-class object, and the entropy split

$$S_{\mathrm{gen}} = \langle L_C \rangle + S_{\mathrm{bulk}}$$

is then a structural consequence instead of an added prescription.

In the bosonic EFT branch, Theorem 5.22 supplies the strict zero-obstruction transport criterion and Theorem 5.26 constructs the fixed-cutoff bosonic collar-sector categories. Compact gauge reconstruction across cutoffs additionally requires the explicit compact-gauge refinement receipt;

from it Theorem 7.3 constructs the refinement functors and compatible forgetful fibers. Crossed-module data handle the genuinely noncentral fixed-cutoff branch separately when $o_\Sigma^{(2)} \neq 0$; the ordinary compact-group theorem does not. If $o_\Sigma^{(2)} = 0$, that branch enters the ordinary theorem only on representations with trivial residual G_Σ-holonomy. This obstruction calculus is a classification and routing theorem. On the receipt-certified branch, DR/Tannaka reconstruction yields some compact group $G = \mathrm{Aut}_\otimes(\mathcal{F})$ from the zero-obstruction transportable sector category. On the realized MAR-admissible branch with the explicit realized one-generation chiral matter plus one-Higgs package, minimal admissibility selects the Standard Model quotient. This differs sharply from approaches in which the gauge group is specified in advance.

The worldsheet relation is similarly reversed. The edge partition function

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}$$

is the closed partition function obtained after gluing the open-edge weights

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Theorem 8.1 (OPH-to-2D-Yang–Mills edge partition theorem)**.** *Assume the compact-group heat-kernel branch supplied by the fixed-cutoff edge-sector law together with the compact-group / Peter–Weyl lift used in the bosonic gauge lane, in the same quadratic-Casimir normalization as the compact heat kernel. Equivalently, assume the open-edge weights satisfy*

$$p_R(t) \propto d_R e^{-tC_2(R)}$$

on a compact gauge group G, and define the closed edge partition function by gluing the two edge boundaries:

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

Then:

(i) *the closed edge partition function is exactly the compact-group heat kernel at the identity,*

$$Z_{\mathrm{edge}}(t) = K_t(1);$$

(ii) *the same sum is therefore the standard compact-group two-dimensional Yang–Mills heat-kernel partition function at identity, equivalently the closed-surface heat-kernel partition sum on that branch;*

(iii) *the Chapman–Kolmogorov law for K_t is the corresponding collar-sewing rule for the OPH edge partition.*

Thus the OPH edge-sector partition reorganizes exactly into the two-dimensional Yang–Mills heat-kernel form before any large-N_{edge} continuation is invoked.

Proof. The fixed-cutoff edge-sector theorem gives the heat-kernel weights $d_R e^{-tC_2(R)}$ on the compact-group branch. Gluing the two edge boundaries contributes a second factor of d_R, so the closed partition function is the displayed sum. Peter–Weyl identifies

$$\sum_R d_R \chi_R(g) e^{-tC_2(R)}$$

with the compact-group heat kernel $K_t(g)$, and evaluating at the identity $g = 1$ gives

$$K_t(1) = \sum_R d_R \chi_R(1) e^{-tC_2(R)} = \sum_R d_R^2 e^{-tC_2(R)} = Z_{\text{edge}}(t),$$

since $\chi_R(1) = d_R$. The Chapman–Kolmogorov law for K_t is exactly the semigroup gluing law, so it gives the collar-sewing rule on the same branch. $\square$

Scope boundary. The imported inputs are exactly the fixed-cutoff edge heat-kernel / Casimir law, the compact gauge-group / Peter–Weyl lift, the quadratic-Casimir heat-kernel normalization on that lift, and the Peter–Weyl heat-kernel identity. The external content is any large-N_{edge} regime, the Gross–Taylor dictionary on that regime, and every further critical-string ingredient. The theorem above proves the OPH-to-2D-Yang–Mills partition reorganization itself; it does not by itself produce a worldsheet genus expansion. The four-dimensional compact-gauge repair-gap theorem is the separate support-visible repair-dynamics result in Theorem A.37.

Theorem 8.2 (Criterion for a controlled large-N_{edge} worldsheet effective description)**.** *Assume Theorem 8.1 and a large-N_{edge} realization of the compact-group heat-kernel branch, with $N_{\text{edge}} \neq N_c = 3$, for which the 't Hooft-style variable*

$$\tau := tN_{\text{edge}}$$

is kept in a compact interval $I \subset (0, \infty)$. Suppose that for every truncation order $G \geq 0$ there exist coefficient functions $F_g : I \to \mathbb{R}$ and constants $C_{G,I}$ such that

$$\log Z_{\text{edge}}\left(\frac{\tau}{N_{\text{edge}}}\right) = \sum_{g=0}^{G} N_{\text{edge}}^{2-2g} F_g(\tau) + R_{G+1}(\tau, N_{\text{edge}})$$

with

$$\left|R_{G+1}(\tau, N_{\text{edge}})\right| \leq C_{G,I} N_{\text{edge}}^{-2G} \qquad (\tau \in I).$$

Then this branch defines a controlled theorem-level worldsheet effective description of the edge dynamics, with control parameters N_{edge}^{-2}, the fixed-τ window I, and the truncation order G.

Proof. Theorem 8.1 puts the edge partition function on the compact-group heat-kernel / two-dimensional Yang–Mills surface. On that surface, the standard Gross–Taylor dictionary reads a genus expansion of the displayed form as a closed-worldsheet rewriting of the free energy. The stated large-N_{edge} assumption supplies that expansion on the fixed-τ window I, together with an explicit truncation bound. Therefore the worldsheet rewriting is controlled by the displayed parameters N_{edge}^{-2}, I, and G. The statement is only an effective description on the declared branch: it does not by itself construct a critical worldsheet CFT, derive modular invariance, worldsheet supersymmetry, anomaly cancellation, GSO projection, or full massless-spectrum matching. $\square$

External inputs. The criterion theorem uses the declared large-N_{edge} regime and the imported Gross–Taylor large-N worldsheet dictionary for two-dimensional Yang–Mills. The external content is the existence of the large-N_{edge} sequence, the fixed-τ window on which the genus expansion holds, and the uniform remainder control. Any further lift to critical superstring structure requires worldsheet supersymmetry, critical dimension, modular invariance, anomaly cancellation, GSO projection, and full massless-spectrum matching beyond the present declared continuation theorem.

Feature	Asymptotic-boundary holography	OPH
Primitive data	Boundary theory at infinity	Finite-capacity screen with observer-patch net
Factorization across cuts	Subtle and often indirect	Explicit edge-center completion with central labels
Status of Λ	External background datum	Fixed globally by N_{CRC} after local null reconstruction; N_{scr} is the observed de Sitter entropy readout
Gauge group	Usually specified as part of the model	Reconstructed from edge sectors, then fixed on the realized MAR branch
String sector	Fundamental dual description	Controlled large-N_{edge} worldsheet effective description of edge dynamics on the stated branch

9 Support Levels and Falsifiability

The recovered core is

$$(D1\text{–}D5) \cup (D7\text{–}D9),$$

namely the relativity chain together with the realized Standard Model structural chain. D6 is the global closure of that same Einstein branch at the cosmic record-capacity fixed point, D10 is the integrated quantitative-closure branch, and D12 collects phenomenological continuations.

9.1 Prediction check by support level

Tier	Representative outputs	What would actually falsify the tier
Phase I recovered core (Theorem 3.17; D1–D5, D7–D9)	Confluence of overlap repair, Lorentz kinematics, the three-dimensional observer-facing spatial chart, the Jacobson-type Einstein branch in the stated scaling regime, receipt-conditional compact gauge reconstruction in the bosonic branch, the controlled four-dimensional Euclidean Yang–Mills form and compact-gauge repair-gap mechanism under the declared compact-gauge continuum/transfer assumptions, the realized Standard Model quotient chain on the explicit realized matter package, the exact hypercharge lattice on that realized matter package, the realized color triplet $N_c = 3$, the generation count $N_g = 3$, and the product-group consequence of no gauge-mediated proton decay	A mathematical failure in the derivation chain, loss of the Yang–Mills form or exact repair-gap branch conditions, or data that require a different realized gauge quotient, different realized hypercharges, a different color or generation count on the admissible branch, or gauge-mediated proton decay

Tier	Representative outputs	What would actually falsify the tier
Phase II zero-input global closure (D6)	$\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$, $S_{\mathrm{dS}} = N_{\mathrm{CRC}}$, and, after the selected OPH scale certificate, $\Lambda_{\mathrm{CRC}} = \frac{3\pi}{GN_{\mathrm{CRC}}}$, $r_{\mathrm{dS}} = \sqrt{3/\Lambda_{\mathrm{CRC}}}$, and $t_\Lambda = r_{\mathrm{dS}}/c$ from the cosmic record-closure fixed point $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$, with $F(N) = \mathrm{Cap}_{\mathrm{read}}(\mathrm{Obs}(\mathrm{nf}(\mathfrak{U}_N)))$, represented in counts by $N_\star = \mathrm{MAR}\arg\max_N(\log\lvert\Omega_N^{\mathrm{sc}}\rvert - N)$; any retained capacity-level neutrino side estimate on that branch value is legacy bookkeeping; the observed cosmic age is a downstream FLRW benchmark	Failure of the readback fixed-point certificate or of the specific global capacity relation would falsify the D6 closure without erasing the recovered core

Tier	Representative outputs	What would actually falsify the tier
Phase II quantitative-closure branch (D10)	forward transmutation data $\alpha_U(P)$, $t_U(P)$, $t_{\mathrm{tr}}(P)$, pixel-closure gauge-coupling consistency, $\alpha_i(m_Z)$, the source-locked anchor $a_0(P) = \alpha_{\mathrm{em}}^{-1}(m_Z^2; P)$, and the declared electroweak transport family $(W, Z, \alpha_{\mathrm{em}}^{-1}(q^2), \sin^2\theta_W(q^2), v)$. The Thomson endpoint $\alpha_{\mathrm{Th}}^{-1} = 137.035999177(21)$ is an empirical comparison coordinate. The source/root audit trunk emits $\alpha_{\mathrm{root}}^{-1} \simeq 136.9948351646$ at $P_{\mathrm{cand}} \simeq 1.6309720957$. Combining that root with $\alpha_U(P_C)$, evaluated at the CODATA-derived comparison pixel, gives the mixed-provenance no-hadron diagnostic $A^{\mathrm{fp}}_{\alpha_U} = 137.0359595008\ldots$. The root-to-CODATA bookkeeping difference is $\Delta^{\mathrm{fp}}_{\mathrm{H,cal}} = 0.04116401237835\ldots = \alpha_U(P_C)C_{24,Q}$, with $C_{24,Q} = 1.00096478597323262538\ldots$; it is not a source-emitted hadronic correction. The endpoint residual table also records $0.0414658610\ldots$ inverse-alpha units at the public endpoint pixel, with $S_{\mathrm{required}} \simeq 0.8954001326$ and $c_Q \simeq 0.6580257599$. The declared empirical route would use separately labeled $e^+e^- \to$ hadrons input; no same-scheme integrated endpoint payload is emitted here. The 24-register count does not determine the nonconstant OPH-QCD hadronic backend; its two-current spectral measure is only the running-α/HVP marginal, while HLbL and rare-decay rows require higher-point and transition spectral sectors	Failure of the integrated quantitative closure once P and the printed running/matching/scheme conventions are imposed, or failure of the explicit OPH-QCD quotient ensemble, source QCD parameter map, Ward current ledger, two-current/higher-point/transition spectral exports, same-scheme remainder, no-target-leak dependency record, empirical source disclosure, and interval-certificate records

Tier	Representative outputs	What would actually falsify the tier
Phase III phenomenological continuations (D12 and beyond)	Flavor ansätze, charged-lepton continuation ansätze beyond exact centered readback, H^3 record-worldline stitching certificates, texture branches, the target-informed weighted-cycle / Majorana-holonomy neutrino candidate, further neutrino mass/mixing refinements, dark-sector response laws, early compact-object/JWST source-release audit lanes, conditional screen-spectrum and CMB/inflation-replacement kernels, H_0/S_8 and growth continuations, heuristic baryogenesis continuations, proton-spin bookkeeping, proton-lifetime estimates beyond the gauge-channel exclusion, black-hole spectroscopy templates, physical black-hole evaporation/ringdown bridge programs, controlled large-N_{edge} string/worldsheet effective descriptions, conjectural critical-superstring extensions, and other downstream phenomenology	The weighted-cycle candidate fails the NuFIT 6.1 normal-ordering $(\sin^2\theta_{23}, \delta_{\mathrm{CP}})$ profile: $\Delta\chi^2 = 20.12$ with the tabulated atmospheric likelihood and 18.44 without it, above the two-parameter 3σ contour value 11.83 [7]. A separate basis audit finds that the claimed shared-basis recovery defined $U_{\nu,\mathrm{shared}} = U_e U_{\mathrm{wc}}$, making $U_e^\dagger U_{\nu,\mathrm{shared}} = U_{\mathrm{wc}}$ an identity rather than an independent flavor derivation. The stored charged-lepton source is open and has a nearly degenerate singular spectrum, so it does not define a stable physical U_e. The convention scan excludes only row, column, and orientation relabelings of the stored candidate; it does not derive or exhaust source-side physical basis placements. This rejects that candidate, without falsifying the recovered core. Other continuation failures retract only their corresponding branches

Proposition 9.1 (No semantic promotion by relabeling)**.** *Let X_r be a finite OPH branch with declared quotient state, finite records, and branch-local diagnostic or calibration fields Y_r. Suppose a proposed physical observable O_r^{phys} is obtained only by renaming Y_r or by applying a deterministic post-processing map $g(Y_r)$, with no additional source-separated carrier, calibration map, detector or readout algebra, error budget, negative controls, and frozen validation target. Then O_r^{phys} has the same information ancestry as Y_r and is not an independent physical observable. It may be recorded only as a finite theorem, diagnostic, or calibration display. Promotion to a physical claim requires a bridge object whose source roots, readout map, residual ledger, and controls are independent of the target physical label being asserted.*

Proof. A deterministic relabeling or post-processing map does not add new information, new source roots, or a new operational readout. Every dependency of $O_r^{\mathrm{phys}} = g(Y_r)$ therefore factors through the same finite records and quotient data that produced Y_r. Any leakage, calibration circularity, or target-derived value present in Y_r is inherited by the relabeled object. Physical promotion asserts more than finite equality inside the regulator: it asserts agreement with an independently specified operational or continuum quantity. That assertion requires extra data, namely a source-separated bridge, a declared readout/calibration map, residual bounds, negative controls, and a frozen target. Without those objects, the proposed observable is only another name for the original finite diagnostic or calibration display. □

Consequently, a capacity count is not an exterior mass without an independent energy readout; an append-only archive is not radiation without a source, propagation, and detector channel; a finite repair spectrum is not a physical normal-mode spectrum without a continuum operator and readout bridge; and an exact finite reconstruction threshold is not a Page time without a physical radiation entropy curve and exterior time calibration.

Phase I is the recovered-core level, Phase II is adjacent but non-core, and Phase III contains phenomenological continuations. The D6 row permits the conditional holonomy reading of FLRW flatness in Lemma 6.148; it does not support an inflation replacement, CMB likelihood, or high-H_0 branch. The conditional screen-spectrum theorem supports the declared finite-screen branch as a MaxEnt Gaussian screen covariance after the quotient-ensemble selector, scalar-readout, positive repair-operator, and fixed expected quadratic release-energy evidence pass; the tilt, scalar release amplitude, primordial radial lift, and physical TT/TE/EE spectra require their separate finite-source, source-provenance, lift, Boltzmann, and likelihood gates. Low-H_0, Planck-like S_8 dark/anomaly rows are diagnostic continuation checks unless a finite covariant collar-packet parent emits the homogeneous anomaly load, perturbative source functions, recipient stress and exchange-current closure for any nonzero exchange branch, active-fiber and physical-clock evidence for any stated Γ_{rec}, a first-principles dependency DAG, pooled global reducers, and frozen source/calculation/statistical hashes before fitting cosmological data. On the exact finite packet-closed quotient branch, the fixed-cutoff consensus surface does define an affine OPH closure map: the schedule-independent settled-form projection pushes forward to a continuous idempotent self-map of the finite packet simplex, its image is exactly the settled-form simplex, and its fixed points are exactly the packets supported on quotient settled forms. That closure statement is not an ensemble selector; physical probabilities require the quotient base measure/action or intrinsic projective prior stated near Axiom 3.7. That finite branch also supplies the grammar from which the cosmic readback map F and the self-closing observer settled-form density Ω_N^{sc} are defined. The readback fixed point gives the D6 target, with the screen-normalized density as its count representation; interpretive strange-loop discussions and the full habitat theorem sit outside these levels and are not needed for the finite packet-quotient fixed-point statement.

The dark-cosmology abundance selector in `cosmology/oph_dark_matter_paper.tex` is an application of the quotient-ensemble surface: it is the finite expectation of a quotient-visible load observable under a declared source release law.

The gamma-ray morphology continuation obeys the same boundary. An OPH gamma component is not an arbitrary residual in a photon map. It is admissible only as a frozen forward projection

$$I_{\mathrm{OPH}} = \Pi_{\gamma,r}(\mathfrak{G}_{\gamma,r})$$

from a proof-carrying gamma source artifact, with a declared quotient/source law, transported-stress or boundary-record route, photon-response bridge, line-of-sight projection, count-space instrument operator, foreground and alternative-template registry, no-data-use DAG, positivity gate, identifiability test, held-out validation, cross-tracer check, and null tests. Transported anomaly stress does not directly emit gamma rays on the neutral branch unless a separate electromagnetic-current theorem is supplied; it can only modulate ordinary photon-production channels or enter through a boundary-record projection. Thus gamma rows are D12 morphology tests, not excess-power claims.

Proposition 9.2 (Boltzmann-transfer and frozen-likelihood gate)**.** *A scalar dark/anomaly parent that emits only $\rho_A(a)$, $\rho_{A,\mathrm{eq}}(a)$, and $B_A(k,a)$ does not determine a physical TT/TE/EE, BAO, lensing, or growth prediction. A physical OPH dark/anomaly source for an Einstein–Boltzmann solver must instead be emitted by a finite covariant collar-packet parent*

$$\mathcal{P}_r[X,g] = (C_r, Z_r, A_r, R_r, G_r, \pi_r[X], L_r[X], Q_r, D_r),$$

where the finite packet states Z_r split the anomaly and any recipient stress channels, π_r is the finite equilibrium packet functional, L_r is the repair generator, Q_r records the energy-momentum reaction channels, and D_r records any causal auxiliary response. Its source certificate must first declare whether the local source is fixed-reference anomalous modular energy or nonlinear CMI stress. The local BW branch fixes the coefficient $15/(8\pi^2)$ for a source variable proved to equal the anomalous local modular energy. Identifying finite collar CMI with that source additionally requires the D12 CMI-to-anomalous-modular-source matching theorem or an explicit continuation hypothesis with a residual ledger; raw $I(A:D \mid B)$ is otherwise a recoverability diagnostic. The same certificate must derive

$$\rho_A, \quad B_A, \quad \Gamma_{\rm rec}, \quad w_A, \quad c_{s,A}^2, \quad \sigma_A, \quad Q_A^\mu$$

from that same parent, satisfy exact total stress-energy closure, build B_A from the anomaly-frame baryon density $n_b^{(A)} = -u_{A\mu} J_b^\mu$, pass finite domain-of-dependence, subluminal-characteristic, retarded-response, stability, local-frame, quotient-invariance, gauge-invariant perturbation, and refinement-convergence checks, and recover the CDM branch when exchange, pressure, sound speed, and anisotropic stress are switched off. If the exchange branch is nonzero, an explicit recipient stress tensor and equal-and-opposite exchange current are hard parts of the certificate. A transition-matrix spectral number remains $\gamma_{\text{repair step}}$ until the physical clock, active fiber, and common-parent response pole have been certified. The likelihood stage is physical only after immutable source, solver, and likelihood hashes are frozen before the data comparison.

Proof. The scalar triple $(\rho_A, \rho_{A,\rm eq}, B_A)$ omits the fluid closure data $w_A, c_{s,A}^2, \sigma_A$, the covariant exchange current Q_A^μ, the repair rate, and the recipient sector required by the Bianchi identity when energy is transferred. Different packet dynamics can share the same scalar tables while producing different metric sources or different gauge-fixed perturbation variables. The CMI functional also has zero first variation at an exact Markov reference, while fixed-reference modular energy is generically linear, so raw CMI cannot be substituted for the BW modular source without the separate matching theorem. A finite covariant packet parent supplies the missing stress moments and their reaction channels; total conservation then enforces $\sum_z \nabla_\mu T_z^{\mu\nu} = 0$ at the source level, and the anomaly-frame baryon density removes the gauge-fixed-density ambiguity in B_A. Frozen hashes are the no-data-use firewall for the transfer and likelihood stage. Thus diagnostic CAMB/CLASS curves or compressed likelihood rows become physical OPH predictions only with those receipts. □

Proposition 9.3 (CMB source provenance and pooled-reducer gate)**.** *Promoting any of*

$$\eta_R, \quad \gamma_{\text{repair step}} \text{ or certified } \Gamma_{\rm rec}, \quad A_\zeta, \quad q_{\rm IR}, \quad \ell_{\rm IR}, \quad B_A(k,a), \quad \rho_A(a), \quad N_{\rm CRC}$$

requires an immutable source-provenance certificate. The certificate is a finite dependency DAG whose nodes record the source report, source hash, parents, no-CMB-data flag, and measurement-use flags for each promoted quantity. A node that simultaneously asserts no CMB data use and a measurement fit, for example `no_cmb_data_used=true` and `fit_to_Planck=true`, fails closed. Nonlinear estimators for tilt, amplitude, inverse precision, rank, condition number, isocurvature leakage, B_A, ρ_A, and $\Gamma_{\rm rec}$ are evaluated only after additive sufficient statistics have been pooled globally with validated units, coordinate grids, coverage, duplicate policy, interpolation policy, and covariance; a transition diagnostic remains $\gamma_{\text{repair step}}$ until the active-fiber, physical-clock, and common-parent response receipts pass. Shard-local nonlinear averages do not satisfy the gate. $N_{\rm CRC}$ is treated as the D6 consensus invariant unless an additive capacity schema proves disjoint coverage before any summation. Official likelihood and CDM-limit checks are global checks; shard-local `any()` rollups do not promote a physical CMB input contract.

Proof. The listed quantities are inputs to later primordial or Boltzmann claims. If one of them is selected by a measurement-aware fit, a shard-local nonlinear average, or an additive count with overlapping coverage, the value fails the source-readback contract. A dependency DAG fixes the measurement-firewall order, and pooled sufficient statistics are the invariant objects under repartition of the run. Therefore only the DAG-plus-pooling certificate is stable under public re-execution; all other rows remain diagnostics. □

Proposition 9.4 (First-principles screen-to-primordial gate)**.** *The modular repair scalar $R_C = I(A : D|B)$ is a repair/readback defect, not the geometric curvature perturbation. On a source-stress-complete rank-one normal-form branch, the geometric scalar is the relational volume readout*

$$q_\zeta = \frac{1}{3}\log(J_X/\bar{J}_X) = \frac{1}{3}\log\frac{dV_h}{a^3 dV_{\bar{h}}},$$

equals uniform-total-density curvature after monopole and dipole removal. If the total effective pressure is barotropic on that branch, then the exact curvature covector obeys

$$\mathcal{L}_u\zeta_a = -\frac{\Theta}{3(\rho+p_{\rm eff})}\Gamma_a^{\rm eff} = 0.$$

The source clock is the unique future-oriented timelike eigenline of the total stress tensor when $\rho + p > 0$, the line is nondegenerate and hypersurface orthogonal, and the density is monotone. Primordial promotion additionally requires source-stress closure, a freeze-tail certificate, the scalar refinement map $R_r = J_{r+1,r}C_{r+1,r}$, simple isolated leading nonconstant eigenvalue with certified gap, conformal precision intertwiner, spatial curvature branch receipt, adiabatic growing-mode receipt, source-side isocurvature and phase-coherence bounds, a declared radial prior with the correct flat or curved shell kernel, the radial null-space report, and a forward-projection residual certificate. The finite code must certify these quantities from source artifacts; it may not choose the eigenmode, sign, dimension, amplitude, or curvature branch by optimizing CMB data. Without those receipts, A_s, n_s, running, isocurvature, phase coherence, and TT/TE/EE spectra are Phase III conditional outputs outside recovered-core prediction claim.

9.2 Recovered-core discriminators

Once the support boundary is made explicit, the sharpest discriminators are the ones that do not depend on auxiliary flavor, dark-sector proposals, heuristic baryogenesis continuations, or string ansätze.

9.2.1 Realized Standard Model gauge structure

On the realized MAR-admissible branch, this paper fixes

$$G_{\rm phys} = \frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3,$$

with the exact one-generation hypercharge lattice. Here the count statements are internal to the same realized branch: the D8 minimal coupled-carrier theorem stack fixes the color triplet and hence $N_c = 3$; CKM phase counting and weak-sector asymptotic freedom then bound N_g; MAR fixes the smallest realized value $N_g = 3$; and Witten parity is only a consistency check on the resulting triplet-doublet package.

$$Y_Q = \frac{1}{6}, \qquad Y_L = -\frac{1}{2}, \qquad Y_u = -\frac{2}{3}, \qquad Y_d = \frac{1}{3}, \qquad Y_e = 1, \qquad Y_H = \frac{1}{2}.$$

Evidence that the realized low-energy gauge structure or realized hypercharges differ from these values would directly contradict the recovered core.

9.2.2 Known-force and charge coverage

This paper does not leave any known long-range or Standard Model gauge force unassigned. The coverage is explicit by tier:

- **Gravity**: D3–D6 recover Lorentz kinematics, the three-dimensional observer-facing spatial chart, the null-stress bridge, and the Jacobson-type Einstein branch, with T_{ab} as the stress-energy source. On the additional pure Einstein–Hilbert/Minkowski action branch, Theorem 7.31 gives two classical transverse-traceless modes; a graviton state requires the quantum-particle receipt.
- **Strong interaction**: D8–D9 recover the $\mathrm{SU}(3)_c$ color factor, the color triplet $N_c = 3$, quark color triplet/antitriplet assignments, and the eight gluon generators $(8, 1, 0)$. Confinement and hadron spectra are separate infrared QCD questions, not missing gauge-charge assignments.
- **Weak interaction**: D8–D9 recover the $\mathrm{SU}(2)_L$ weak doublet structure, the one-Higgs branch, and the charged weak carriers $W^{\pm}$ from the broken $\mathrm{SU}(2)_L$ generators. D10 supplies the quantitative running/matching readout for v and the W/Z validation rows.
- **Hypercharge and electromagnetism**: Theorem 7.17, Proposition 7.24, and Corollary 7.28 fix the $\mathrm{U}(1)_Y$ lattice, the unbroken generator $Q = T_3 + Y$, integer charge for color singlets, and the electromagnetic connection label A_Q. Corollary 7.29 gives the Maxwell equations $dF_Q = 0$ and $d * F_Q = g_Q^2 * J_Q$ only after the low-energy Maxwell action is supplied. Theorem 7.31 then gives two classical transverse modes; a photon particle requires Corollary 7.32. The Thomson-limit fine-structure endpoint is a D10 quantitative closure.

9.2.3 Product-group corollary

Because the realized gauge structure is a product group up to the finite central quotient fixed above, its connected adjoint has no mixed $(3, 2, \pm 5/6)$ generator or connection component. This algebraic statement is not a particle-spectrum claim. Hence

$$\tau_p^{(\mathrm{gauge})} = \infty$$

for gauge-mediated proton decay. Observation of proton decay specifically attributable to gauge-boson exchange would therefore falsify the realized product-group branch.

9.2.4 Relativity branch

The relativity claim is sharp about its scope: given the support-visible geometric-modular theorem and the stated null-bridge conditions in the intended scaling regime, and the same realized cap-label-preserving MaxEnt family satisfying the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on that branch, this paper predicts Lorentz kinematics on the screen, the three-dimensional observer-facing spatial chart, and a Jacobson-type Einstein branch. What is being tested here is the existence of that scaling regime and its branch conditions, not a fixed-cutoff matrix-algebra identity.

9.3 What does *not* count as a contradiction of the recovered core

The following do *not* falsify Theorem 3.17 if they fail as stated:

1. the uniform $\mathbb{Z}_6$ center-label ensemble and the associated $\varepsilon = 1/6$ flavor ansatz;
2. charged-lepton continuation ansätze beyond the exact centered-readback / common-shift frontier, texture exponents, or legacy capacity-level neutrino side estimates;
3. modular-anomaly dark-sector response ansätze, including MOND/RAR-style scaling claims;
4. heuristic baryogenesis continuations based on extra suppression-counting assumptions; a baryogenesis theorem requires a concrete out-of-equilibrium mechanism, derived CP-odd source terms, justified defect/sphaleron counting, washout control, and a freeze-out computation of the final asymmetry;
5. strong-CP continuations; the available corpus does not derive the bare QCD angle θ_{QCD}, the physical anomaly-invariant combination $\bar{\theta}$, or a proof that $\bar{\theta} = 0$;
6. proton-spin ansätze or proton-lifetime estimates beyond the structural exclusion of gauge-mediated decay;
7. discrete-horizon spectroscopy templates, physical black-hole evaporation claims, Page-curve/island closure, or QNM/ringdown predictions;
8. application-level material, topological-phase, plasma, or device claims that require their own quotient-intrinsic source laws, material actions, repair ledgers, public receipts, and promotion gates;
9. controlled large-N_{edge} string/worldsheet effective descriptions.

These are all post-Phase-I, non-core branches. Some are Phase-II implemented estimates or conditional continuation checks, while the rest are Phase-III continuations. None is part of this paper's recovered relativity-plus-Standard-Model core.

10 Common Objections and Clarifications

10.1 Is the use of P circular?

The pixel area P is fixed by the outer/inner closure condition in the synthesis paper. That condition matches the outer pixel detuning to the inner electromagnetic observation scale emitted by the same cell. Within the printed quantitative implementation, that P first fixes $M_U(P)$ and $E_{\mathrm{cell}}(P)$, then a one-dimensional pixel-closure solve fixes $\alpha_U(P)$, equivalently the internal transmutation data $t_U(P)$ and $t_{\mathrm{tr}}(P)$, and only then do $t_2(P)$, $t_3(P)$, $v(P)$, and the running electroweak outputs appear. Quantities algebraically entangled with that quantitative branch are on the Phase-II implementation surface: they are forward-emitted there, and comparison with observation checks that printed implementation instead of enlarging the recovered-core claim set.

Operationally, changing P moves the quantitative family

$$(\alpha_U, \alpha_i(m_Z), a_0, \alpha_{\mathrm{em}}^{-1}(q^2), \sin^2\theta_W(q^2))$$

through that forward solve. It does not alter the parameter-free structural outputs such as the gauge quotient or hypercharge lattice. Downstream matter-sector continuations, including charged-lepton continuation ansätze beyond the exact centered-readback / common-shift frontier, are outside this SM/GR derivation paper's recovered-core theorem package.

10.2 Does a fixed UV cell size break Lorentz invariance?

The UV regulator is placed on the screen algebra. The emergent bulk spacetime has no microscopic bulk lattice in this formulation. Lorentz kinematics arise from the continuum modular action on caps, summarized in Corollary 6.40.

Spherical geometry is the bridge that makes this possible. A support-visible observer cut is charted by S^2. Round caps on that chart carry the modular flows used by the BW branch, and orientation-preserving conformal maps of the same sphere give

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1).$$

The Lorentz group therefore enters as the symmetry of observer-facing cap geometry. A finite echosahedral or cellulated carrier sits on the regulator side; the Lorentz claim concerns the support-visible scaling limit of the spherical cap chart extracted from that carrier. The observer rest-space chart is

$$H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3), \qquad \dim H^3 = 3,$$

so the spatial dimension follows from the same Lorentz branch. Finite point-cloud dimension fits belong to separate evidence gates.

This is conceptually similar to other continuum limits in statistical mechanics and lattice field theory. A regulator may break a symmetry at finite cutoff while the infrared fixed point restores it exactly. Here the restoration is stronger than a generic RG expectation because the modular-flow theorem identifies the symmetry group itself:

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1).$$

Residual Lorentz violation would therefore have to arise from a failure of the support-visible modular-flow theorem. A UV cell size by itself is insufficient. The UV-side burden on the Lorentz branch is therefore the support-visible scaling theorem on the prime geometric subnet, not the mere existence of a finite cell size.

10.3 Why use type-I algebras if continuum QFT uses type III?

Type I is a regulator statement. At every finite cutoff the patch and collar algebras are finite-dimensional matrix algebras, so entropy, recovery maps, and

$$K = -\log \rho$$

are literal finite-dimensional objects. The Lorentz branch is *not* a claim that the continuum observer algebra stays in that class. The refinement picture is

$$\text{finite type-I regulator net} \;\longrightarrow\; \text{scaling-limit observer net } \mathcal{A}_\infty,$$

and the limit can leave the regulator class. Axiom 3.7 controls the realized state-side branch across refinement; it does *not* by itself prove that the refinement limit is type I or that the emitted scaling-limit pair lies in the BW / canonical cap phase with standard geometric modular action.

On the geometric-subnet branch of Theorem 6.13, the scaling-limit cap algebras may be non-type-I and, in the continuum-QFT case of interest, are expected to be type III. Then there is generally no cap density matrix $\rho_C \in \mathcal{A}_\infty^{\mathrm{geo}}(C)$ and no bounded operator $K_C = -\log \rho_C$ inside the limit algebra. The correct continuum object is the modular automorphism group of the pair $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$, which on that branch acts geometrically and typically outerly.

There is therefore no contradiction between using type-I regulators and aiming at a type-III continuum. The finite regulator provides the collar decomposition, recovery control, and carried errors. What the fixed-cutoff MaxEnt package adds is control of the realized state-side branch through one common finite-dimensional multiplier family under refinement. Theorem 6.13 does not require the continuum algebra to be type I and does not require a full-algebra unregularized common spectral floor. It extracts the support-visible prime geometric cap pair, proves the geometric modular automorphism statement on that observer-facing limit, and leaves only the false stronger full-algebra statement unclaimed.

This is analogous to ordinary lattice field theory: one computes at finite regulator in a type-I algebra and then asks which continuum algebraic phase the scaling limit realizes. The additional structural point here is that the realized state-side branch is controlled under refinement. Theorem 6.13 extracts the support-visible cap pair on the prime geometric subnet and proves BW rigidity at automorphism level from the finite cap-normal certificate: regularized modular transport, weak-$*$/GNS extraction, support-readable modular covariance, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric 2π-KMS convergence with wrong-scale controls. Only stronger claims about off-support full-algebra directions, certificate production from bare finite consensus, or unique microscopic phase representatives lie outside that theorem.

10.4 UV branch and scaling-limit scope

At fixed cutoff, Axiom 3.7 yields a quasi-local finite-range interacting branch, and Propositions 4.20 and 4.29 show that the physical normal form is unique only modulo quotient-level OPH-stable equivalence, while the terminal expectation functionals on the declared fixed-point / quotient-local physical algebras are schedule-independent on that carrier even when microscopic representatives differ by gauge labels globally or by sector labels on one regional quotient-local glued state. The invariant fixed by the axiom language is the class $[\mathfrak{U}]_{\mathrm{OPH}}$ of the physical branch modulo implementation hiding and inert ancillary stabilization, not a unique microscopic representative. The genuinely noncentral topological case is also closed at fixed cutoff by the higher-gauge package of Theorems 5.17–5.20 and Proposition 5.28, so the fixed-cutoff topological UV surface is closed on the ordinary, central-defect, and crossed-module branches alike. Beyond fixed cutoff, Theorem 6.13 closes the support-visible continuum modular/geometric lift on the prime geometric subnet. On the bosonic compact-gauge lane, Theorems 5.22 and 5.26 supply the zero-obstruction fixed-stage category and symmetric braiding; the monoidal refinement ladder, compatible forgetful fiber, colimit, and realized cofinal witness additionally require Definition 7.2 before Theorems 7.3, 7.6, and 7.35 apply. The consensus paper also classifies that D1 lane explicitly: on each fixed finite patch net, the accepted relation induces the quotient maps locRep_λ and Rep_λ, normal-form computation is finite-state and decidable with the Lyapunov step bound, the layered carrier gives the finite $H_B \wedge H_{\mathrm{fib}}$ witness, the automatic approximate-stability inputs are the collar-local splice and record controls, and long-run noisy approximate consensus requires the separate fair-block contraction certificate. Computational universality for growing patch-net families is an expressive-power question in the consensus paper, not a premise for the Lorentz, local Einstein, compact-gauge, or realized Standard Model branches.

10.5 Why is charge quantization possible without a simple GUT?

The framework uses the global quotient and anomaly structure instead of a simple-group embedding. Once the realized matter content is fixed, the subgroup acting trivially on all states is exactly $\mathbb{Z}_6$, and the anomaly equations force the sixth-integer hypercharge lattice. Integer electric charge for color singlets is then a property of the realized quotient structure. A simple unified gauge group is not required.

11 Discussion

The framework is strongest where the outputs are discrete or structurally rigid:

1. the Lorentz branch on the support-visible extracted prime geometric cap pair;
2. the scaling-limit Einstein branch under the E0 dependency discharge, the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations, the null modular bridge, and the bounded-interval kernel;
3. the Standard Model gauge quotient chain on the explicit realized matter package;
4. exact hypercharges and structural electroweak force content on the realized one-generation matter package with one Higgs doublet;
5. the realized color triplet $N_c = 3$ and generation count $N_g = 3$ on the realized MAR-admissible branch;
6. the compact two-fixed-point quantitative layer, kept explicit as a secondary layer outside Phase I.

The local MaxEnt branch gives a finite-range interacting fixed-cutoff dynamics, Propositions 4.20 and 4.29 fix uniqueness of the UV branch only modulo OPH-stable equivalence, and Theorems 5.17–5.20 with Proposition 5.28 close the genuinely noncentral topological branch at fixed cutoff. The continuum BW/geometric lift is closed in the support-visible sense by Theorem 6.13 once the finite cap-normal certificate is carried. The unregularized full-algebra common-floor route is deliberately not claimed, because Proposition 6.17 shows that off-support directions can collapse while every finite stage remains faithful and Markov. The observer-facing theorem uses regularized modular transport, weak-$*$/GNS extraction of the support-visible prime cap pair, support-readable modular covariance, BW framing, oriented cross-ratio rigidity, and geometric 2π-KMS normalization with wrong-scale controls; this is the content needed for the Lorentz, null-stress, and local Einstein branches. Producing that finite cap-normal certificate from bare finite consensus is excluded by the underdetermination no-go of Theorem 6.32; on towers carrying the computable incidence/mesh/cross-ratio/normalization receipts it is produced from repair normal forms by Theorem 6.31. The parent Einstein branch-entry obligation is the production or empirical selection of those receipts, together with the D4-standardness and event-manifold receipt families of §6.2.1 and §6.3. With the stress, entropy-bridge, and uniform small-diamond packets of §6.6, the derivation chain is complete as the composed conditional theorem 6.127. Realized-branch nonemptiness, a repair tower that carries the finite receipts on its own output, is the branch-entry gate; every structural receipt family carries a realized witness at finite or one-particle level, and the margin-convergence, limit-clause, cap-interior, and physical-receipt obligations are work in progress (Remark 6.129).

Shared excitation dictionary and flavor theorem boundaries. The D10 quantitative-closure branch gives integrated gauge-coupling closure on the displayed carrier. The pixel fixed-point equation fixes the arithmetic bridge from P through $\alpha_U(P)$, the transmutation data, and the electroweak source anchor. The public Ward-projected Thomson endpoint is downstream of a low-energy transport layer: charged-lepton vacuum polarization is explicit, while the OPH-QCD hadronic precision backend remains a separate source payload. That backend must provide the QCD quotient ensemble, source QCD parameter map, Euclidean slab/vacuum-transfer construction, hadronic Hilbert quotient, Ward-normalized current ledger, two-current spectral export, same-scheme finite remainder, and no-target-leak DAG. Its two-current spectral measure is sufficient only for the running-α/HVP marginal; HLbL and rare-decay long-distance rows require higher-point and transition spectral data from the same backend. The exact W/Z chart is a compare-only validation sidecar on the public validation surface; source spectral measure payloads, same-scheme remainders, and interval certificates document that surface, but empirical hadron closure does not promote the fine-structure endpoint to a source-only theorem. On the declared runtime surface the running/matching packet is a declared-convention contract with displayed scope; a theorem deriving every coefficient, threshold, and conversion is outside that contract. The separate D6 cosmological-parameter package closes at the cosmic record-capacity fixed point $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$. On the flavor side the manuscript treats one shared OPH excitation dictionary as the only proof-facing family datum; lane-specific integers are not introduced independently. For each realized same-label refinement arrow e on the three-generation bundle, let ω_e be the same-label overlap holonomy scalar, let

$$d_e := 1 - \omega_e$$

be the derived edge defect, let g_e be the same-label gap scalar, and define

$$q_e := \sqrt{g_e d_e}, \qquad \eta_e := \log q_e - \frac{1}{3}\sum_f \log q_f, \qquad \mu_e := \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

These are the common descendants of overlap holonomy and edge-defect data. They exist once the same-label gap and overlap witnesses are fixed, and they are the shared family-side input used by the charged, quark, and neutrino continuations.

To include the charge-side bookkeeping explicitly, let s_e denote the realized sector-charge step carried by the same-label arrow e. For any lane-specific readout γ on the realized same-label transport graph, let

$$m_{\gamma,e} \in \mathbb{Z}_{\geq 0}$$

be the multiplicity with which γ traverses the elementary same-label excitation arrow e. Equivalently, along the realized transport path one has

$$s_\gamma = \sum_e m_{\gamma,e} s_e.$$

The associated excitation weight and excitation action are

$$A_\gamma := -\sum_e m_{\gamma,e} \log q_e = -\log\Big(\prod_e q_e^{m_{\gamma,e}}\Big), \qquad Y_\gamma^{\mathrm{exc}} := \prod_e q_e^{m_{\gamma,e}}.$$

So overlap holonomy, edge defects, gap data, and sector-charge transport determine the suppression law before any base choice is made. What earlier D12 language called a “texture exponent” or “defect count” is therefore not an extra primitive beyond the structural branch. The primitive flavor-side integers are the transport multiplicities $m_{\gamma,e}$. A one-number exponent in “units of $\log 6$” is only the compressed summary

$$n_\gamma^{(6)} := A_\gamma / \log 6,$$

and it becomes a literal defect count only after the additional uniform sixfold center-label collapse compatible with the realized

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

quotient. Without that extra collapse, this layer contains the multiplicity vector $m_{\gamma,\bullet}$ and the excitation action A_γ. A standalone flavor integer belongs only to the extra collapse.

The imported realized-branch input here is the D9 result that the same realized branch carries the color triplet $N_c = 3$ and the generation count $N_g = 3$. This shared dictionary feeds the downstream particle continuations. Their lane-specific theorem objects, exact witnesses, and quantitative closures are developed in Ref. [4]. On the compact-paper surface, the dictionary serves as the common family-side input beneath those continuations. Old base-6 exponents are compare-only compressions of excitation action and have no role as primitive flavor data.

Non-hadron output lane. For reader-facing outputs, the non-hadron surface is explicit and lane-split. The gauge and gravity chains fix connection and metric carrier roles. Under the additional action/background hypotheses of Theorem 7.31, their reduced quadratic kernels have

$$K_X^{\mathrm{phys}} = Z_X(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_X,$$

which is a classical or perturbative mode receipt rather than a zero-GeV particle-mass output. No photon, gluon, or graviton mass row is promoted until the quantum-particle receipt of Definition 7.30 passes. The electroweak quantitative-closure chain

$$\text{Axioms 1–5} \longrightarrow D7\text{–}D10$$

has an exact frozen authoritative repair sidecar

$$(M_W, M_Z) = (80.377,\ 91.18797809193725)\,\mathrm{GeV}$$

on the frozen authoritative repair surface, but that exact pair is compare-only beneath the source spectral measure payload, same-scheme remainder, and interval certificate. The D11 Higgs/top branch closes as one source-only split theorem on the declared D10/D11 quantitative surface. On the emitted D10 repair tuple $(\eta_{\mathrm{source}}, \beta_{EW}, \lambda_{EW}, \tau_{2,\mathrm{tree}}^{\mathrm{exact}}, \delta n_{\mathrm{tree}}^{\mathrm{exact}})$, define

$$\rho_{HT} = \log(1 + \tau_{2,\mathrm{tree}}^{\mathrm{exact}}),$$

$$R_T = -\tau_{2,\mathrm{tree}}^{\mathrm{exact}}\eta_{\mathrm{source}}^2 + \left(1 + \frac{\beta_{EW}}{28}\right)\eta_{\mathrm{source}}^6 + \frac{\eta_{\mathrm{source}}^8}{14} + \frac{\eta_{\mathrm{source}}^9}{27},$$
$$R_H = \eta_{\mathrm{source}}^5 - \frac{3}{25}\eta_{\mathrm{source}}^6 + \frac{\lambda_{EW}\eta_{\mathrm{source}}^6}{18} + \frac{\eta_{\mathrm{source}}^8}{2\beta_{EW}}.$$

Then the forward split coordinates are

$$\pi_y = \frac{\eta_{\mathrm{source}} + \left(\frac{3}{2} + \frac{\beta_{EW}}{4}\right)\rho_{HT} + R_T}{\sqrt{\pi}}, \qquad \pi_\lambda = \frac{\eta_{\mathrm{source}} - \left(\frac{4}{3} - \frac{\beta_{EW}}{54}\right)\rho_{HT} + R_H}{\sqrt{\pi}},$$

and the declared D11 Jacobian reads out

$$\delta y_t(\mu_t) = \pi_y\, y_t^{\mathrm{core}}(\mu_t), \qquad \delta\lambda(\mu_t) = -\frac{16}{9}\pi_\lambda\, \lambda^{\mathrm{core}}(\mu_t).$$

On that same declared surface,

$$m_H = 125.1995304097179\ \mathrm{GeV}, \qquad m_t^{D11} = 172.3523553288312\ \mathrm{GeV}.$$

At the precision quoted by the PDG Higgs listing, the Higgs row lands on the 2025 Higgs average. The same surface emits a companion top coordinate. It is not a separate public quark-mass row. The quark target audit uses a distinct PDG 2025 cross-section extraction coordinate. The bridge to the auxiliary direct-top PDG row is closed as a corpus-limited codomain no-go; the auxiliary row is compare-only. A source-side extraction-response kernel is outside the emitted corpus. The one-scalar companion seed

$$\sigma_{D11,\mathrm{HT}} = \frac{\alpha_U \cos(2\theta_{W0})}{\sqrt{\pi}}$$

is on disk only as the lower-rank fixed-ray companion branch with $\pi_y = \pi_\lambda = \sigma_{D11,\mathrm{HT}}$. The exact inverse slice on the same D11 Jacobian is compare-only and does not define the theorem surface. The same D11 Jacobian also has a compare-only exact inverse slice at the canonical Higgs/top reference pair

$$(m_H, m_t) = (125.1995304097179,\ 172.3523553288312)\,\mathrm{GeV}.$$

Detailed particle-spectrum continuation statements are carried by Ref. [4]. That companion paper develops the quark source-spread non-identifiability theorem and target-audit boundary, the charged-lepton determinant-line boundary, a target-informed weighted-cycle neutrino comparison candidate, the H^3 record-worldline stitch certificate for localized observer-visible record tokens, and the hadron execution boundary. After target and exact-witness ancestry are removed, the compatible ordered quark-spread data form a two-parameter positive fiber with a free $(\mathbb{R}_{>0})^2$ rescaling action. The source package therefore does not select the two family spreads. Selected-frame descent makes a supplied spread datum representative-independent, but cannot fix its value. Numeric quark rows remain outside the source-only prediction surface. The stored dimensionful quark matrices are mass-texture audit witnesses; a physical Yukawa claim additionally requires one common renormalization scale, threshold transport, the running Higgs expectation value, and dimensionless normalization. The neutrino candidate descends from a hand-written flavor template and fails the NuFIT 6.1 correlated $(\sin^2\theta_{23}, \delta_{\mathrm{CP}})$ profile [7]; it has no theorem or prediction status. The H^3 stitch certificate is conditional on a declared hyperboloid atlas, common clock line, real transverse interface, sector/gauge transport, ID-independent assignment gap, and refinement contraction check; it is not a species, mass, charge, or scattering-amplitude derivation. The hadron backend is a source-backend boundary with empirical closure policy documented. The compact paper promotes no source-only hadron masses. Empirical hadron closure values use a separate $e^+e^- \to$ hadrons payload class. These particle-continuation statements do not belong to the compact-paper claim surface.

The obstruction persists at the axiom level without adding any premise. MAR orders admissible packages by $C(\mathfrak{S}) = (\chi_{\mathrm{cpl}}, N_{\mathrm{nonab}}, N_c, N_g)$; Yukawa eigenvalues are not entries of this order. Starting from any generic admissible pair, independent positive rescalings of the centered up/down log spectra preserve the gauge representations, anomalies, hypercharges, Higgs content, generation count, CKM frames, CP capability, and MAR score, while changing the quark masses. The resulting packages are physically inequivalent because the physical-equivalence relation preserves Yukawa invariants. Hence Axioms 1–5 and fixed P admit a continuous family of equal-MAR-score quark spectra. Interpreting "minimal" as the smallest positive Yukawa does not repair the problem: the positive family has infimum zero and no positive minimum (and admitting zero selects massless/degenerate quarks). A unique numerical spectrum therefore cannot be obtained from the stated MAR definition.

Local unification surface and exact-release frontier. The local quantitative bridge can be stated without inflating the recovered-core claim. On the bosonic side, the declared pixel input P

fixes the D10/D11 trunk

$$P \longmapsto \alpha_U(P) \longmapsto \bigl(t_U(P), t_{\mathrm{tr}}(P)\bigr) \longmapsto v(P),$$

after which the D10 source basis fixes the declared electroweak transport package and the D11 source-only split theorem with

$$\rho_{HT} = \log\bigl(1 + \tau_{2,\mathrm{tree}}^{\mathrm{exact}}\bigr), \qquad (\pi_y, \pi_\lambda) \longmapsto (m_t, m_H)$$

by the declared D11 Jacobian, again with no inverse readback of the internal transmutation data from measured low-energy couplings. The D10/P/fine-structure chain is the declared arithmetic bridge on this quantitative surface. The electroweak W/Z and running-family rows carry separate public validation artifacts: the source spectral measure payload, same-scheme remainder, and interval certificate. The same D11 Jacobian then emits the Higgs/top branch. On the gravity side, the same D10 pixel law packages the shared edge entropy,

$$\bar{\ell}_{\mathrm{SU}(2)}(t_{2,\mathrm{run}}) \;+\; \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\mathrm{run}}) = P/4.$$

Together with $a_{\mathrm{cell}} = P\ell_\star^2$, the Newton area-law dictionary gives $G_{\mathrm{geom}} = \ell_\star^2$; the pixel P cancels. The exact local release burden consists of two theorem-sized objects: one strict classical-regime clause above the BW/Lorentz support package and one target-free D10 chart-identity theorem for exact W/Z. The familiar-unit readout is not a third theorem-sized burden. It is the explicit display package attached to the theorem-side data $\ell_\star^2$, P, c, and $\bar{\ell}_{\mathrm{shared}}$. On the gravity side, Proposition 5.42 gives

$$\bar{\ell}_{\mathrm{shared}} = \bar{\ell}_{\mathrm{SU}(2)}(t_{2,\mathrm{run}}) \;+\; \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\mathrm{run}}) = P/4,$$

and Proposition 5.43 gives the local SI readout

$$G_{\mathrm{SI}} = \frac{c^3\ell_\star^2}{\hbar}$$

from the selected scale certificate. The same proposition fixes the full familiar-unit package:

$$L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\,\widehat{L}(P), \qquad t_{\mathrm{loc}} = \frac{\sqrt{a_{\mathrm{cell}}}}{c_\star}\,\widehat{T}(P),$$

$$E_{\mathrm{loc}} = \frac{\hbar c_\star}{\sqrt{a_{\mathrm{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\mathrm{loc}} = \frac{\hbar c_\star}{k_B\sqrt{a_{\mathrm{cell}}}}\,\widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. Meters and seconds are therefore read from the single local ruler $\sqrt{a_{\mathrm{cell}}}$ together with the structural Lorentz output $c_\star$, while GeV and Kelvin are downstream familiar-unit displays of the inverse local ruler through $\hbar$ and k_B. On that declared extension surface the local release values are

$$c = 299792458\,\mathrm{m/s},$$
$$G = 6.674299995910528 \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2},$$

$$(M_W, M_Z, M_H) = (80.377,\ 91.18797809193725,\ 125.1995304097179)\,\mathrm{GeV}.$$

The structural result is the common invariant null cone and speed $c_\star$. The displayed decimal $c = 299792458\,\mathrm{m/s}$ is exact by the SI definition of the metre and is therefore a unit convention, not a predicted magnitude. The displayed G row is emitted by the selected scale certificate instead of by back-solving $a_{\rm cell}/P$ against the rounded benchmark 6.6743×10^{-11}. The same certificate turns the dimensionless $\Lambda_{\rm CRC}\ell_\star^2$ capacity relation into an SI static-patch display.

Particle-spectrum continuation boundaries beyond the compact SM/GR ledger are stated in Ref. [4].

The paper does not claim complete closure of all low-energy observables.[1]

12 Finite Quotient Ensembles and Simulation Claim Tiers

Finite quotient ensemble theorem surface. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any promoted physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\rm phys}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\rm phys}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\rm phys} = z_{r,b}B(\widetilde{\mathcal{H}}_r)^{G_r}z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

[1] Questions outside this theorem package do not alter the specific claim set established here. Even a separate habitat theorem for OPH state-and-law data would not enlarge this SM/GR derivation paper's Phase-I scope unless the actual closure map, invariant admissible sector, and stability hypotheses were also proved at the same tier.

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_\# \nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any map $N : Q \to Q_{\mathrm{nf}}$, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_\# \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_{\#}\mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#}\mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then

$$\|(c_{nr})_{\#}\mu_n - \mu_r\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0}\left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r\right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#}\mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \qquad \sum_{x : \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x,y)$ descends to Q_r only when

$$P_Q(q,q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x,y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q,q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q,q') = \min\left\{1, \frac{w(q')R(q',q)}{w(q)R(q,q')}\right\}$$

gives detailed balance

$$\mu(q)R(q,q')a(q,q') = \mu(q')R(q',q)a(q',q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound γ_*, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \leq \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \leq 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Vacuum promotion gate. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. Vacuum promotion requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \geq 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\text{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. Continuum promotion additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance is not automatically a primordial curvature spectrum. The map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has a null space unless a radial prior, finite parametric family, or source-stress bridge is declared. OPH primordial promotion requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, screen-to-radial-lift, radial-null-space, and forward-projection receipts. Observable CMB promotion further requires frozen source, solver, likelihood, no-data-use, pooled-reducer, and falsification-rule receipts.

Claim tiers and required receipts. Every ensemble-facing run records immutable ensemble id, claim tier, regulator id, representative schema hash, gauge action hash, canonicalizer hash, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler hash, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt, not to the ensemble definition. The claim tiers are

$$\begin{aligned}
E0 &: \text{ seed noise, proposal noise, repair jitter,}\\
E1 &: \text{ conventional reference ensemble,}\\
E2 &: \text{ OPH-native quotient ensemble,}\\
E3 &: \text{ OPH vacuum,}\\
E4 &: \text{ OPH primordial field,}\\
E5 &: \text{ observable cosmological prediction.}
\end{aligned}$$

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

13 Conclusion

Observer-Patch Holography starts from overlap consistency of observer patches and yields, within one framework, several effective structures usually treated separately in modern fundamental physics. This SM/GR derivation paper supports a unique schedule-independent quotient normal form for overlap repair from each fixed initial physical quotient state, with same-boundary uniqueness only under the additional unique-extension condition, and hence a quotient-level physical UV branch only modulo boundary redundancy, implementation hiding, and inert ancillary stabilization. It also supports a Lorentz branch on the extracted prime geometric cap pair, a Jacobson-type Einstein branch whose OPH5 recovered-core dependencies are discharged by Theorem 6.134, the global closure of that same gravity lane at the cosmic record-capacity fixed point, the Standard Model gauge quotient with exact hypercharges and structural electroweak force content on the realized matter package on the realized MAR one-Higgs branch in the bosonic internal-gauge sector, and the realized color triplet $N_c = 3$ and generation count $N_g = 3$ on that branch. After an independently supplied Maxwell action, the electromagnetic connection branch gives Maxwell's equations and two classical transverse modes; a photon particle remains conditional on the quantum receipt. The paper also gives a conditional support-visible compact-gauge Yang–Mills form together with repair-gap accounting of the Yang–Mills mass gap under the separately named finite and continuum receipts, and a separate integrated quantitative-closure branch. Phenomenological continuations are discussed separately from the recovered core.

The framework therefore presents general relativity and the Standard Model as effective descriptions arising from the five-axiom basis together with the support-visible BW scaling theorem, geometric-subnet construction, and categorical gauge reconstruction. The local quantitative claim surface is equally explicit: the declared pixel fixed point P controls the electroweak/Higgs mass trunk and the shared cell/edge identity used by the Newton area law, the selected no-G scale certificate controls the classical-regime gravity normalization through $\ell_\star^2 = 3\pi/B_\star$, the invariant causal speed $c_\star$ is the structural Lorentz output, and the familiar-unit package reads meters, seconds, GeV, and Kelvin from the single local ruler $\sqrt{a_{\text{cell}}} = \sqrt{P}\,\ell_\star$ with $\hbar$ and k_B used only as display conventions. The exact SI coordinate $c = 299792458\,\text{m/s}$ remains a unit convention, not a predicted decimal. Separate continuation boundaries concern sharper quantitative control, a fuller fermionic gauge branch, and theorem-side local exact-release objects beyond the corpus-backed surfaces stated here.

A Supplemented Gravity and Structural Gauge Details

This appendix carries the gravity-side and structural gauge items that support the compact paper's technical surface. They sharpen the compact paper's Lorentz, Einstein, cosmological-constant, and product-group claim surfaces without widening the recovered core beyond its stated branch conditions.

A.1 Lorentz and Einstein Bridge

Theorem A.1 (Conformal group isomorphism)**.** *The orientation-preserving conformal group of S^2 is isomorphic to the connected Lorentz group:*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

Theorem A.2 (Support-visible geometric modular flow on caps on the extracted prime geometric subnet)**.** *Assume the OPH axioms, the derived fixed-cutoff collar package, and the support-visible*

BW scaling theorem on the extracted prime geometric subnet. Then the scaling-limit modular automorphism group of the cap pair is geometric conformal dilation on that subnet, with the standard 2π normalization supplied by the finite cap-normal certificate. If the emitted scaling-limit cap algebra is type I, this may be written as $K_C = 2\pi B_C + Z_C$ with Z_C central; in the generic continuum case the theorem is the automorphism statement on a non-type-I cap algebra with outer geometric modular action. The collar replacements used in this branch are exact only at exact Markovity or in controlled fixed-collar families with $\delta^{\mathrm{M}} \to 0$; small CMI supplies a Fawzi–Renner recovered comparison state, not a dimension-free one-shot exact Markov normal form.

Definition A.3 (BW-branch observer-relative time reading)**.** *On the branch satisfying the hypotheses of the preceding geometric modular-flow theorem, OPH uses the modular automorphism parameter t of the extracted cap pair as that cap observer's relative time parameter. This is a declared BW-branch reading of the derived geometric modular flow; it is not an additional proof that arbitrary operational clocks, global time, or the full problem of time have been derived from the axioms.*

BW-side UV scaffold. The UV data on this branch are not a separate cap-isotropy or Euclidean-regularity selector. The fixed-cutoff cap algebras are type-I regulators, but the scaling-limit observer algebra may leave that class, and MaxEnt alone does not select the BW / canonical cap phase. The UV package is the realized transported geometric cap-local system together with the carried-collar schedule derived from the transported fixed-local-collar Markov/faithfulness datum on each fixed local collar model, followed by support-readable modular covariance and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair.

Null-strip completion boundary. On the D4 side, the fixed-cutoff strip package is more specific than the generic inherited-strip summary. The null cuts first transfer the same cut-center data as the spatial collar branch, which fixes the central sector-pair decomposition of the strip algebra. The stronger left/right tensor decomposition used by the null modular bridge then requires the extra inherited strip-split condition on the multiplicity spaces, together with the exact-or-controlled Markov hypotheses on one fixed inherited strip model. On that same fixed strip model, the renormalized half-line family is endpoint-Lipschitz, hence defines the weak tail generator, and on the scaling-limit geometric-cap branch the half-line blow-up net carries the derived half-sided modular pair. Borchers–Wiesbrock then supplies the positive null-translation generator on its Stone domain together with the affine half-line modular relation $K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$, and the same half-line family fixes the generator/charge identification internally. Bounded-interval formulas are downstream of the E0.5 affine/projective kernel in the Einstein bridge.

Lemma A.4 (Null data ambiguity)**.** *If a symmetric tensor X_{ab} satisfies*

$$X_{ab}k^a k^b = 0$$

for every null vector k, then $X_{ab} = \phi\, g_{ab}$ for some scalar ϕ.

Corollary A.5 (Null modular data determine Einstein only up to the metric term)**.** *Null modular data determine T_{ab} only up to ϕg_{ab}. Consequently the Einstein equation is fixed locally only up to Λg_{ab}.*

Theorem A.6 (Jacobson-type rest-frame relation)**.** *In the local Lorentzian scaling regime, once the realized cap-label-preserving MaxEnt family satisfies the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap variations, the half-line generator/charge identification* *of the null bridge, the bounded-interval transport input used in the local Lorentzian regime, and the internal small-ball bridge yield the rest-frame first-variation relation that drives the compact paper's local Einstein branch; on the same scaling branch, if that rest-frame relation holds for all local observer four-velocities and all reference states, the compact paper upgrades it to the full tensor equation by the explicit local quadratic-polarization argument of Corollary 6.107.*

Theorem A.7 (Newton coupling from the scale certificate and edge entropy density)**.** *The geometric coupling read by the Newton area law is*

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

On the OPH gravity row the scale is supplied first by $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, as an independent scale certificate rather than a consequence of $P_\star$ and $N_\star$ alone. With $a_{\mathrm{cell}} = P\ell_\star^2$ and $\bar{\ell}_{\mathrm{shared}} = P/4$, this gives

$$G_{\mathrm{geom}} = \ell_\star^2, \qquad G_{\mathrm{SI}} = \frac{c^3 \ell_\star^2}{\hbar}.$$

Here $\bar{\ell}_{\mathrm{shared}}$ is a shared-cut density. It is not the logarithm of an independent cell Hilbert-space dimension and does not select a fixed primitive observer capacity.

A.2 Cosmological-Constant / Screen-Capacity Closure

Lemma A.8 (Vacuum energy blindness)**.** *For any null vector k,*

$$T_{kk}^{\mathrm{vac}} = T_{ab}^{\mathrm{vac}} k^a k^b = -\rho_{\mathrm{vac}}\, g_{ab} k^a k^b = 0.$$

Vacuum-energy contributions therefore lie in the kernel of the null map $T_{ab} \mapsto T_{kk}$.

Proposition A.9 (Structural separation)**.** *Within the structural split used here:*

1. *local modular/null data fix T_{ab} only up to ϕg_{ab}; and*
2. *the dimensionless Λ-capacity relation is fixed by global screen capacity, not by local null data.*

So the large vacuum-energy bookkeeping of EFT is not itself the local quantity determining curvature on this branch.

Theorem A.10 (No local Λ prediction)**.** *Within the null-modular reconstruction used here:*

1. *all T_{kk} data are unchanged under $T_{ab} \to T_{ab} + \phi g_{ab}$;*
2. *therefore Λ cannot be fixed by local overlap consistency alone; and*
3. *the dimensionless Λ-capacity relation requires the global screen-capacity closure that closes the same Einstein branch beyond the local null data; the zero-input closure is the cosmic record-closure fixed point $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$, with $N_\star$ as its count-density representation, and SI curvature values additionally use the selected scale certificate.*

Corollary A.11 (Benchmark cosmological readout on the D6 branch)**.** *On the same D6 branch as Corollaries 6.132–6.142, inserting the observed value $\Lambda \approx 1.09 \times 10^{-52}\,\mathrm{m}^{-2}$ gives*

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}} \approx 1.66 \times 10^{26}\ \mathrm{m}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c} \approx 17.5\ \mathrm{Gyr},$$

$$N_{\mathrm{patch}} = \left(\frac{r_{\mathrm{dS}}}{\ell_P}\right)^2 \approx 1.05 \times 10^{122}, \qquad N_{\mathrm{scr}} = S_{\mathrm{dS}} = \pi N_{\mathrm{patch}} \approx 3.31 \times 10^{122},$$

$$\Lambda \ell_P^2 \approx 2.85 \times 10^{-122}.$$

Proof. Immediate from Corollary 6.142. □

Observed-age benchmark boundary. The branch quantity t_Λ is the de Sitter static-patch timescale. The usual cosmic age t_0 is a compare-only FLRW benchmark, with no additional D6 theorem output. On the flat ΛCDM benchmark

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_\Lambda}}\sinh^{-1}\left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\right)$$

one gets the standard comparison value $t_0 \approx 13.8\,\mathrm{Gyr}$ at $H_0 \approx 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_\Lambda \approx 0.685$.

Lemma A.12 (FLRW curvature as visible scalar holonomy)**.** *On a homogeneous-isotropic spatial slice with constant sectional curvature K, small spatial loop holonomy obeys*

$$\mathrm{Hol}_{\square_{uv}} = \exp\Big(K A_\square J_{uv} + O(A_\square^{3/2})\Big)\,.$$

On a visibly separated OPH refinement system, the refinement-limit scalar spatial holonomy vanishes if and only if $K = 0$. Thus a flat FLRW branch is the zero-visible-spatial-holonomy branch.

Flatness boundary. This holonomy statement names a conditional cosmology bridge only. It does not add a D6 theorem output and does not solve the inflationary flatness or horizon problems. Selecting $K = 0$ requires an additional continuation theorem or premise: a direct flatness theorem, a conditional cosmological-minimal-holonomy selector on a clocked FLRW boundary, or an explicit flat-branch assumption. Minimal Admissible Realization (MAR) acts on low-energy gauge/matter packages and is not a cosmological flatness selector.

Theorem A.13 (D5–D6 cosmological-capacity closure stack)**.** *Assume the compact paper's local Einstein branch, the cosmic record-capacity fixed point*

$$N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}}),$$

its observed-branch de Sitter entropy readout

$$N_{\mathrm{CRC}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard static-patch formulas

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Then the cosmological-constant package is one local/global theorem stack: local null data fix the Einstein branch only modulo Λg_{ab}, and with the selected scale certificate the same branch has the global display

$$G_{ab} + \frac{3\pi}{GN_{\mathrm{CRC}}}\,g_{ab} = 8\pi G\,\langle T_{ab}\rangle,$$

while the D6 closure itself fixes the entropy and dimensionless capacity relations

$$S_{\mathrm{dS}} = N_{\mathrm{CRC}}, \qquad A_{\mathrm{dS}} = 4GN_{\mathrm{CRC}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c},$$

and the observed cosmic age is a downstream FLRW benchmark, with no additional theorem output.

Scope boundary. The D6 hypotheses are exactly the D5 local Einstein branch, the cosmic record-capacity fixed point $N_{\mathrm{CRC}} = F(N_{\mathrm{CRC}})$, its observed-branch de Sitter entropy readout $N_{\mathrm{CRC}} = S_{\mathrm{dS}}$, the standard de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed by the readback closure. The capacity closure fixes $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$; the SI static-patch scale additionally requires the selected scale certificate. CMB kernels, inflation-replacement claims, H_0/S_8 branches, dark/anomaly growth kernels, and baryogenesis continuations require separate continuation theorems or likelihood contracts.

Theorem A.14 (Conditional OPH screen-spectrum theorem)**.** *Let $(\mathcal{S}_r)$ be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar q_r, removal of the background and dipole sector, and a target-free positive quadratic repair operator K_r. If $K_r \to K$, the finite scalar release energy satisfies $E_r/d_r \to A_q$, and local MaxEnt is imposed at fixed expected quadratic release energy, then q_r converges in finite harmonic distributions to the centered Gaussian screen field with covariance $A_q K^{-1}$. Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is $t^{\theta/2}\,dt/\Gamma(1+\theta/2)$, then*

$$K_{\mathrm{asy}} = (-\Delta_{S^2})^{1+\theta/2}, \qquad C^q_{\ell,\mathrm{asy}} = A_q[\ell(\ell+1)]^{-1-\theta/2},$$

as the large-ℓ asymptotic model. The theorem-grade finite-ℓ conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell+2+\theta/2)}{\Gamma(\ell-\theta/2)},$$

with

$$C^q_\ell = A_q\frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}.$$

If q is additionally certified as the thin-shell pullback of a homogeneous curvature field with $\Delta^2_\zeta(k) = A_\zeta(k/k_\star)^{-\theta}$, the corresponding inverse amplitude relation is

$$A_\zeta = \frac{A_q^{\mathrm{shell}}\Gamma(3/2+\theta/2)}{\pi^{3/2}Z_\star^2(k_\star D_\star)^\theta\Gamma(1+\theta/2)}.$$

The numerical value of θ, the expected quadratic release energy, and the lift normalization remain conditional on their finite certificates. A Fawzi–Renner/Markov-collar observable estimate gives an upper bound on release observables, not an amplitude equality. No physical TT/TE/EE statement follows without independent Boltzmann transfer and likelihood closure.

Screen-spectrum boundary. The theorem licenses a Phase III conditional continuation only. A derived screen spectrum requires the quotient-normal-form scalar field, positive repair operator, operator-tilt, and scalar-release energy receipts. A derived primordial curvature spectrum additionally requires the radial lift receipt: the heuristic $\ell \simeq kD_\star$ assignment is only a scaffold, while the exact thin-shell power-law lift uses the gamma-ratio operator above and finite-width windows require their own Bessel-kernel error certificate.

Separate pixel-side benchmarks. The companion pixel numbers

$$a_{\mathrm{cell}} \approx 1.63\,\ell_\star^2, \qquad \ell_{\mathrm{UV}} = \sqrt{a_{\mathrm{cell}}} \approx 1.28\,\ell_\star, \qquad \bar{\ell} \approx 0.408$$

belong to the separate pixel-closure/local-readout package, outside the D6 cosmological-parameter corollary itself. After the selected scale certificate is supplied, $\ell_\star$ is displayed as the usual Planck length.

A.3 Product-Group Structural Corollaries

Theorem A.15 (Factorization equivalence)**.**

Factorizing edge weights $\Longleftrightarrow$ *Additive boundary Laplacian* $\Longleftrightarrow$ *Product gauge group.*

If

$$H_\partial = H_\partial^{(1)} + H_\partial^{(2)} + H_\partial^{(3)} \qquad \textit{with} \qquad [H_\partial^{(i)}, H_\partial^{(j)}] = 0,$$

then

$$p(R_1, R_2, R_3) \propto \prod_{i=1}^{3} d_{R_i} e^{-t_i C_2(R_i)}.$$

Applied to the assumed transportable refinement-directed edge-sector colimit on the realized branch, together with the rigid symmetric C^*-tensor and faithful bosonic fiber-functor conditions of the compact-gauge theorem, Tannaka–Krein reconstruction first yields some compact group G. Under the MAR admissibility package, and once the admissible class includes one connected abelian charge factor, the selected connected Lie realization is $SU(3) \times SU(2) \times U(1)$ up to finite quotient.

Corollary A.16 (No leptoquarks, hence no gauge-mediated proton decay)**.** *With product gauge group, the adjoint representation of the full connected gauge group is*

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0),$$

equivalently the derived nonabelian adjoint is

$$(8, 1, 0) \oplus (1, 3, 0).$$

There are therefore no gauge generators in mixed representations $(3, 2, \pm 5/6)$*, i.e. no simple-GUT X, Y bosons on the realized branch. The structural claim is exactly*

$$\tau_p^{(\text{gauge})} = \infty,$$

meaning that the gauge-boson proton-decay channel is absent. Scalar-mediated or higher-dimensional baryon-violating operators are outside this structural corollary.

Proton-spin continuation benchmark. On the realized D8–D9 branch the color factor is SU(3), so the structural OPH input for proton-spin bookkeeping is

$$C_F = \frac{4}{3}, \qquad C_A = 3.$$

If one adds the QCD-dependent continuation ansatz that quark/gluon spin sharing equilibrates according to these Casimirs, then

$$\Delta\Sigma \approx \frac{C_F}{C_F + C_A} = \frac{4}{13} \approx 0.308.$$

Against a representative lattice benchmark $\Delta\Sigma \approx 0.286$, this lands within about 8%, but it is only a deferred continuation benchmark. A first-principles OPH derivation of proton spin would require the nonperturbative light-quark/hadron completion plus an explicit map from OPH data to renormalized proton matrix elements.

A.4 Controlled Worldsheet Effective Description

Exact OPH-to-2D-Yang–Mills bridge. On the compact-group heat-kernel branch, Theorem 8.1 proves the exact identity

$$Z_{\text{edge}}(t) = K_t(1)$$

by gluing the open-edge weights $p_R(t) \propto d_R e^{-tC_2(R)}$ into the closed partition sum

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

This is the precise theorem-level sense in which the OPH edge-sector partition reorganizes into the two-dimensional Yang–Mills heat-kernel surface, and the Chapman–Kolmogorov law supplies the matching collar-sewing rule. This bridge is a two-dimensional heat-kernel partition identity on the stated compact-group branch. The four-dimensional compact-gauge repair-gap theorem is the separate support-visible result in Theorem A.37.

Large-N_{edge} boundary. The large-N_{edge} worldsheet interpretation is carried only when one fixes a distinct large-N_{edge} sequence, with $N_{\text{edge}} \neq N_c = 3$, a fixed-$\tau$ window for

$$\tau = tN_{\text{edge}},$$

and the uniform genus-remainder control of Theorem 8.2. On that branch the edge free energy satisfies the compact paper's theorem-level criterion for a controlled genus expansion, so the standard Gross–Taylor rewriting becomes a controlled worldsheet effective description of edge dynamics.

External items. The exact bridge theorem uses the compact-group heat-kernel branch and the quadratic-Casimir normalization declared in the compact paper. The continuation theorem uses the declared large-N_{edge} regime and the imported Gross–Taylor large-N worldsheet dictionary for two-dimensional Yang–Mills. It does not identify a critical worldsheet CFT. Worldsheet supersymmetry, critical dimension, modular invariance, anomaly cancellation, GSO projection, and full massless-spectrum matching use separate continuation inputs beyond the compact recovered-core chain.

A.5 Support-Visible Yang–Mills Gap from Repair Dynamics

Assumption A.17 (Support-visible compact-gauge Yang–Mills branch)**.** *The compact-gauge branch used in this subsection is the ordinary or central zero-obstruction compact-gauge sector with compact simple structure group G, a four-dimensional Euclidean scaling chart, and a cofinal tail carrying the compact-gauge refinement receipt of Definition 7.2. It also has a reflection-positive ordinary vacuum, topological angle $\theta = 0$, gauge-invariant local finite-constraint MaxEnt/Gibbs refinement, no additional relevant dimension-four pure-gauge operator on the branch besides the positive quadratic curvature invariant, active exact-Markov repair collars, bounded-color collar covers, and repair completeness. It is carried on the separated cofinal regulator system of Theorem 4.22. No continuum cylinder state, GNS space, transfer generator, or vacuum projection is included in this assumption; those objects are constructed, or separately certified, below.*

Definition A.18 (Finite support-visible compact-gauge cylinder system)**.** *At regulator r, make the following construction twice: for the finite four-dimensional Euclidean slab $\Lambda_r^{(4)} = (V_r^{(4)}, E_r^{(4)})$ cut out by the scaling chart, and for its time-zero spatial slice $\Lambda_r^{(0)} = (V_r^{(0)}, E_r^{(0)})$. In the formulas below, $\Lambda_r = (V_r, E_r)$ denotes either choice and the superscript is restored whenever the two must be* *distinguished. Let D_r be the finite collection of repaired collar records and zero-obstruction sector labels retained by the compact-gauge patch carrier. In the gauge-register presentation, let*

$$\widetilde{X}_r \subseteq G^{E_r} \times D_r$$

be the closed set satisfying the finite Gauss, overlap, and repaired-record constraints, let $\mathcal{G}_r = G^{V_r}$ act by endpoint gauge transformations, and let $\sim_{\mathrm{ov}}$ be the closed equivalence relation of support-visible overlap/presentation indistinguishability. The finite-stage support-visible configuration space is

$$X_r := (\widetilde{X}_r/\mathcal{G}_r)/{\sim_{\mathrm{ov}}}.$$

Thus X_r is compact Hausdorff. "Finite stage" means that Λ_r and D_r are finite; X_r need not be a finite set when G is a compact Lie group. A finite quantum-link truncation gives the same commuting Euclidean readout algebra after restriction to its joint cylinder spectrum; no noncommutative support-quotient claim is made here.

For each active collar C, let

$$\rho_{C,r} : X_r \longrightarrow Y_{C,r}$$

be the continuous complete repaired readback. On the classical compact-holonomy branch used by the repair lemma, the complementary conditional fiber contains every record or holonomy coordinate actually resampled by collar repair. It is either finite with its uniform conditional law or a standard atomless probability space with its conditioned MaxEnt/Haar law. The complete readback retains exactly the fixed repaired datum, not every pre-repair holonomy; otherwise nonconstant Wilson cylinders would incorrectly lie in the repair kernel.

For a bounded cell region $O \subset \Lambda_r$, let $\mathfrak{C}_r^{G,\mathrm{sv}}(O)$ be the unital *-algebra generated by

(a) *contracted Peter–Weyl spin-network functions, Wilson-loop characters, and any explicitly retained gauge-invariant boundary-carrier contractions supported in O;*

(b) *the overlap-sector projectors whose collars lie in O; and*

(c) *continuous functions of the complete repaired readbacks $\rho_{C,r}$ for those collars.*

Its uniform closure

$$\mathcal{A}_r^{G,\mathrm{sv}}(O) := \overline{\mathfrak{C}_r^{G,\mathrm{sv}}(O)}^{\|\cdot\|_\infty} \subseteq C(X_r)$$

is the finite-stage local cylinder algebra. Equivalently, start from the freely presented readout-generator *-algebra and quotient by the kernel of its evaluation homomorphism on X_r, then complete. That kernel is a closed two-sided overlap-trivial ideal fixed before a state is chosen. The relation $\sim_{\mathrm{ov}}$ identifies exactly the configurations on which all declared spin-network, sector, and repaired-readback generators agree. These generators are self-adjoint, contain the constants, and therefore separate points of X_r. Stone–Weierstrass gives

$$\mathcal{A}_r^{G,\mathrm{sv}}(\Lambda_r) = C(X_r).$$

Consequently, for any selected stage measure π_r, their GNS cylinder closure is the full $L^2(X_r, \pi_r)$ after the finite π_r-null support reduction.

Write the resulting spaces and algebras as

$$(X_r^{(4)}, \mathcal{A}_r^{G,\mathrm{sv},(4)}(O)) \quad \textit{and} \quad (X_r^{(0)}, \mathcal{A}_{r,0}^{G,\mathrm{sv}}(B)).$$

Restriction of a Euclidean history to the time-zero slice gives a continuous map

$$\tau_{0,r} : X_r^{(4)} \longrightarrow X_r^{(0)}$$

and the corresponding pullback embeds time-zero cylinders into four-dimensional cylinders.

For $r \preceq s$, the deterministic coarse shadow multiplies the ordered fine-edge holonomies lying over each coarse edge, restricts collar readbacks, and sends a fine sector label to its coarse shadow. Gauge covariance makes this a map

$$p_{sr} : X_s \longrightarrow X_r, \qquad p_{tr} = p_{sr} \circ p_{ts} \quad (r \preceq s \preceq t).$$

The four-dimensional and time-zero coarse shadows satisfy the finite commuting square

$$\tau_{0,r} p_{sr}^{(4)} = p_{sr}^{(0)} \tau_{0,s}.$$

On local cylinders it gives the unital *-map

$$\iota_{rs}^{O} : \mathcal{A}_r^{G,\mathrm{sv}}(O) \longrightarrow \mathcal{A}_s^{G,\mathrm{sv}}(O_s), \qquad \iota_{rs}^{O}(a) = a \circ p_{sr}.$$

On the compact-gauge refinement branch, the sector-projector part must agree with the separately supplied block-multiplicity injection $\jmath_{rs}^{B}$ of Definition 7.2; it is not inferred from the deterministic coarse-shadow map. The holonomy and readback parts are its declared compatible extension. The maps preserve isotony and obey

$$\iota_{rt}^{O} = \iota_{st}^{O_s} \iota_{rs}^{O}.$$

If every coarse configuration on the selected branch has a fine extension, p_{sr} is surjective and ι_{rs}^{O} is injective. Without that finite extendability receipt, the C^*-inductive limit quotients the common refinement-null kernel; the extraction proof below does not assume prelimit injectivity of the full cylinder map. That broader quotient extraction is not, by itself, the receipt-certified sector ladder of Theorem 7.3. Write $\mathcal{A}_r^{G,\mathrm{sv}} := \mathcal{A}_r^{G,\mathrm{sv}}(\Lambda_r)$. Choose a countable cofinal regulator sequence and a countable bounded-region exhaustion of the four-dimensional chart. This is the cylinder family used below. For a non-countable regulator presentation, the same construction uses the full directed family and a subnet instead of this cofinal sequence.

Proposition A.19 (Projective compact-gauge cylinder extraction)**.** *For every regulator s, let $\mu_s^{(4)}$ be the declared finite quotient-Gibbs probability measure on $X_s^{(4)}$, let*

$$\pi_s := (\tau_{0,s})_{\#} \mu_s^{(4)}$$

be its stationary time-zero marginal on $X_s^{(0)}$, and put $\mu_s^{(0)} := \pi_s$. For $d \in \{4, 0\}$, write

$$\omega_s^{(d)}(a) := \int_{X_s^{(d)}} a \, d\mu_s^{(d)}, \qquad \omega_{s\downarrow r}^{(d)} := \omega_s^{(d)} \circ \iota_{rs}^{(d)}.$$

The statements below hold for both $d = 4$ and $d = 0$; the superscript is suppressed in the displayed proof. Then the following statements hold.

(i) *The marginals of any one fine state are exactly projective: for $q \preceq r \preceq s$,*

$$\omega_{s\downarrow r} \circ \iota_{qr} = \omega_{s\downarrow q}.$$

(ii) *There is a cofinal subsequence, or a cofinal subnet in the general directed case, and states* $\overline{\omega}_r \in S(\mathcal{A}_r^{G,\mathrm{sv}})$ *such that*

$$\omega_{s_\alpha \downarrow r} \underset{\alpha}{\to} \overline{\omega}_r \quad \textit{weak-*}$$

on every fixed local cylinder algebra. The limit family is projective:

$$\overline{\omega}_r = \overline{\omega}_t \circ \iota_{rt} \qquad (r \preceq t).$$

(iii) *The compatible local states define a state* ω_∞ *on the algebraic inductive limit and extend uniquely to its local* C^**-completion*

$$\mathcal{A}_\infty^{G,\mathrm{cyl}}(O) := \overline{\varinjlim_r \mathcal{A}_r^{G,\mathrm{sv}}(O_r)}.$$

The resulting local algebras are isotone and their union is dense in the global cylinder algebra.

(iv) *Let* $\mathcal{N}_\omega(O)$ *be the null ideal of the limiting cylinder measure,*

$$\mathcal{N}_\omega(O) := \{a \in \mathcal{A}_\infty^{G,\mathrm{cyl}}(O) : \omega_\infty(a^*a) = 0\}.$$

Because the Euclidean cylinder algebra is commutative, this is a closed two-sided ideal. On

$$\mathcal{A}_\infty^{G,\mathrm{supp}}(O) := \mathcal{A}_\infty^{G,\mathrm{cyl}}(O)/\mathcal{N}_\omega(O)$$

the induced state is faithful. The local GNS spaces glue isometrically, their local cylinder images have dense union, and their Hilbert direct limit is the GNS space $(K^{(d)}, \Pi^{(d)}, \mathbf{1}^{(d)})$ *of* $\omega_\infty^{(d)}$*. Denote the four-dimensional Euclidean cylinder GNS space by* $K^E := K^{(4)}$ *and the time-zero repair GNS space by* $K := K^{(0)}$*. Each is a faithful cyclic GNS pair on its support quotient; the time-zero vector is identified as the physical vacuum only by the transfer/OS receipt below.*

If, in addition, the finite four-dimensional quotient presentation satisfies the fiber-sum identity stated after Axiom 3.7, then $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$*. The time-zero commuting square gives* $(p_{sr}^{(0)})_\# \pi_s = \pi_r$*, so both original state families are projective. Without that finite receipt, the bars on* $\overline{\omega}_r$ *are essential: compactness extracts a projective cluster family but does not turn the originally chosen coarse Gibbs measures into an exactly projective family. In the continuous compact-holonomy presentation, the corresponding receipt is the pushforward identity* $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$*, expressed through disintegration rather than a finite sum.*

Proof. Functoriality of the coarse shadows gives

$$(\omega_s \circ \iota_{rs}) \circ \iota_{qr} = \omega_s \circ \iota_{qs},$$

which proves (i) without a continuum assumption.

For each fixed cylinder algebra, its state space is weak-* compact by Banach–Alaoglu. The algebras in the chosen cylinder exhaustion are separable: compact metrizable G has a countable Peter–Weyl test algebra, and only finitely many readbacks and sector projectors occur at one stage. Their state spaces are therefore weak-* metrizable. Starting with the first cylinder, extract a subsequence on which its marginals converge; from that subsequence extract one for the second cylinder, and continue. Choose the k-th diagonal index beyond the k-th regulator stage. The resulting diagonal subsequence is cofinal and converges on every cylinder. In the general directed presentation, compactness of the product of the local state spaces gives the same conclusion with a cofinal subnet: unresolved early coordinates may be filled by arbitrary states because every fixed coordinate is genuinely resolved on a cofinal tail. For $q \preceq r$, weak-* continuity of precomposition by ι_{qr}, together with (i), gives

$$\overline{\omega}_r \circ \iota_{qr} = \lim_{\alpha} \omega_{s_\alpha \downarrow r} \circ \iota_{qr} = \lim_{\alpha} \omega_{s_\alpha \downarrow q} = \overline{\omega}_q.$$

This proves (ii).

If $a \in \mathcal{A}_r^{G,\mathrm{sv}}$ represents an element $[a]_r$ of the algebraic inductive limit, set

$$\omega_\infty([a]_r) := \overline{\omega}_r(a).$$

Projectivity makes this independent of the representative. Positivity and normalization hold at the finite stage containing any given algebraic element. Moreover, for every $t \succeq r$,

$$|\overline{\omega}_r(a)| = |\overline{\omega}_t(\iota_{rt}(a))| \leq \|\iota_{rt}(a)\|.$$

Taking the infimum along the directed tail gives the C^*-inductive-limit norm bound, including when the maps are noninjective. Hence the state extends uniquely to the completion. The same construction region by region proves isotony and density, giving (iii).

By the Riesz representation theorem, a commutative local limiting state is integration against a Radon probability measure μ_O. Its null ideal consists precisely of the continuous functions vanishing on the measure support, so the quotient is canonically $C(\operatorname{supp} \mu_O)$ and the induced state is faithful. If $O \subseteq O'$ and $\iota_{OO'}$ is the local inclusion, state compatibility gives

$$a \in \mathcal{N}_\omega(O) \quad \Longleftrightarrow \quad \iota_{OO'}(a) \in \mathcal{N}_\omega(O').$$

Thus the local inclusions descend to injective maps of the support quotients, which therefore form an isotone faithful local net. For $r \preceq t$, projectivity makes

$$V_{rt} : K_{\overline{\omega}_r} \longrightarrow K_{\overline{\omega}_t}, \qquad V_{rt}[a]_r = [\iota_{rt}(a)]_t$$

well defined and isometric, because

$$\|V_{rt}[a]_r\|^2 = \overline{\omega}_t(\iota_{rt}(a^*a)) = \overline{\omega}_r(a^*a).$$

The maps compose, preserve $\mathbf{1}$, and intertwine left multiplication. Their Hilbert direct limit is therefore the GNS completion of the algebraic cylinder union and has dense local images. This proves (iv). Finally, suppressing the $d = 4$ superscript, the finite fiber-sum identity gives

$$Z_s = \sum_{x \in X_r} \sum_{x' : p_{sr}(x') = x} m_s(x') e^{-S_s(x')} = \alpha_{sr} Z_r,$$

and hence

$$(p_{sr})_\# \mu_s(x) = \frac{\alpha_{sr} m_r(x) e^{-S_r(x)}}{\alpha_{sr} Z_r} = \mu_r(x).$$

This is equivalent to $\omega_s^{(4)} \circ \iota_{rs}^{(4)} = \omega_r^{(4)}$; the commuting time-zero square gives the corresponding identity for $d = 0$. $\square$

Remark A.20 (Two different quotients)**.** *The overlap-trivial presentation ideal in Definition A.18 is not the GNS null ideal $\mathcal{N}_\omega(O)$. The former removes data that no support-visible readout can distinguish before a state is selected; the latter removes directions that have zero norm in the extracted state. Faithful finite-stage states can have a nonfaithful weak-* limit, so quotienting only the overlap-trivial ideal would not prove faithfulness of the limiting pair.*

Assumption A.21 (Renormalized four-dimensional Yang–Mills identification receipt)**.** *On the selected four-dimensional cluster family, compatible renormalized holonomy cylinders obey uniform local Cauchy and regularity bounds. Their infinitesimal rectangle tests converge, in the declared distributional sense, to a connection/curvature pair*

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \qquad F = dA + A \wedge A.$$

The renormalized local actions converge to the unique reflection-even positive dimension-four pure-gauge density on this branch, while the declared irrelevant remainders vanish. This is the regularity/universality receipt that identifies the projective generalized-holonomy state with the four-dimensional Yang–Mills cylinder state. The notation $e^{-S_E[A]}DA/G$ below abbreviates that cylinder family; it is not a literal infinite-dimensional Lebesgue measure.

Theorem A.22 (Conditional OPH four-dimensional Euclidean Yang–Mills form)**.** *Under Assumptions A.17 and A.21, on an extracted cylinder family from Proposition A.19 that lies on the declared local four-dimensional scaling branch, the continuum gauge-sector Euclidean action is*

$$S_E[A] = \frac{1}{4g^2}\int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu}\rangle \, d^4x, \qquad F = dA + A \wedge A,$$

with compact simple structure group G. The extracted gauge-quotient cylinder family is denoted

$$d\mu_{\mathrm{YM}}(A) = Z^{-1} e^{-S_E[A]}\, DA/G$$

in the OPH support-visible GNS representation. When Assumption A.31 also holds, its support-visible continuum transfer semigroup is the corresponding Euclidean Yang–Mills semigroup.

Proof. The proof spine is included here to make the branch target explicit. The present Yang–Mills branch takes compact simple G as branch data. When G is imported from OPH reconstruction, that import is conditional on the compact-gauge refinement receipt; on that tail the zero-obstruction transportable bosonic sector category and its faithful forgetful fiber functor reconstruct G. At fixed cutoff, the finite gauge-register / quantum-link presentation gives support-visible link holonomies and plaquette holonomies. In the refinement limit, the zero-obstruction gluing law makes infinitesimal rectangle holonomies multiplicative and path-local. The regularity and Cauchy bounds in Assumption A.21 upgrade this generalized-holonomy limit to a local connection A on the four-dimensional scaling chart, with infinitesimal plaquette defect

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \qquad F = dA + A \wedge A.$$

The Euclideanized MaxEnt/local-Gibbs branch supplies a local finite-range action density built from support-visible gauge-invariant collar data. Gauge quotienting permits only class functions of the curvature and its covariant derivatives. Four-dimensional scaling, Euclidean rotation invariance, locality, and reflection positivity leave one relevant dimension-four positive quadratic invariant in the pure gauge sector:

$$\langle F_{\mu\nu}, F_{\mu\nu}\rangle.$$

The possible topological density $\langle F \wedge F\rangle$ is reflection odd and belongs to a separate topological-angle sector; it is absent on the ordinary reflection-positive zero-obstruction vacuum branch used here. Higher curvature powers and covariant-derivative terms are irrelevant under the declared continuum scaling; their disappearance in the extracted state is the remainder bound in Assumption A.21. Normalizing the unique positive quadratic invariant defines the coupling g. Definition A.18 identifies the finite-stage sources of the gauge cylinders, and Proposition A.19 proves their projective weak-$*$ / GNS extraction. Identification of its transfer generator is the separate dynamic statement in Theorem A.32; it is not a consequence of weak-* compactness alone. □

Prize-facing proof separation. The proof has two separate claims. Theorem A.22 conditionally identifies the four-dimensional Euclidean Yang–Mills form on the certified support-visible compact-gauge branch. The repair-dynamics theorem proves a spectral gap for that Hamiltonian.

Standing compact-gauge setup. Fix a compact simple gauge group G carried by an OPH compact-gauge zero-obstruction vacuum branch on a cofinal tail satisfying Definition 7.2, realized at fixed cutoff by the declared compact-gauge patch-carrier architecture. For each regulator r, let

$$(\mathcal{H}_r, \Omega_r, H_r), \qquad T_r(t) = e^{-tH_r},$$

be the physical Euclidean Hilbert space, vacuum, Hamiltonian, and transfer semigroup. Let $X_r := X_r^{(0)}$ be the support-visible time-zero compact-gauge quotient configuration space, and let π_r be the stationary time-zero measure from Proposition A.19, and set

$$K_r := L^2(X_r, \pi_r).$$

Write $\omega_r(a) := \int_{X_r} a\, d\pi_r$ for its state functional. All statements in K_r are automatically made after the finite π_r-null support reduction. Let $\mathcal{C}_r$ be the finite family of active repair collars. For each $C \in \mathcal{C}_r$, use the complete repaired visible datum $\rho_C := \rho_{C,r} : X_r \to Y_{C,r}$, and let

$$E_C : K_r \to K_r$$

be conditional expectation onto the ρ_C-measurable functions. This subsection stays on the ordinary or central zero-obstruction vacuum branch where the compact-gauge reconstruction ladder is carried by the explicit refinement receipt used in the compact paper.

Proposition A.23 (Local exact repair equals conditional expectation)**.** *For each active collar C, the exact-Markov repair map on the support-visible quotient is the π_r-preserving conditional expectation E_C.*

Proof. On the exact-Markov branch, repair preserves exactly the repaired visible datum ρ_C, changes only complementary invisible fiber data, and acts on the quotient-first physical algebra rather than on representatives. Let Φ_C be the Heisenberg repair map and let $\mathcal{N}_C$ be the repaired local fixed algebra. Exact repair semantics require $\mathcal{N}_C$ to be exactly the ρ_C-measurable subalgebra. Then

$$\Phi_C(a) = a \quad (a \in \mathcal{N}_C), \qquad \Phi_C(\mathcal{A}_r^{G,\mathrm{sv}}) \subseteq \mathcal{N}_C, \qquad \omega_r \circ \Phi_C = \omega_r,$$

and Φ_C is $\mathcal{N}_C$-bimodular. Hence, for $a \in \mathcal{N}_C$ and $x \in \mathcal{A}_r^{G,\mathrm{sv}}$,

$$\omega_r(a^*\Phi_C(x)) = \omega_r(a^*x).$$

Since $\Phi_C(x) \in \mathcal{N}_C$, this characterizes its class in K_r as the orthogonal projection of x onto the ρ_C-measurable subspace, uniquely modulo π_r-null functions. That projection is the π_r-preserving conditional expectation E_C. $\square$

Lemma A.24 (Fiber-homogeneous orbit condition)**.** *Fix an active collar C and a repaired value $y \in Y_{C,r}$. Let $F_C(y) = \rho_{C,r}^{-1}(y)$, with conditional law $\nu_{C,y}$. Assume this is either a finite uniform probability space or a standard atomless probability space, and that the primitive collar relaxation is invariant under every $\nu_{C,y}$-preserving automorphism. Then the complete conditional fiber, including every holonomy coordinate actually resampled by repair, is homogeneous for the local repair receipt.*

Proof. This is the complete-fiber homogeneity receipt. Quotienting removes declared implementation labels, but the proof must also check that no remaining source constraint or relaxation rate distinguishes points in the conditional fiber. Under that check, the conditioned state and primitive relaxation carry the stated full measure-preserving symmetry. □

Lemma A.25 (Scalar relaxation on a homogeneous conditional fiber)**.** *Let (F, ν) be either a finite uniform probability space or a standard atomless probability space. Let E_F be expectation onto constants, and let D_F be a bounded positive self-adjoint Markov relaxation generator with*

$$\ker D_F = \mathrm{Ran}(E_F)$$

that commutes with every measure-preserving automorphism of (F, ν). Then $D_F = c_F(I - E_F)$ for a scalar $c_F > 0$.

Proof. For a finite uniform fiber this is the symmetric-group argument. For a standard atomless fiber, restrict first to step functions on a partition into n equal-measure pieces. Automorphisms within pieces and permutations of pieces force every bounded operator in the commutant to be scalar on the mean-zero step functions. Compatibility under equal-measure refinements gives one scalar, and such step functions are dense in $L^2_0(F, \nu)$. Thus the commutant on L^2_0 is scalar. Positivity and the stated kernel make that scalar strictly positive. □

Assumption A.26 (Finite ground-state-transform and cross-fiber receipt)**.** *For each regulator, the reflection-positive finite transfer matrix supplies a unitary $U_r : \mathcal{H}_r \to K_r$, with $U_r\Omega_r = \mathbf{1}_r$, whose ground-state-transformed generator decomposes as*

$$U_r H_r U_r^{-1} = \sum_{C \in \mathcal{C}_r} D_C.$$

Each D_C is the bounded positive self-adjoint detailed-balance Markov generator acting on the complete complementary conditional fiber of collar C, including every record or holonomy coordinate changed by repair, with fixed algebra exactly the $\rho_{C,r}$-measurable algebra. If the fiberwise scalar supplied by Lemma A.25 over repaired value y is $c_C(y)$, the finite receipt also verifies the cross-fiber equality $c_C(y) = c_C$ on the support of π_r. A Gibbs state alone does not imply this transfer-matrix/decomposition receipt.

Theorem A.27 (Exact Euclidean-consensus law under homogeneous fibers)**.** *Under the fiber-homogeneous orbit condition for every active collar and Assumption A.26, there are positive constants $c_C > 0$ such that the ground-state transformed physical Euclidean generator is exactly*

$$L_r^{\mathrm{EC}} = \sum_{C \in \mathcal{C}_r} c_C(I - E_C),$$

and therefore

$$U_r e^{-tH_r} U_r^{-1} = e^{-tL_r^{\mathrm{EC}}} \qquad (t \geq 0),$$

for a unitary $U_r : \mathcal{H}_r \to K_r$ with $U_r\Omega_r = \mathbf{1}_r$.

Proof. The finite receipt supplies the ground-state transform and a decomposition into primitive local pieces D_C. Each such piece is supported on collar C, preserves the repaired visible datum $\rho_{C,r}$, and relaxes the complete complementary conditional fiber. Thus

$$\ker D_C = \mathrm{Ran}(E_C).$$

Lemma A.24 gives full hidden-fiber permutation symmetry. Applying Lemma A.25 fiberwise gives $D_{C,y} = c_C(y)(I - E_{C,y})$. The cross-fiber part of the finite receipt makes $c_C(y) = c_C$, so

$$D_C = c_C(I - E_C)$$

for a collar-type constant $c_C > 0$. Summing over the active collars gives the displayed generator. Positivity and self-adjointness give the transfer-semigroup identity. □

Lemma A.28 (Uniform active-collar rate floor)**.** *Assume finite local combinatorial type, a branch-homogeneous local constraint family, and refinement-stable exact repair semantics. Then the local Euclidean repair rate depends only on active collar type, the active collar-type set is finite across the cofinal refinement family, and there is $c_* > 0$, independent of r, such that*

$$c_C \geq c_* \qquad \textit{for every active collar } C \in \mathcal{C}_r.$$

Proof. Active collars are exactly the collars on which complementary invisible fiber data are genuinely relaxed, so $D_C \neq 0$ on $\ker(E_C)$ and $c_C > 0$. Finite local combinatorial type gives a finite list of bounded collar patterns, including visible interfaces, hidden fiber cardinalities, constraint templates, and repair maps. Branch homogeneity makes the conditioned MaxEnt relaxation scalar a function of this finite type data. Refinement stability replaces collars by copies of the same bounded type templates with the same normalized local Euclidean repair semantics. Taking the minimum over the finite active type list gives $c_* > 0$. □

Proposition A.29 (Special commuting-color finite-stage repair gap)**.** *Assume the active collars admit a bounded-color decomposition*

$$\mathcal{C}_r = \bigsqcup_{a=1}^{q} \mathcal{C}_{r,a}, \qquad q < \infty,$$

independent of r, and that exact quotient-local gluing makes the corresponding color expectations commute. Let $P_{0,r}$ project onto constants in K_r. Then

$$L_r^{\mathrm{EC}} \geq c_*(I - P_{0,r}), \qquad \Delta_{\mathrm{rep},r} \geq c_* > 0.$$

Proof. For each color a, define the parallel color expectation

$$E_{r,a} := \prod_{C \in \mathcal{C}_{r,a}} E_C.$$

Same-color collars are disjoint, so the factors commute, and exact quotient-local gluing makes the family $\{E_{r,a}\}_{a=1}^{q}$ a commuting family of orthogonal projections on the vacuum branch. Repair completeness says that the only support-visible observables fixed by every local repair are constants:

$$\bigcap_{a=1}^{q} \mathrm{Ran}(E_{r,a}) = \mathbb{C}\,\mathbf{1}_r.$$

Set

$$\widetilde{L}_r := \sum_{a=1}^{q} c_*(I - E_{r,a}).$$

For commuting projections, $I - \prod E_C \leq \sum_C (I - E_C)$, so $\widetilde{L}_r \leq L_r^{\mathrm{EC}}$. Since the $E_{r,a}$ commute, they are simultaneously diagonalizable. On every nonconstant joint eigenspace at least one color eigenvalue is zero. Thus $\widetilde{L}_r \geq c_*(I - P_{0,r})$, and the same lower bound holds for L_r^{EC}. □

Remark A.30 (Generic repair-gap certificate)**.** *The commuting-color proposition is a special certificate, not the generic repair-generator route. For overlapping collars one must instead compute*

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a lower bound $\gamma_* > 0$, then

$$L_r^{\mathrm{EC}} \geq \gamma_* \kappa_r (I - P_{0,r}).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator; a uniform refinement lower bound is a separate receipt.

Assumption A.31 (Finite transfer and vacuum compatibility receipt)**.** *On the cofinal family selected in Proposition A.19, suppose the following finite-stage identities are certified.*

(T1) *The quotient ensembles obey the finite fiber-sum identity, or its continuous push-forward/disintegration analogue, so $(p_{sr}^{(4)})_\# \mu_s^{(4)} = \mu_r^{(4)}$ and, by time-zero restriction, $(p_{sr}^{(0)})_\# \pi_s = \pi_r$. Hence pullback gives coherent isometries*

$$J_{rs}^K : K_r \longrightarrow K_s, \qquad J_{rs}^K f = f \circ p_{sr}^{(0)}.$$

(T2) *There are coherent physical refinement isometries $J_{rs}^{\mathcal{H}} : \mathcal{H}_r \to \mathcal{H}_s$ satisfying*

$$J_{rs}^K \mathbf{1}_r = \mathbf{1}_s, \qquad J_{rs}^{\mathcal{H}} \Omega_r = \Omega_s, \qquad U_s J_{rs}^{\mathcal{H}} = J_{rs}^K U_r.$$

(T3) *With one refinement-independent physical Euclidean-time normalization, the finite semigroups are coherent:*

$$J_{rs}^K e^{-tL_r^{\mathrm{EC}}} = e^{-tL_s^{\mathrm{EC}}} J_{rs}^K, \qquad J_{rs}^{\mathcal{H}} e^{-tH_r} = e^{-tH_s} J_{rs}^{\mathcal{H}} \quad (t \geq 0).$$

(T4) *Let $\iota_{0,r}$ pull a time-zero observable into the four-dimensional cylinder algebra and let $\tau_t^{(r)}$ be finite Euclidean time translation. The finite time-zero Markov/transfer identity holds on the cylinder core:*

$$\omega_r^{(4)}\Big((\iota_{0,r} f)^* \tau_t^{(r)}(\iota_{0,r} g)\Big) = \langle f, e^{-tL_r^{\mathrm{EC}}} g\rangle_{K_r} = \langle U_r^{-1} f, e^{-tH_r} U_r^{-1} g\rangle_{\mathcal{H}_r}.$$

The reflection-positive finite OS time-zero quotient is therefore the same finite $(\mathcal{H}_r, \Omega_r, H_r)$, through U_r, rather than an unrelated Hilbert space.

These are finite commuting-diagram receipts. They are not consequences of the sector-category refinement functors or of weak- state compactness.*

Theorem A.32 (Continuum exact transfer identification)**.** *Under Assumption A.31, the finite repair and physical Hilbert spaces have Hilbert direct limits K and $\mathcal{H}$. Their semigroups induce strongly continuous self-adjoint contraction semigroups with nonnegative generators L^{rep} and H, and the finite unitaries induce a unitary $U : \mathcal{H} \to K$ such that*

$$U e^{-tH} U^{-1} = e^{-tL^{\mathrm{rep}}} \qquad (t \geq 0),$$

and hence

$$UHU^{-1} = L^{\mathrm{rep}}.$$

The constant vectors and physical vacua define limit vectors $\mathbf{1} \in K$ *and* $\Omega \in \mathcal{H}$*, with*

$$U\Omega = \mathbf{1}, \qquad P_0^K = |\mathbf{1}\rangle\langle\mathbf{1}|, \qquad P_0^{\mathcal{H}} = |\Omega\rangle\langle\Omega|, \qquad UP_0^{\mathcal{H}}U^{-1} = P_0^K.$$

If $L_r^{\mathrm{EC}} \geq c_*(I - P_{0,r})$ *for one* $c_* > 0$*, then*

$$L^{\mathrm{rep}} \geq c_*(I - P_0^K), \qquad H \geq c_*(I - P_0^{\mathcal{H}}).$$

Proof. The fiber-sum identity makes J_{rs}^K isometric, since

$$\|f \circ p_{sr}^{(0)}\|_{K_s}^2 = \int_{X_r} |f|^2 \, d(p_{sr}^{(0)})_{\#}\pi_s = \|f\|_{K_r}^2.$$

It also makes $\overline{\omega}_r^{(0)} = \omega_r^{(0)}$, so these J_{rs}^K are exactly the time-zero GNS isometries $V_{rs}^{(0)}$ in Proposition A.19. Thus their Hilbert direct limit is canonically the support-visible GNS space K constructed there, not a second continuum repair space. Take the Hilbert direct limits under J_{rs}^K and $J_{rs}^{\mathcal{H}}$. On the dense finite-stage images define

$$S^K(t)J_r^K f := J_r^K e^{-tL_r^{\mathrm{EC}}} f, \qquad S^{\mathcal{H}}(t)J_r^{\mathcal{H}}\psi := J_r^{\mathcal{H}} e^{-tH_r}\psi.$$

The finite coherence identities make these definitions independent of the representative. They are contraction semigroups. Strong continuity holds on every finite-stage image and therefore, by density and uniform contractivity, on the whole direct limits. For each fixed t, the finite-stage inner-product identity makes the bounded extension symmetric on a dense union and hence symmetric everywhere; a bounded everywhere-defined symmetric operator is self-adjoint. Positivity also passes from the finite semigroups. Thus the spectral theorem gives nonnegative self-adjoint generators L^{rep} and H.

Define $UJ_r^{\mathcal{H}}\psi := J_r^K U_r \psi$. The finite unitary square makes this well defined and isometric; the same construction with U_r^{-1} shows that its range is dense and closed, so it is unitary. The finite transfer identity of Theorem A.27 gives $US^{\mathcal{H}}(t)U^{-1} = S^K(t)$ on a dense union and hence everywhere. Uniqueness of self-adjoint semigroup generators gives $UHU^{-1} = L^{\mathrm{rep}}$.

Unitality and the physical refinement identity make the constant and vacuum vectors coherent. For $f \in K_r$,

$$P_{0,s}J_{rs}^K f = \langle \mathbf{1}_s, J_{rs}^K f\rangle \mathbf{1}_s = J_{rs}^K P_{0,r} f,$$

so the rank-one vacuum projections induce the displayed limit projections; the same holds on the physical side and the unitary identifies them.

The finite operator inequality implies

$$\|e^{-tL_r^{\mathrm{EC}}}(I - P_{0,r})\| \leq e^{-c_* t}.$$

Passing this norm bound through the direct-limit definition gives $\|e^{-tL^{\mathrm{rep}}}(I - P_0^K)\| \leq e^{-c_* t}$. The spectral theorem is equivalent to $L^{\mathrm{rep}} \geq c_*(I - P_0^K)$, and conjugation by U^{-1} gives the physical bound. ∎

Remark A.33 (Asymptotic alternative)**.** *If the exact semigroup identities in Assumption A.31 are not available, the valid replacement is generalized Mosco convergence of both finite transfer forms, strong cylinder-core convergence of* U_r *and* U_r^{-1}*, and strong convergence of the vacuum projections. The Mosco theorem then supplies the two limit semigroups and the same intertwining conclusion. Weak-* convergence of cylinder states by itself supplies none of those dynamic statements.*

Assumption A.34 (OS regularity and noncollapse receipt)**.** *The selected finite cylinder family carries reflection-compatible refinement maps and exact finite reflection positivity. Its renormalized local Schwinger functions have uniform bounds sufficient to pass Euclidean covariance, locality, OS regularity, and clustering to the extracted family. Moreover, there are gauge-invariant local cylinders A_r, transported from one fixed cylinder test, such that*

$$\inf_r \operatorname{Var}_{\pi_r}(A_r) > 0, \qquad \sup_r \langle A_r \mathbf{1}_r, L_r^{\mathrm{EC}} A_r \mathbf{1}_r \rangle < \infty.$$

The centered vectors $A_r\mathbf{1}_r - \omega_r(A_r)\mathbf{1}_r$ converge under the same cylinder/Hilbert identifications used in the extraction and transfer receipts. This is the separate OS/nontriviality receipt. It is not part of the weak- extraction theorem.*

Theorem A.35 (Osterwalder–Schrader reconstruction on the compact-gauge branch)**.** *Under Assumptions A.17, A.21, A.31, and A.34, with the extraction of Proposition A.19, the continuum support-visible compact-gauge cylinder family is Euclidean invariant, reflection positive, regular on gauge-invariant local cylinder observables, and cyclic for the vacuum sector. Thus Osterwalder–Schrader reconstruction gives a four-dimensional quantum Yang–Mills theory $(\mathcal{H}, \Omega, H, \mathcal{A}^G_{\mathrm{loc}})$ on the support-visible gauge-invariant local algebra, with $H \geq 0$ and e^{-tH} equal to the Euclidean transfer semigroup of Theorem A.22.*

Proof. For every finite list of positive-time cylinders, finite reflection positivity says that the associated reflected Gram matrix is positive semidefinite. Its entries converge along the extraction subnet, and the cone of positive semidefinite matrices is closed, so reflection positivity passes to the limit. Euclidean covariance, locality, regularity, and clustering pass by the compatible transformation maps and uniform bounds in Assumption A.34; weak-* compactness alone would not supply those bounds. The vacuum vector is cyclic for the GNS closure of the gauge-invariant local cylinder algebra. The finite time-zero identity (T4) passes to the limit on the dense cylinder core: the left side converges by four-dimensional cylinder extraction and the right side by the coherent semigroup limit. Consequently the OS time-zero inner product and translation semigroup are unitarily identical to the direct-limit $(\mathcal{H}, \Omega, e^{-tH})$ of Theorem A.32; the notation does not identify two unrelated Hamiltonians. The Osterwalder–Schrader reconstruction theorem then produces the Hilbert space, vacuum, local algebra, positive Hamiltonian, and Euclidean transfer semigroup [50, 51]. ☐

Proposition A.36 (Nontriviality of the support-visible compact-gauge theory)**.** *Under Assumptions A.31 and A.34, The support-visible compact-gauge local algebra on the zero-obstruction vacuum branch strictly contains the vacuum scalars and admits a non-vacuum finite-energy local excitation.*

Proof. The compact-gauge witness and physical-UV landing theorem supplies finite support-visible gauge-invariant local observables, for example nonconstant Wilson/plaquette cylinders. The uniform positive variance in Assumption A.34 says that the chosen centered cylinder does not enter the GNS null ideal in the limit, so it gives a nonzero vector orthogonal to the vacuum. The uniform form bound and lower-semicontinuity of the certified transfer-form limit give this vector finite continuum energy. Thus the support quotient has not erased the excitation used to prove nontriviality. ☐

Theorem A.37 (Conditional positive compact-gauge Yang–Mills mass gap from OPH repair dynamics)**.** *Let G be a compact simple gauge group carried by a support-visible compact-gauge OPH vacuum branch satisfying the standing setup, Assumptions A.31 and A.34, the renormalized Yang–Mills identification receipt of Assumption A.21, the finite ground-state-transform and cross-fiber receipt of Assumption A.26, and all hypotheses of Theorem A.27, Lemma A.28, and Proposition A.29.*

The theory reconstructed in Theorem A.35 is nontrivial by Proposition A.36, and its continuum support-visible Hamiltonian H satisfies

$$H \geq c_*(I - P_0^{\mathcal{H}}),$$

where $P_0^{\mathcal{H}}$ projects onto the physical vacuum. Therefore

$$\mathrm{Spec}(H) \cap (0, c_*) = \varnothing, \qquad \Delta_{\mathrm{YM}} \geq c_* > 0.$$

Moreover the Yang–Mills gap is the repair gap:

$$\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}.$$

Proof. Proposition A.29 gives

$$L_r^{\mathrm{EC}} \geq c_*(I - P_{0,r})$$

at every finite stage. Theorem A.32, using the explicit finite transfer/vacuum receipt, transports the bound to the support-visible continuum:

$$L^{\mathrm{rep}} \geq c_*(I - P_0^K).$$

Using $UHU^{-1} = L^{\mathrm{rep}}$, conjugation by U^{-1} gives the Hamiltonian lower bound. The spectral-gap statement follows immediately. Since U is unitary and maps the vacuum to the constant sector, it preserves the nonzero spectrum, so the Yang–Mills gap and repair gap are equal. □

Exact gap accounting. The theorem gives an identity stronger than a phenomenological estimate:

$$\mathrm{Spec}(H) \setminus \{0\} = \mathrm{Spec}(L^{\mathrm{rep}}) \setminus \{0\}.$$

Thus

$$\Delta_{\mathrm{YM}} = \inf\big(\mathrm{Spec}(H) \setminus \{0\}\big) = \inf\big(\mathrm{Spec}(L^{\mathrm{rep}}) \setminus \{0\}\big) = \Delta_{\mathrm{rep}}.$$

The finite-stage projection argument proves positivity of that same quantity.

Relation to the Clay/Jaffe–Witten statement. The Clay problem asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$, for each compact simple G, satisfying axiomatic properties at least as strong as the stated Wightman or Osterwalder–Schrader references and possessing a positive mass gap [49, 48]. Theorem A.22 supplies the four-dimensional Euclidean Yang–Mills form on the OPH support-visible compact-gauge branch, and Theorem A.35 supplies the support-visible OS reconstruction on the gauge-invariant local algebra. Proposition A.36 supplies nontriviality. Theorem A.37 supplies the exact spectral gap accounting on that same branch. Proposition A.19 proves the algebra/state/GNS extraction only. Full Clay admissibility additionally requires the finite transfer/vacuum and OS/noncollapse receipts in Assumptions A.31 and A.34, together with the renormalized Yang–Mills identification, finite ground-state-transform/cross-fiber, and uniform-gap receipts; weak-* compactness is not a substitute for those inputs. The standalone Clay note is only a focused presentation of this same theorem surface; it does not enlarge the branch beyond the data stated here.

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